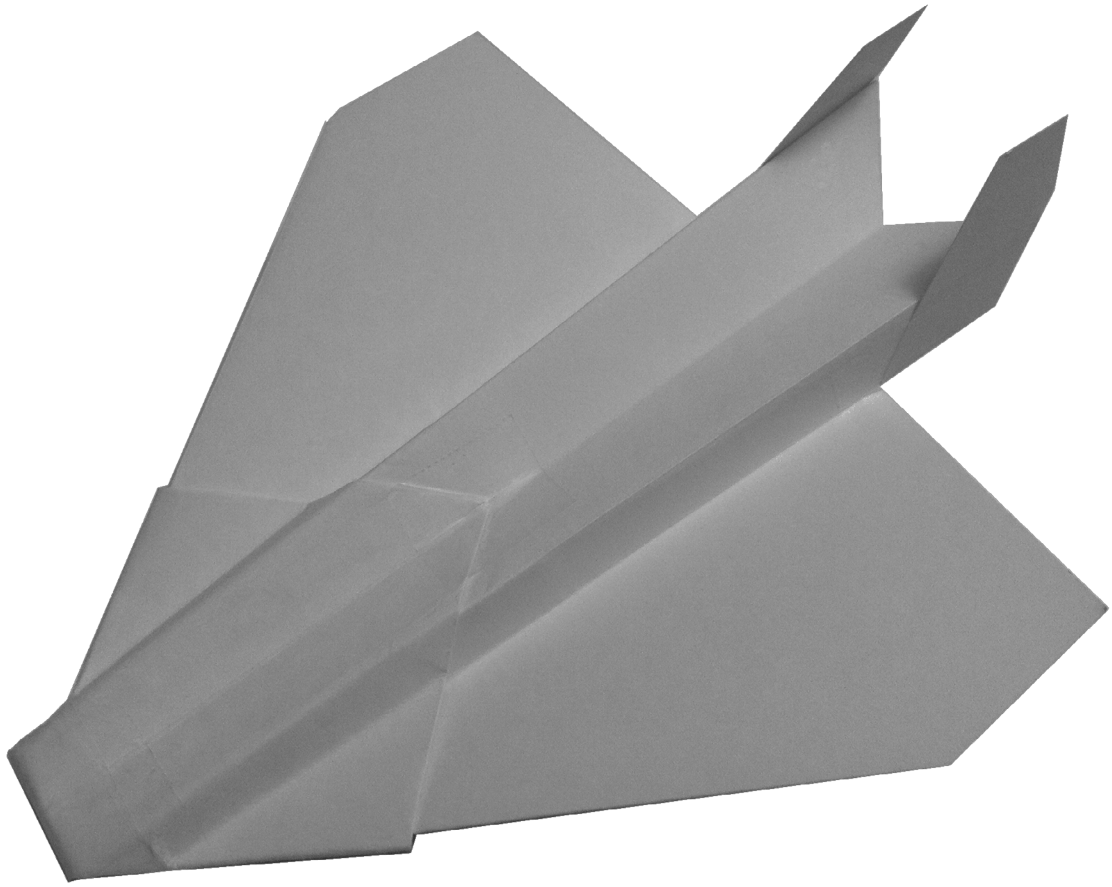
AFS PAPER AIRPLANES

STEP BY STEP INSTRUCTIONS & PRINTABLE TEMPLATES

Andrew F. Scharhag

AFS PAPER AIRPLANES

Incredible Designs You Can Make at Home



Andrew F. Scharhag

DEDICATION

To Robert H. Buhrmaster. Thank you for my exposure to the wonderful world of flight. Your steadfast character and great achievements will always be an inspiration to me. Thank you "Paw-Paw."

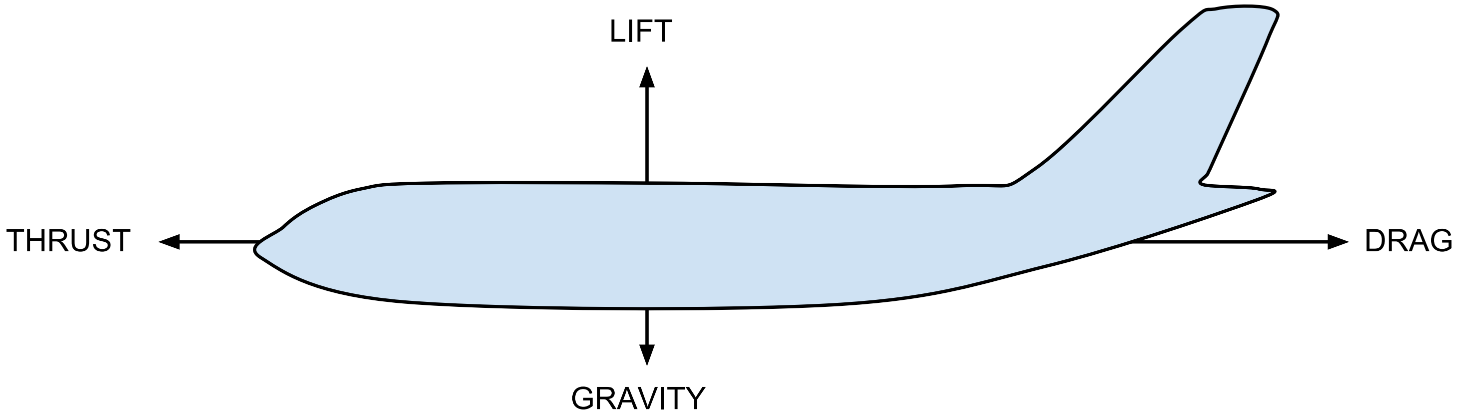
ACKNOWLEDGMENTS

If you are passionate about fantastic flights these great resources will fit your needs and inspire. This book would not have happened without these amazing resources and people.

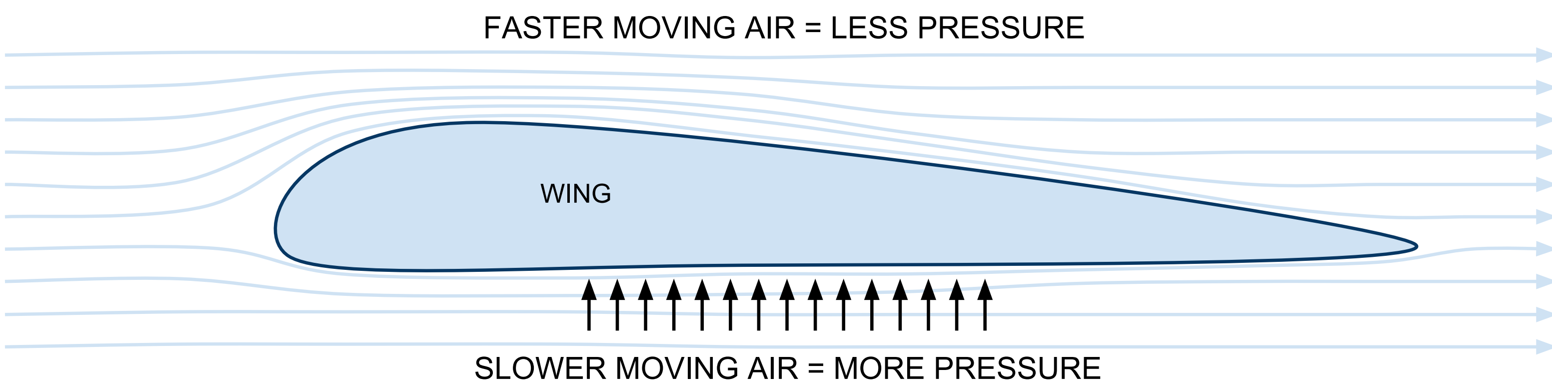
- The Ultimate Paper Airplane- Richard Kline and Floyd Fogleman talk about the story of their airfoil and provide some great plane designs. They were featured on 60 Minutes in 1973 and were granted 2 US patents.
- WhiteWings.com- These balsa and paper gliders can do amazing things that no paper gliders can.
- Origami Omnibus- Making paper airplanes requires folding paper. I know of no greater resource for the art of folding paper than the Origami Omnibus by Kunihiko Kasahara.
- Thanks to all those who provided amazing user feedback during the book's creation. It wouldn't have happened without you!

FLIGHT

First things first. In order to make great paper airplanes let's first take a quick look at how flight works. There are four basic terms that are used to communicate the forces acting on a plane in flight. They are thrust, lift, drag and gravity. Thrust is the forward push (your throw). Lift is the upward push generated by the plane's design as it moves through the air. Drag is the force that the air exerts on the plane slowing it down. And gravity is pulling the plane back to earth.

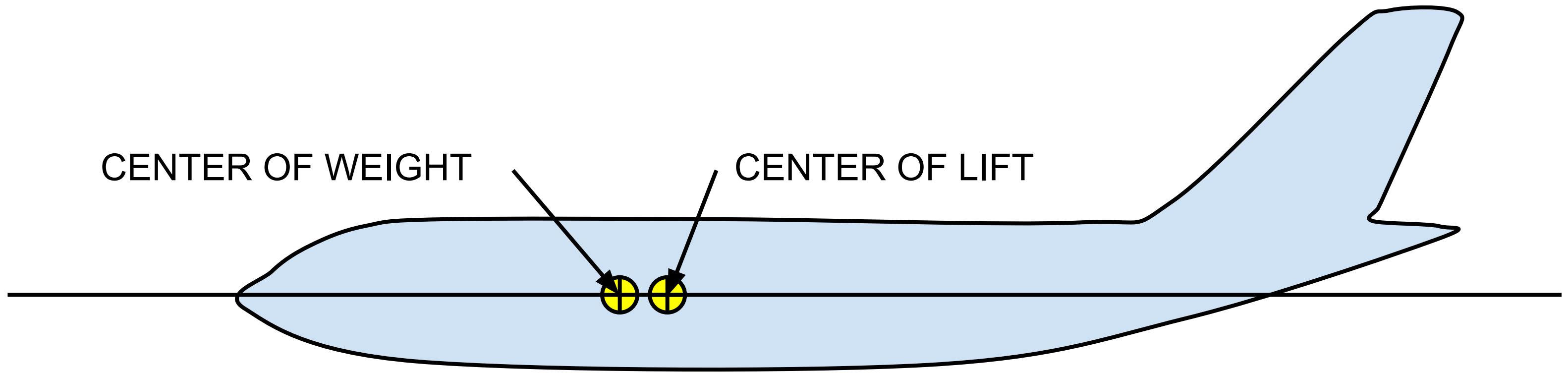


So how can an object moving through the air generate a force strong enough to help keep it from falling down right away? How can it produce enough lift to equal or overcome the force of gravity pulling it down? The key to this lies in air pressure and what is called Bernoulli's Principle. The principle describes how when a liquid speeds up its motion, the result is a simultaneous decrease in pressure or a decrease in the fluid's potential energy. As far as we are concerned here, the theoretical result is that when air travels faster over a surface the pressure it exerts on that surface decreases. If you can make the air travel faster across the upper side of the wing than the lower side, the air pressure will be lower on the top than the bottom producing lift.

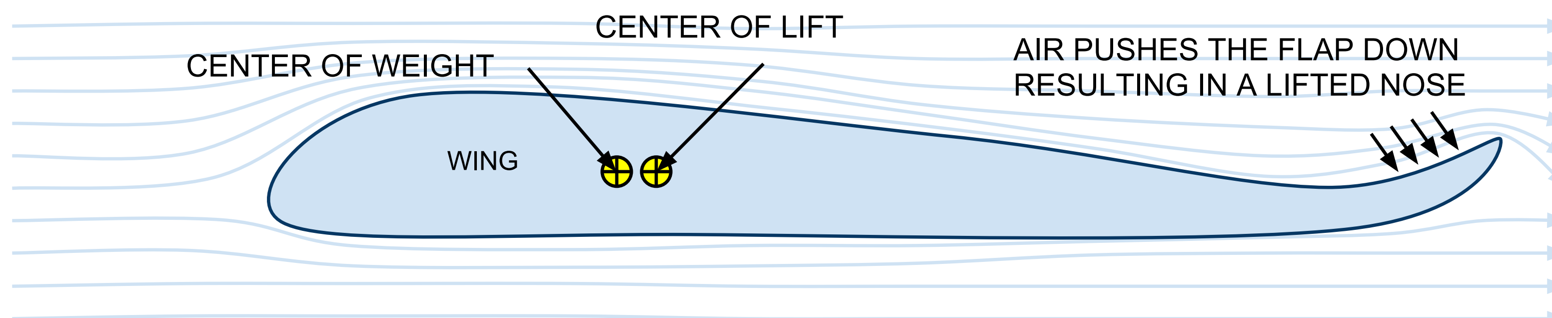


The key then, for great flights of your paper airplane, is to be able to control how these forces act upon the plane during flight. The more you can control these forces the more you can control the plane's flight.

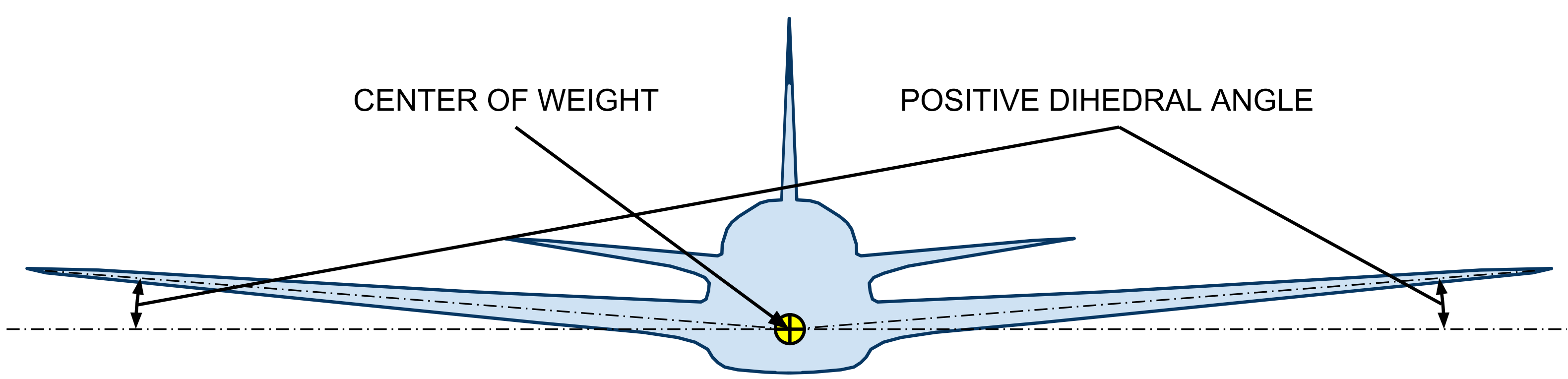
Next, let's quickly touch on the concepts of center of weight and center of lift. For great paper airplane flights we want the center of weight of our plane to be slightly ahead of the center of lift. This helps keep the plane flying forward during its flight. To help illustrate this, take a look at the diagram below. Imagine a scenario where the plane were simply stationary in the air above your head and started falling. If the center of weight is forward of the center of lift (where the wings produce lift) then the nose will fall first and the plane can recover from the dive. Alternately, if the center of weight were not far enough forward the plane may start to fall backwards and would not be able to fly. Additionally, you want to ensure the center of weight is not too far forward or the plane will simply nose dive to the ground.



Another important aspect is control of your plane. Assuming you can make your plane fly straight by properly balancing the prior forces, you may want to make it do other things. By incorporating control surfaces into a paper plane that you can adjust before flight you will be able to control the flight. This is accomplished through the addition of flaps. The paper airplanes in this book use flaps along the trailing edge of the plane. As the plane moves forward through the stationary air the air pushes against the flap pushing it back rotating the plane around its center of weight. You could think of it like a seesaw rotating around the center of weight. Flaps can be used to make the nose of the plane go up or down, or they can be used to make the plane turn right or left if they are positioned on the vertical tail.



One final concept is dihedral wing position. This refers to the angle that the plane wing makes relative to the fuselage. A positive dihedral angle is angled upward from the fuselage relative to horizontal while a negative dihedral angle is angled below horizontal. Negative dihedral angle is also called anhedral angel. Positive dihedral angle can be beneficial for a plane's stability. It helps them stay stable in flight by balancing the force of lift generated from the wings such that the plane self-corrects if it gets off level to one side or the other.



By understanding these basic principles you will be better able to control the planes you make. I believe the most entertaining paper airplane flights are ones where the plane shoots away from the ground ascending higher than a two story building and then levels off into a gentle steady glide where it slowly descends as a real plane might until it comes to a gentle rest on the ground far from where it was thrown. The best flights are the ones that seem to take forever to come back down almost as if they are being piloted. The planes in this book are designed to perform these types of flights. To that end, they are designed to be sturdy enough to make it up to the required altitude but also light enough to meander their way back to earth slowly in an entertaining manner.

DESIGNS

So how can a piece of paper be designed to take off, reach a high altitude, level off and then begin a slow descent back to earth without being able to adjust anything about the plane during the flight? This dilemma is at the core of designing paper airplanes that perform well. Upon takeoff the plane must be able to fly straight in the direction thrown and produce little to no lift, flying like a dart or ball. Then, when it reaches a maximum altitude, it must be able to transition into a plane that produces enough lift to allow it to descend slowly back down. Low lift at high speed and high lift at low speed. This is fundamentally pposed to what one might expect paper to be able to do. This fundamental contradiction makes the pursuit of great paper airplane design challenging and rewarding. There's always a way to improve your design. I believe that these types of flights are the most exhilarating and entertaining to watch. For example, as of June 2013, the world record for time aloft by a paper airplane is held by Takuo Toda at 27.9 seconds but he is working to beat his own record. Stop for a second and start counting... Now imagine a plane traveling thought the air for that long. It's quite a long time!

One fun thing about designing with paper is the endless possibilities that can be quickly designed, constructed and tested. While paper is great because it is so malleable, it can also be frustrating for the same reason. The designs in this book are fashioned with the goal of creating consistently interesting flights. With that goal in mind the first and most fundamental aspect of good construction is proper paper. Paper can be bought in various weights. The weights have varying strength. Basically, the heavier the paper the stronger it is. For example, you could think of tissue paper which is very light and very weak as compared to stock paper, such as a note or index card. For paper airplane construction the best weight paper to use is one that is strong enough to maintain its shape during flight and yet still effectively foldable. I have found that the best compromise is at 32lb/ream or, in metric, 120 grams/square meter. At this weight the paper is foldable but still stiff enough for great flights. While 32lb/ream paper is a great for flights, there is another factor to consider. Imagine if you made a folded paper plane 5 feet across. As you can imagine, the 32lb/ream paper is not strong enough to support its own weight and droops all over the place making it unable to fly. On the other end of the spectrum if you were to make a paper plane 1 inch across it would probably be very difficult to construct. I have found that the best compromise in this situation is to select half of an 8.5in x 11in piece of paper. The result is a 4.25in x 5.5in piece that will be stiff enough to hold its shape during your throw, light enough to glide well once it reaches the top of its trajectory and not too small to fold.

Now we have the right material. The next step is design. There are a few unique characteristics that I have included in these designs:

- The wings merge into the fuselage. In order to get great glides, these planes have an open fuselage design. The "M" and "W" fuselage shapes increase wing area and increase the amount of lift the plane creates during flight. Additionally, the shapes are designed to increase the amount of rigidity the plane has during flight. One of the strongest factors working against great paper airplane flight is poor paper rigidity. When you throw the plane it needs to maintain its shape. The "M" and "W" surfaces that run the length of the plane also serve to help the plane maintain straight flight. Air gets corralled along the length of the plane.
- Having a tail of some type can be vital. It allows for control surfaces so you can make your plane go right or left. Although vertical tails can be great for controlling a plane's turning they also reduce aerodynamic efficiency. When was the last time you saw a bird with a vertical tail? They don't exist.
- These designs include tape to bolster the rigidity of the plane. The tape makes the plane's flights more entertaining and makes the planes last longer.

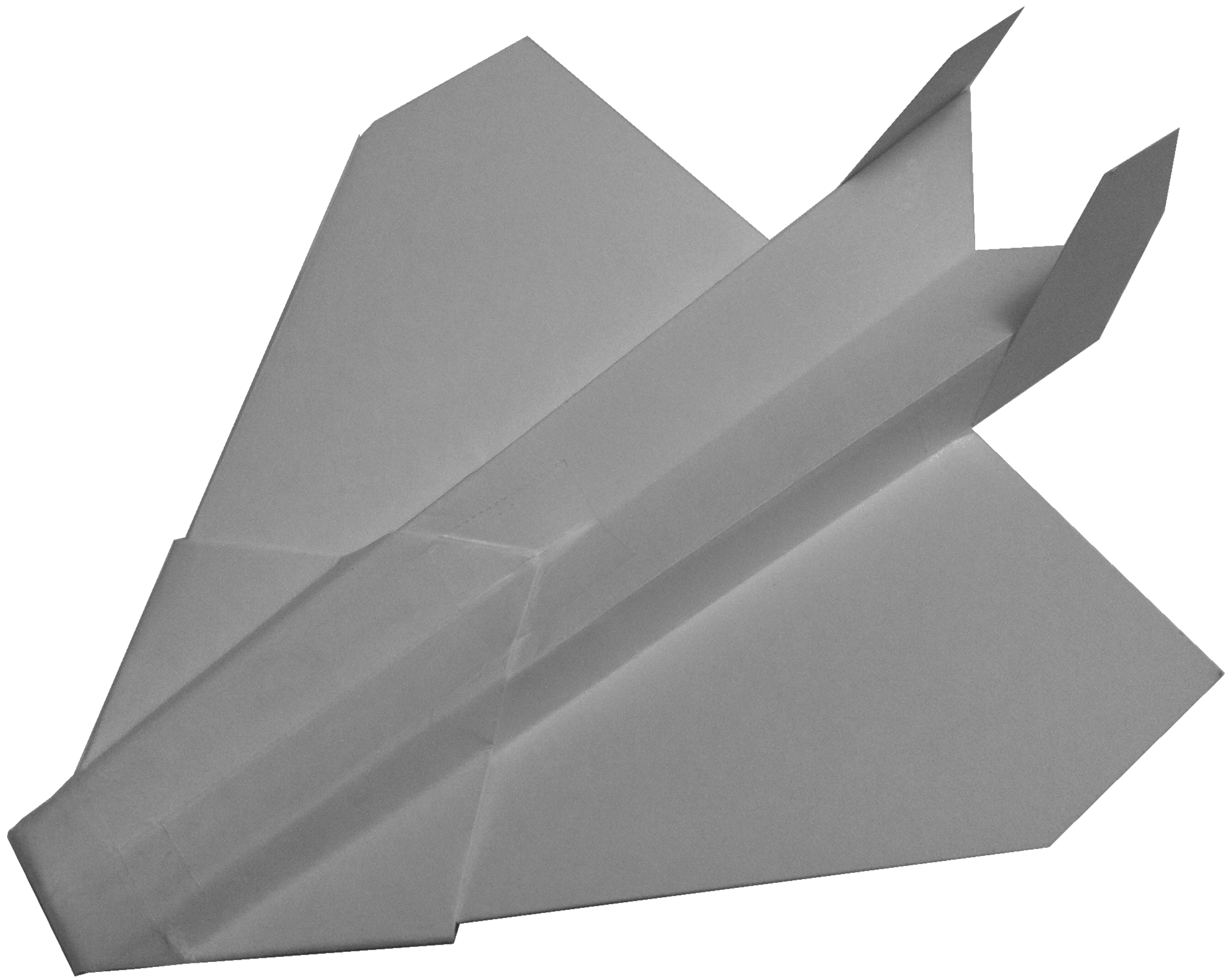
P11 EAGLE

I recommend making this design first. While construction takes a bit longer than others, it is the most forgiving when it comes to construction errors. The "M" shaped fuselage helps to funnel air along the fuselage and helps the plane's stability in flight. The dual tails are stiffened by the tape that is on either side of them and this helps the plane maintain consistent flights. The flaps on the rear of the wings allow you to control turning and elevation.

Internal link to build instructions: "Build it!"

External (internet) link for more info behind the name:

http://en.wikipedia.org/wiki/McDonnell_Douglas_F-15_Eagle



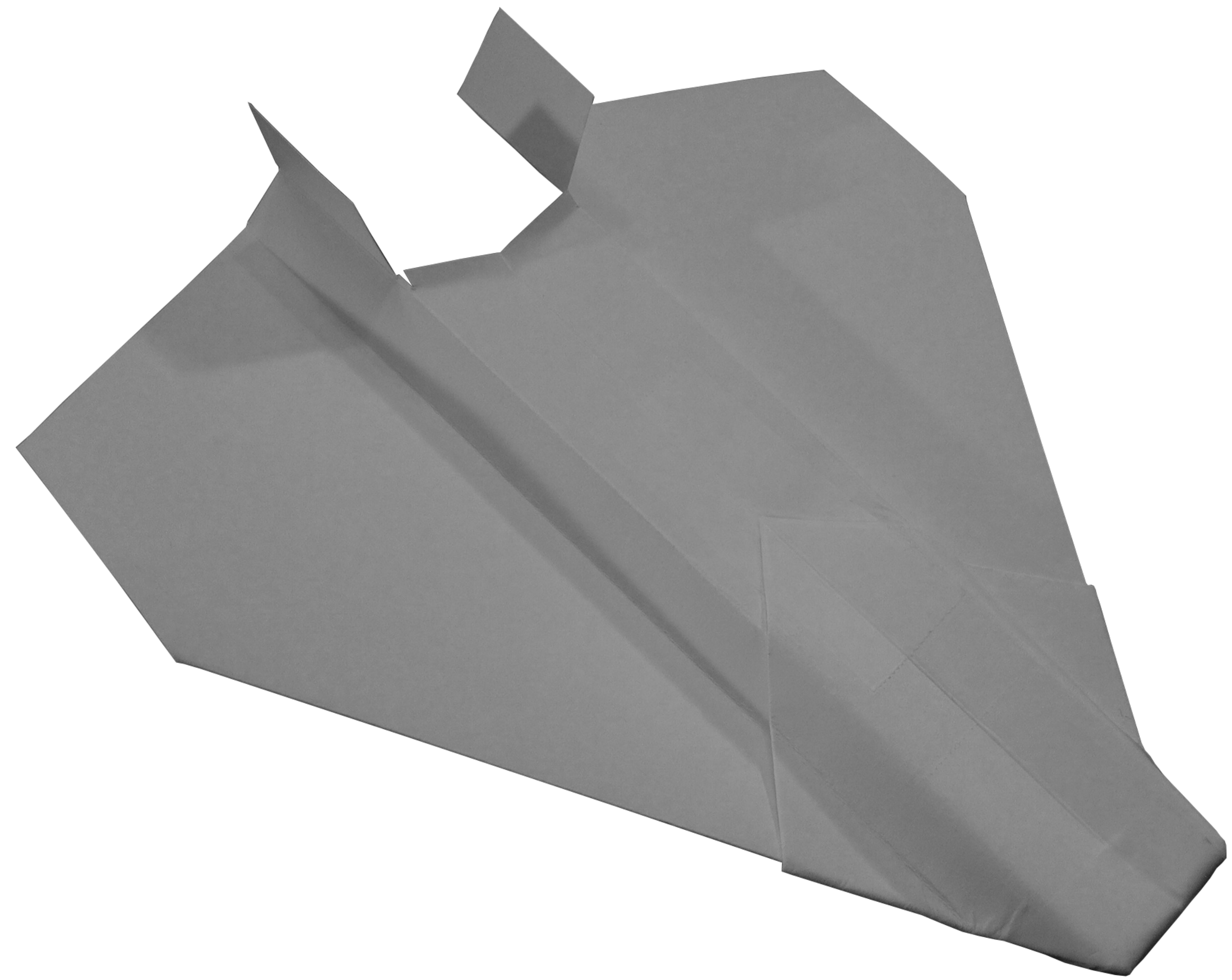
P12 RAPTOR

With increased wing area this plane has increased lift capabilities. The fuselage utilizes the "M" shape. The flaps are positioned slightly closer to the center of gravity as an effort to reduce their level of effect on flight. The result is that flap adjustments result in less dramatic changes in flight.

Internal link to build instructions: "Build it!"

External (internet) link for more info behind the name:

http://en.wikipedia.org/wiki/Lockheed_Martin_F-22_Raptor



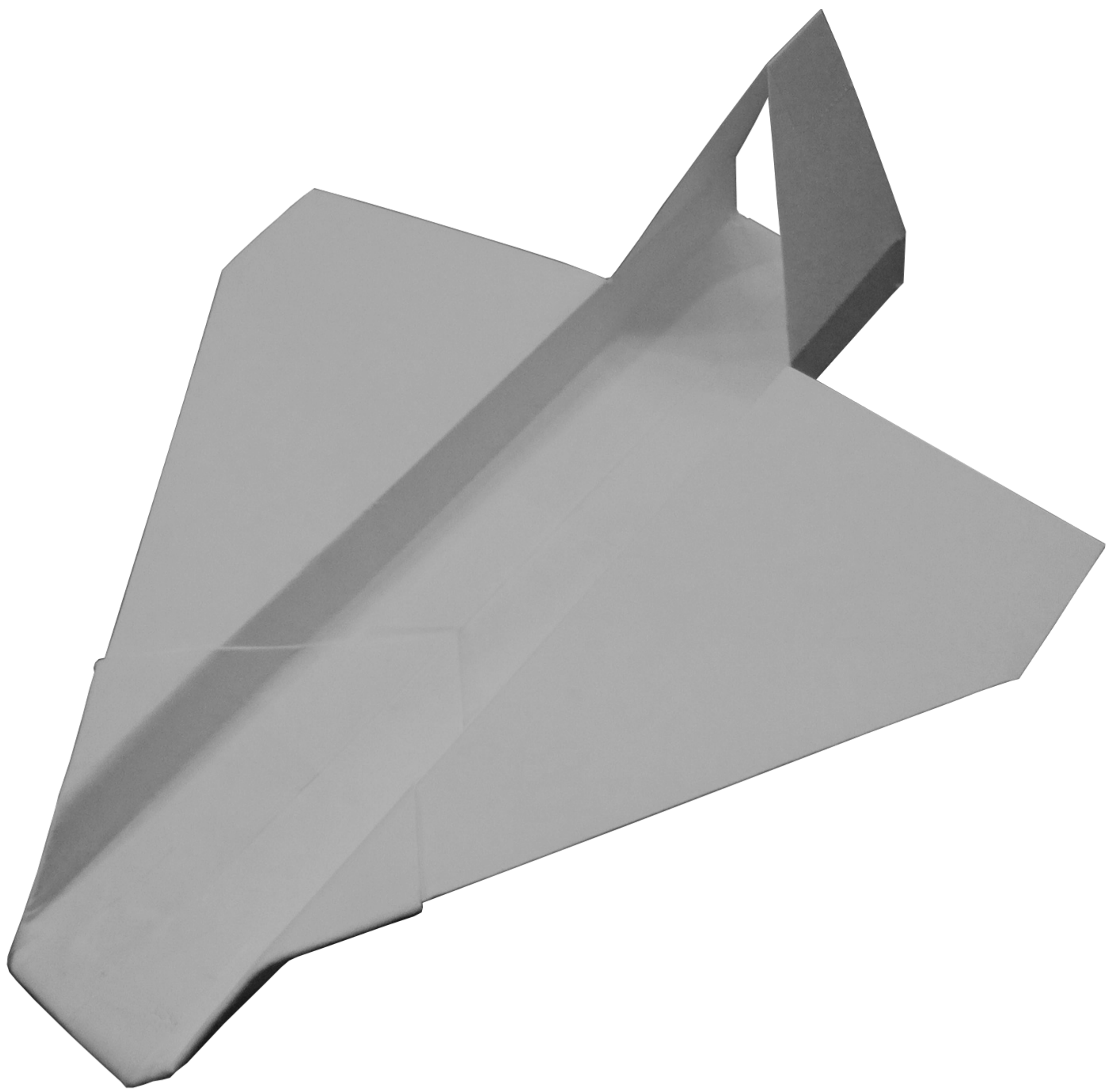
P13 SPARTAN

This design is similar to P12 however the fuselage is inverted vertically and it has a lengthier fuselage. If you look at birds head-on you will notice that their wings attach to their bodies along their spinal cord or back. When they fly the weight of their body “hangs” below their wings as opposed to balancing above. Also if you look at most cargo planes you will see the same thing. A good example of this is the Antonov An-225 Mriya which is the world’s largest plane. This configuration can allow for better stability and softer turning with heavy payloads. The tail helps hold the wings at the proper angle relative to the fuselage during flight.

Internal link to build instructions: “Build it!”

External (internet) link for more info behind the name:

http://en.wikipedia.org/wiki/Alenia_C-27J_Spartan



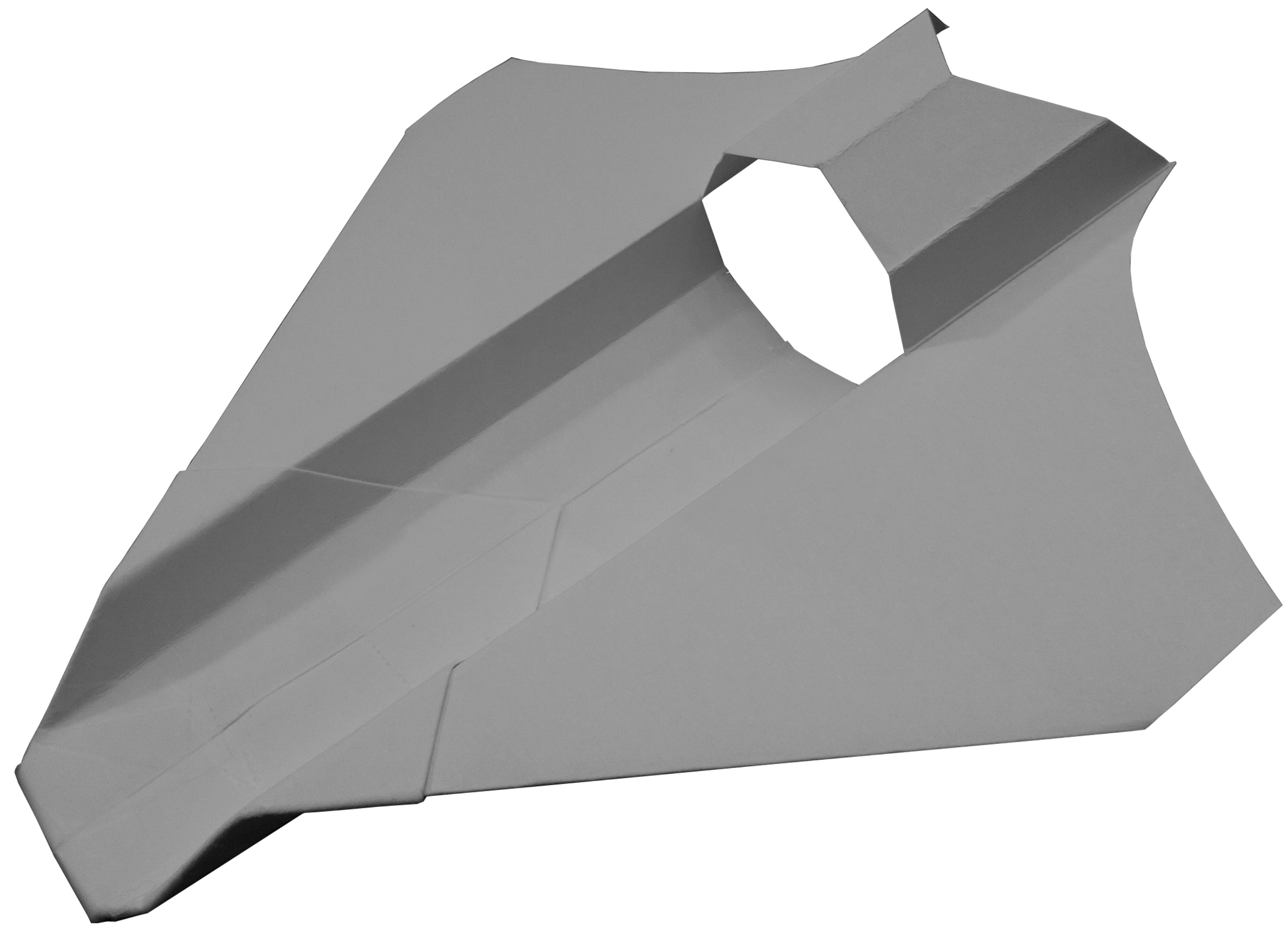
P14 SKYMASTER

This design utilizes an inverted fuselage and joined tail. By creating a hole in the fuselage the structural integrity can be a bit compromised relative to the other planes, however the large tail can allow for good control. Weight is reduced on the rear of the plane to help compensate for this plane’s lower level of rigidity.

Internal link to build instructions: “Build it!”

External (internet) link for more info behind the name:

http://en.wikipedia.org/wiki/O-2_Skymaster



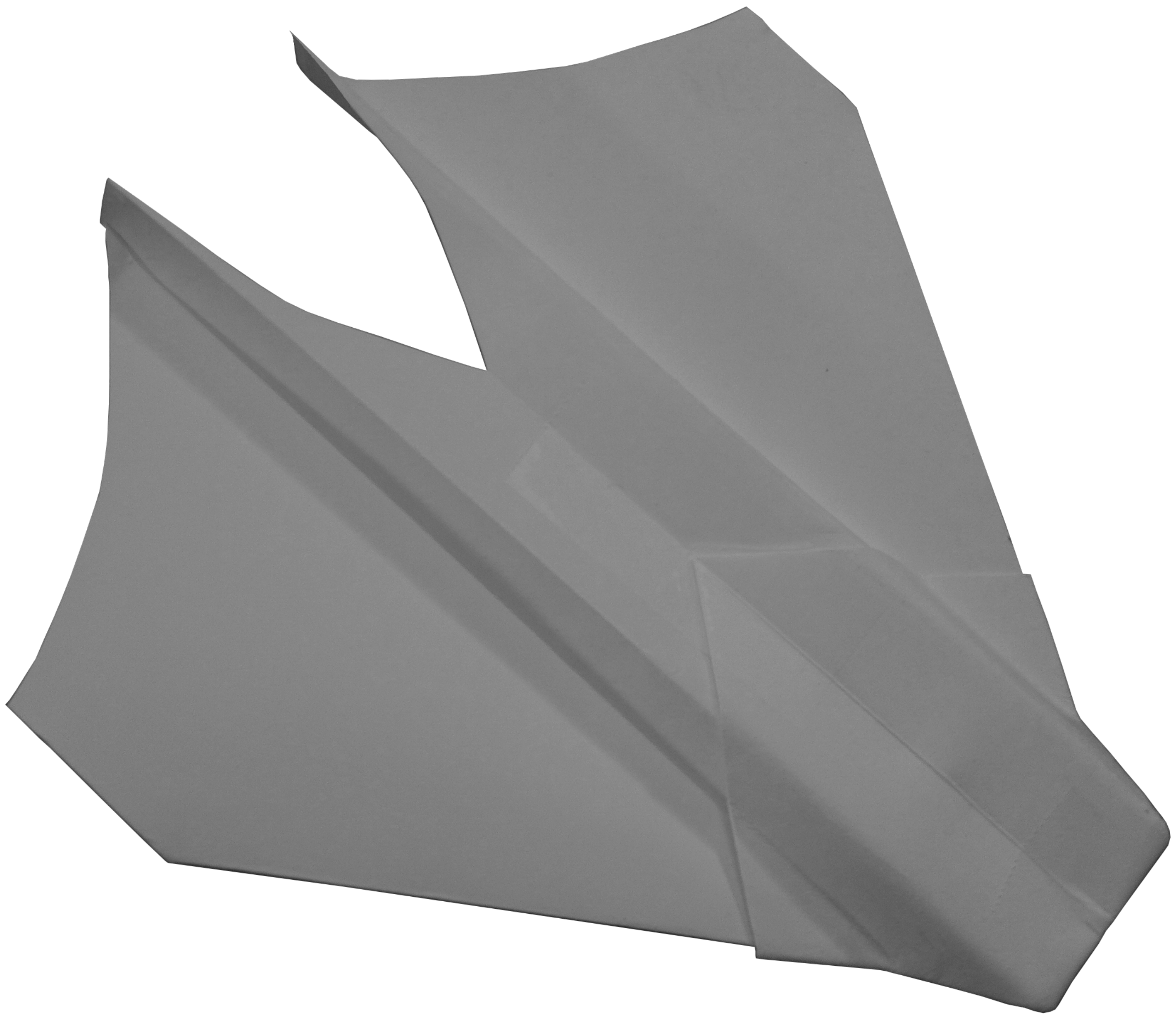
P15 MANTA

Ready for a challenge? This design resulted from the idea that the "M" shaped fuselage could act as the rear stabilizer eliminating the need for a tail section. However, it is relatively unstable compared to the other planes. Fine-Tuning it can be very difficult. Because this design does not have a vertical stabilizer it ends up with less drag above and aft of the center of gravity. As a result it has a tendency to need more of a reflex action (flaps in upward position) in order to hold the nose in a stable slightly up position. The flaps on the rear of the wings create this reflex action and the design of the fuselage helps to channel air to the flaps.

Internal link to build instructions: "Build it!"

External (internet) link for more info behind the name:

http://en.wikipedia.org/wiki/X-44_MANTA



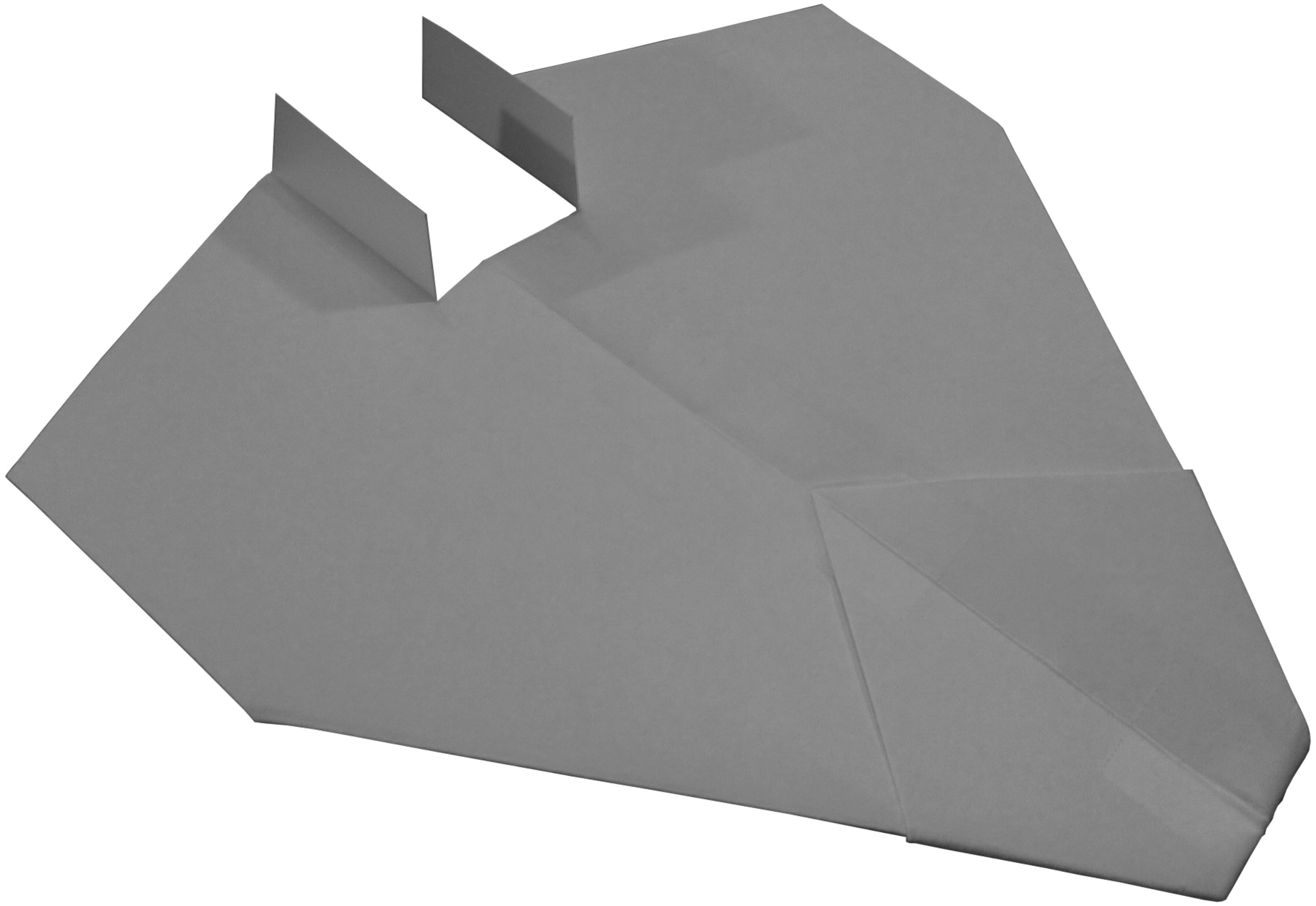
P16 NASP

If the tail can be eliminated, why not the fuselage? This design keeps a flat body making the plane one big wing. The twin tails help with stability. Because of its small head-on profile it creates less drag as it moves through the air. As a result, it can fly faster and farther than the other planes. However, controlling its flights can be difficult.

Internal link to build instructions: "Build it!"

External (internet) link for more info behind the name:

http://en.wikipedia.org/wiki/Rockwell_X-30



CONSTRUCTION INSTRUCTIONS

In an eBook, pictures can move from page to page depending on your viewing device. Be sure to make note of the figure reference in each text instruction. Corresponding pictures will always be ***below*** the written instructions.

These planes are designed to simply introduce the reader to concepts of flight and design. My hope is that you as the reader will expand on these designs and create your own. Think of these designs as a starting point to a limitless world of paper airplane design.

My intention in this book is to make designs that can be made easily without the need to go to a specialty store. Below are the required tools and materials to make each of the designs in this book.

You Will Need

1. Basic 8.5in x 11in printer paper. (While the designs in this book will fly great with regular printer paper, I recommend 32lb/ream, 8.5in x 11in paper. The stronger paper will hold its shape better during flight allowing for higher and more exciting flights.)
2. A pair of scissors that are sharp and small enough to make precise cuts
3. Clear adhesive tape
4. A paper clip (The planes are designed to include #1 Size Paper Clips, 3.2cm x 0.8cm)
5. You may want to have a straight edge or ruler if you do not trust your ability to make precise folds.

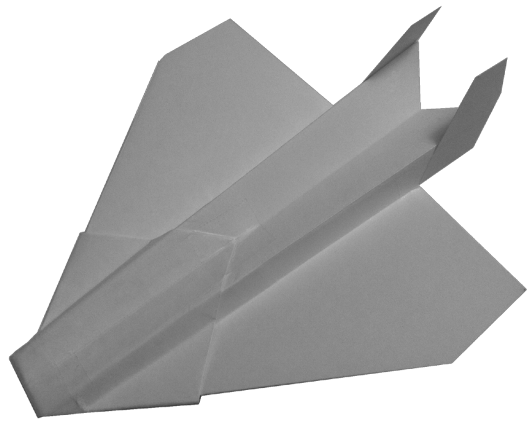
Template Instructions

- Cut out the rectangular template. Leave the Dashed Line cuts for later.
- The template is ~4.25in x ~5.5in. That is equivalent to half of an 8.5in x 11in sheet of printer paper. FYI, your PDF print settings may enlarge or shrink the document slightly. This template assumes the setting "Shrink to Printable Area" which is the most likely scenario.
- Once you become comfortable making the planes from this template you can simply cut regular printer paper in half in order to start.
- While the templates are symmetrical, during construction, if you need to, ***always choose symmetry*** of your plane over following the printed lines.
- ——— Solid Line = Mountain Fold (the printed solid line is visible on the top of the "mountain")
- •••••••• Dotted Line = Valley Fold (the printed dotted line is visible at the bottom of the "valley")
- ------- Dashed Line = Cut

Construction Tips

- Don't worry about tearing the tape off of the spool at exactly the right size. You can trim it after it is on the plane in most situations.
- If you are in a situation where you need to wrap tape around a corner fold you can do this easily with scissors. Simply cut a slit in the tape up to where the tape meets the corner and then you can wrap both of the separate sides of the tape around the corner.
- Although the designs in this book work well as they are, they can also be mixed and matched. You could for example take the nose and wings of one plane and add the tail and aft sections of another to it to create a completely new design. I encourage you to experiment with different combinations of the designs in this book and to continue to expand with your own designs.

P11 EAGLE



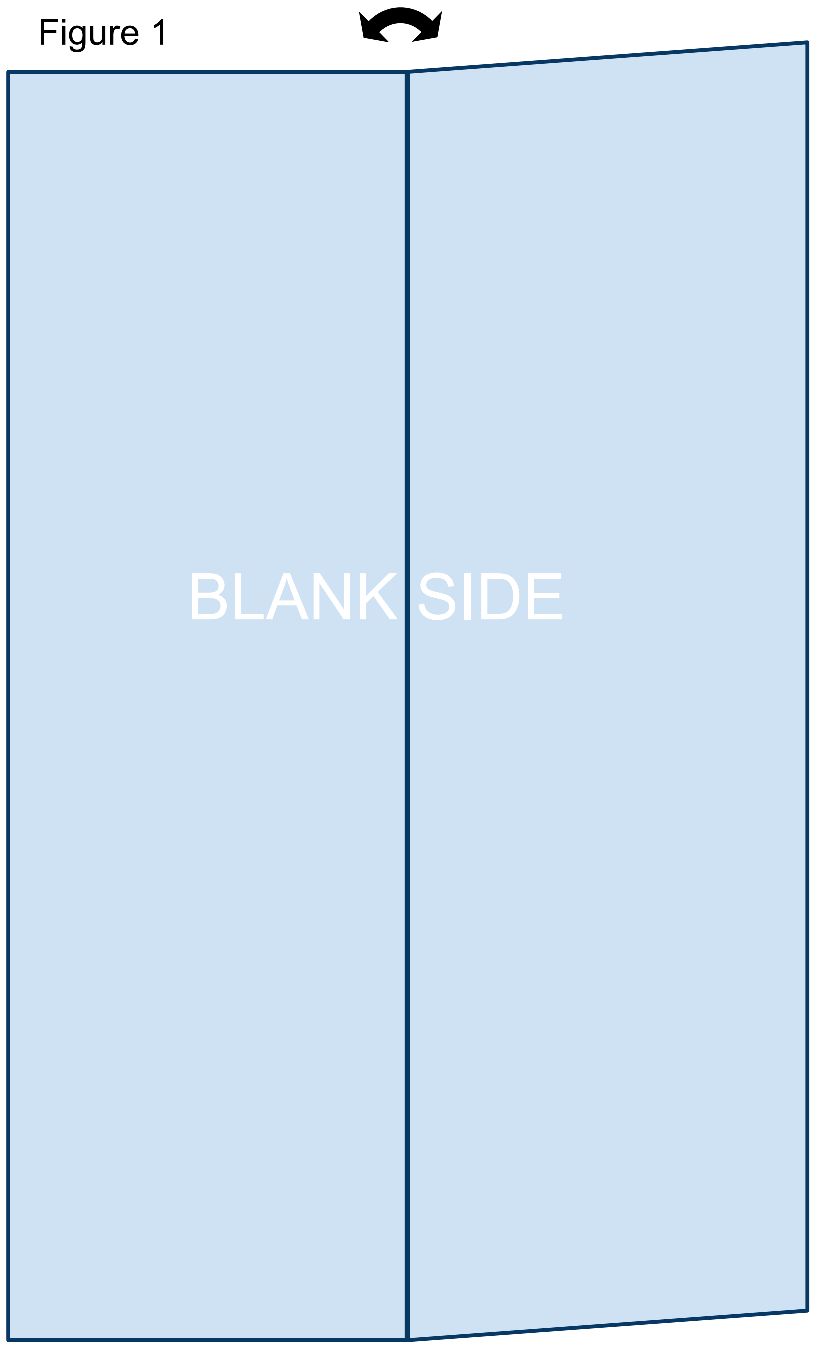
The printable template linked below is 4.25in x 5.5in. That is equivalent to half of an 8.5in x 11in sheet of printer paper. While the designs in this book will fly great with regular printer paper, I recommend 32lb/ream paper. The stronger paper will hold its shape better during flight allowing for higher and more exciting flights. Once you become comfortable making the planes from this template you can simply cut regular printer paper in half in order to start. Print the PDF template located at the bottom of the web page: https://sites.google.com/site/afspaperairplanes/p11

In case your reading device is not connected to the internet, doing a Google search for "AFS Paper Airplanes" should get you to the webpage.

1

Cut out the template and place the paper on the table so that the printed information is on the underside with the printed triangle facing away from you. You will be looking at the blank side. Bring the right side up and fold the paper in half the long way. Unfold leaving a visible crease as shown in Figure 1 below.

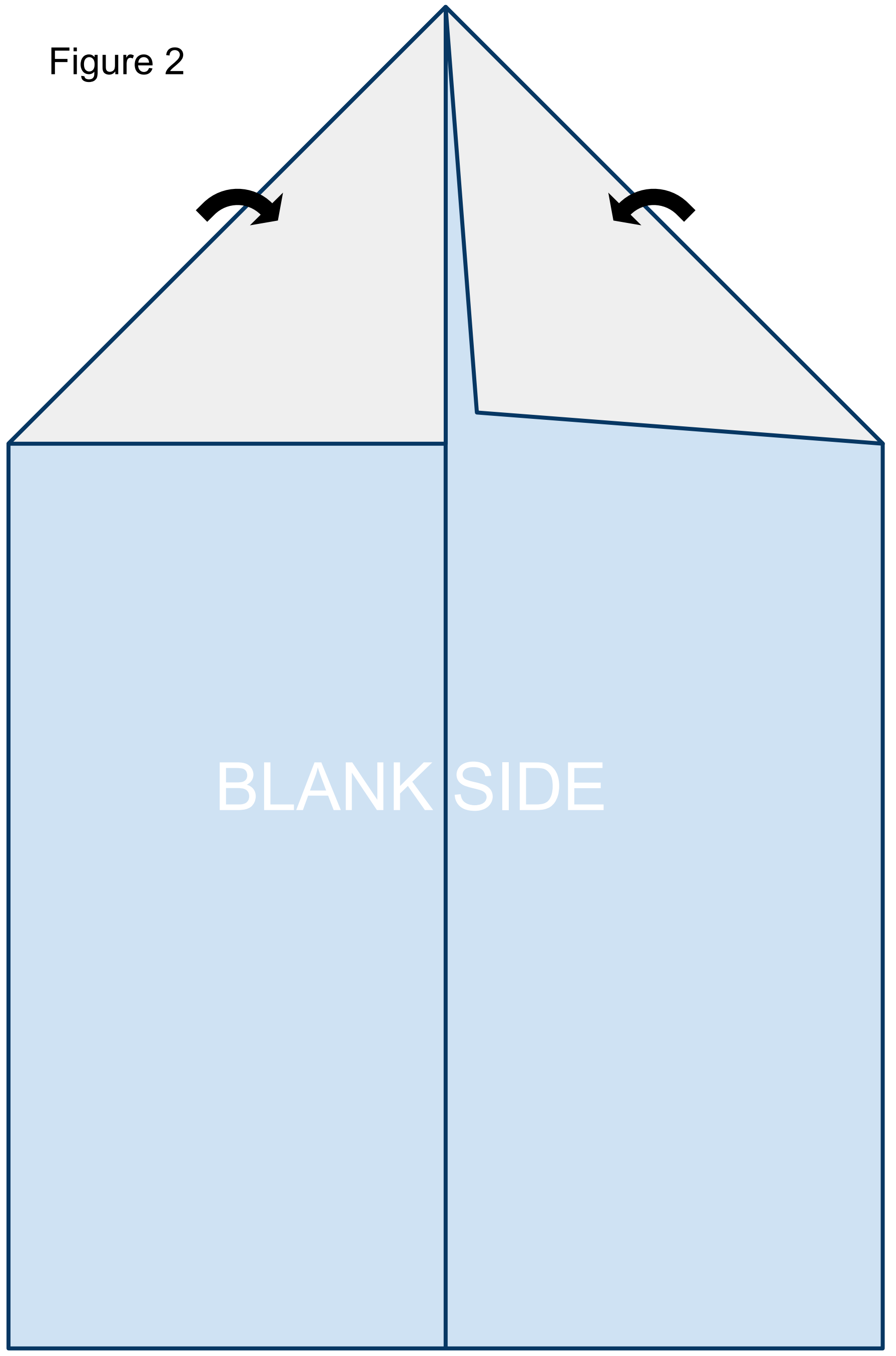
Figure 1



2

Fold the leading corners back. Line them up along the crease you created in step 1 as shown in Figure 2 below.

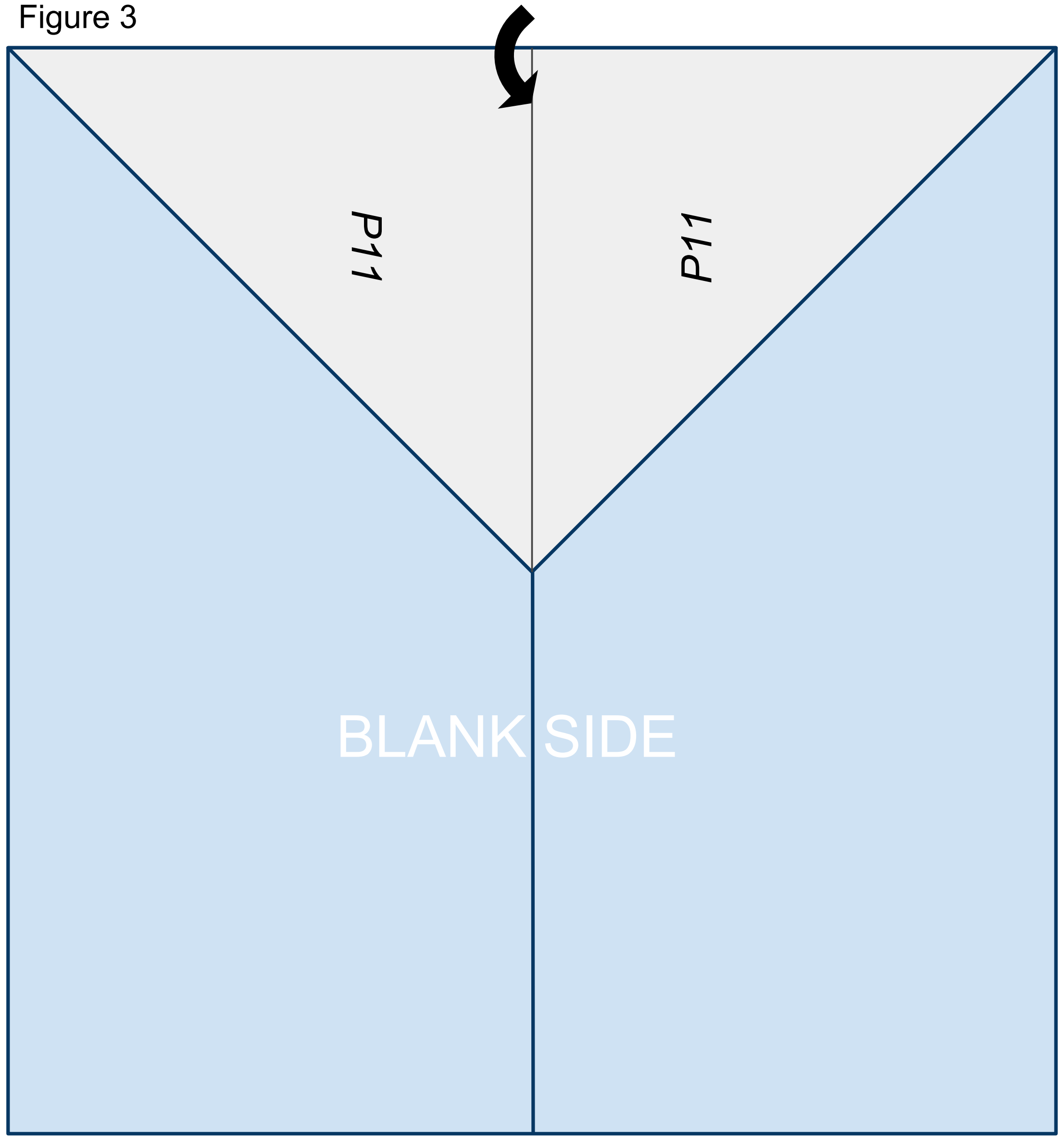
Figure 2



3

Fold the nose of the plane created in step 2 back along the solid line printed on the front of the paper so that it lines up on the center crease created in step 1 as shown in Figure 3 below.

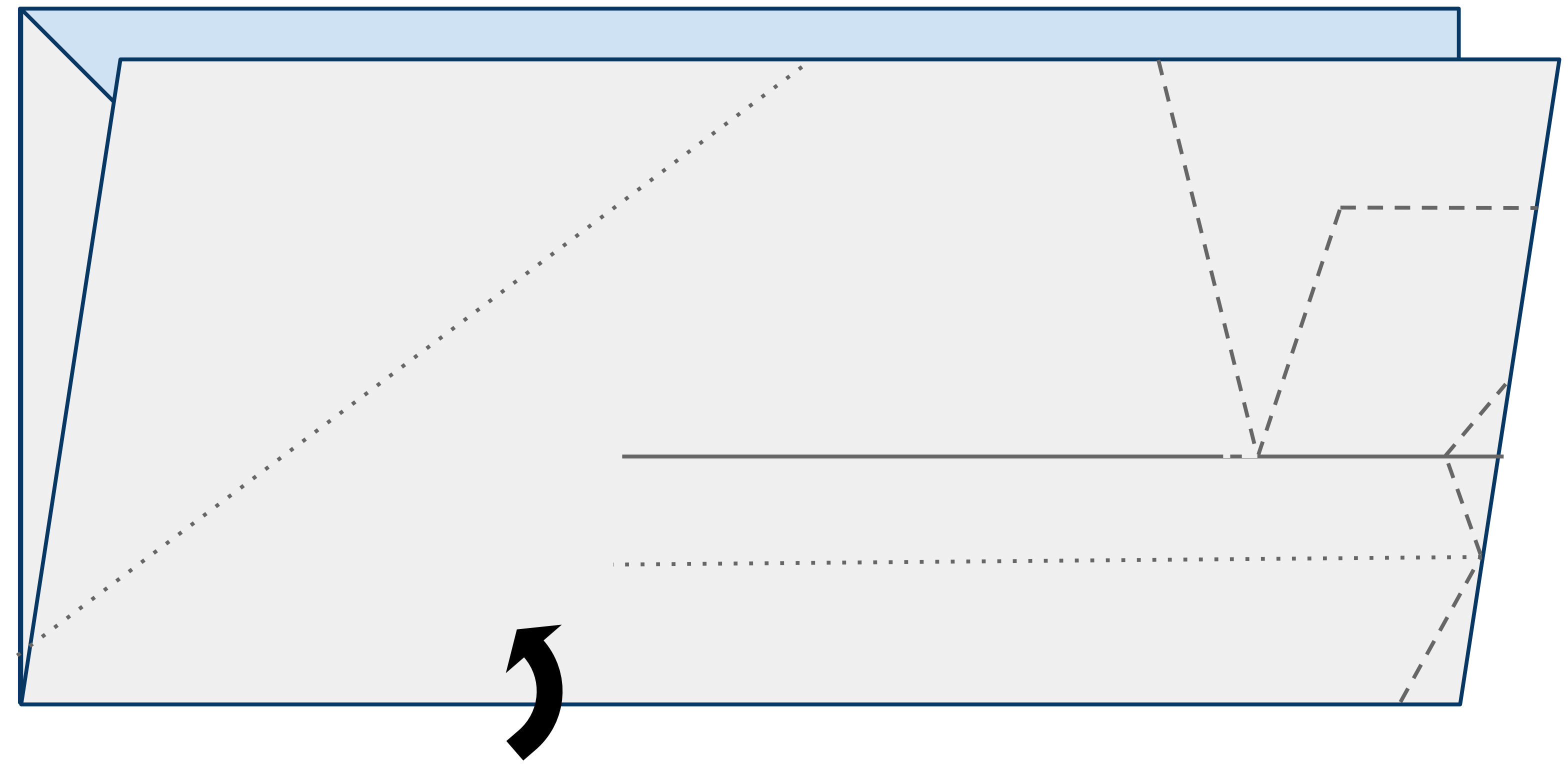
Figure 3



4

Turn the paper to the left (counterclockwise) and fold the paper upward in half hiding the triangle created in the previous steps refolding along the fold from step 1 as shown in Figure 4 below.

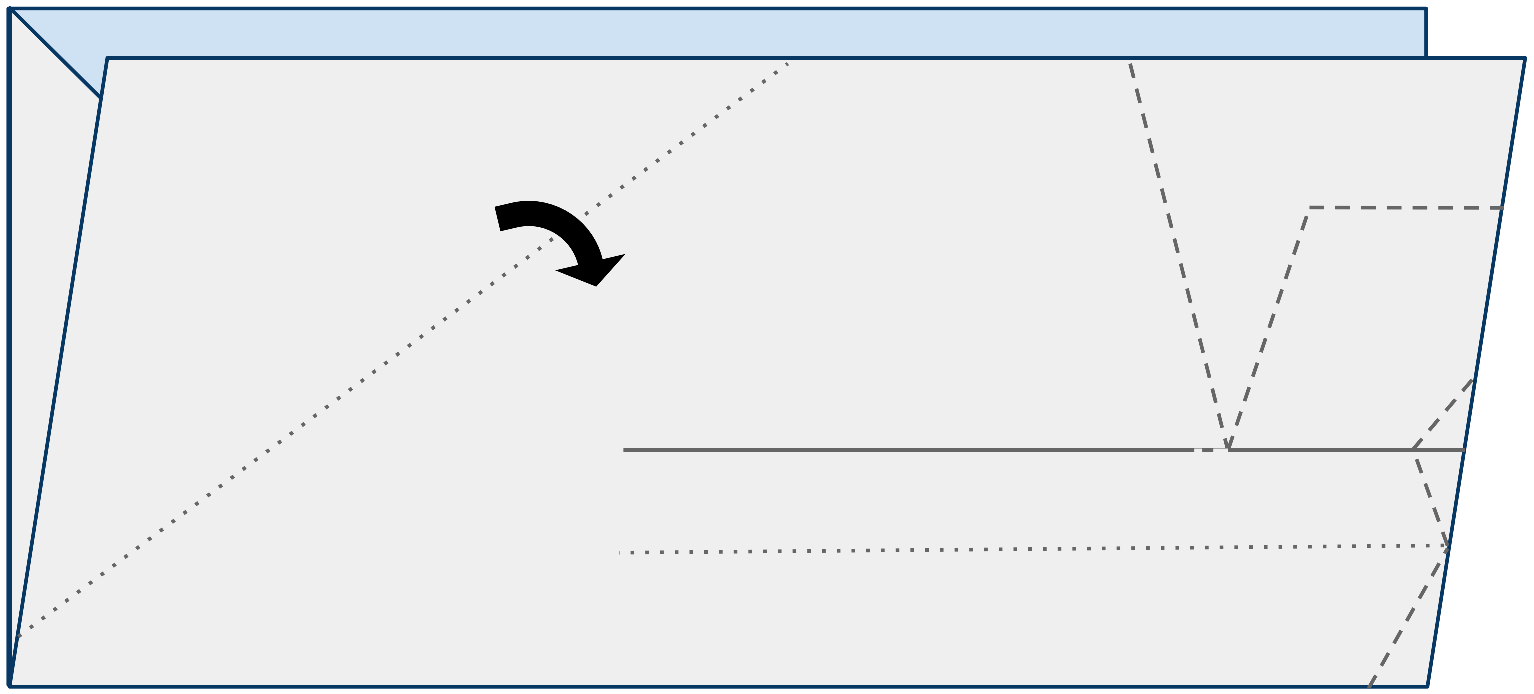
Figure 4



5

Fold the leading edge of the near side of the plane down along the dotted line as shown in Figure 5 below.

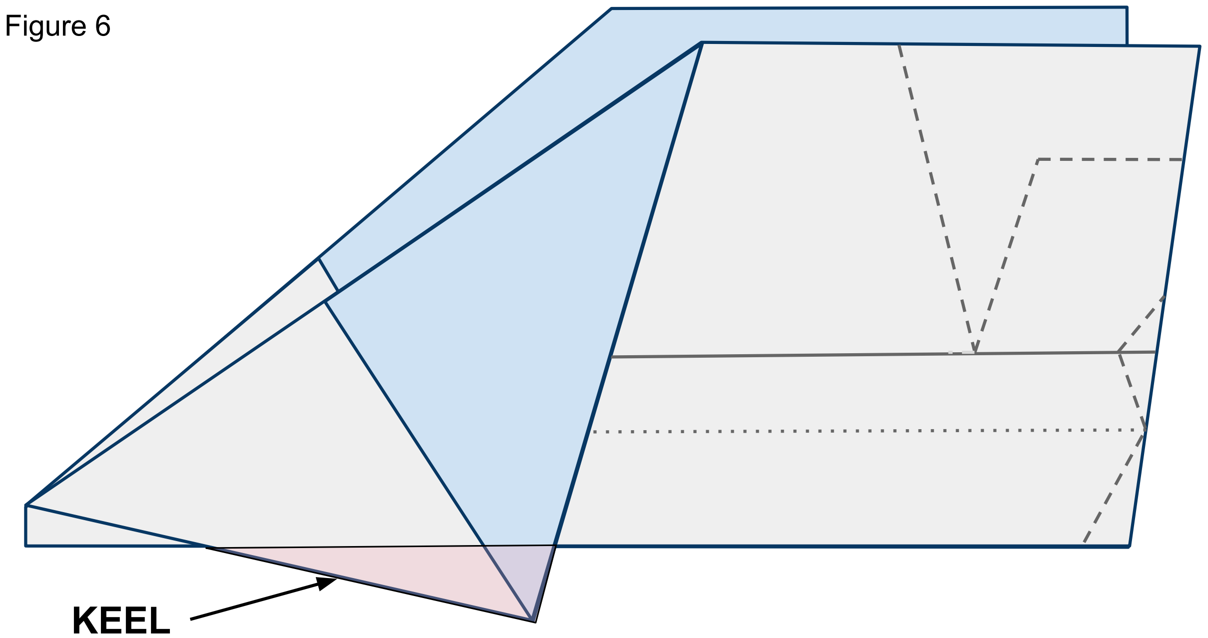
Figure 5



6

Do the same to the other side of the plane. It is important to match these 2 folds exactly so that your plane is symmetrical and will fly straight. The area that is created below the body and designated in pink will be called the keel.

Figure 6



7

Place tape along the keel as shown in Figure 7 below and cut a slit in the tape along the red line. Now fold the tape around and under both sides of the plane and keel securing them together. See Figure 7b for what the far side of the plane looks like.

Figure 7

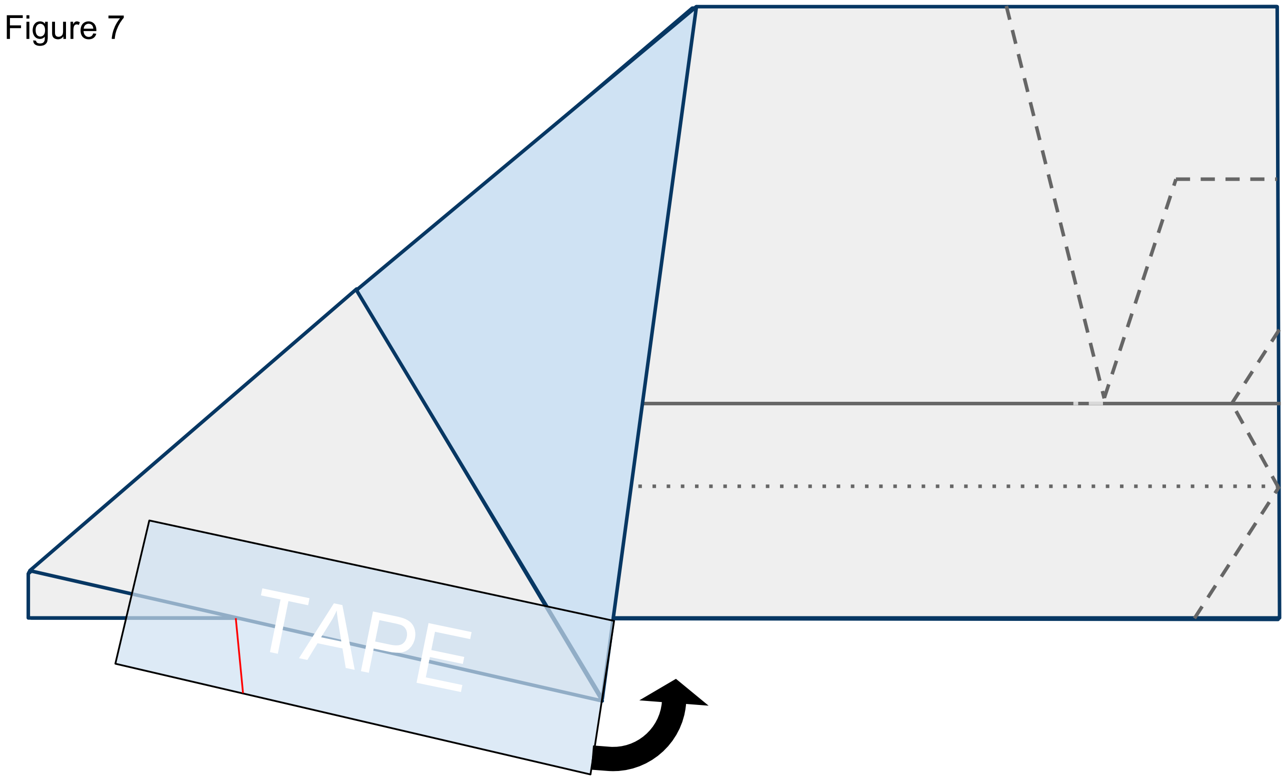
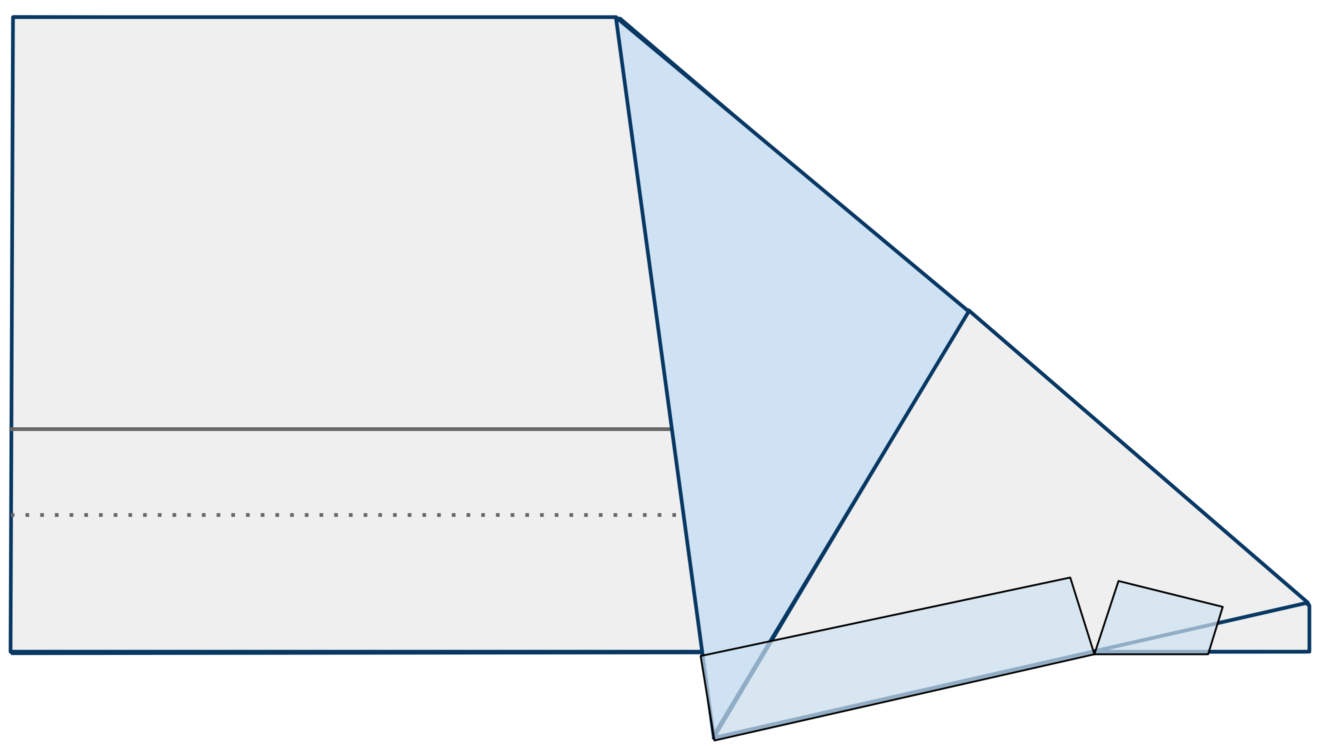


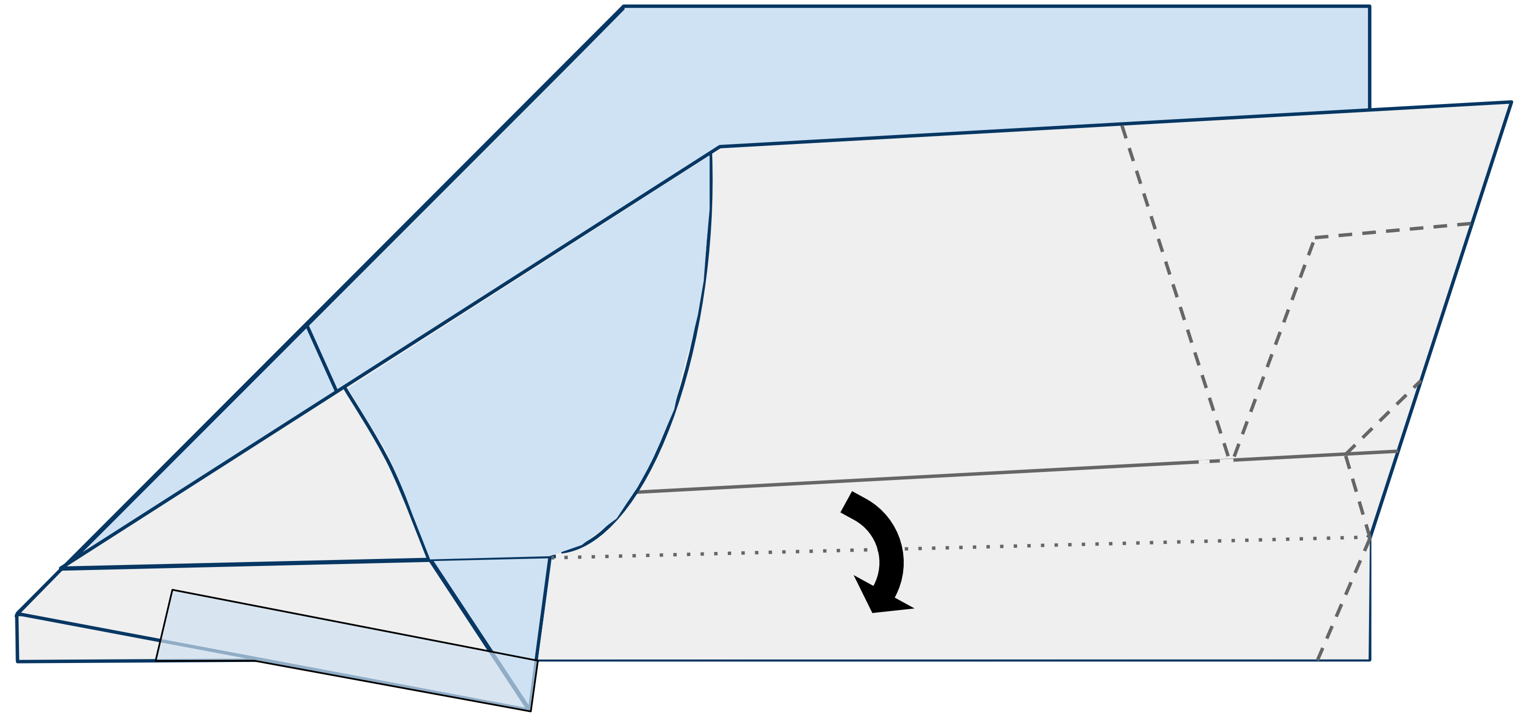
Figure 7b



8

Fold down the near side wing along the dotted line as shown in Figure 8 below.

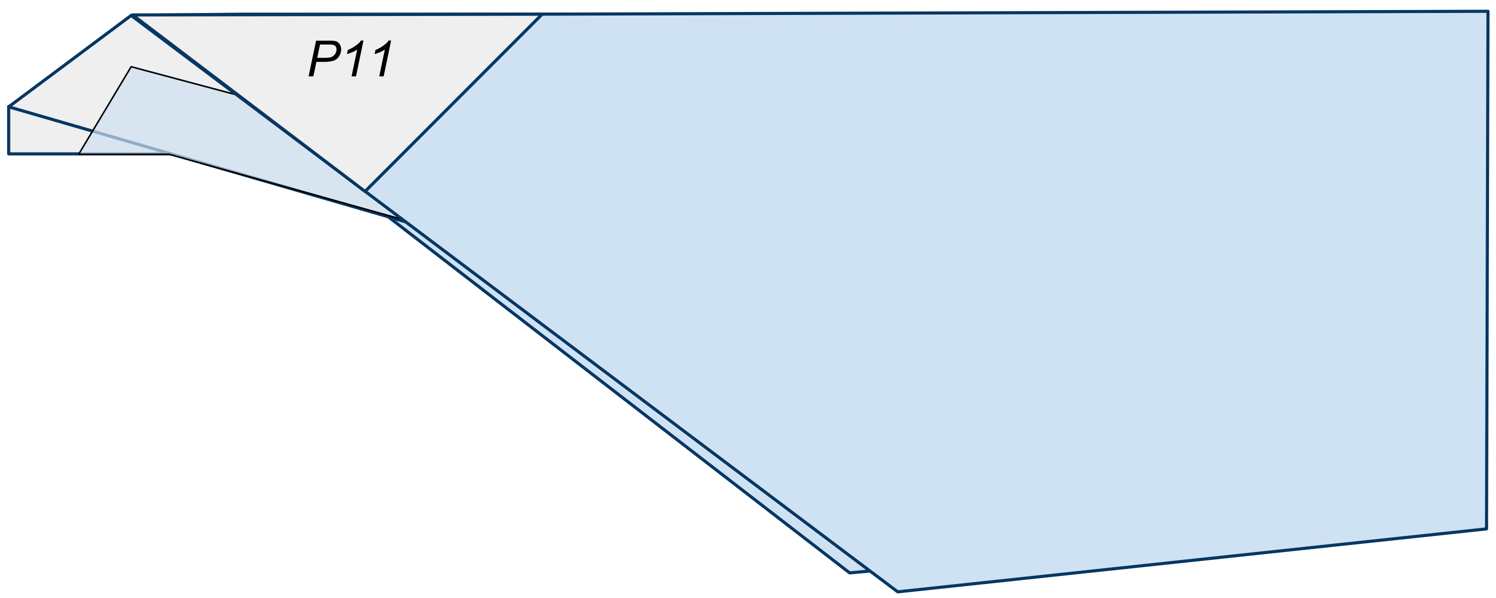
Figure 8



9

Fold down the far side wing along the dotted line also as shown in Figure 9 below. Insure that these folds are as symmetrical as possible. If you need to choose, always choose symmetry over following the lines printed on the paper.

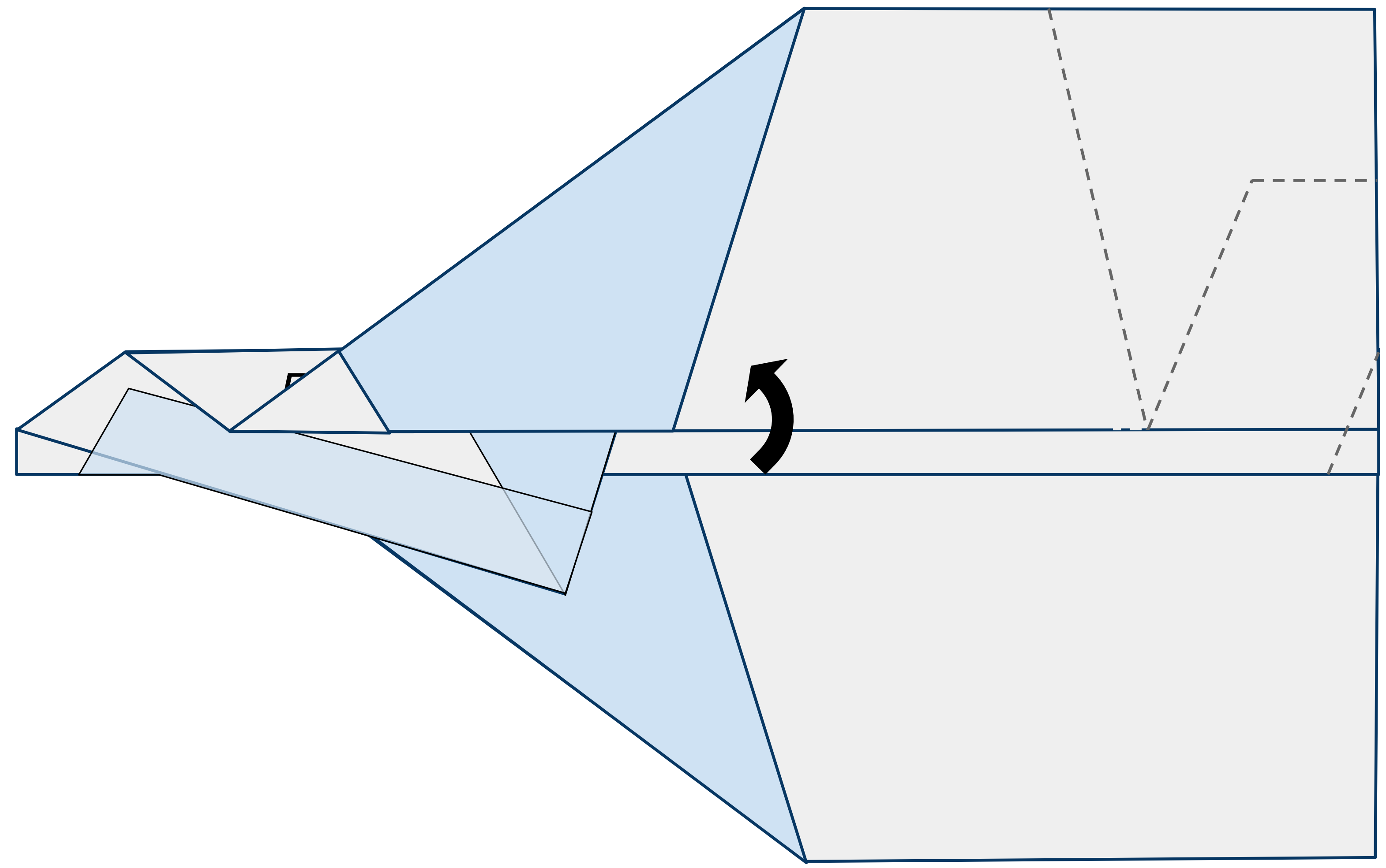
Figure 9



10

Fold the near side wing up along the printed solid line as shown in Figure 10 below.

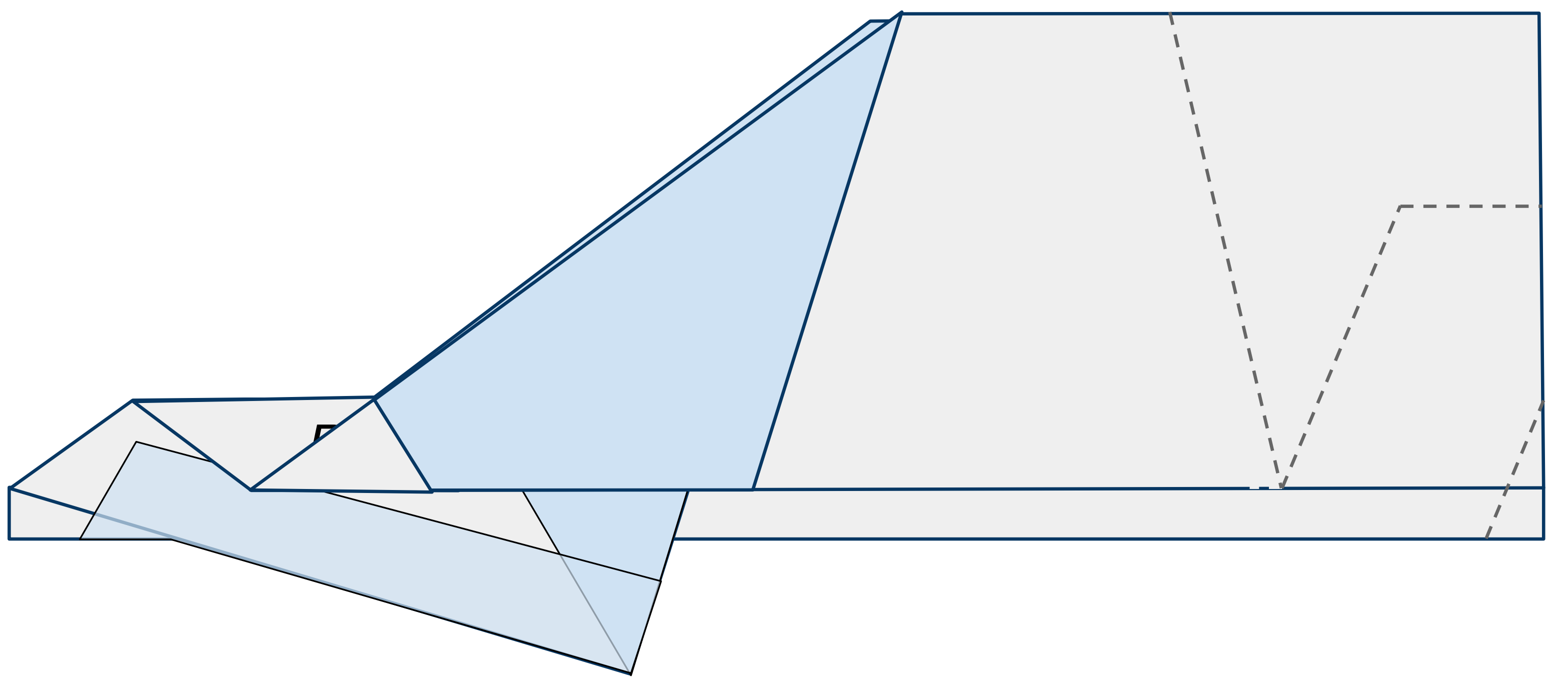
Figure 10



11

Do the same to the other side. Fold the far side wing up along the sold line on the other side of the plane as shown below in Figure 11 below.

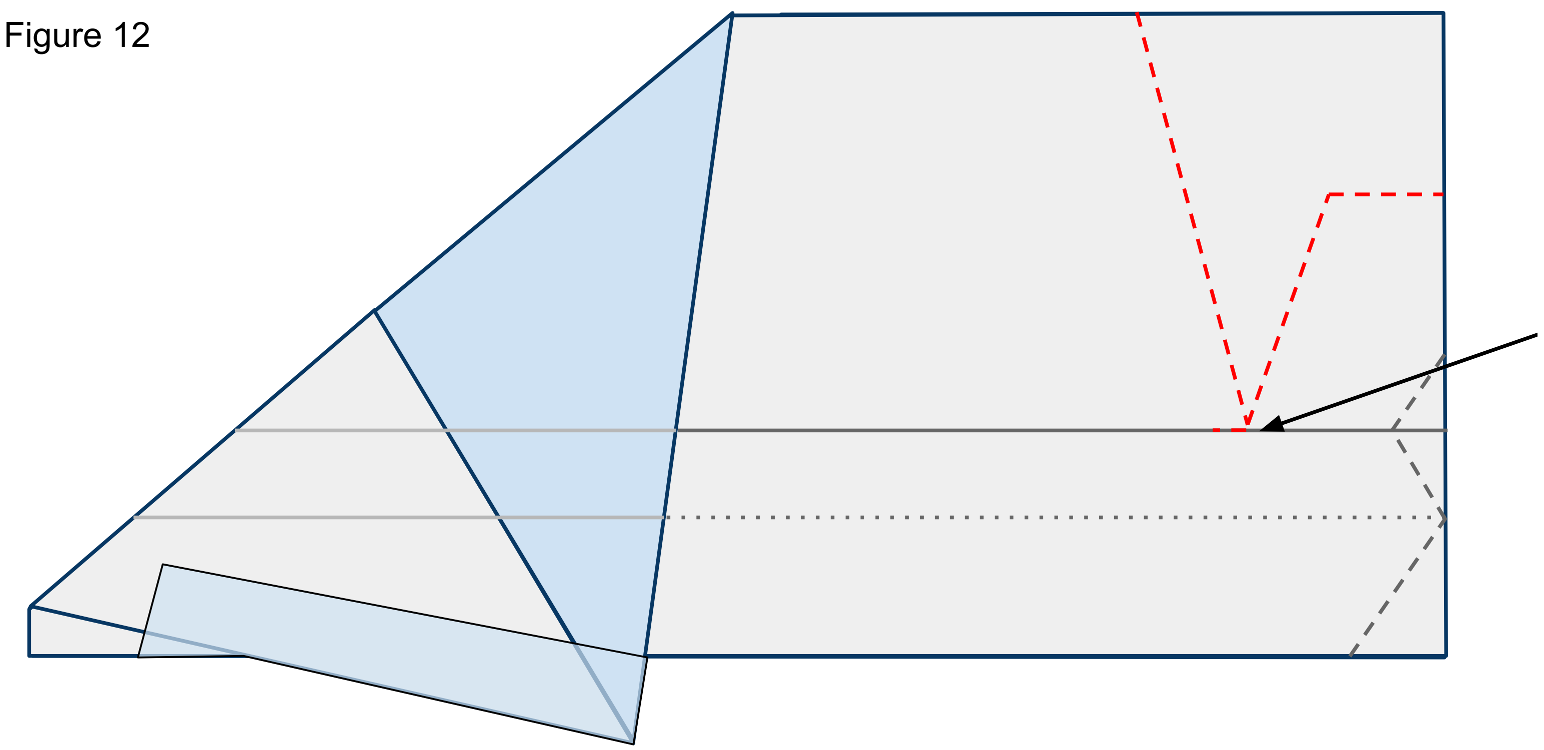
Figure 11



12

Open up the wings that were folded in steps 8-11 and lay the plane flat on your surface as shown in Figure 12 below. Cut thought both layers of paper along the **red** dashed line. Don't forget the small slit along the wing fold. This will create your plane's flaps.

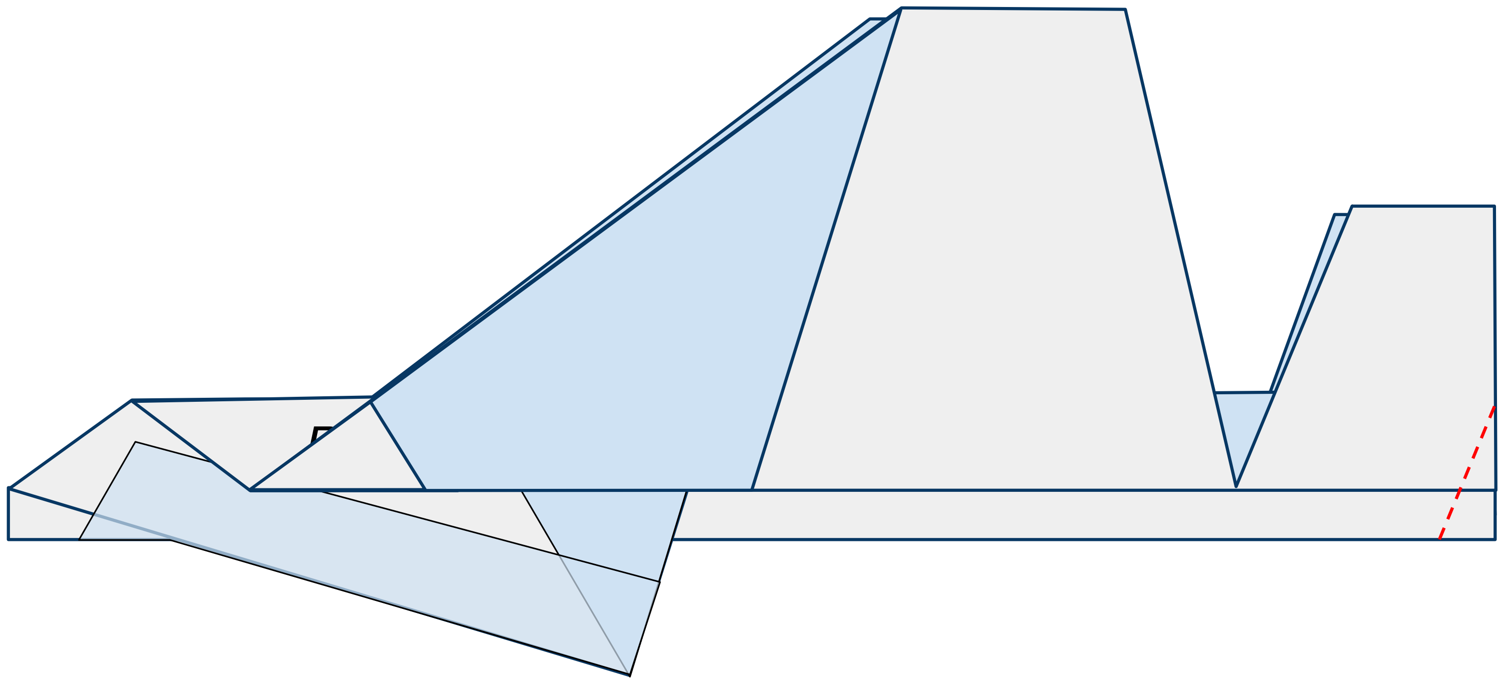
Figure 12



13

Refold the wings back into their folded positions. Place the plane flat on the work surface as shown in Figure 13 below. Now cut along the **red** line cutting through all 6 layers of paper.

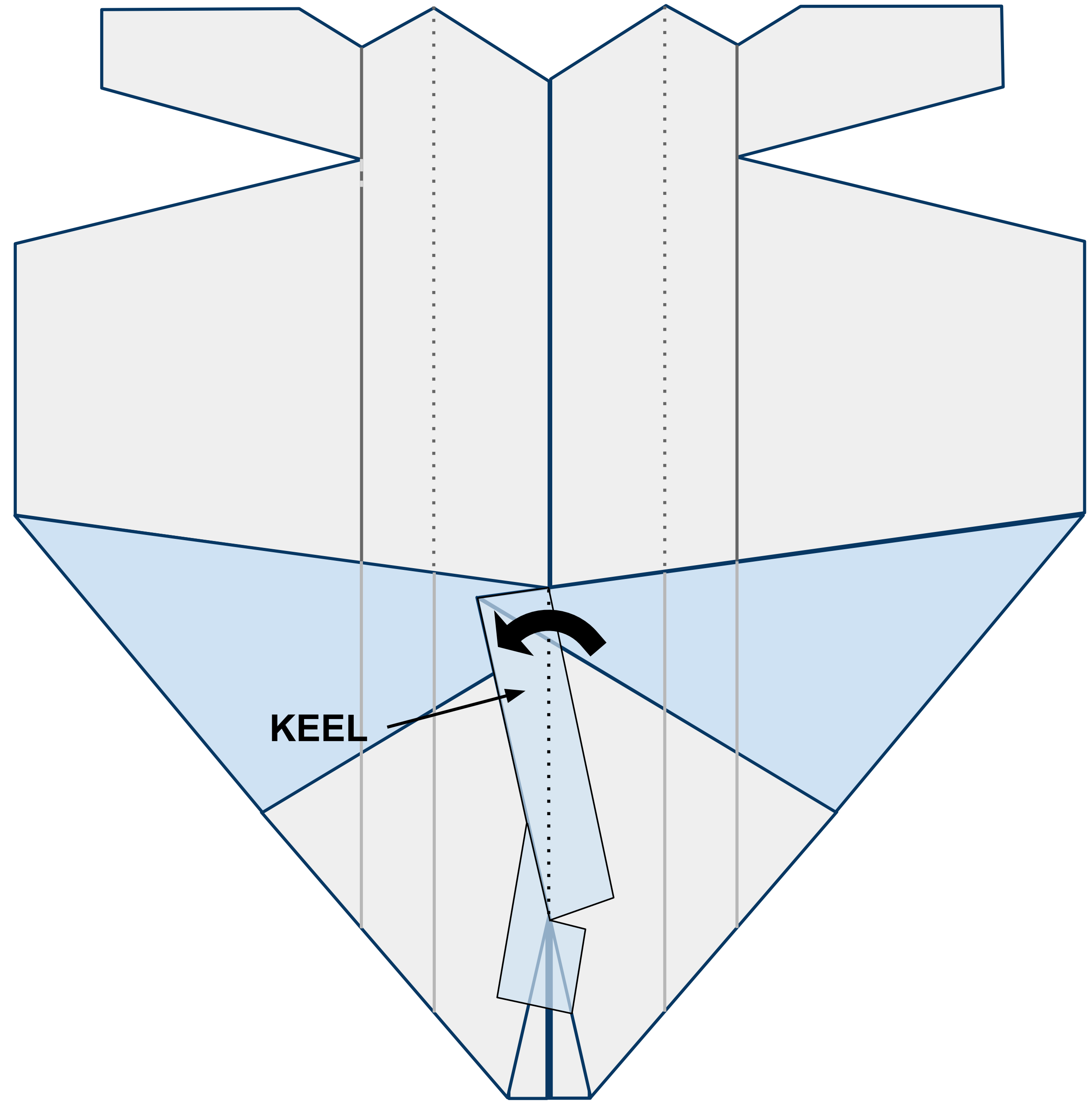
Figure 13



14

Open the plane up flat on your work surface with the nose toward you and looking at the underside of the body. When you do this the keel will want to stand up off the table. Bend it to the left along the center line and give it a strong crease as shown in Figure 14 below.

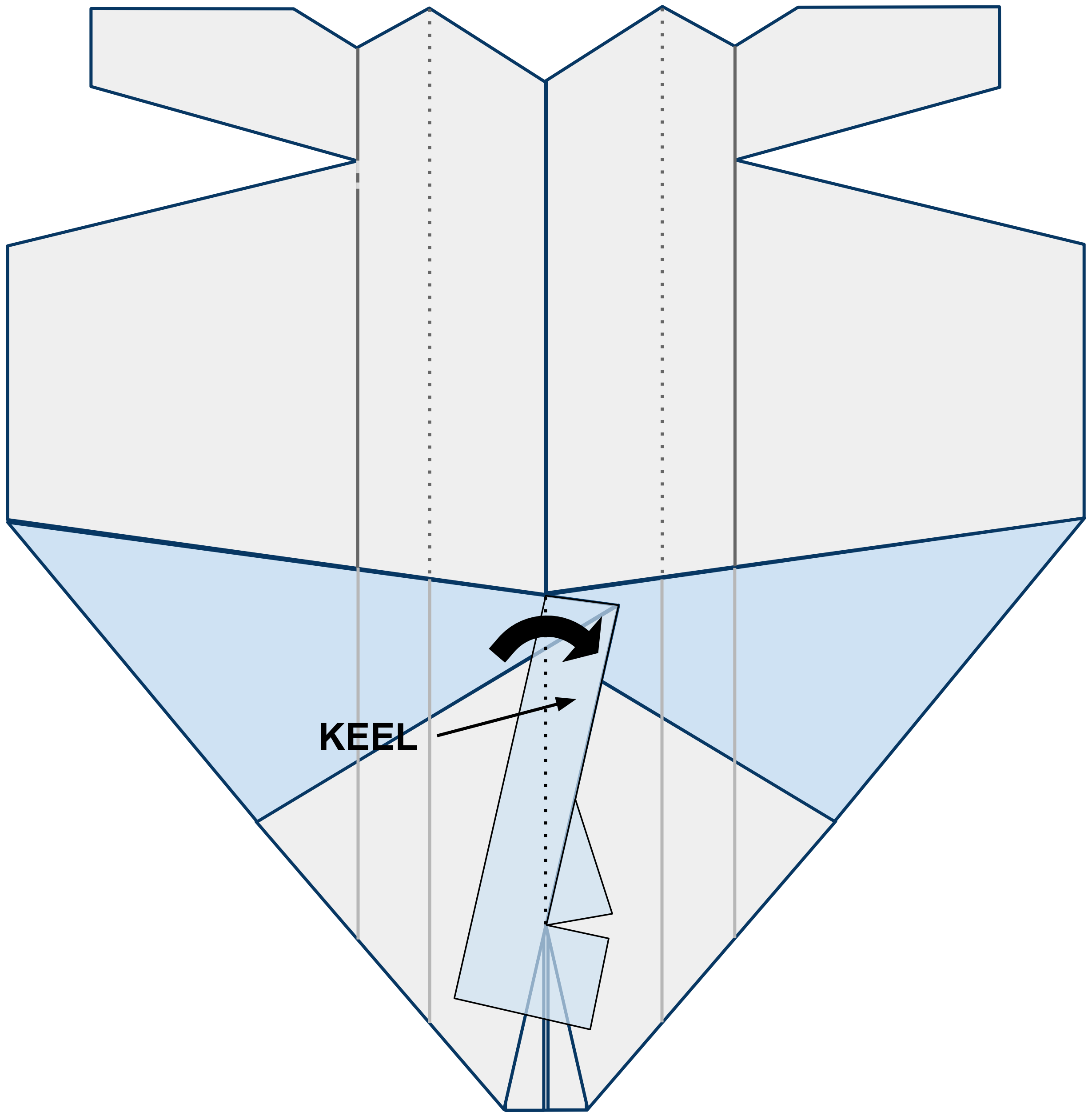
Figure 14



15

Now fold the keel to the right side again creating a strong crease as shown in Figure 15 below. After making this crease, allow the keel to remain upright for the remainder of construction. This is where you will hold the plane when launching it.

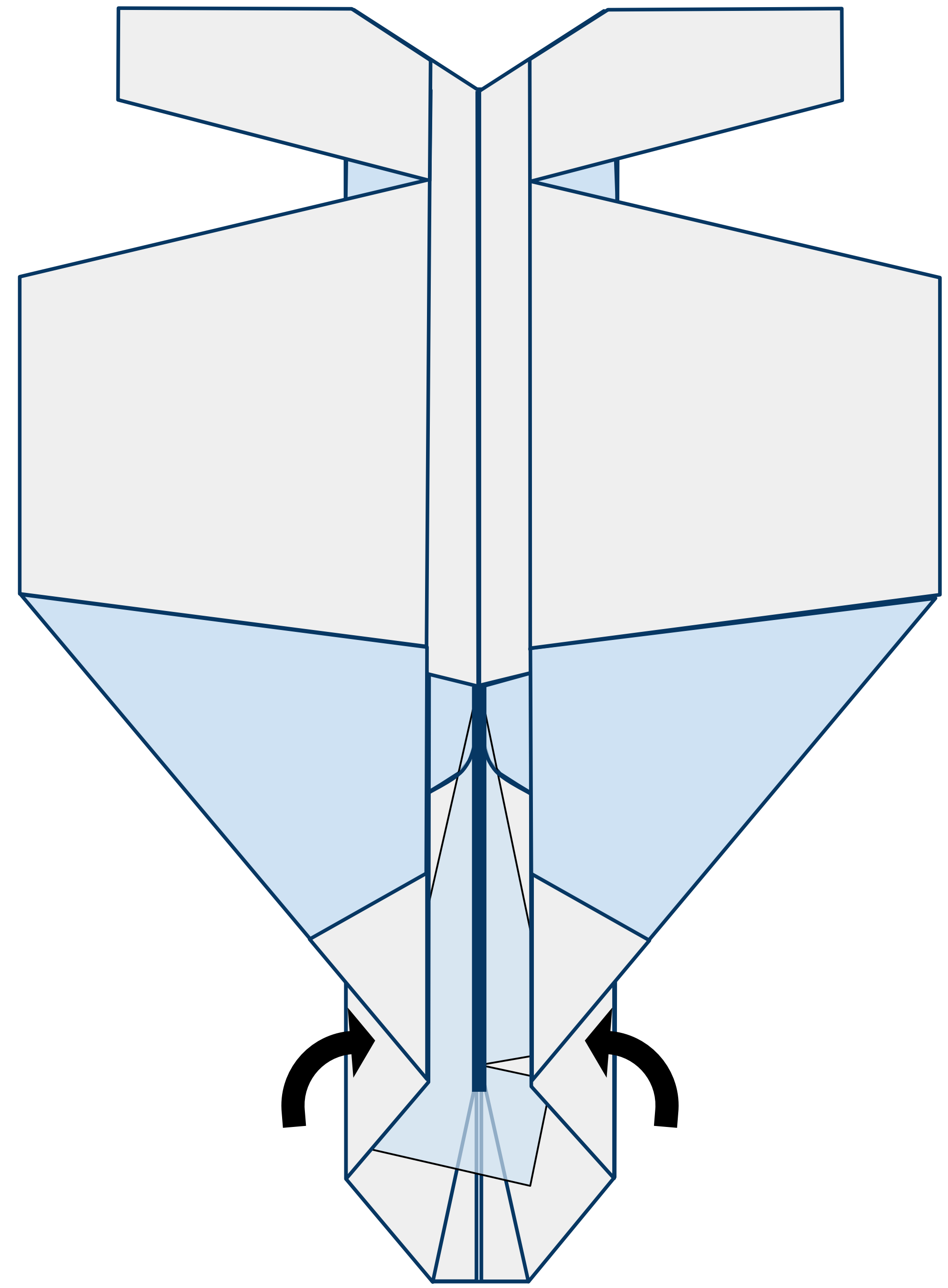
Figure 15



16

Allowing the keel to remain vertical, bring the wings back in along their creases created in steps 8-11 as shown in Figure 16 below. The keel is shown standing vertically as the thick **blue** line in the picture.

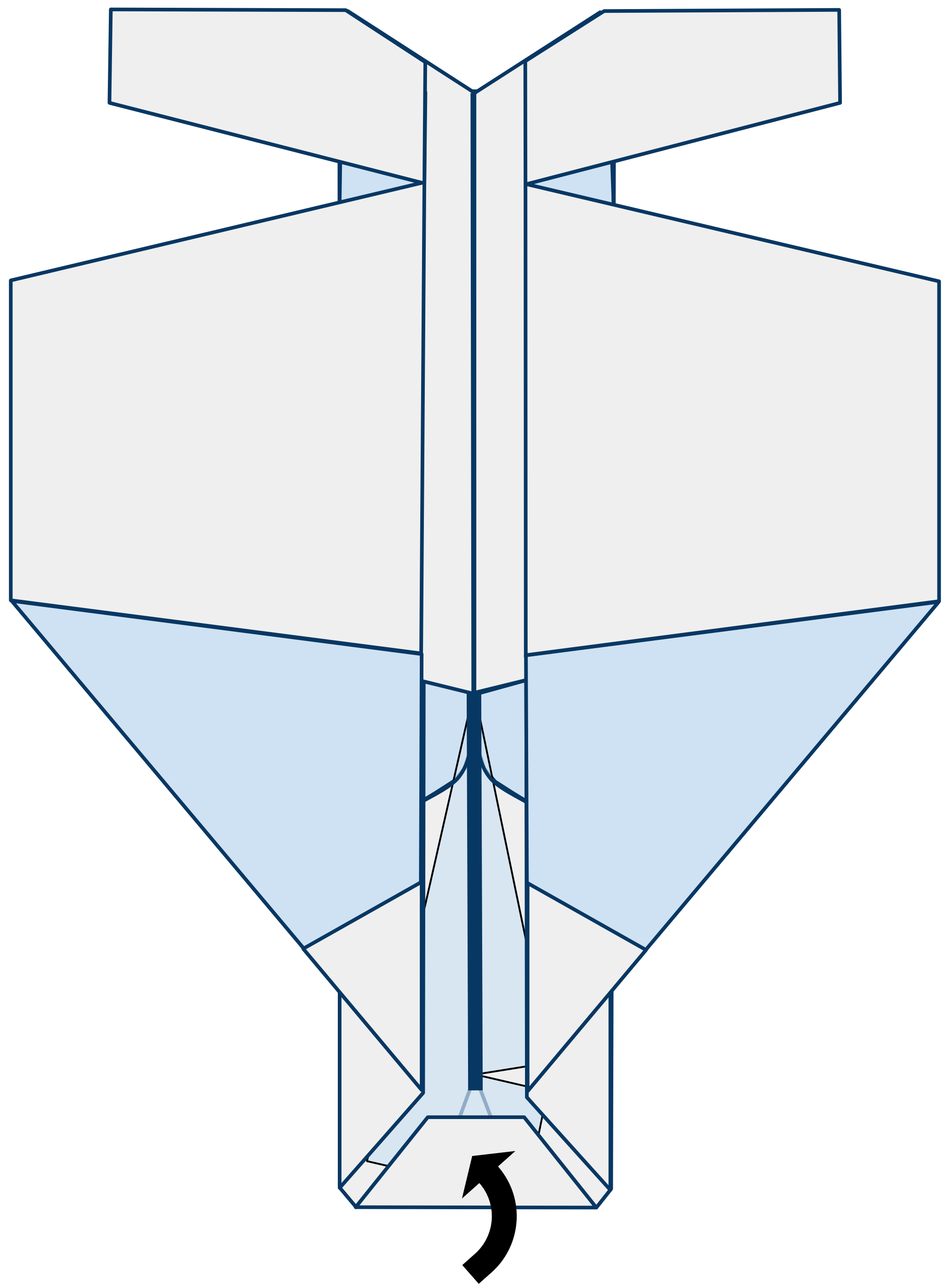
Figure 16



17

Fold the nose of the plane up and in as shown below in Figure 17 below. The keel is shown standing vertically in the picture.

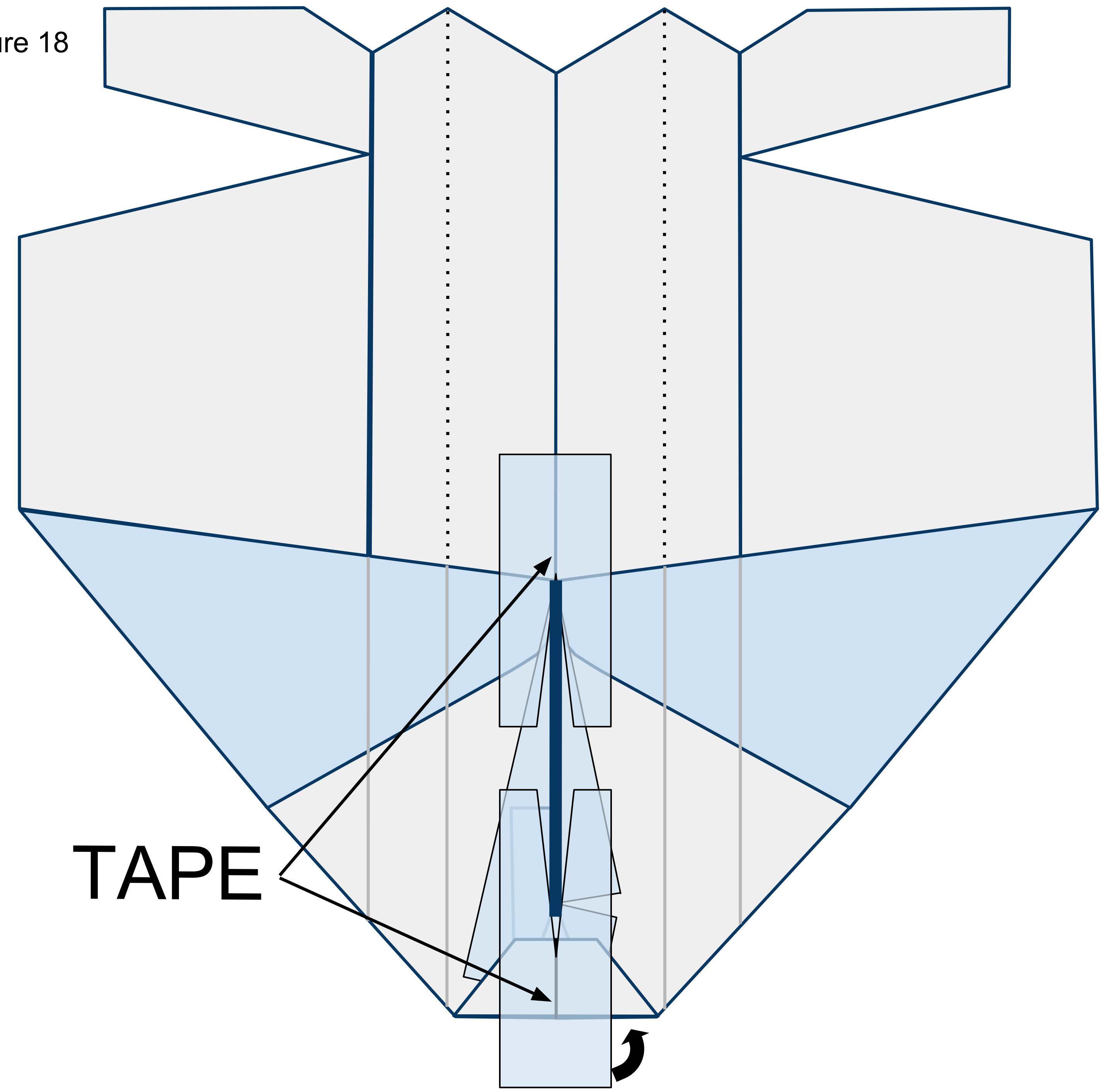
Figure 17



18

Open the wings up flat and place 2 pieces of tape with slices cut in them on the underside of the plane in front of and behind the keel as shown in Figure 18 below. The keel is shown standing vertically in the picture. Wrap the tape around the nose to the top side of the plane.

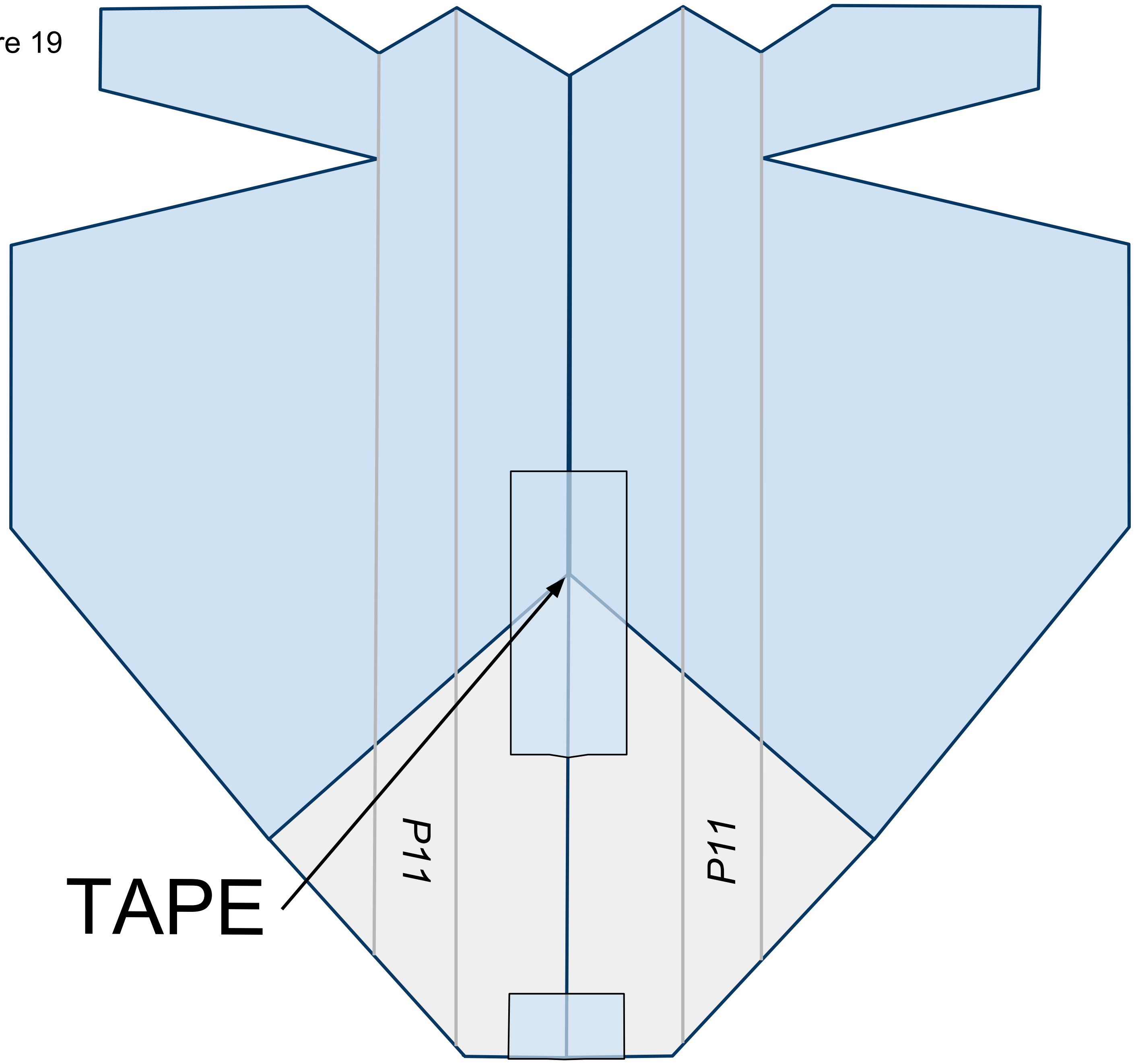
Figure 18



19

Turn the plane over laying the keel to one side. Place tape along the fuselage as shown in Figure 19 below.

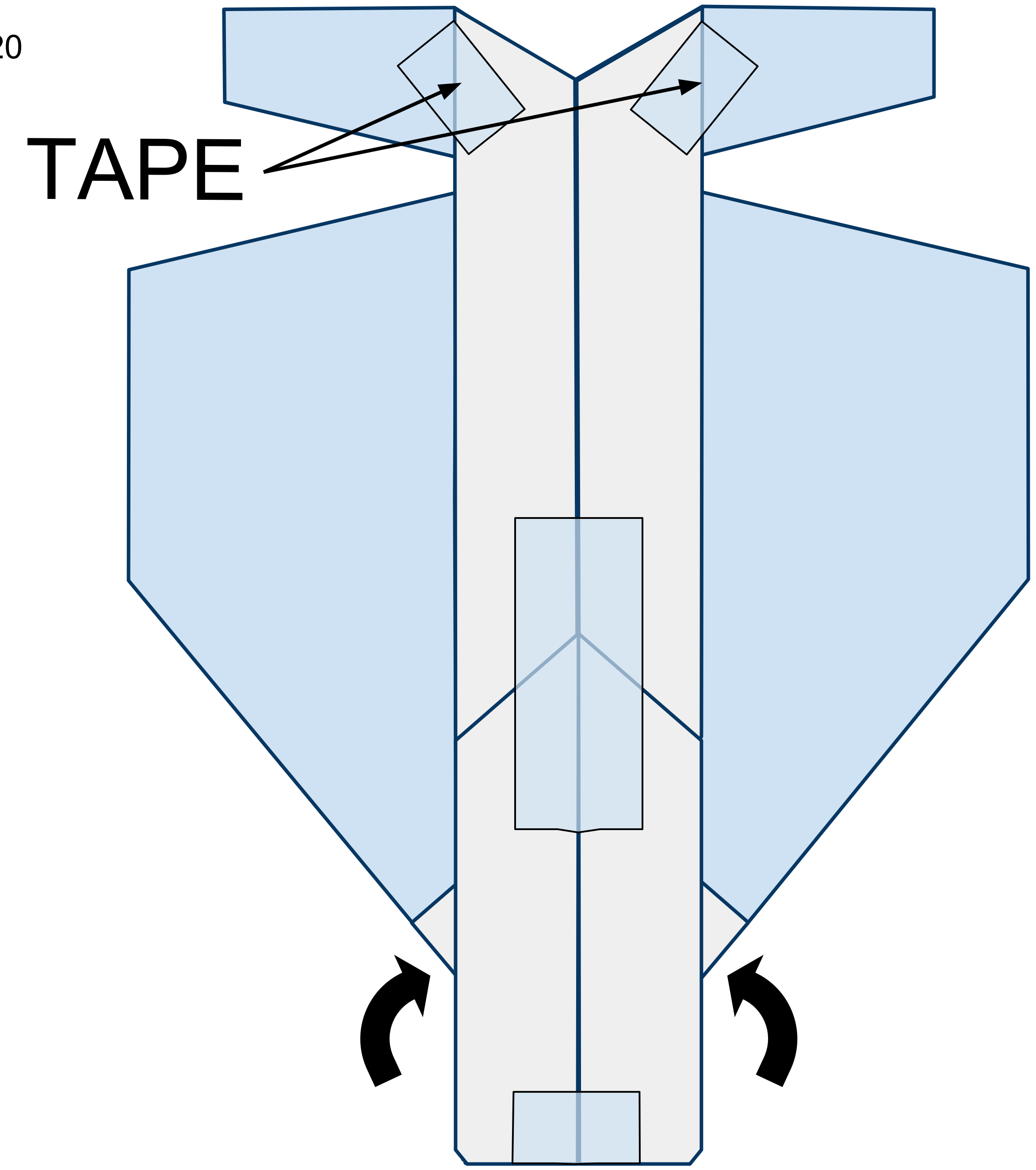
Figure 19



20

Refold the wings under the fuselage following the existing creases and place tape against the tail sections as shown in Figure 20 below.

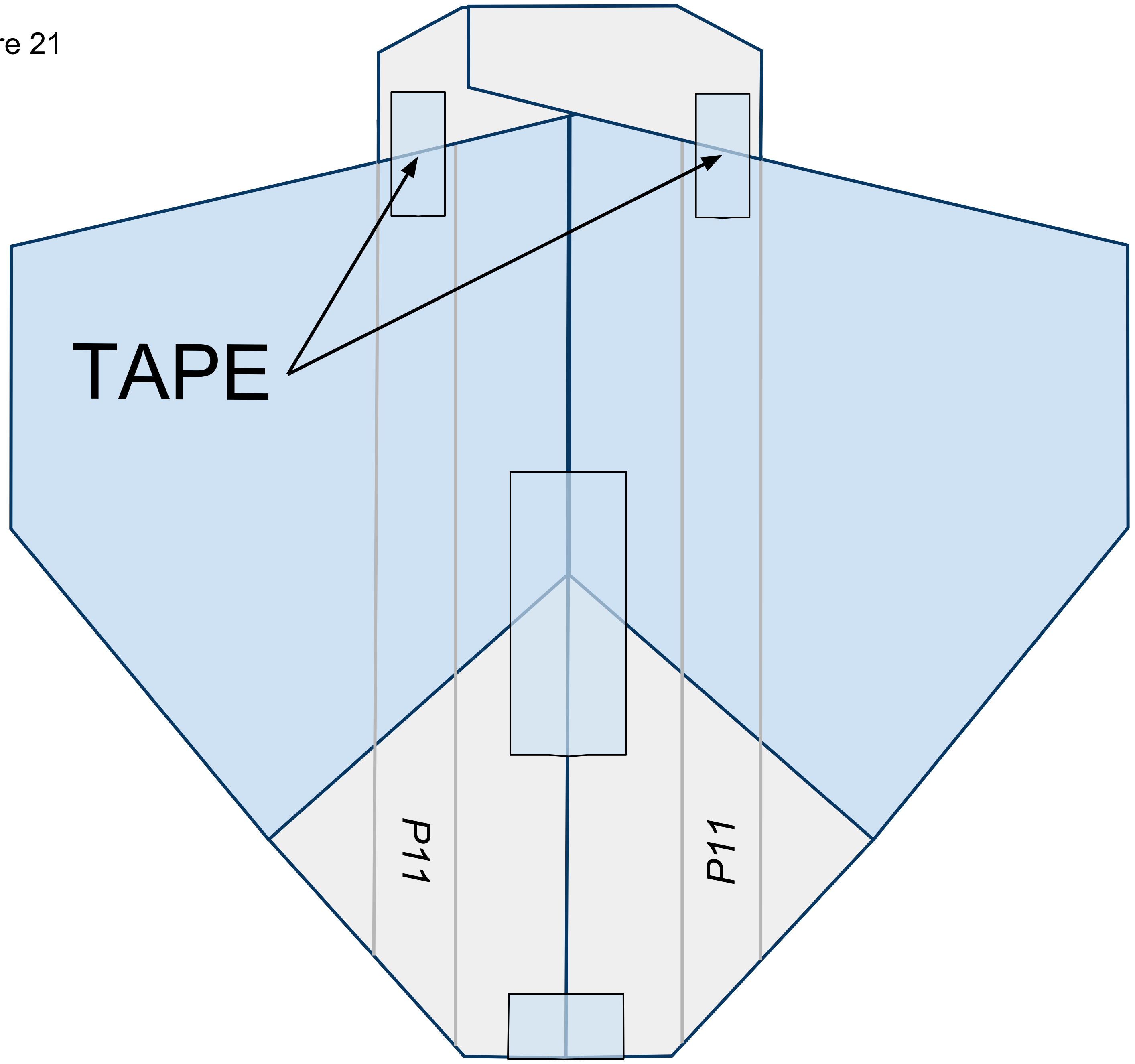
Figure 20



21

Unfold the wings flat on the work surface and place tape across the tail section as shown below Figure 21 below.

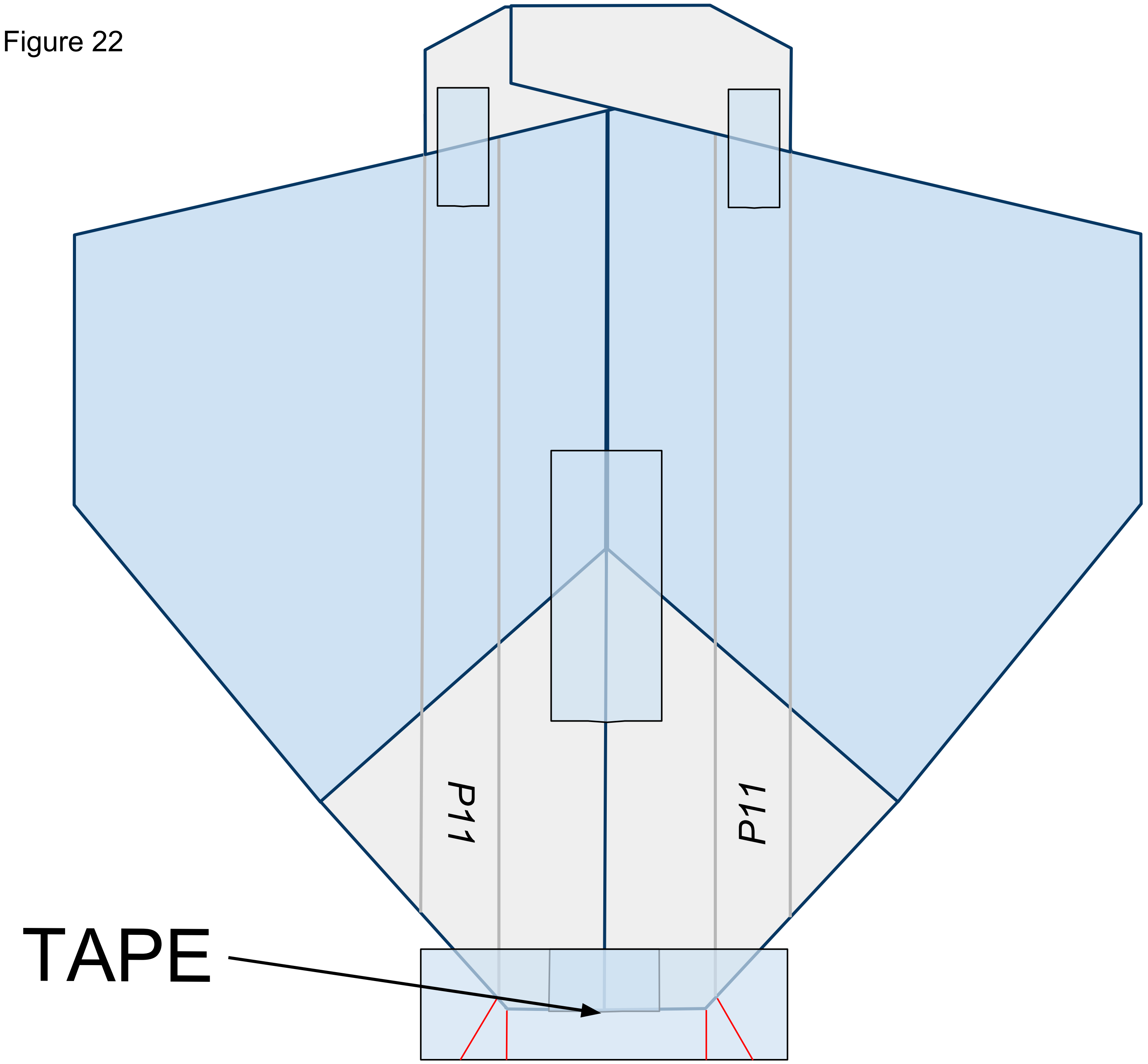
Figure 21



22

Place a piece of tape across the nose of the plane as shown in Figure 22 below. Cut slits along the **red** lines as shown. These slits will help you fold the tape under the plane in the next step.

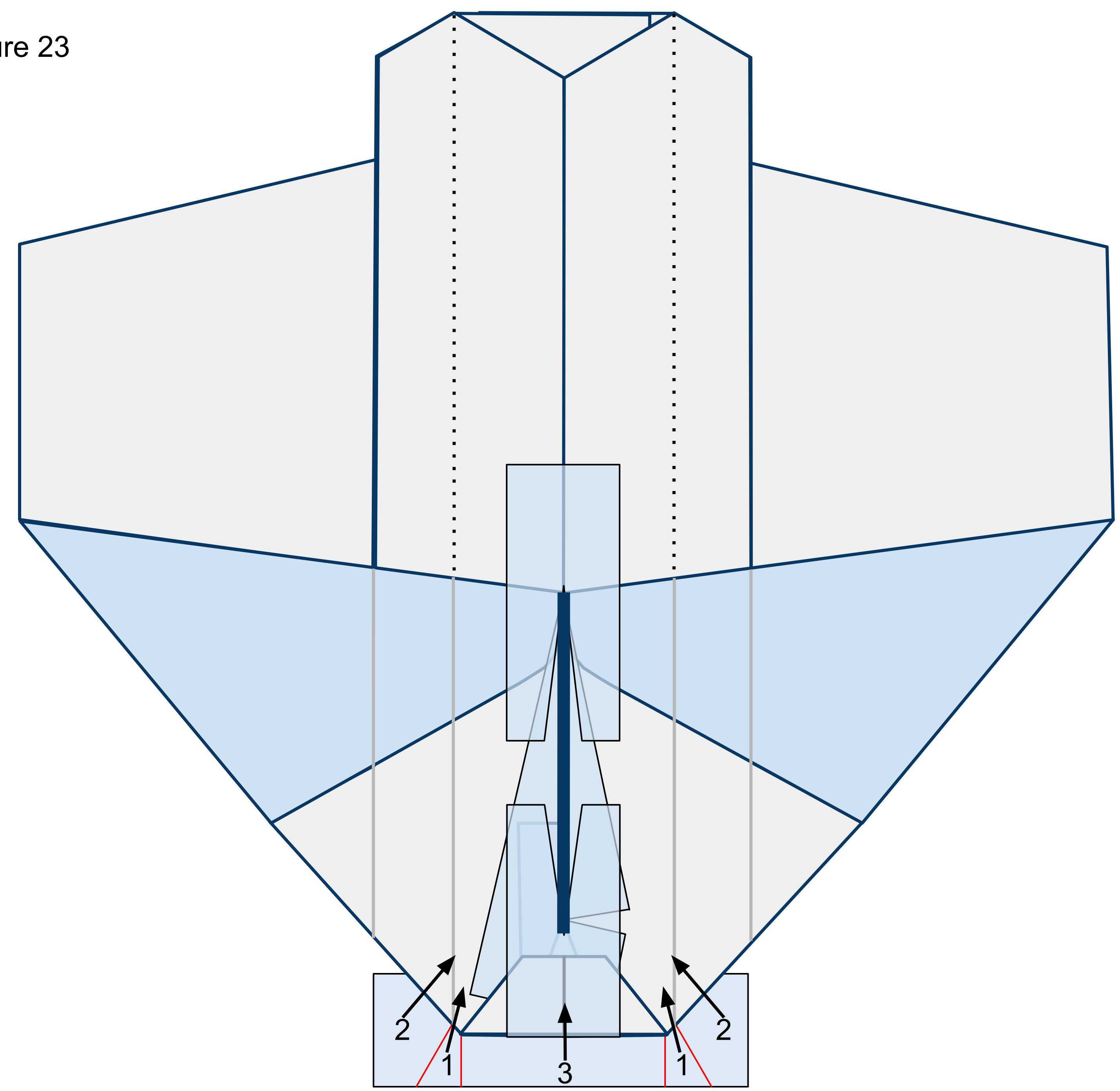
Figure 22



23

Flip the plane over so you can see the top side. Fold the flaps of tape, in number order, up and in as shown in Figure 23 below. They will overlap each other once folded.

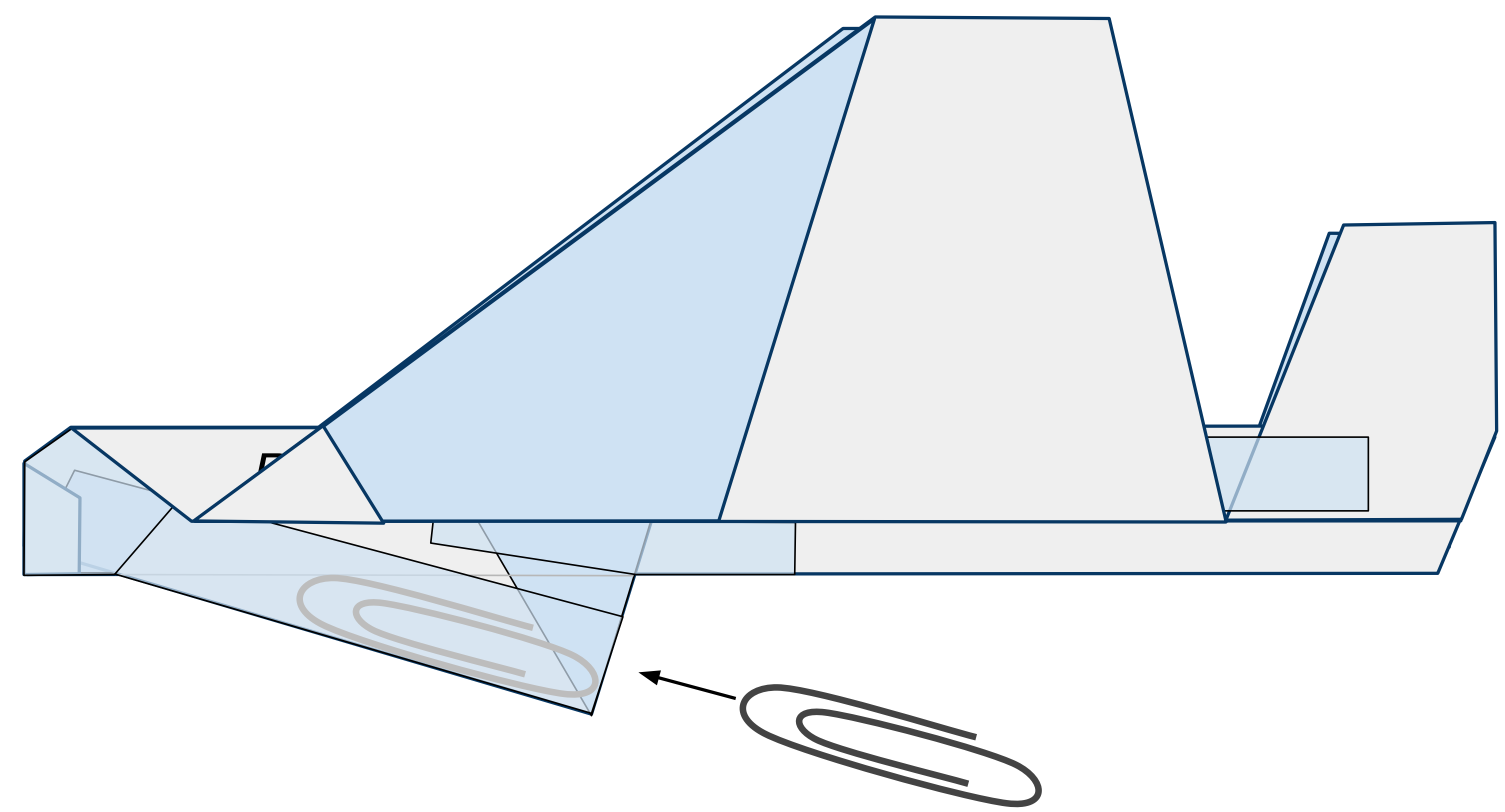
Figure 23



24

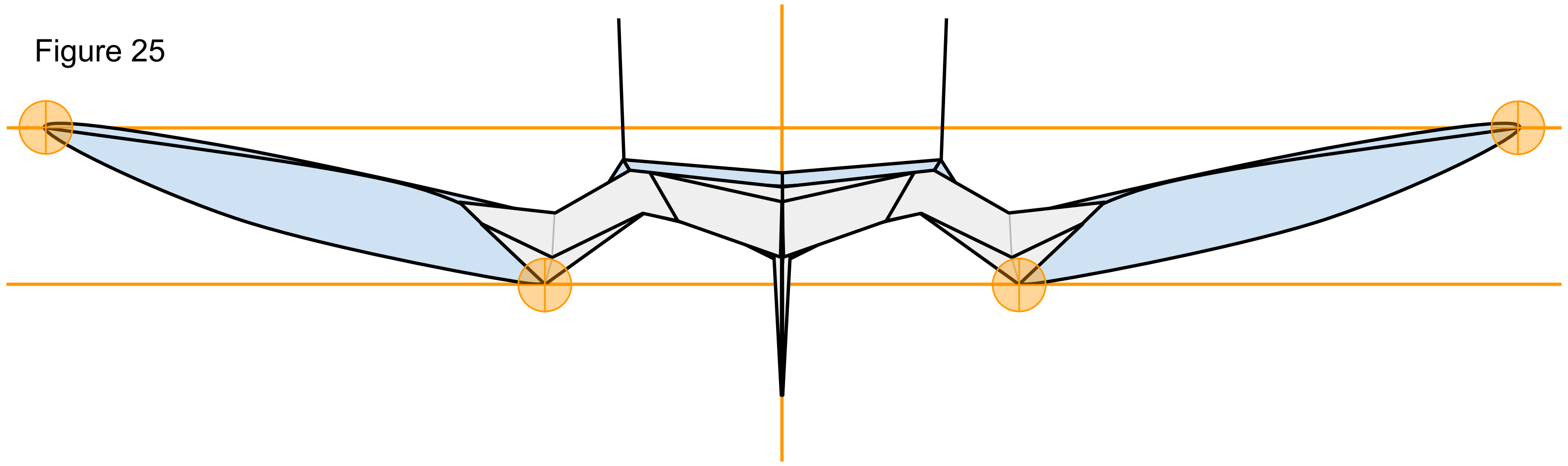
Turn the nose to the left refold the wings keeping the tails up. Insert a #1 size paper clip ***just*** inside the keel through the opening behind it as shown in Figure 24 below. Then seal the opening with a piece of tape.

Figure 24



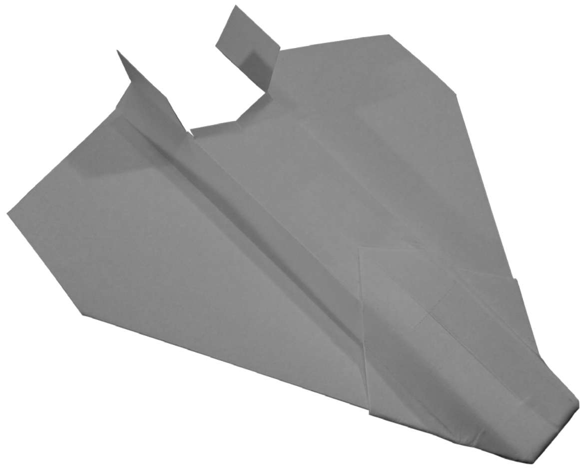
Now that you have constructed the plane ensure that it has the proper shape. Take a look at your plane head-on from the nose. The plane should be symmetrical. Take a look at Figure 25 below. Note the points of interest. Note the curved shape of the paper on the underside of the wings. To achieve this, carefully stick one side of a pair of scissors up into the pocket on the underside of the wing from behind and gently mold the paper into the proper shape. Don't crease it.

Figure 25



We are almost ready to launch! I recommend taking a quick look at the "PREFLIGHT" section to get some good tips. All paper planes fly erratically on their first flights. They require refining to make them great. To do this, take a look at the "FINE-TUNING" section.

P12 RAPTOR



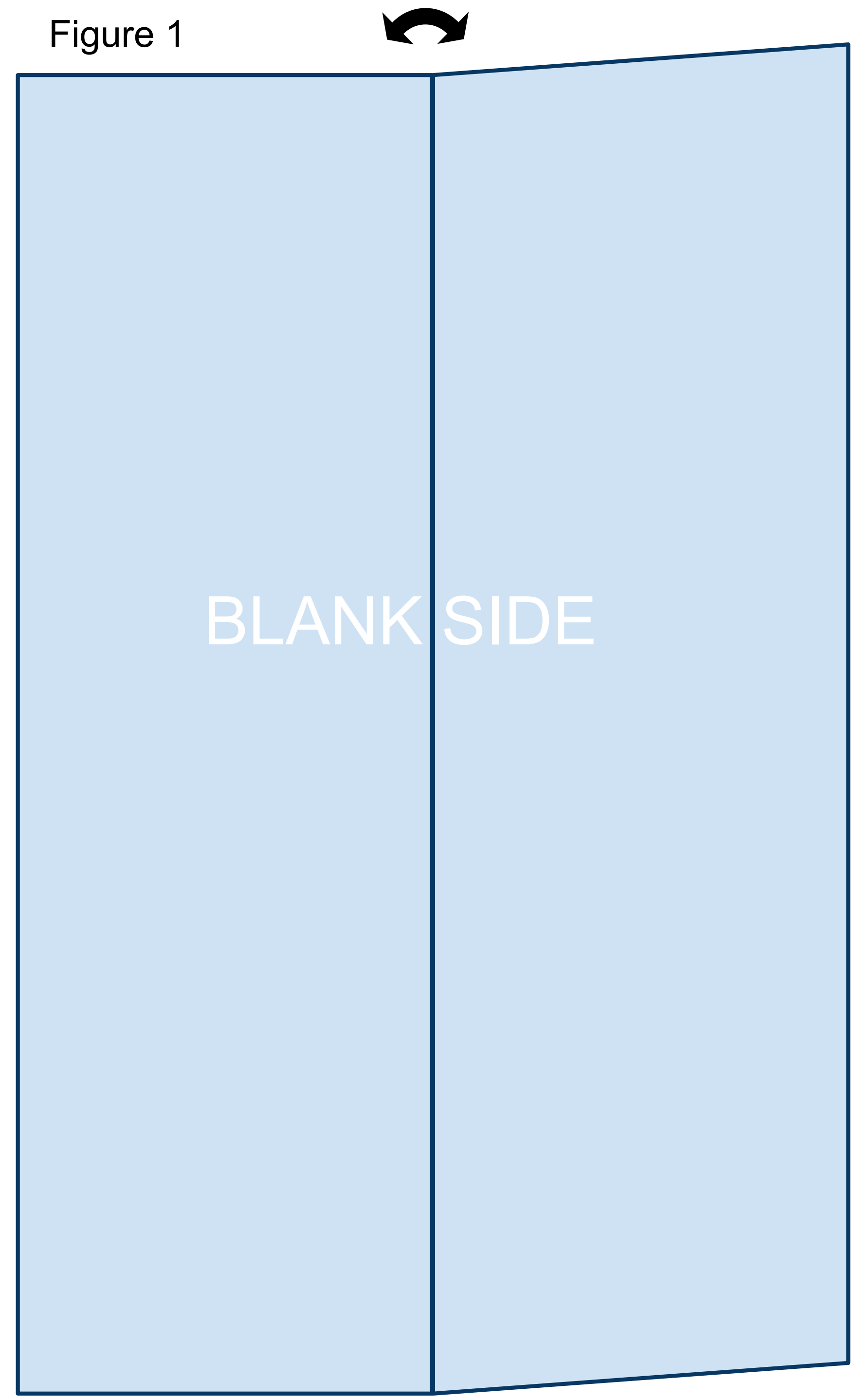
The printable template linked below is 4.25in x 5.5in. That is equivalent to half of an 8.5in x 11in sheet of printer paper. While the designs in this book will fly great with regular printer paper, I recommend 32lb/ream paper. The stronger paper will hold its shape better during flight allowing for higher and more exciting flights. Once you become comfortable making the planes from this template you can simply cut regular printer paper in half in order to start. Print the PDF template located at the bottom of the web page: https://sites.google.com/site/afspaperairplanes/p12

In case your reading device is not connected to the internet, doing a Google search for "AFS Paper Airplanes" should get you to the webpage.

1

Cut out the template and place the paper on the table so that the printed information is on the underside and you are looking at the blank side. Fold the paper in half the long way and unfold leaving a visible crease as shown in Figure 1 below.

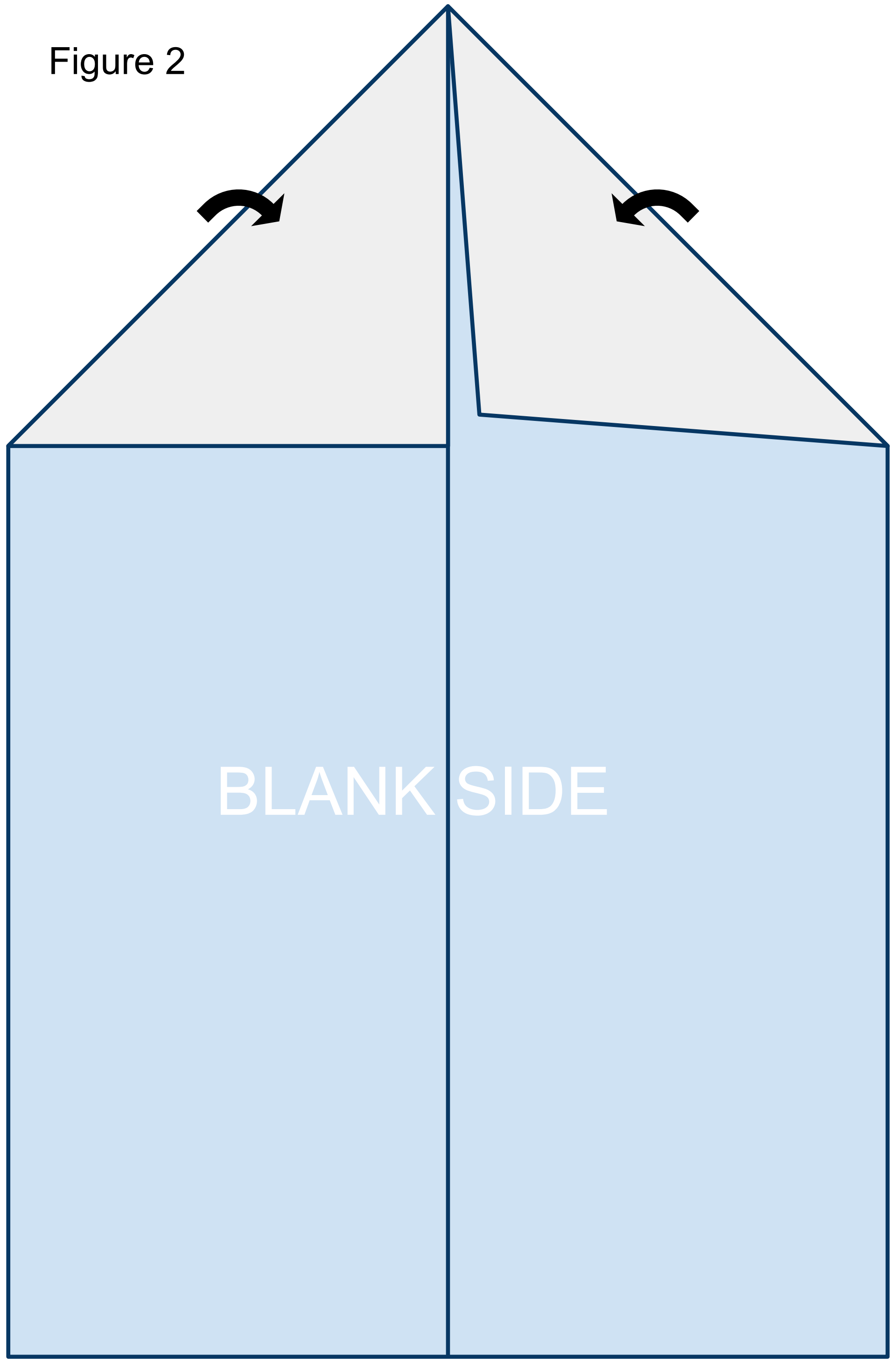
Figure 1



2

Fold the leading corners back. Line them up along the crease you created in step 1 as shown in Figure 2 below.

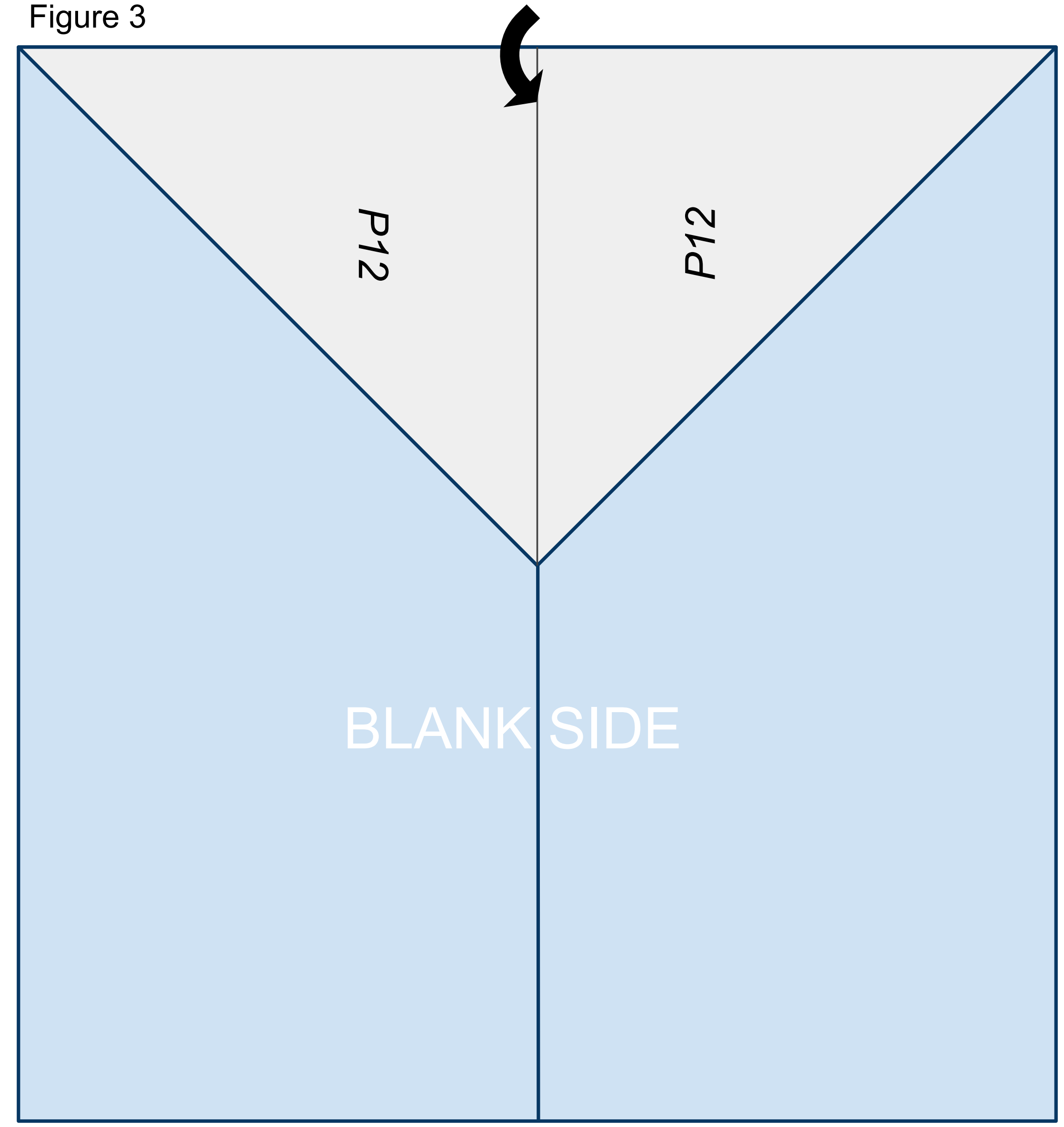
Figure 2



3

Fold the nose of the plane created in step 2 back along the solid line printed on the front of the paper so that it lines up on the center crease created in step 1 as shown in Figure 3 below.

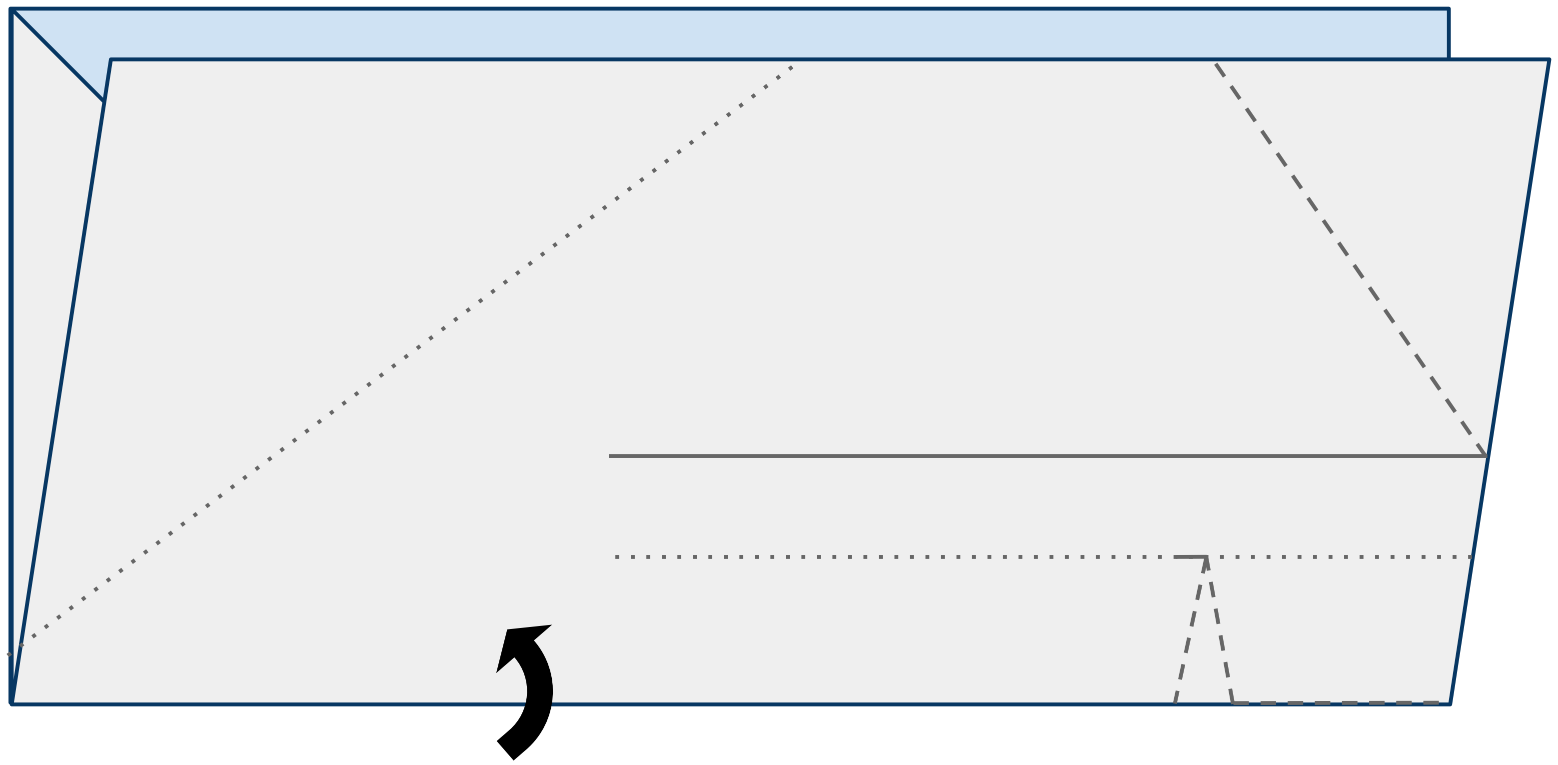
Figure 3



4

Turn the paper to the left (counterclockwise) and fold the paper upward in half hiding the triangle created in the previous steps refolding along the fold from step 1 as shown in Figure 4 below.

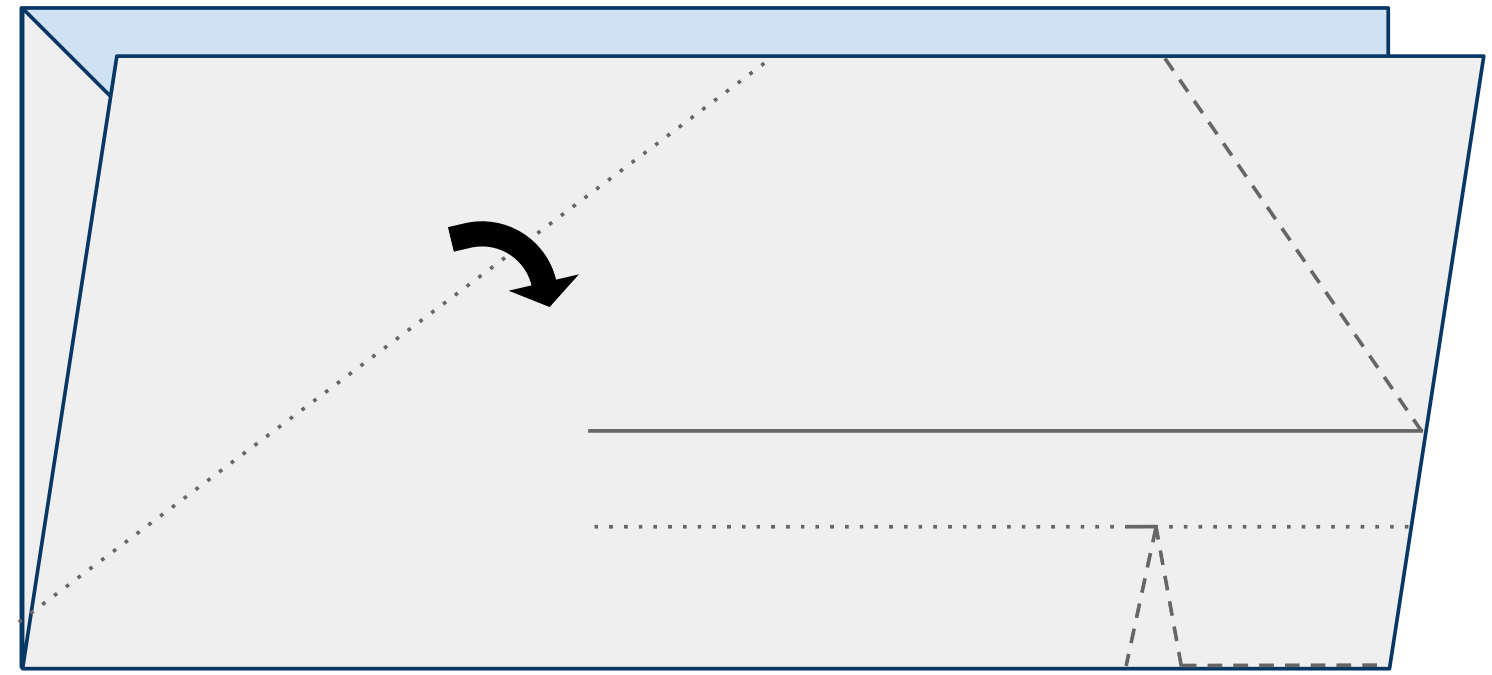
Figure 4



5

Fold the leading edge of the near side of the plane down along the dotted line as shown in Figure 5 below.

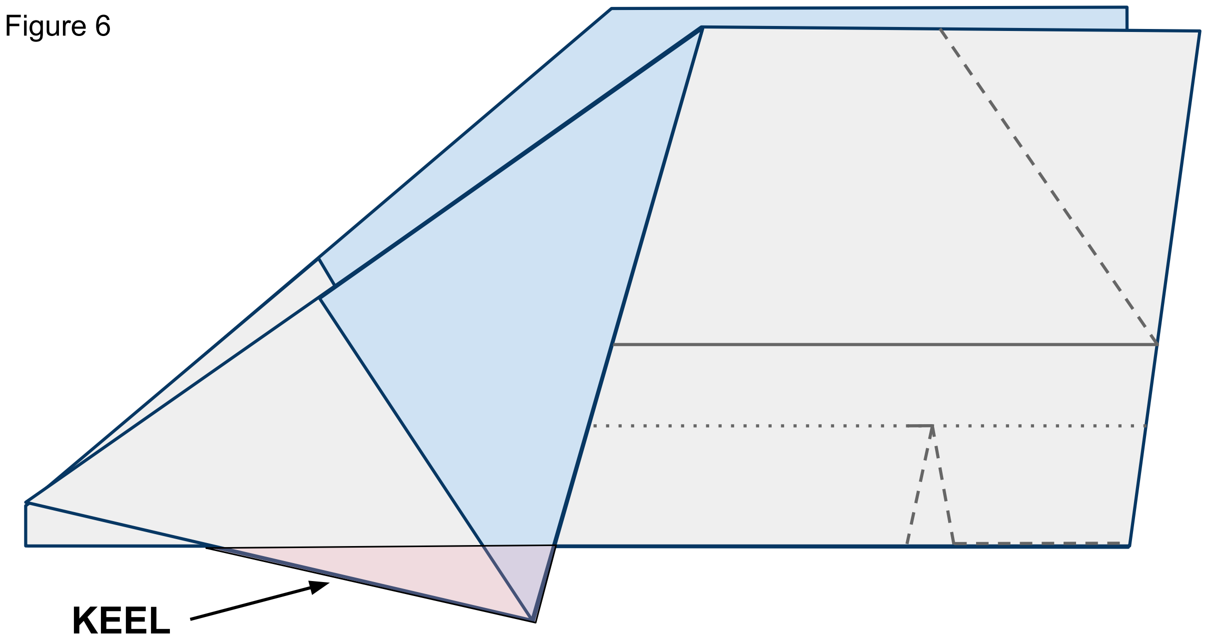
Figure 5



6

Do the same to the other side of the plane. It is important to match these 2 folds exactly so that your plane is symmetrical and will fly straight. The area that is created below the body and designated in pink will be called the keel.

Figure 6



7

Place tape along the keel as shown in Figure 7 below and cut a slit in the tape along the red line. Now fold the tape around and under both sides of the plane and keel securing them together. See Figure 7b for what the far side of the plane looks like.

Figure 7

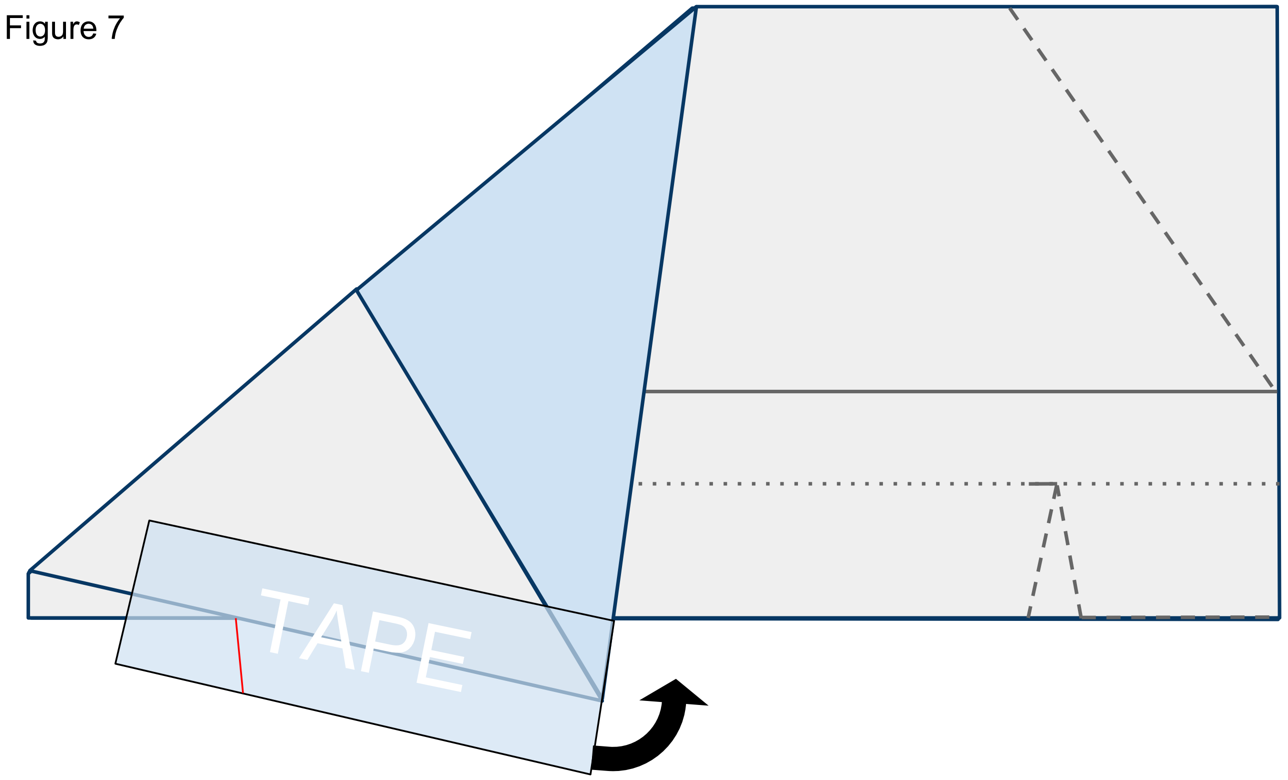
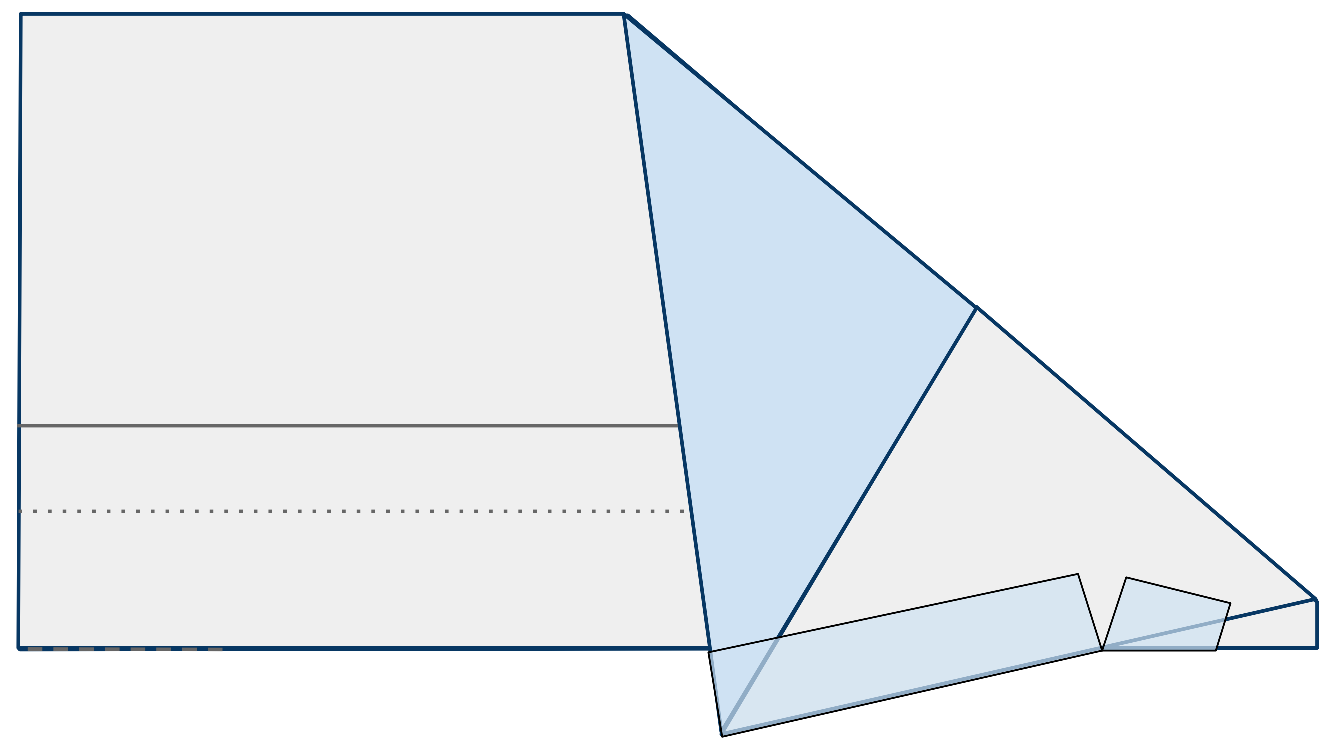


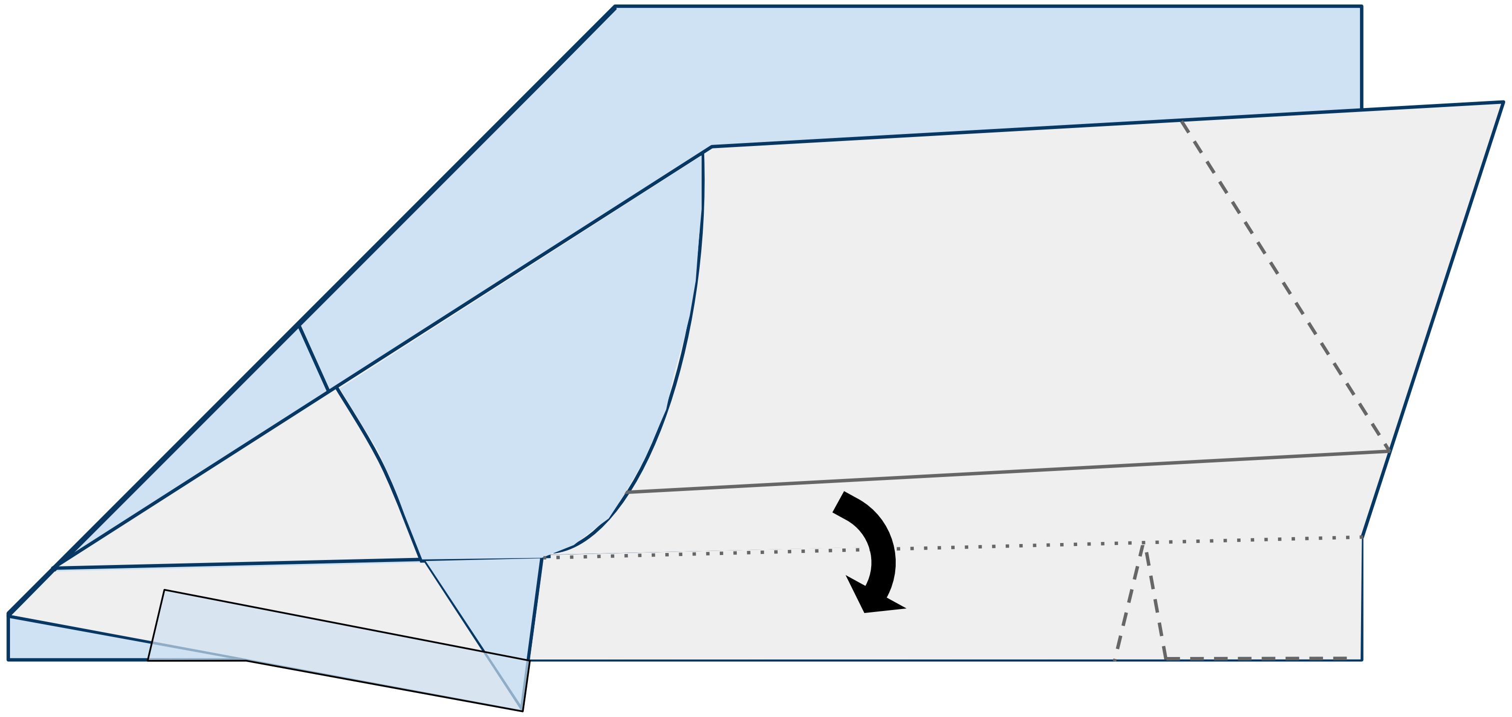
Figure 7b



8

Fold down the near side wing along the dotted line as shown in Figure 8 below.

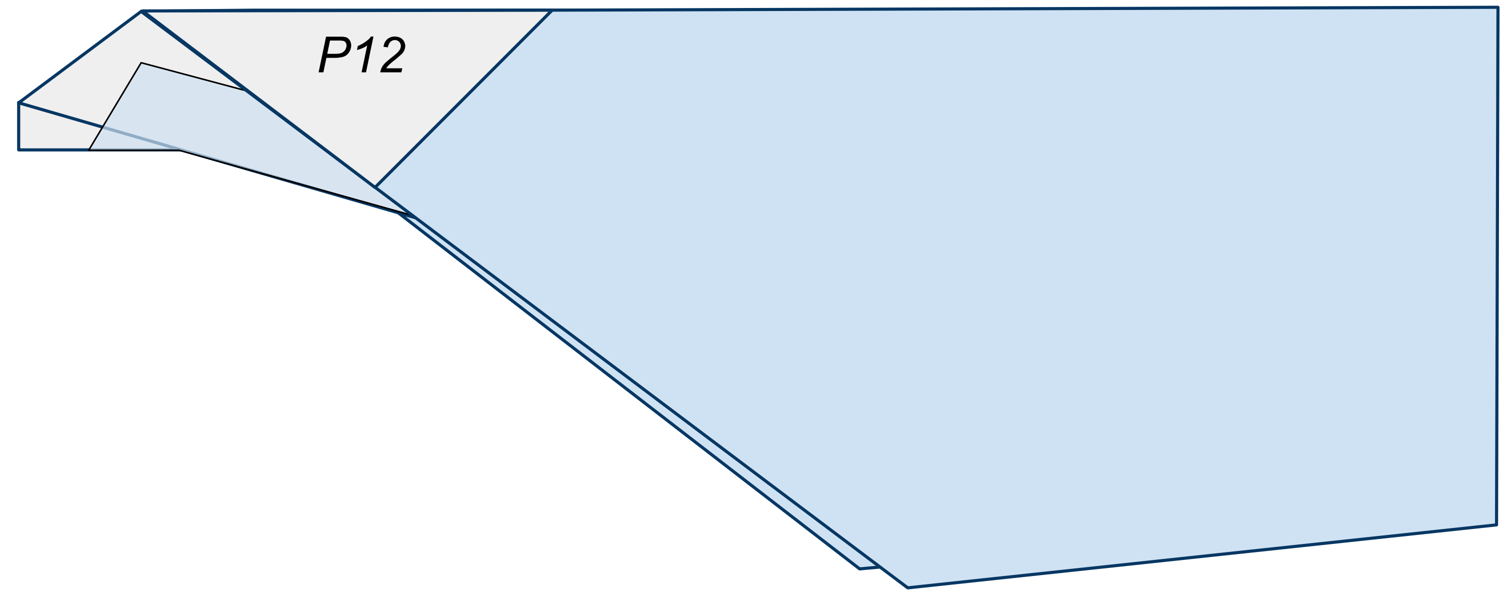
Figure 8



9

Fold down the far side wing along the dotted line also as shown in Figure 9 below. Insure that these folds are as symmetrical as possible. If you need to choose, always choose symmetry over following the lines printed on the paper.

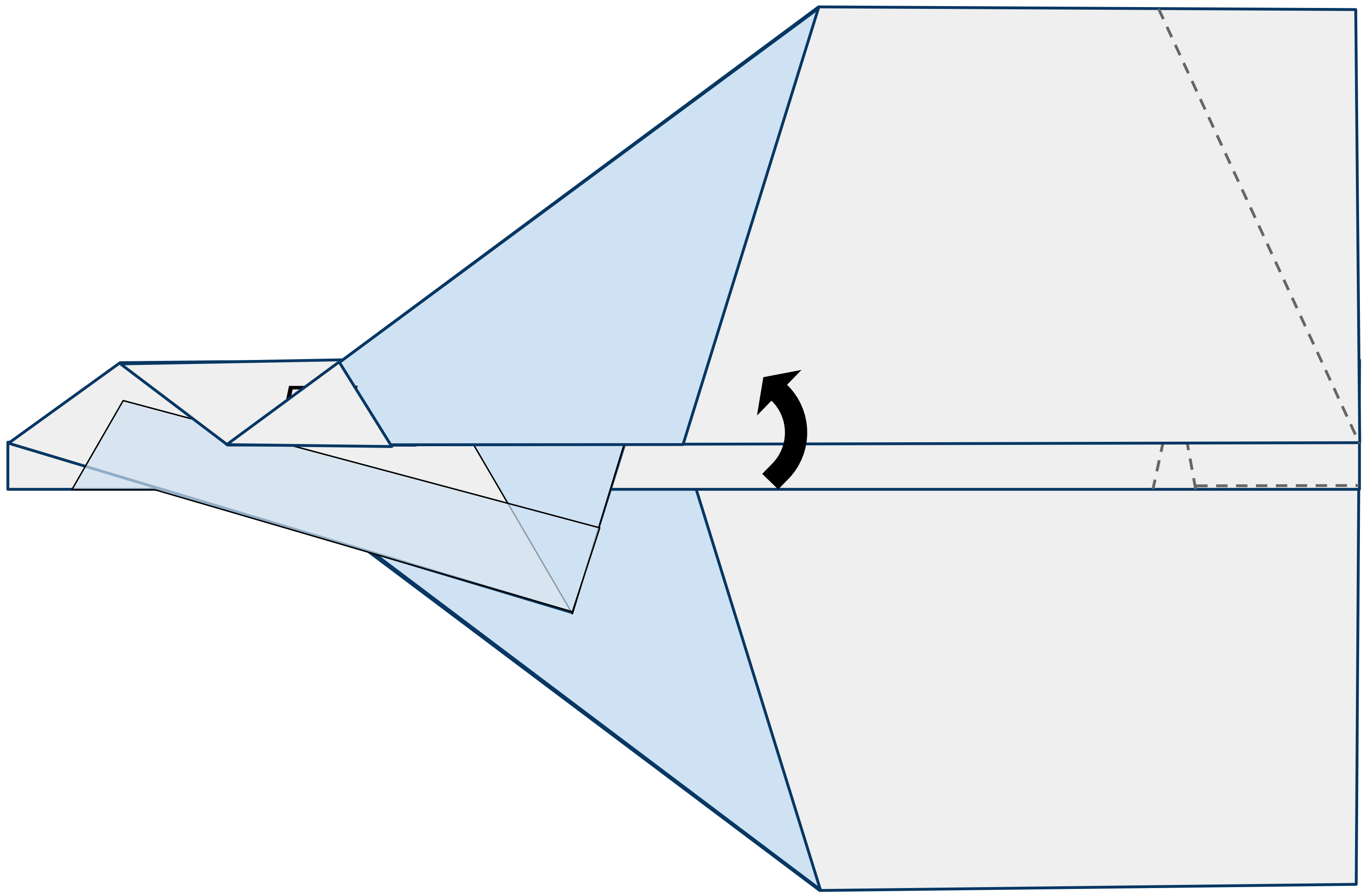
Figure 9



10

Fold the near side wing up along the printed solid line as shown in Figure 10 below.

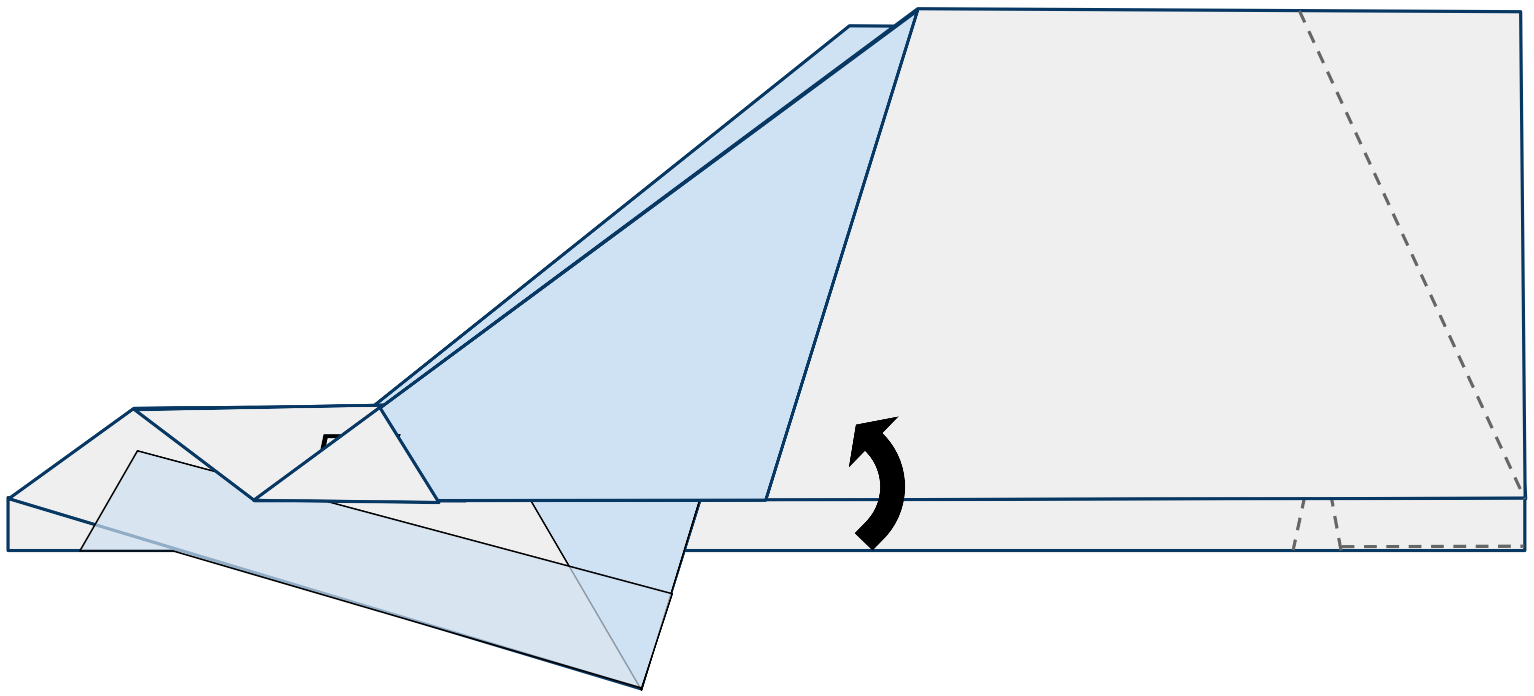
Figure 10



11

Do the same to the other side. Fold the far side wing up along the sold line on the other side of the plane as shown in Figure 11 below.

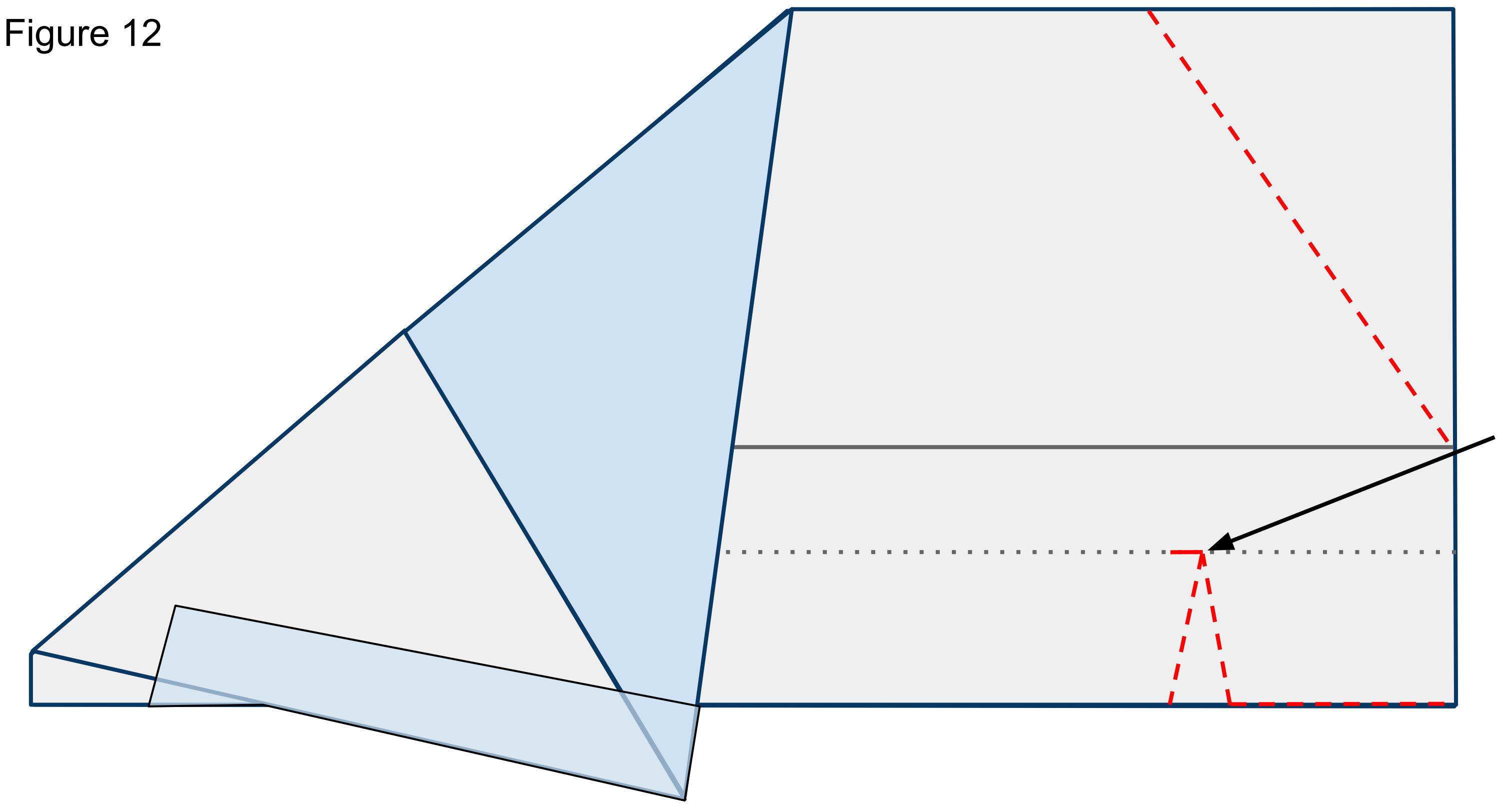
Figure 11



12

Open up the wings that were folded in steps 8-11 and lay the plane flat on your surface as shown in Figure 12 below. Cut thought both layers of paper along the red dashed lines. Don't forget the small red slit that creates the flaps.

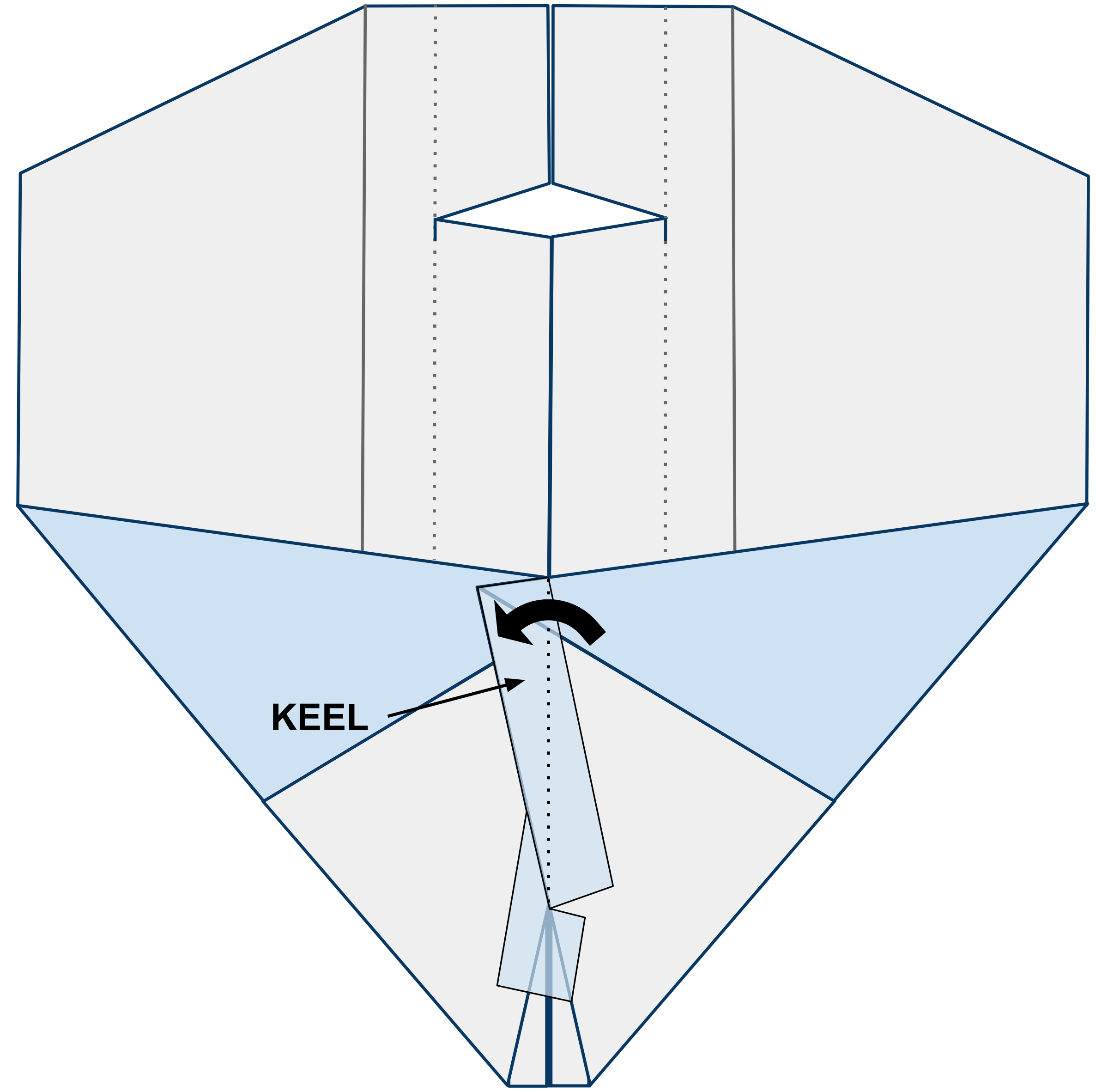
Figure 12



13

Open the plane up flat on your work surface with the nose toward you and looking at the underside of the body. When you do this the keel will want to stand up off the table. Bend it to the left along the center line and give it a strong crease as shown in Figure 13 below.

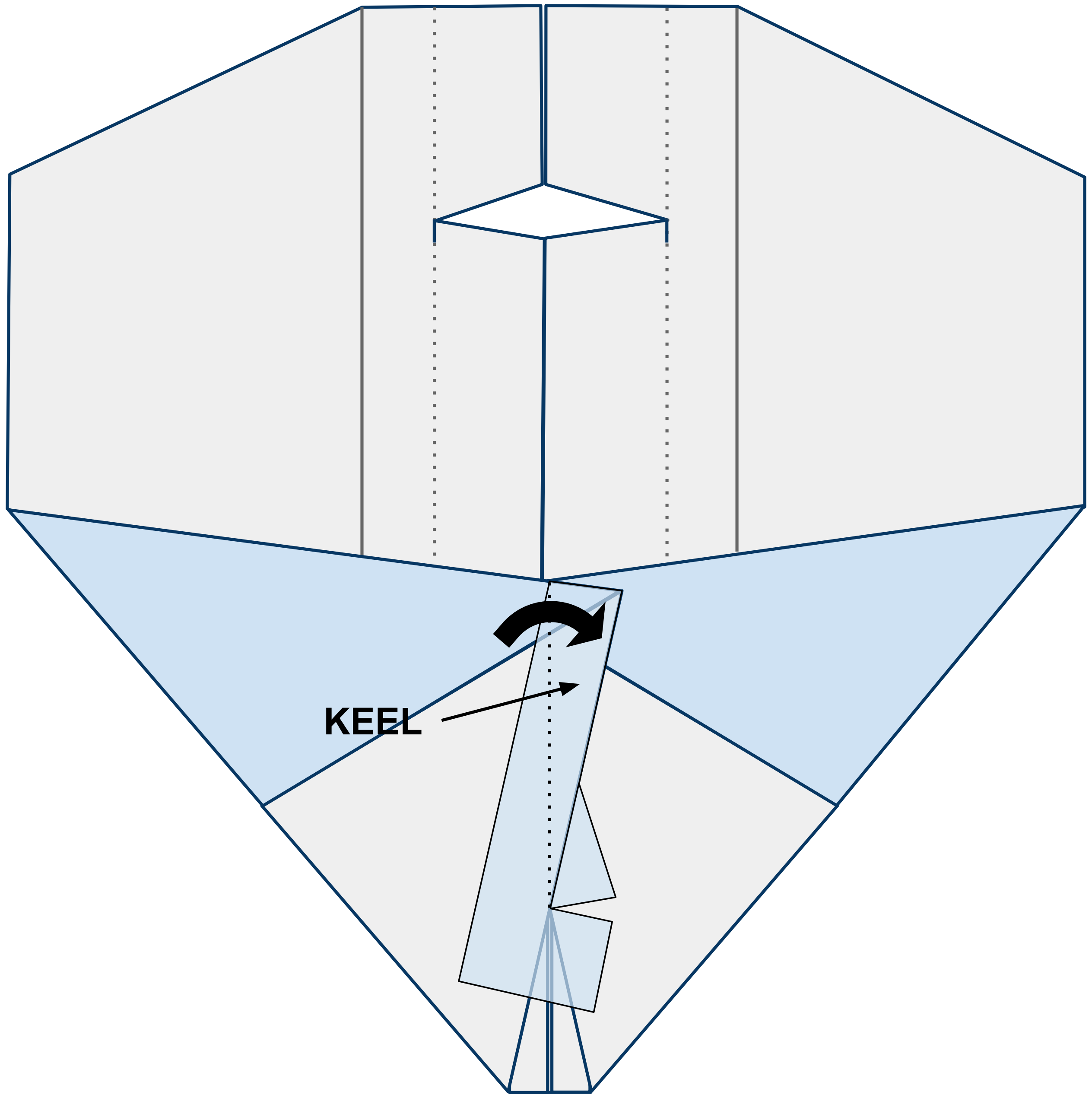
Figure 13



14

Now fold the keel to the right side creating a strong crease as shown in Figure 14 below. After making this crease, allow the keel to remain upright for the remainder of construction. This is where you will hold the plane when launching it.

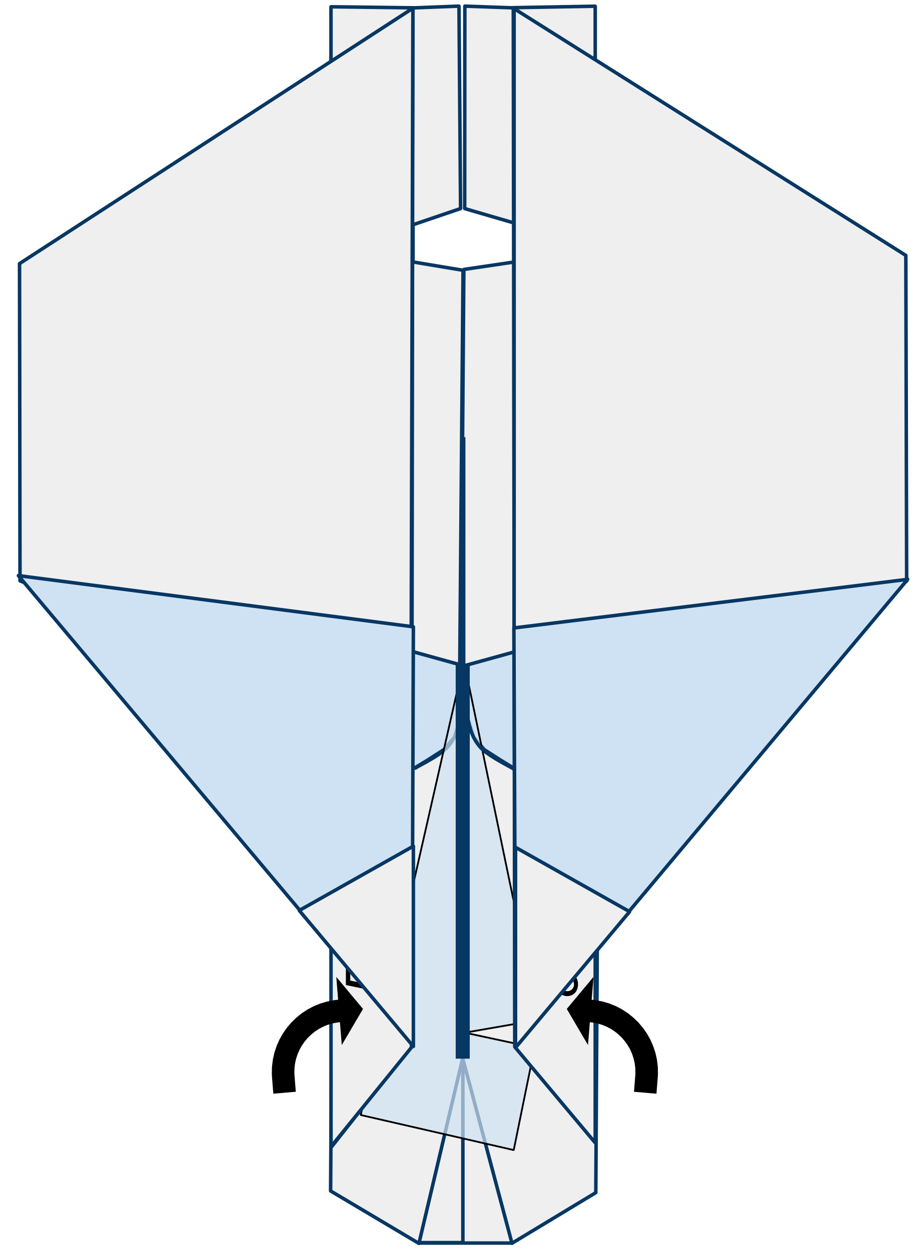
Figure 14



15

Allowing the keel to remain vertical, bring the wings back in along their creases created in steps 8-11 as shown in Figure 15 below. The keel is represented by the thick **blue** line.

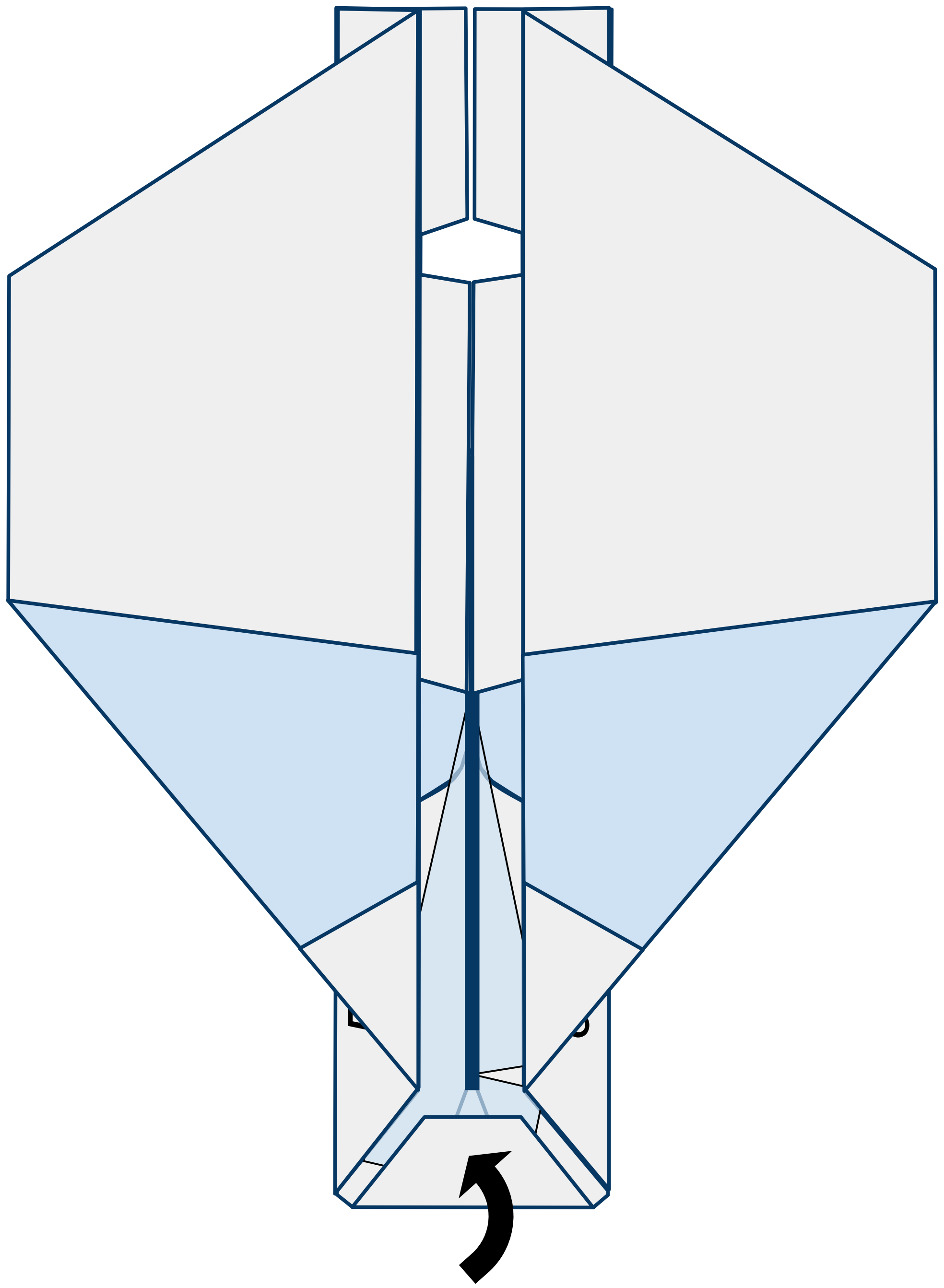
Figure 15



16

Fold the nose of the plane up and in as shown in Figure 16 below.

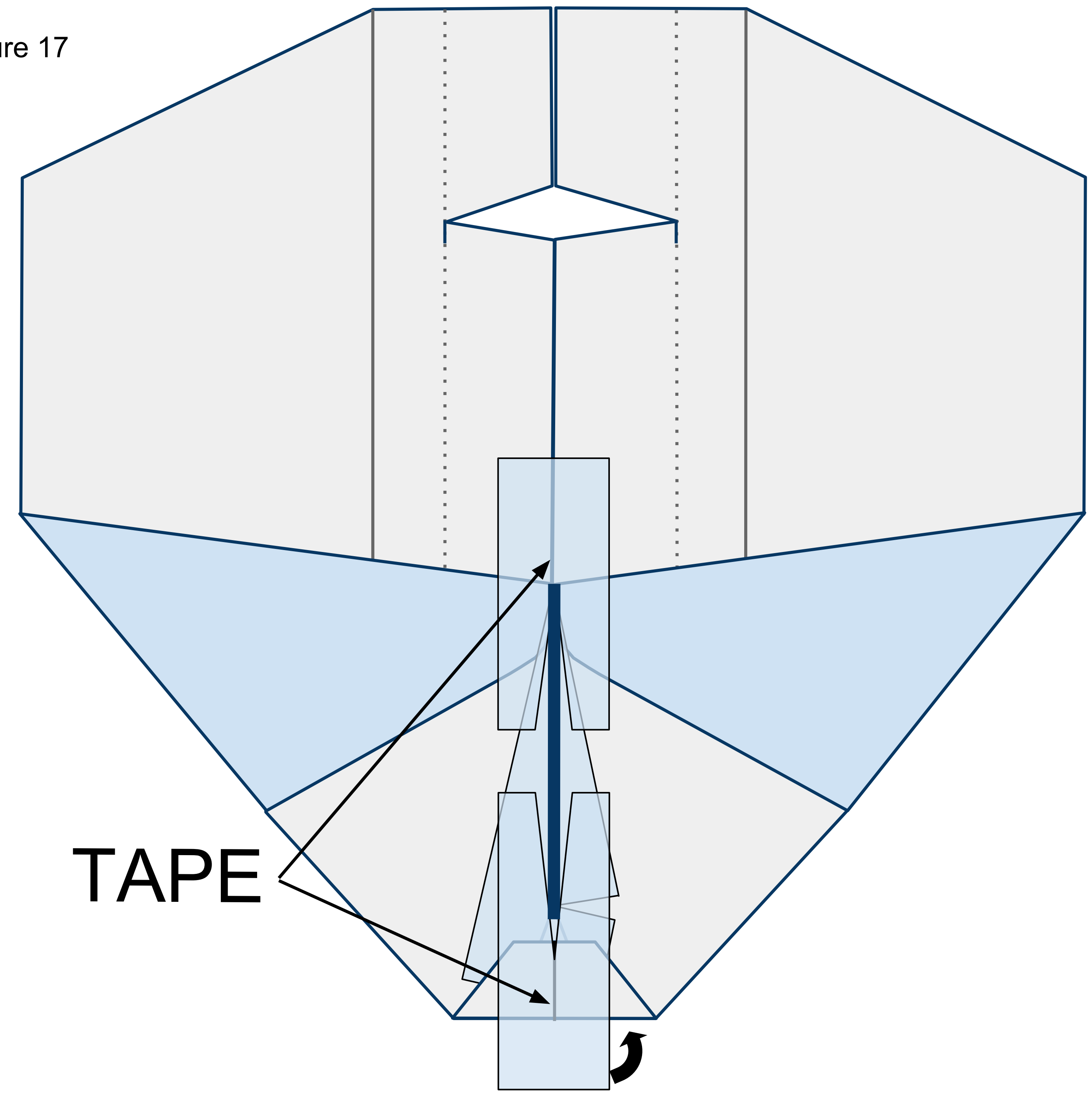
Figure 16



17

Unfold the wings flat. Place 2 pieces of tape with slices cut in them on the underside of the plane in front of and behind the keel as shown in Figure 17 below. The keel is shown standing vertically in the picture. Wrap the tape around the nose to the top side of the plane.

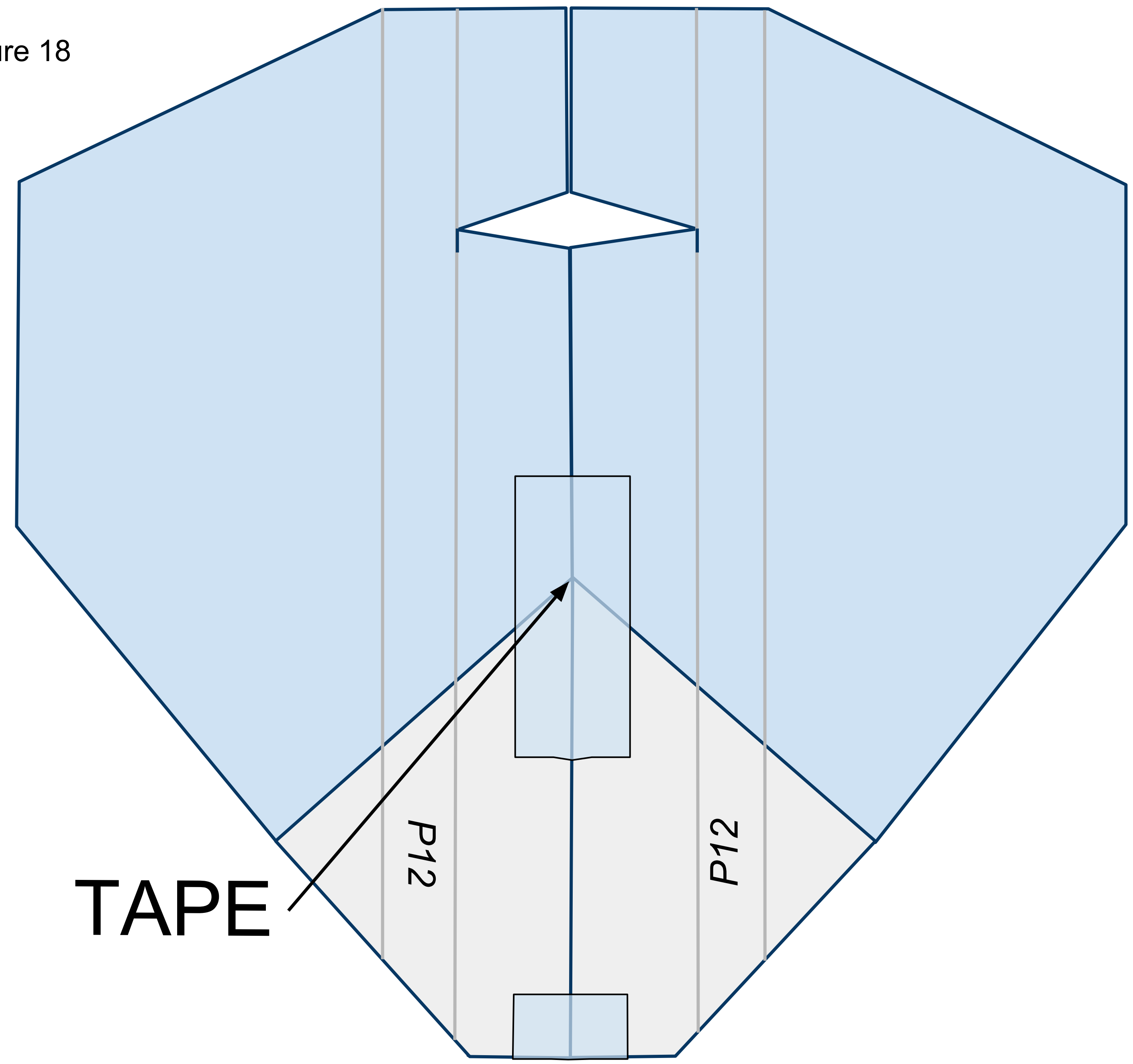
Figure 17



18

Turn the plane over laying the keel to one side. Place tape along the fuselage as shown in Figure 18 below.

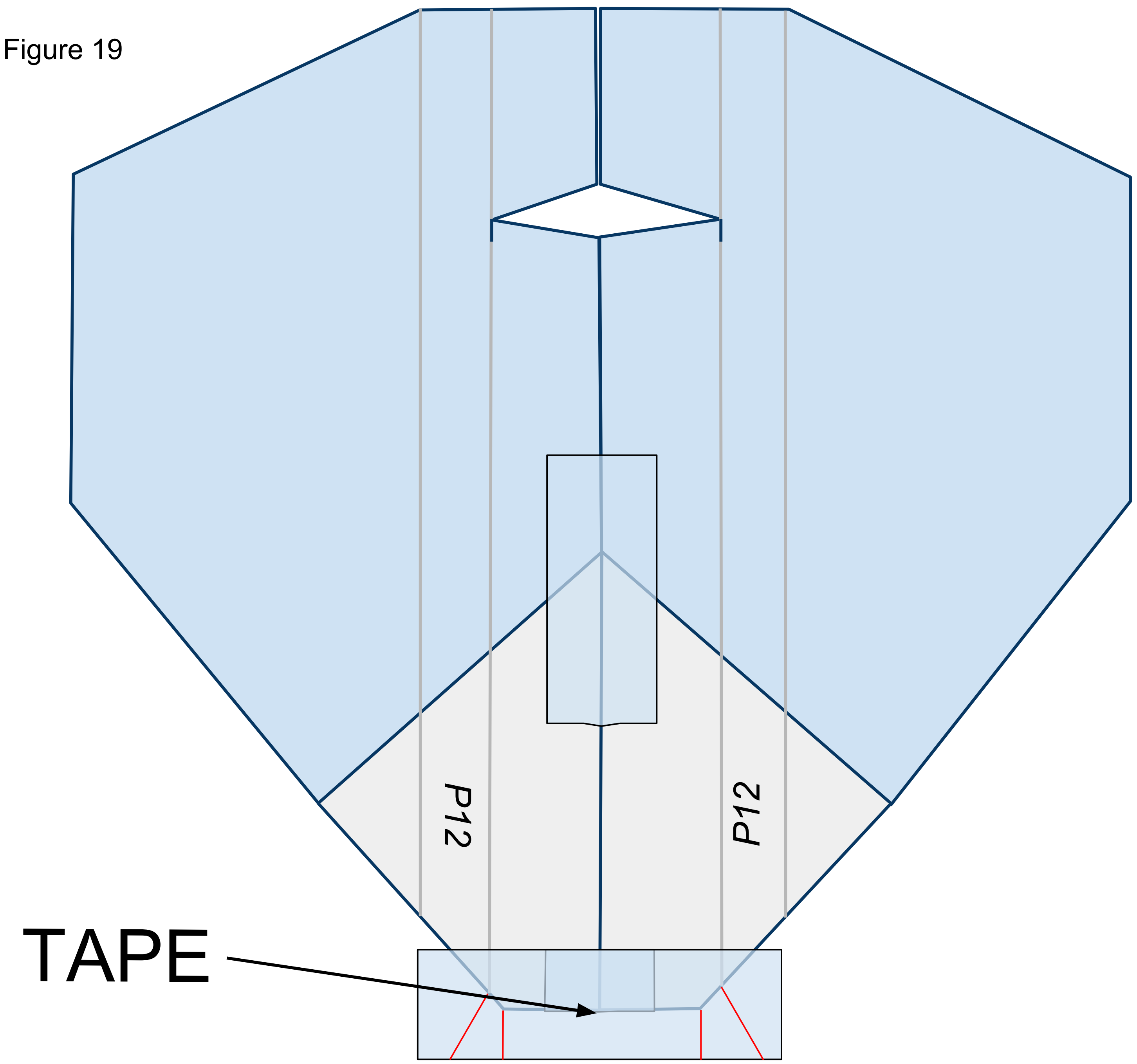
Figure 18



19

Place a piece of tape across the nose of the plane as shown in Figure 19 below. Cut slits along the **red** lines as shown. These slits will help you fold the tape under the plane in the next step.

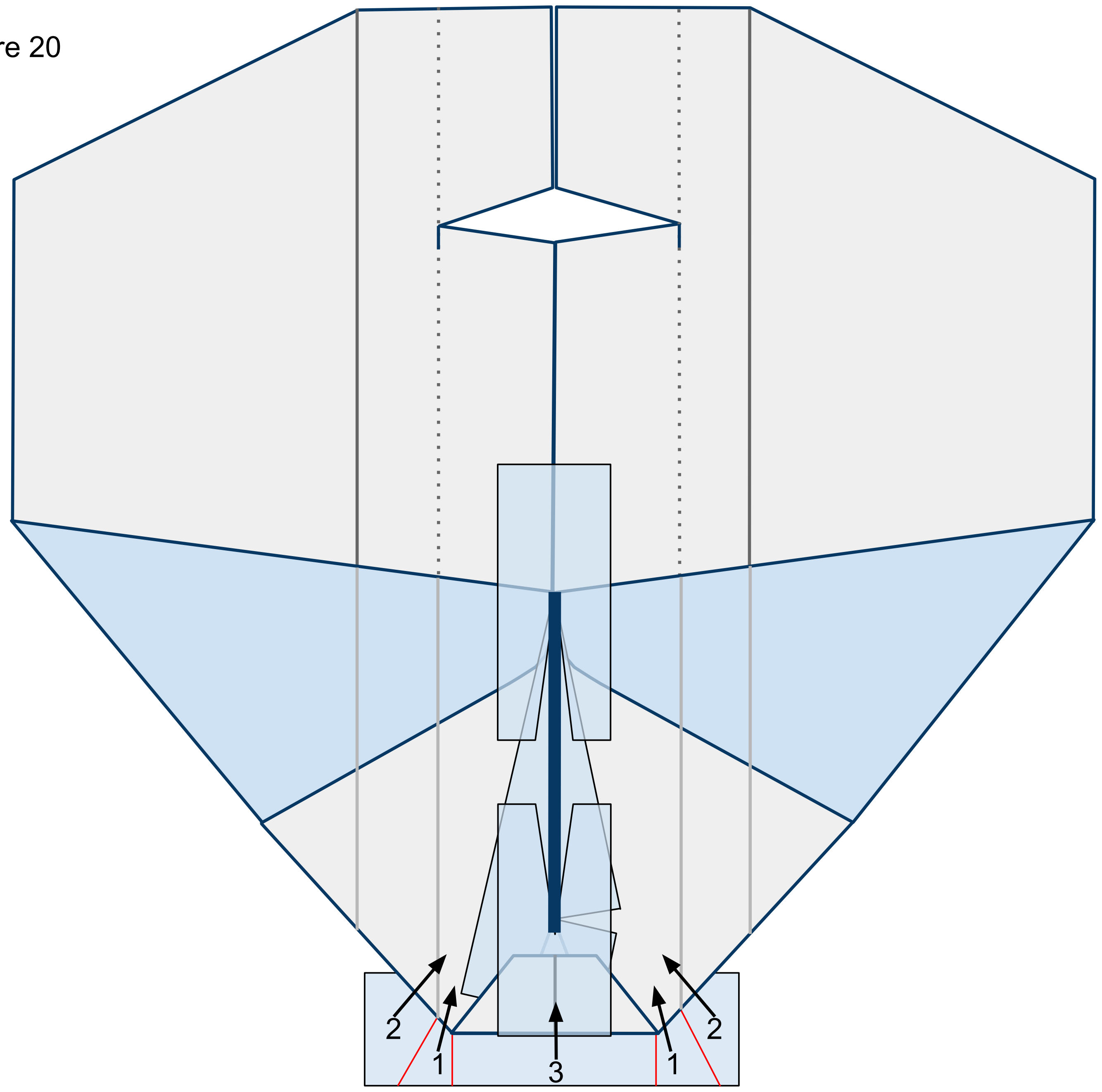
Figure 19



20

Flip the plane over so you can see the top side. Fold the flaps of tape, in number order, up and in as shown in Figure 20 below. They will overlap each other once folded.

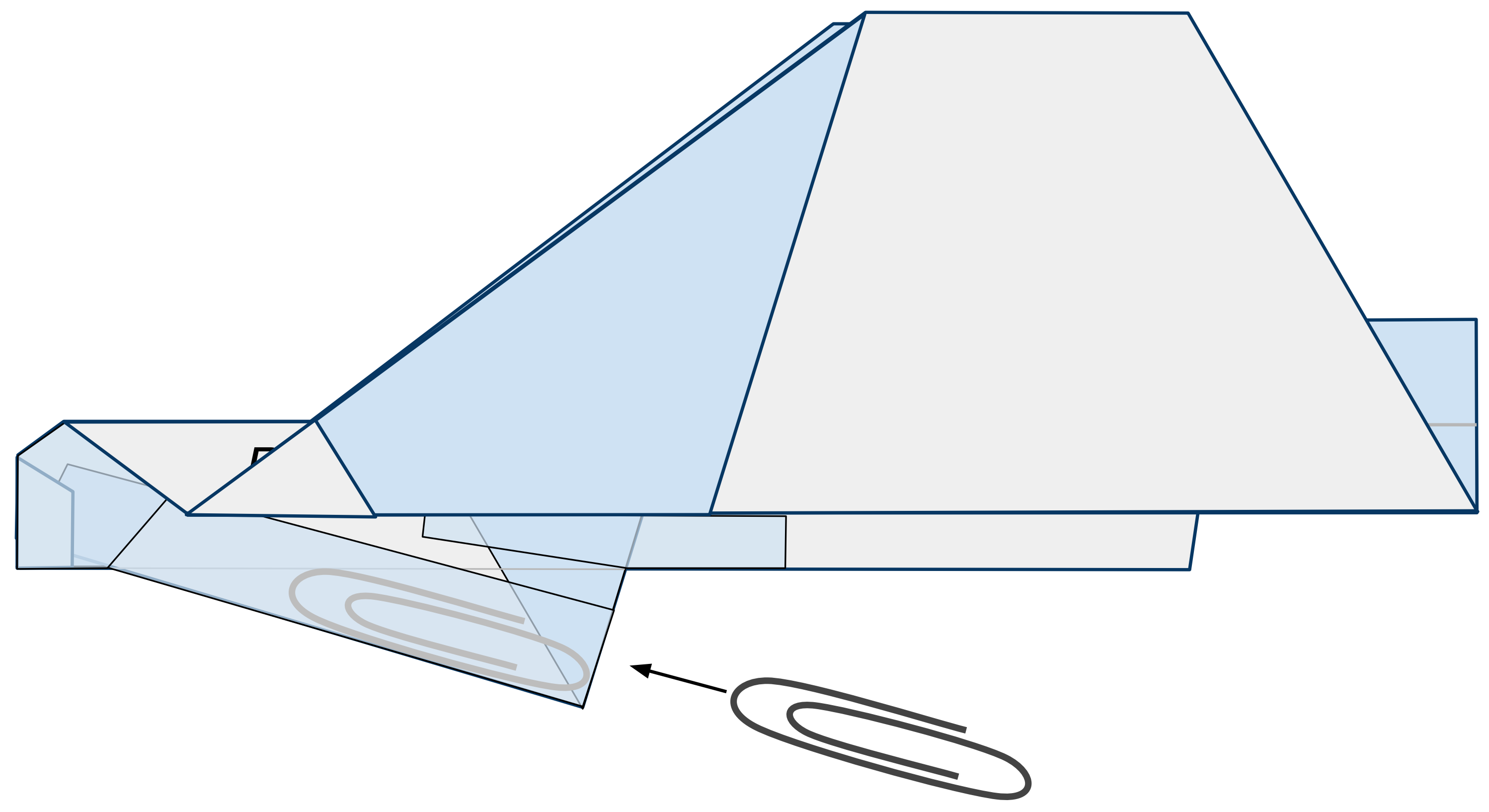
Figure 20



21

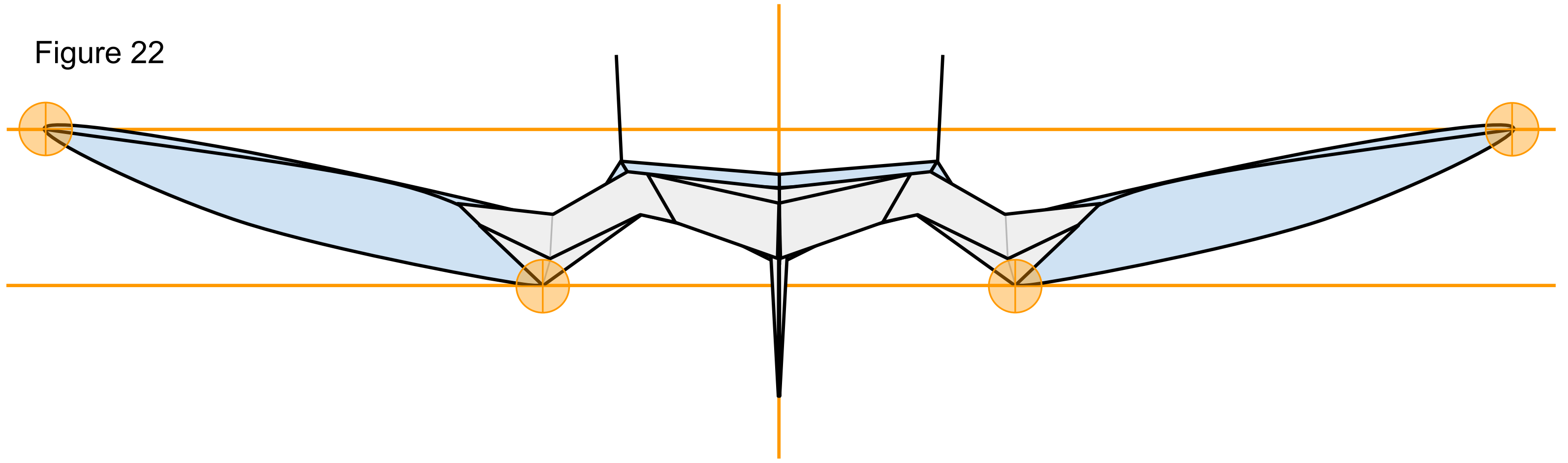
Turn the nose to the left refold the wings keeping the tail up. Insert a #1 size paper clip ***just*** inside the keel through the opening behind it as shown in Figure 21 below. Then seal the opening with a piece of tape.

Figure 21



Now that you have constructed the plane ensure that it has the proper shape. Take a look at your plane head-on from the nose. The plane should be symmetrical. Take a look at Figure 22 below. Note the points of interest. Note the curved shape of the paper on the underside of the wings. To achieve this, carefully stick one side of a pair of scissors up into the pocket on the underside of the wing from behind and gently mold the paper into the proper shape. Don't crease it.

Figure 22



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P13 SPARTAN

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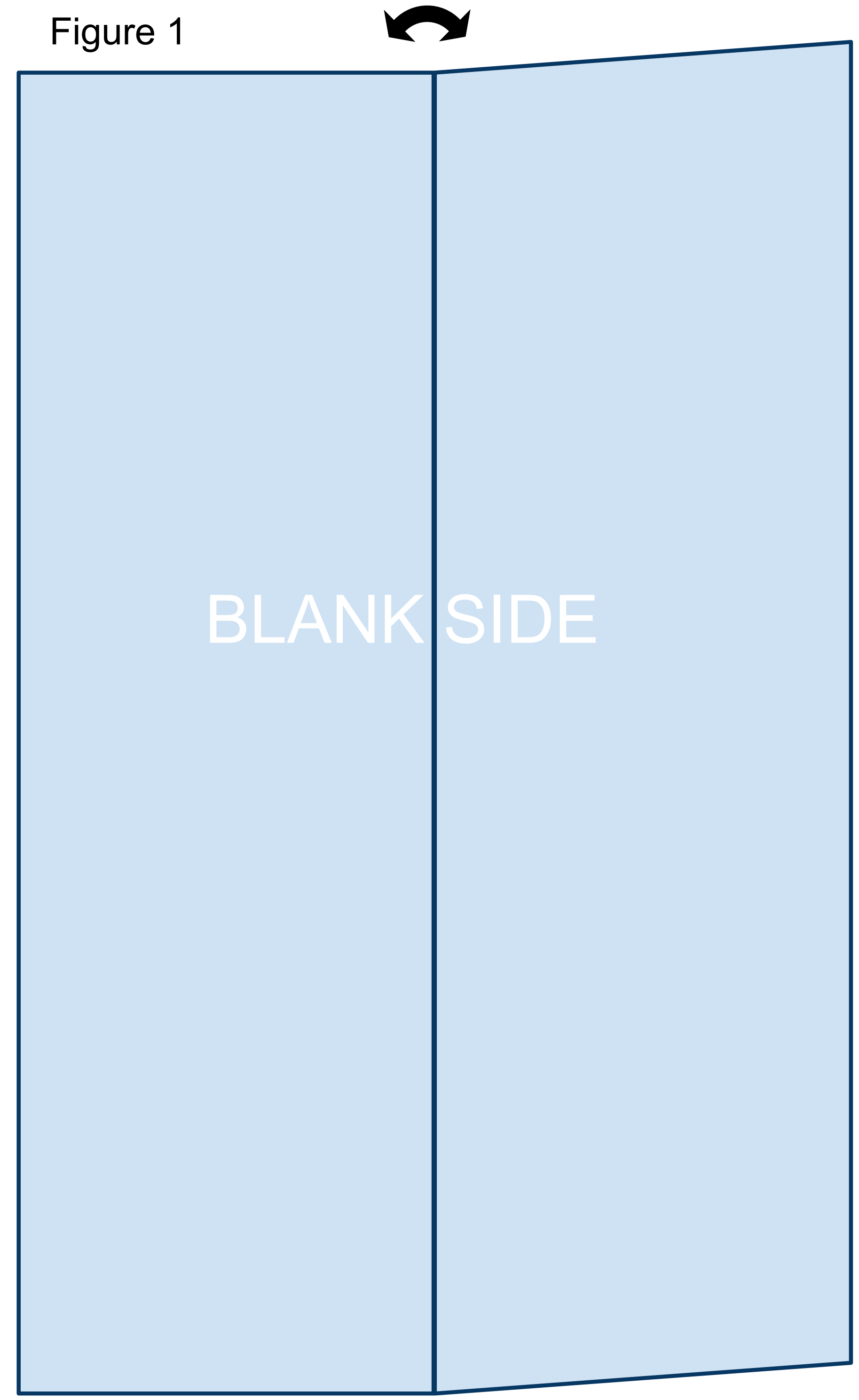
1

Cut out the template and place the paper on the table so that the printed information is on the underside with the printed triangle facing away from you. You will be looking at the blank side. Bring the right side up and fold the paper in half the long way. Unfold leaving a visible crease as shown in Figure 1 below.

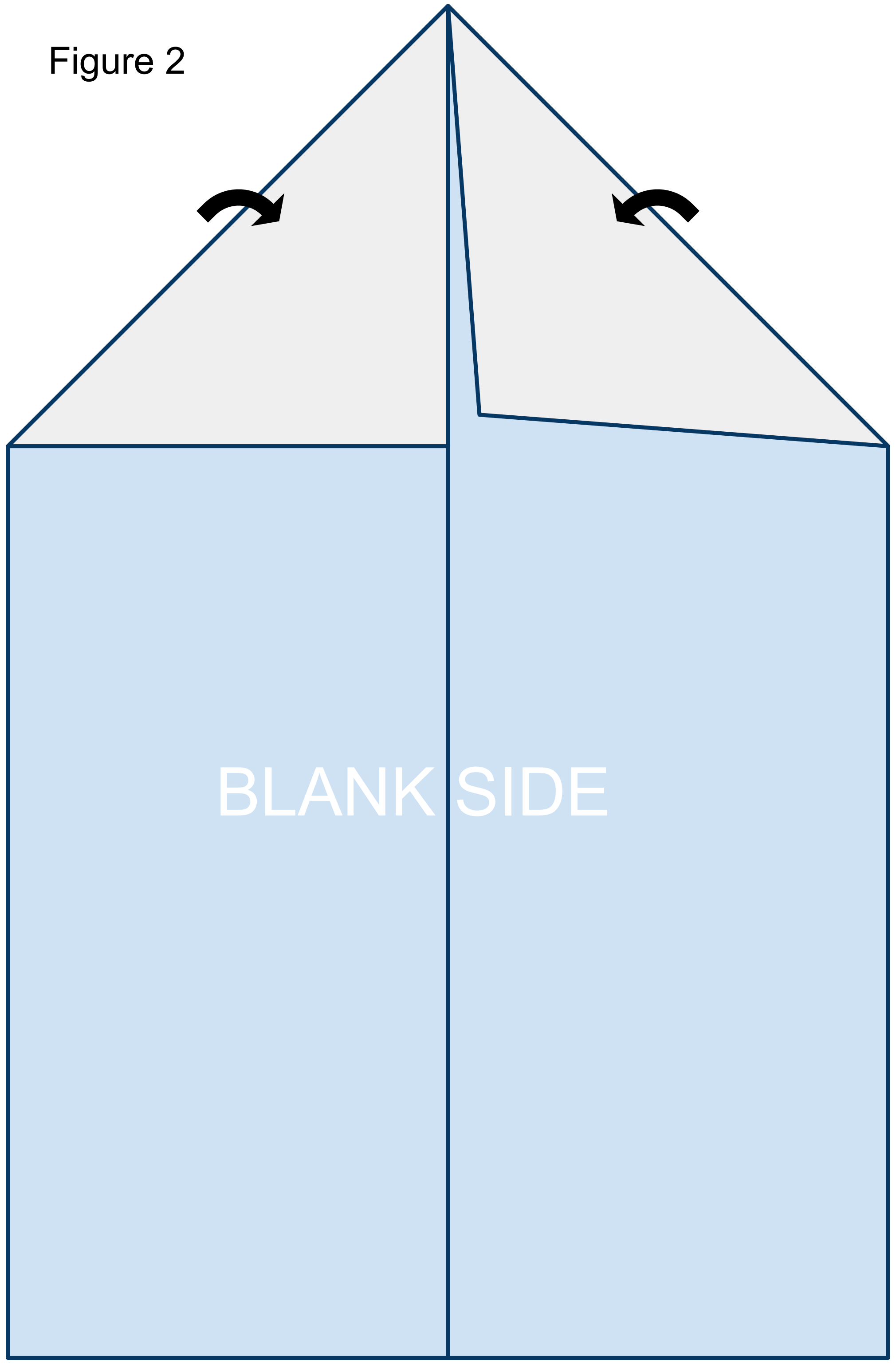
Figure 1



2

Fold the leading corners back. Line them up along the crease you created in step 1 as shown in Figure 2 below.

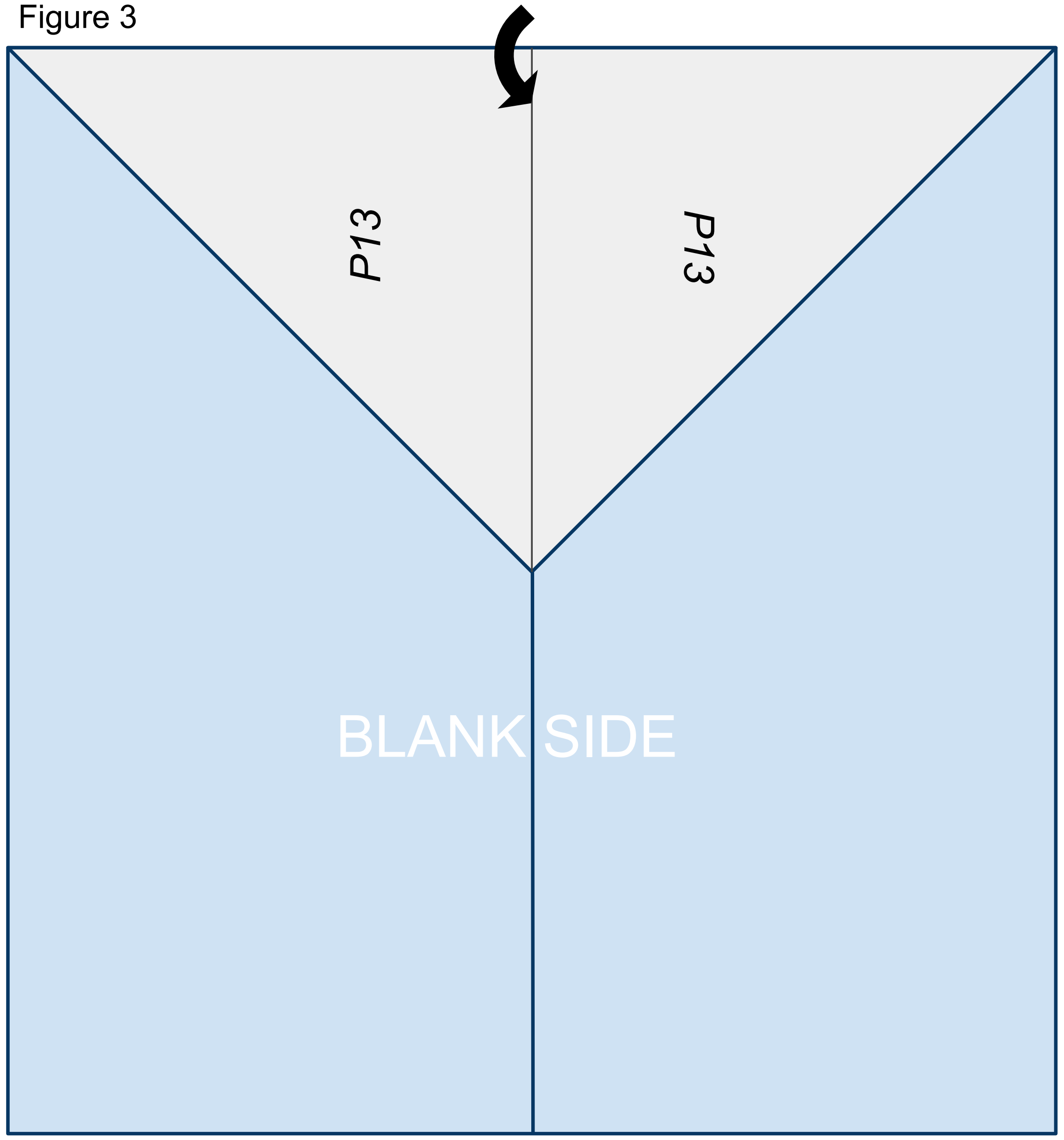
Figure 2



3

Fold the nose of the plane created in step 2 back along the solid line printed on the front of the paper so that it lines up on the center crease created in step 1 as shown in Figure 3 below.

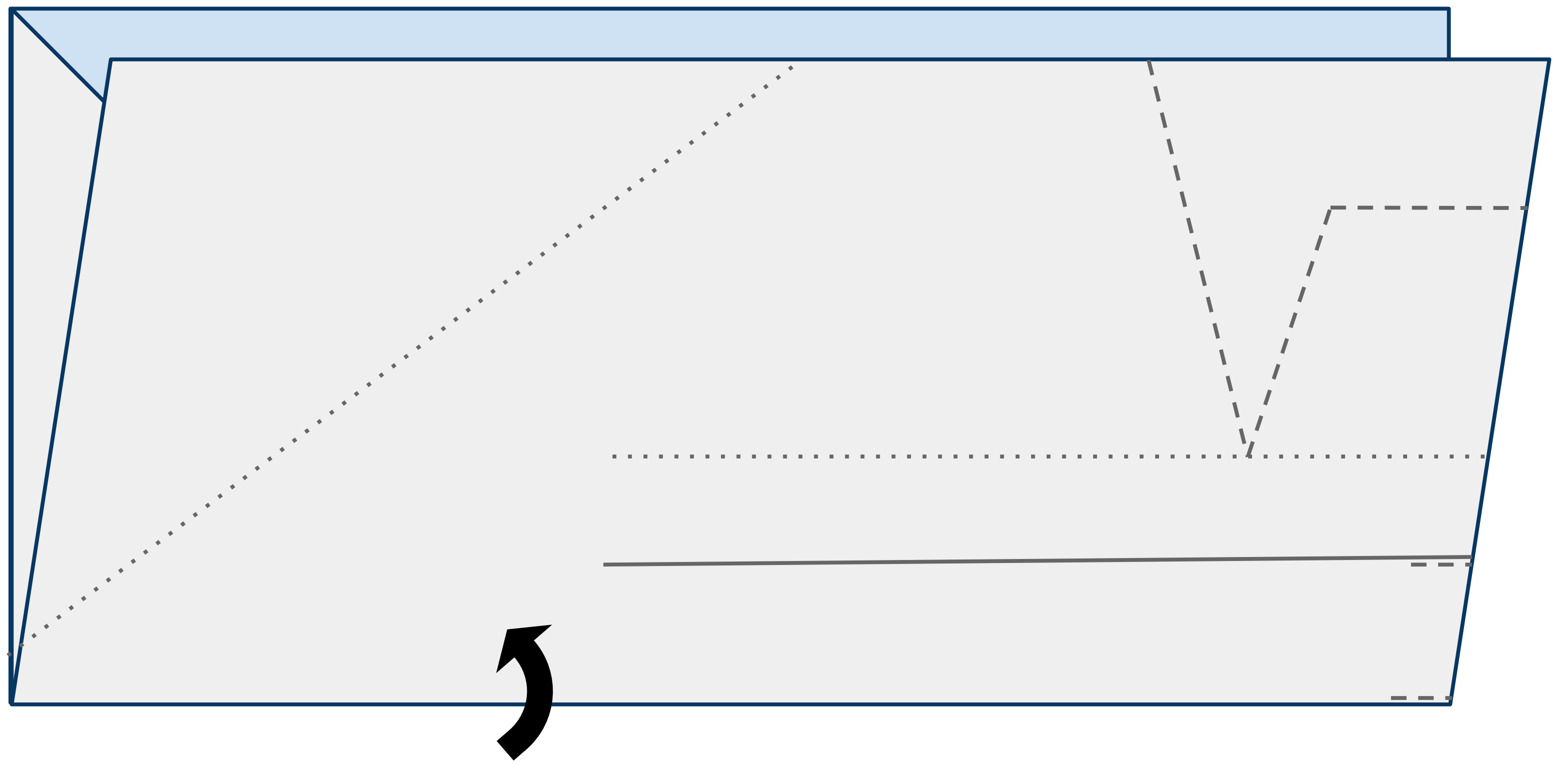
Figure 3



4

Turn the paper to the left (counterclockwise) and fold the paper upward in half hiding the triangle created in the previous steps refolding along the fold from step 1 as shown in Figure 4 below.

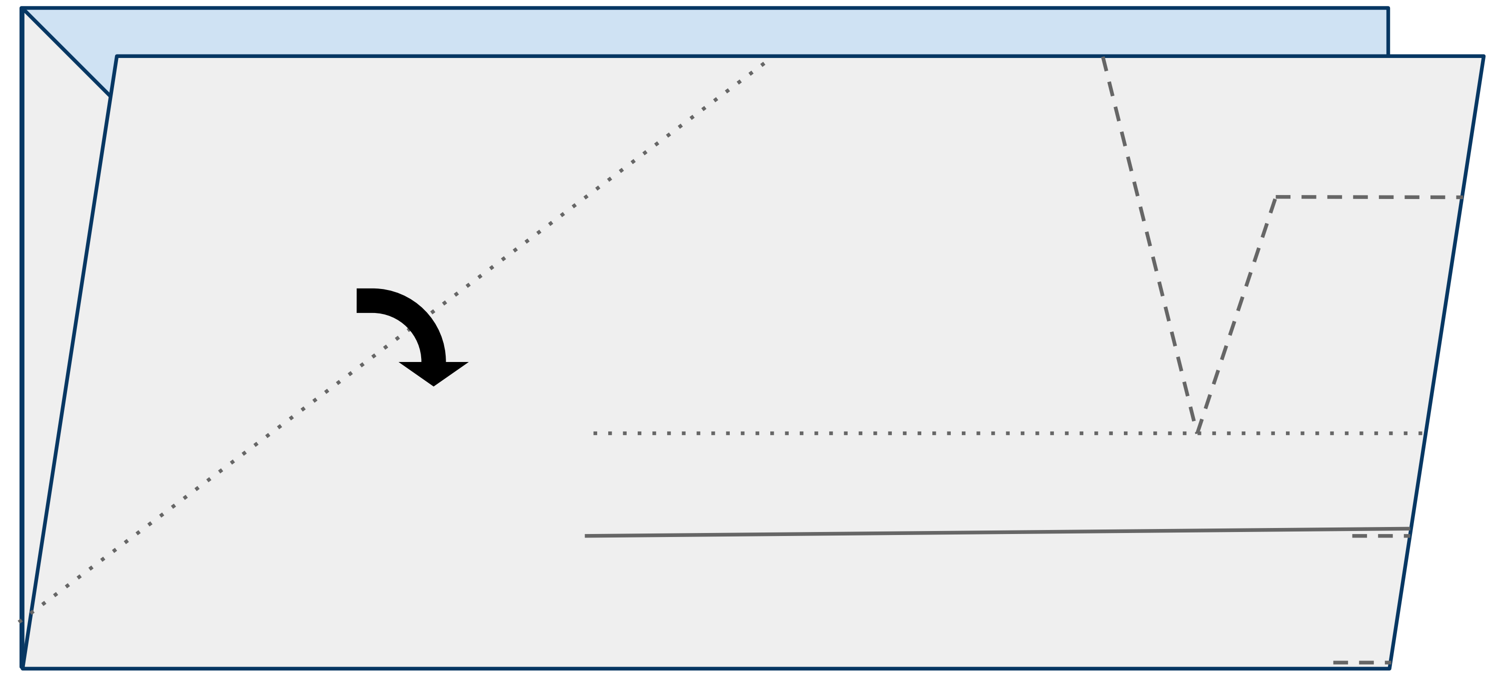
Figure 4



5

Fold the leading edge of the near side of the plane down along the dotted line as shown in Figure 5 below.

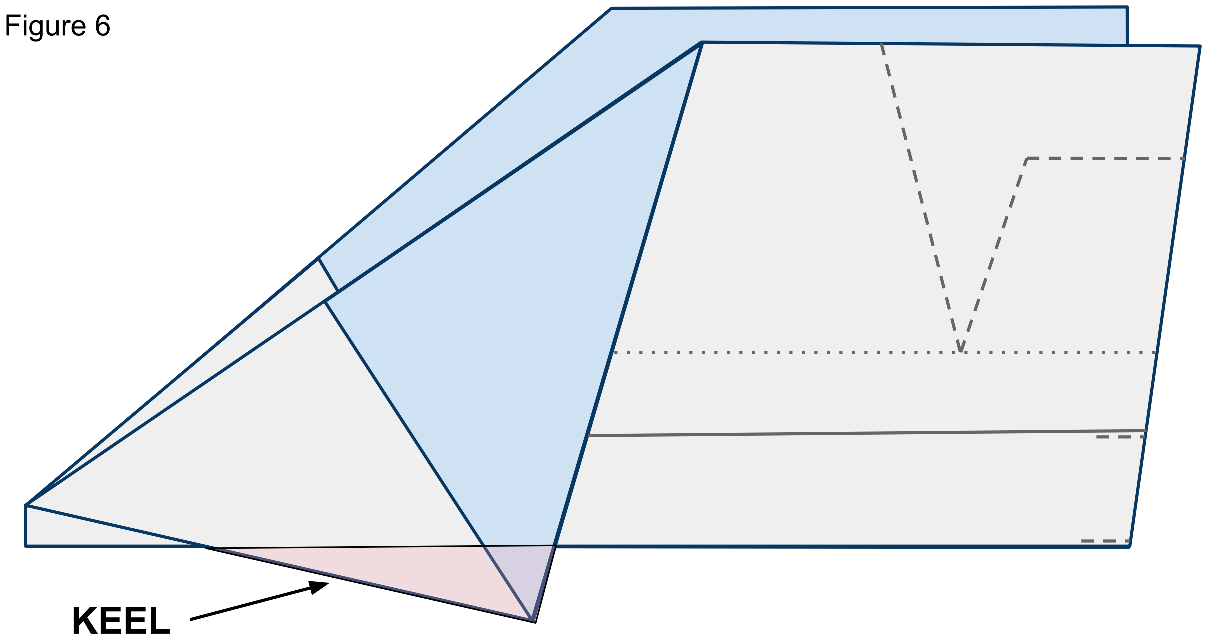
Figure 5



6

Do the same to the other side of the plane. It is important to match these 2 folds exactly so that your plane is symmetrical and will fly straight. The area that is created below the body and designated in pink will be called the keel.

Figure 6



7

Place tape along the keel as shown in Figure 7 below and cut a slit in the tape along the red line. Now fold the tape around and under both sides of the plane and keel securing them together. See Figure 7b for what the far side of the plane looks like.

Figure 7

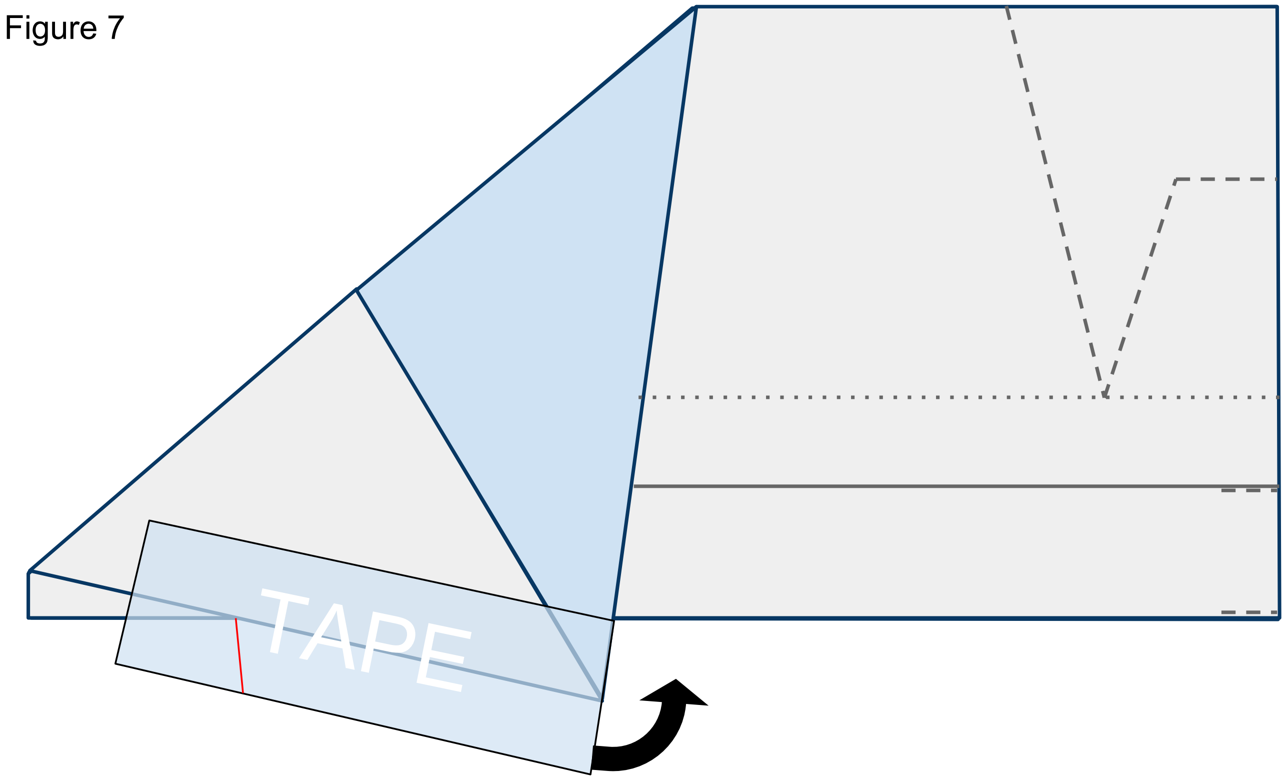
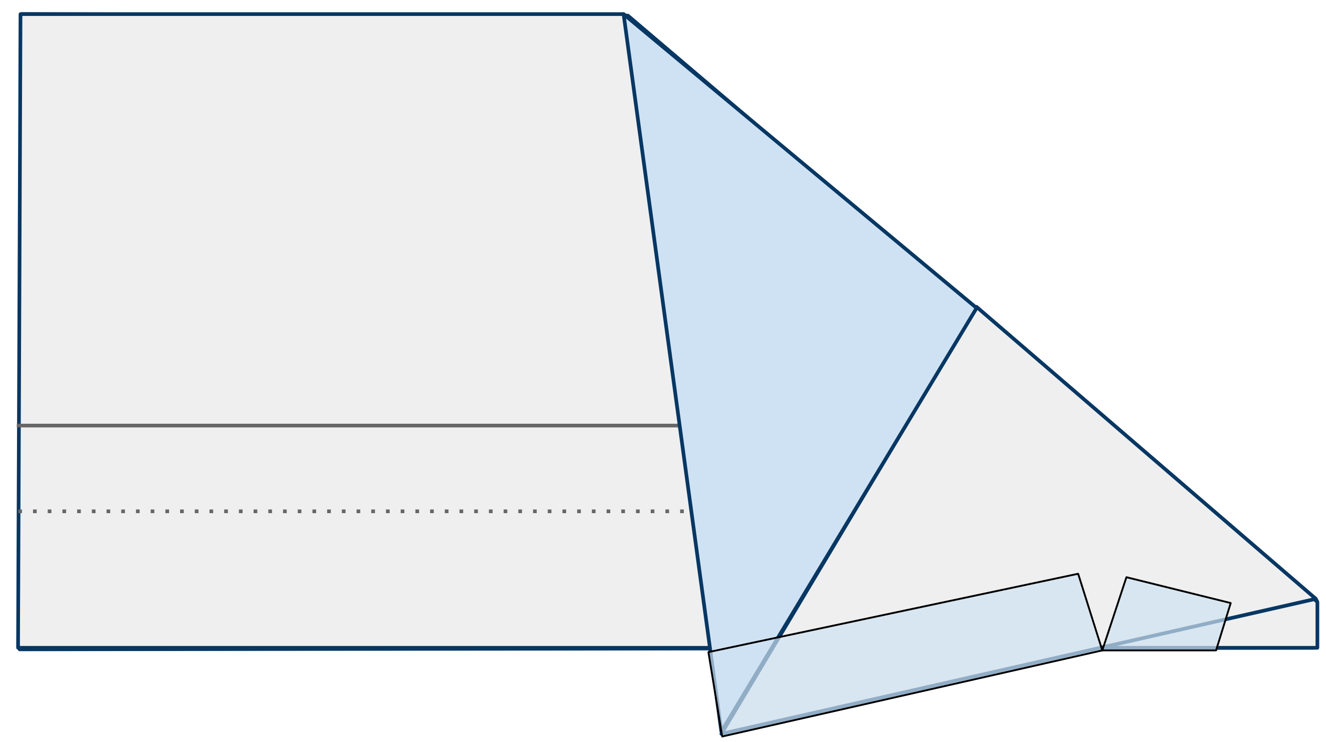


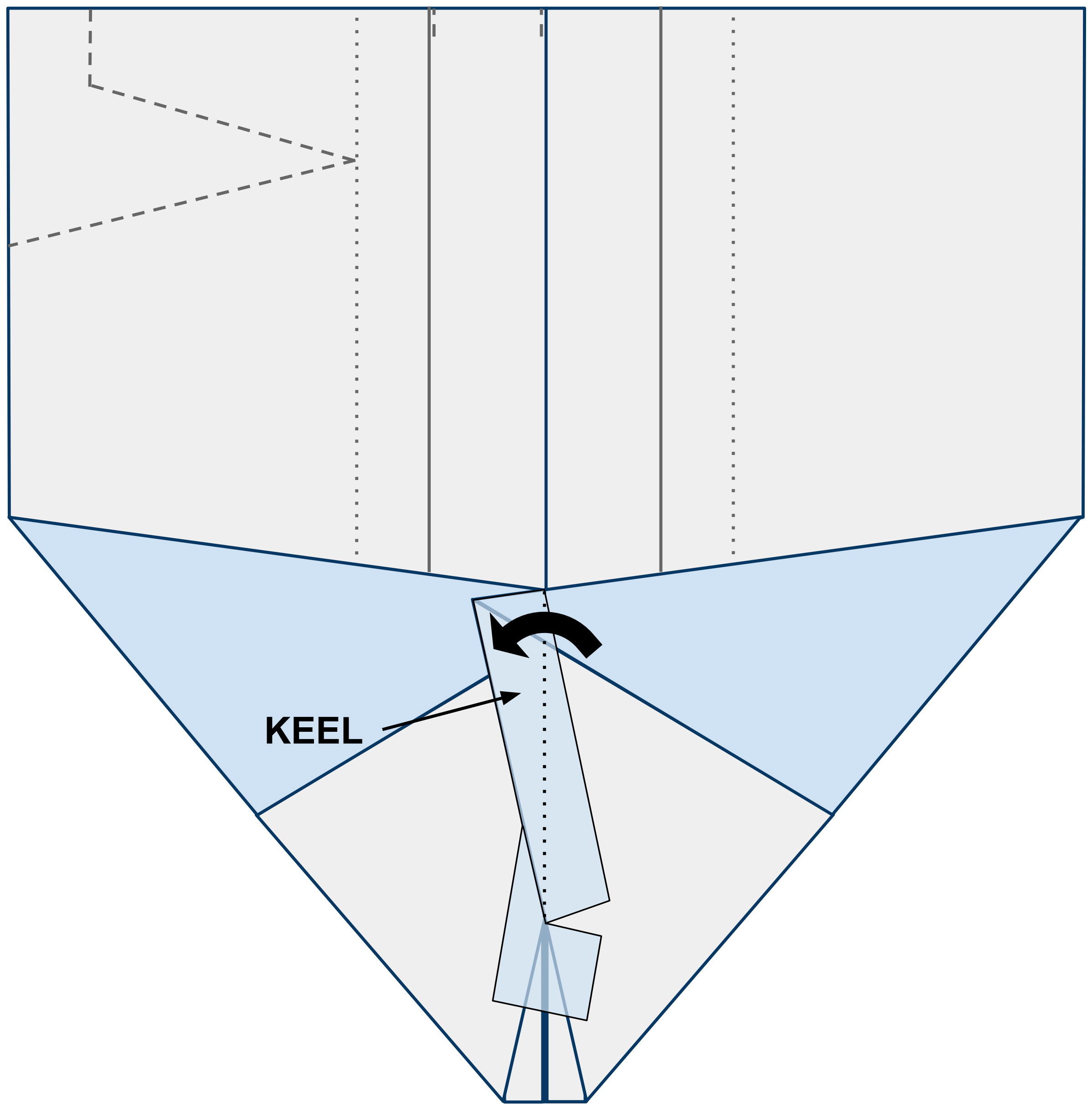
Figure 7b



8

Open the plane up flat on your work surface with the nose toward you and looking at the underside of the body. When you do this the keel will want to stand up off the table. Bend it to the left along the center line and give it a strong crease as shown in Figure 8 below.

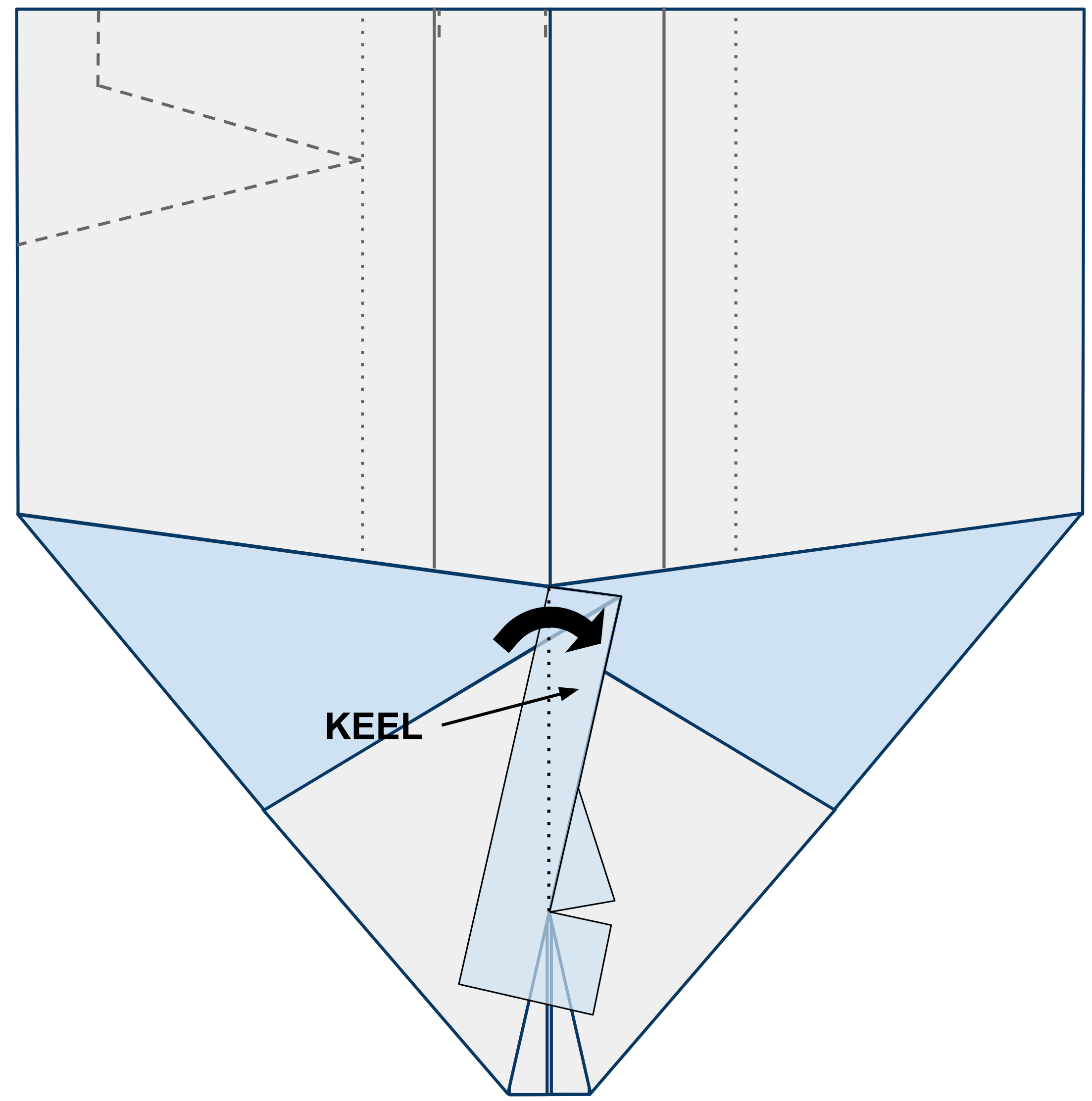
Figure 8



9

Now fold the keel to the right side creating a strong crease as shown in Figure 9 below. After making this crease, allow the keel to remain upright. This is where you will hold the plane when launching it.

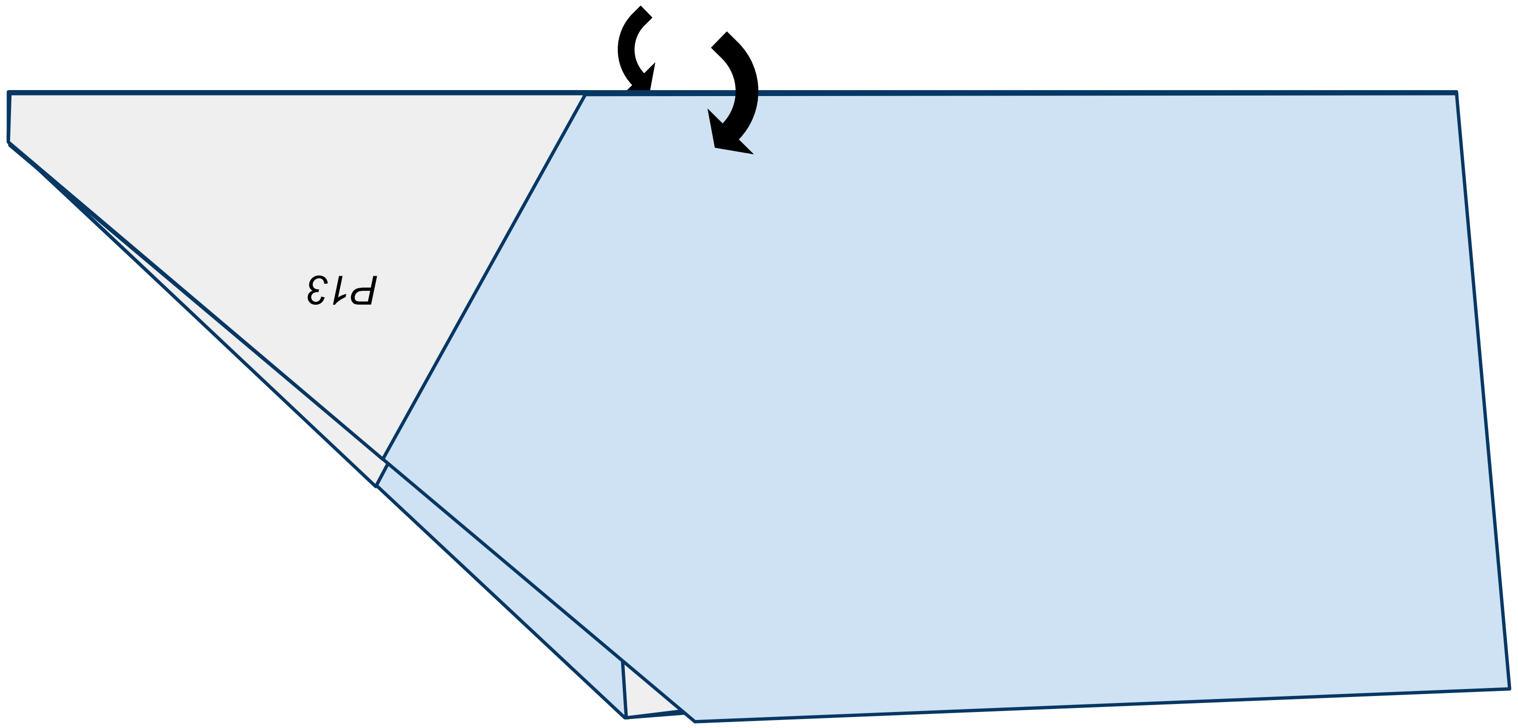
Figure 9



10

Rotate the nose of the plane to the left clockwise and fold the plane in half down the middle covering up the keel. Fold along the fold created in step 1 along the center line of the plane as shown in Figure 10 below.

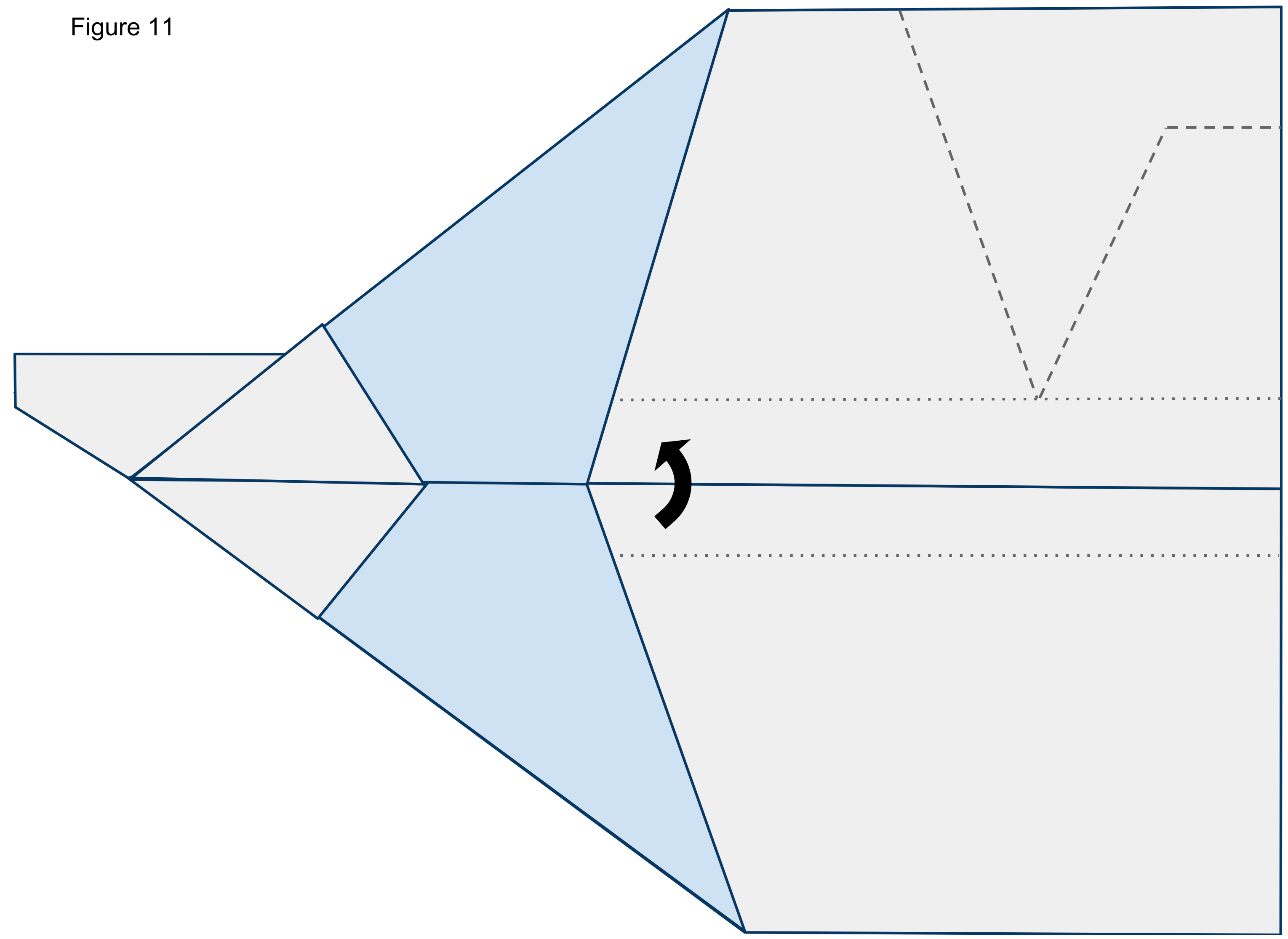
Figure 10



11

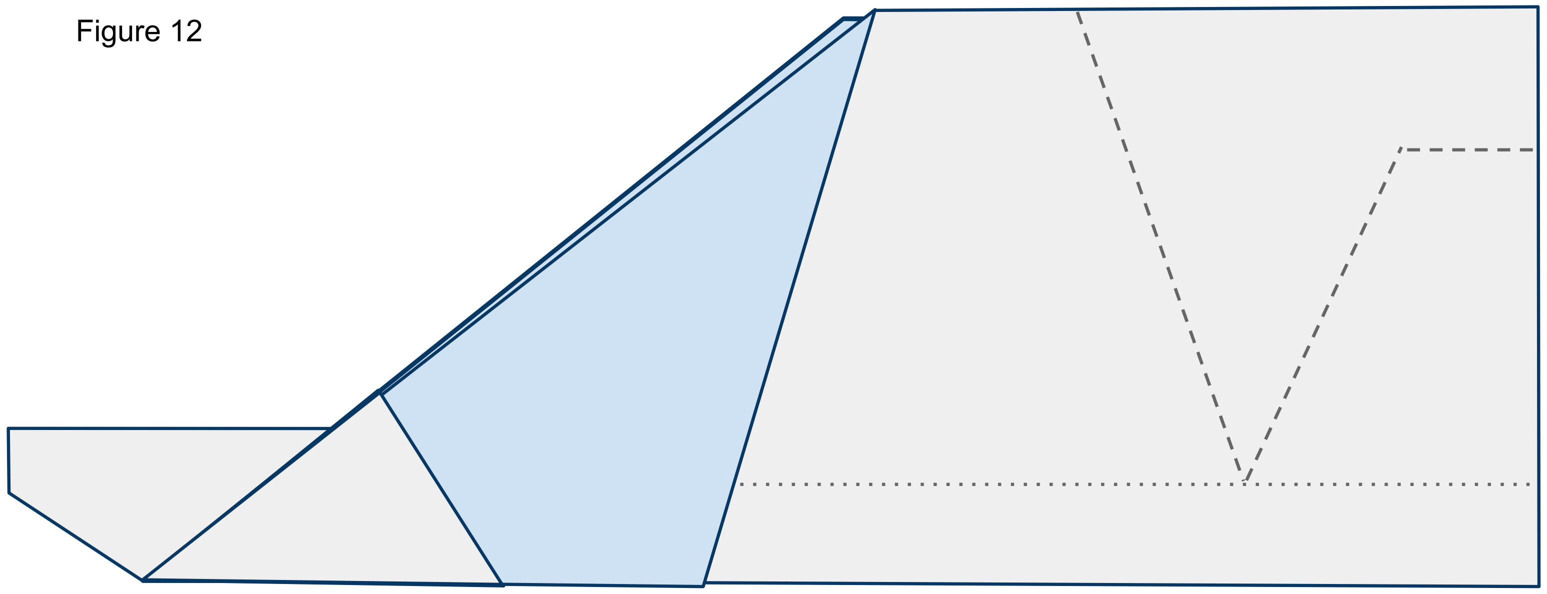
Now fold the near side wing back up along the solid line shown in Figure 11 below.

Figure 11



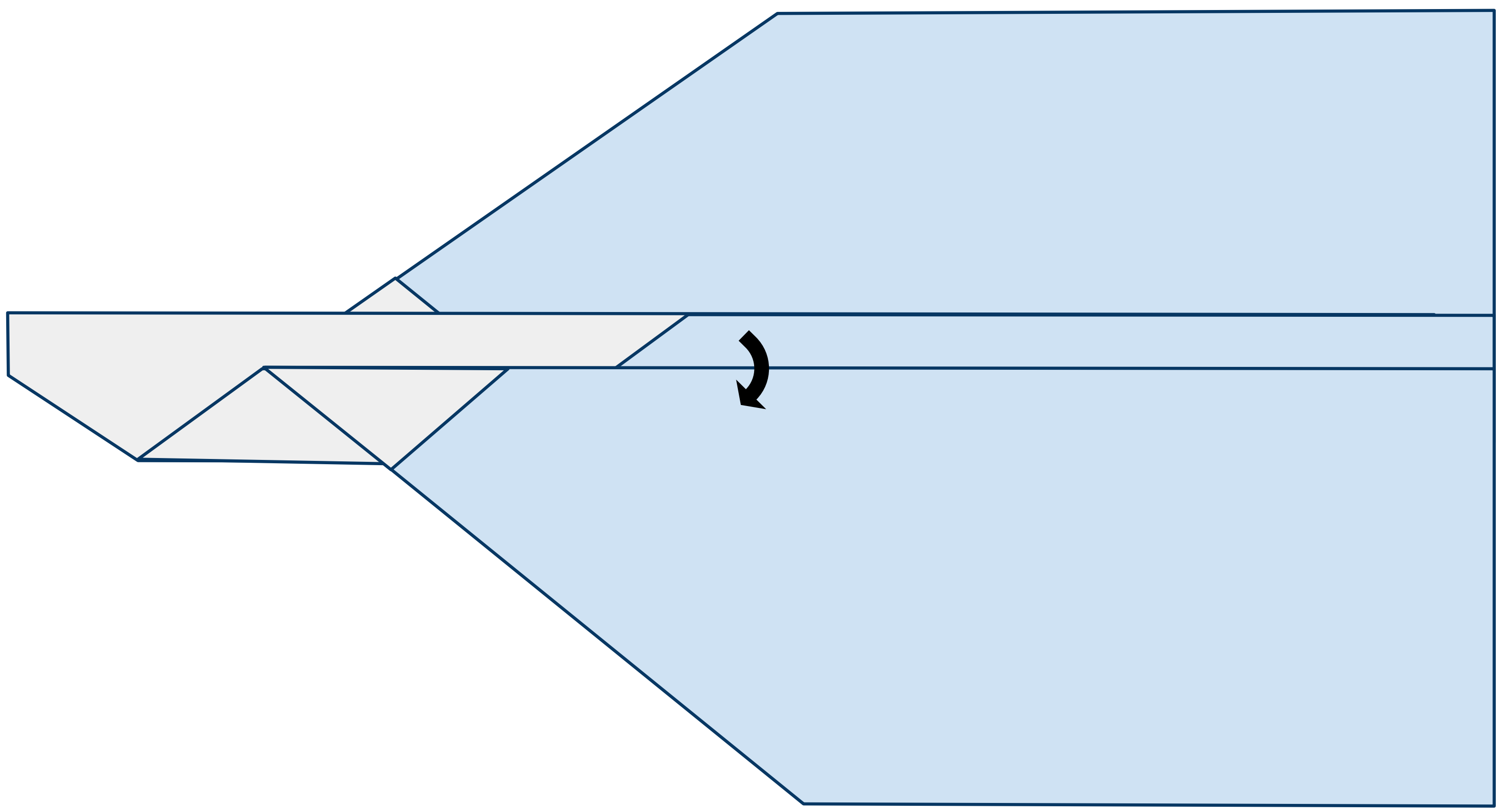
Do the same to the other side as shown in Figure 12 below. Remember that matching symmetry of the plane is more important that lining up the printed lines on the page.

Figure 12



Now fold the near side wing down along the dotted line as shown in Figure 13 below.

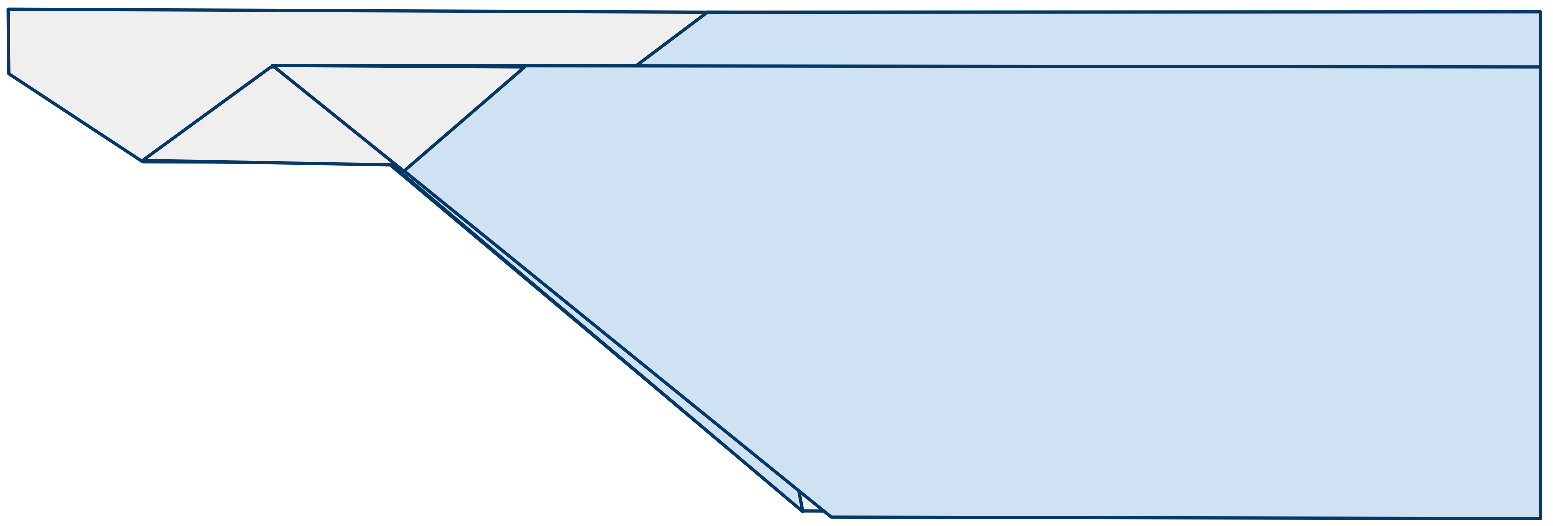
Figure 13



14

Do the same to the other side matching the plane symmetrically as shown in Figure 14 below.

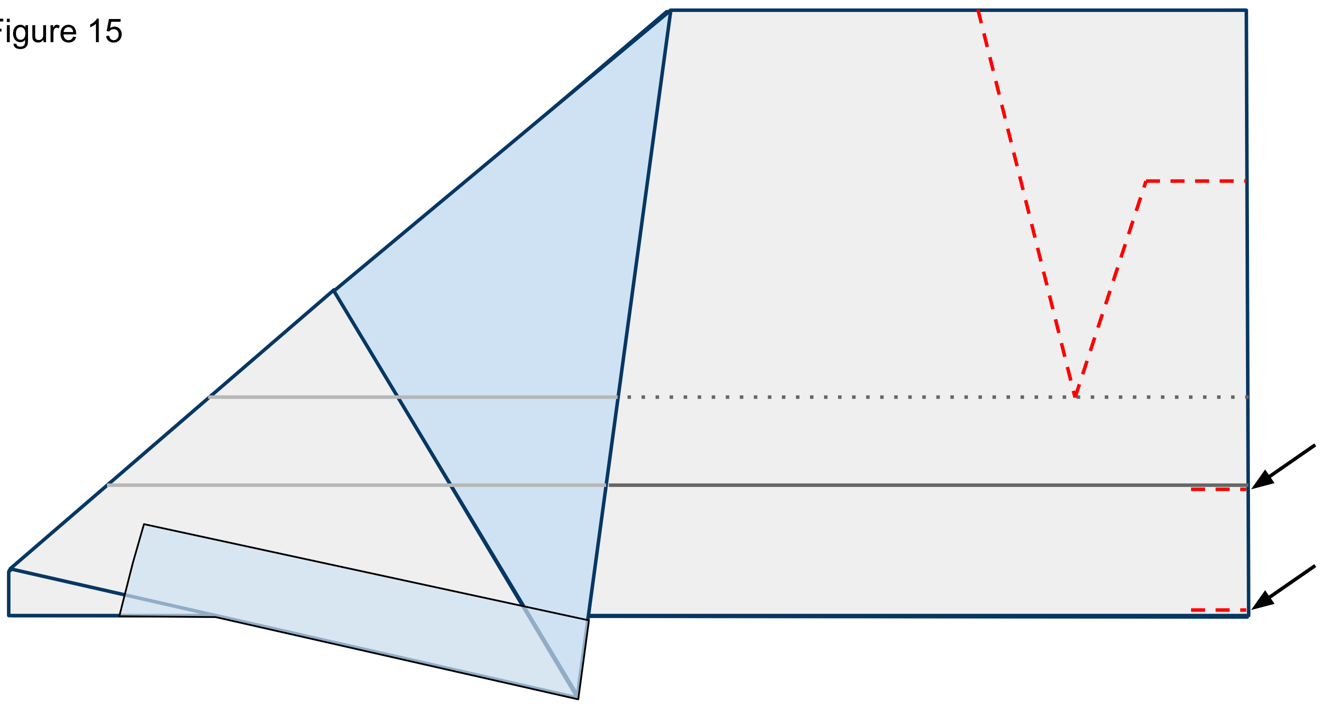
Figure 14



15

Open the wings back up and lay the plane flat on the work surface. Now cut through both layers of paper along the **red** dashed lines as indicated by Figure 15 below. Don't forget the slits in order to create your flaps.

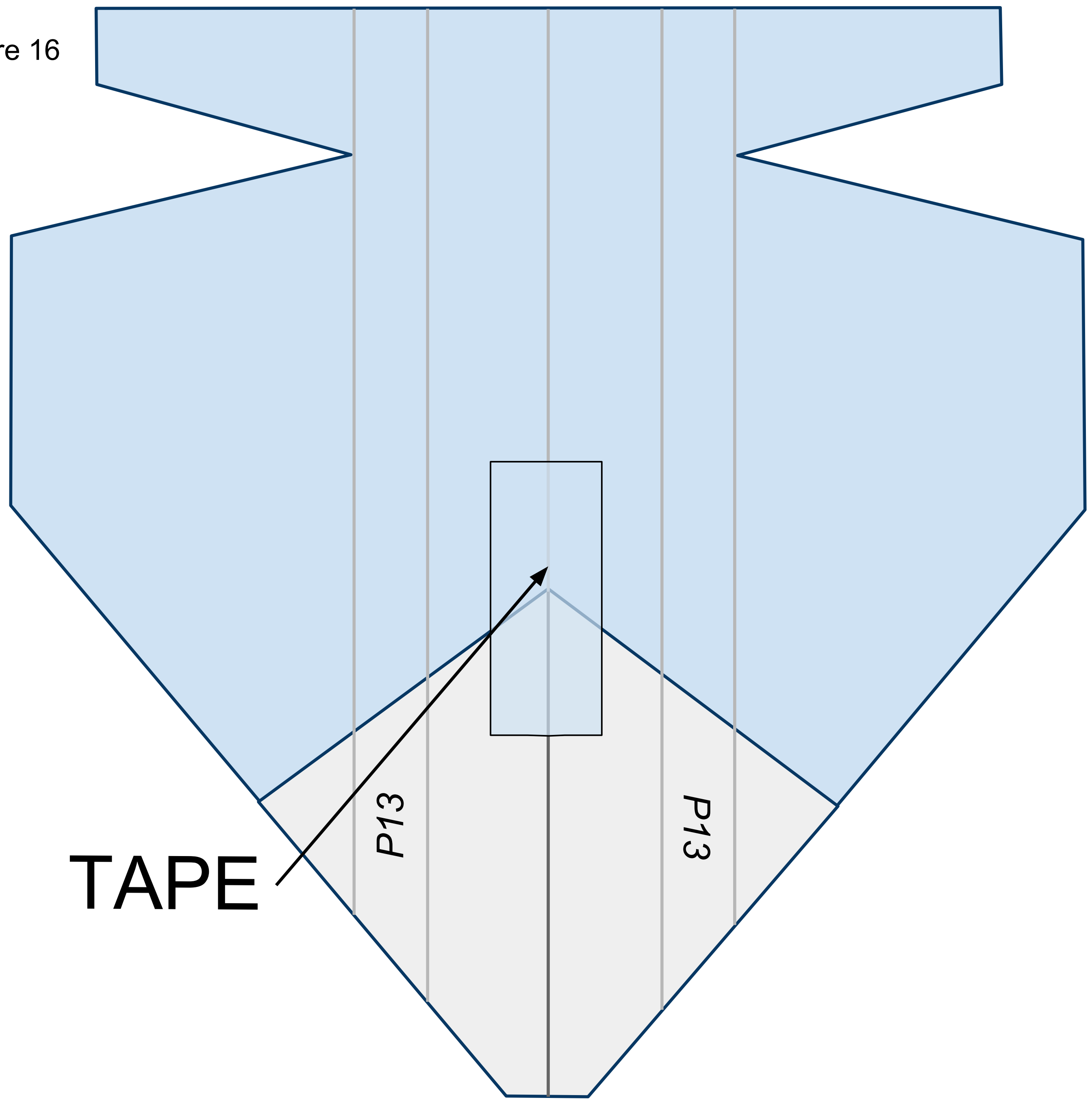
Figure 15



16

Turn the nose of the plane toward you counterclockwise and lay the plane flat with the keel on the underside. Lay the keel to one side or the other. Place tape along the fuselage as shown in Figure 16 below.

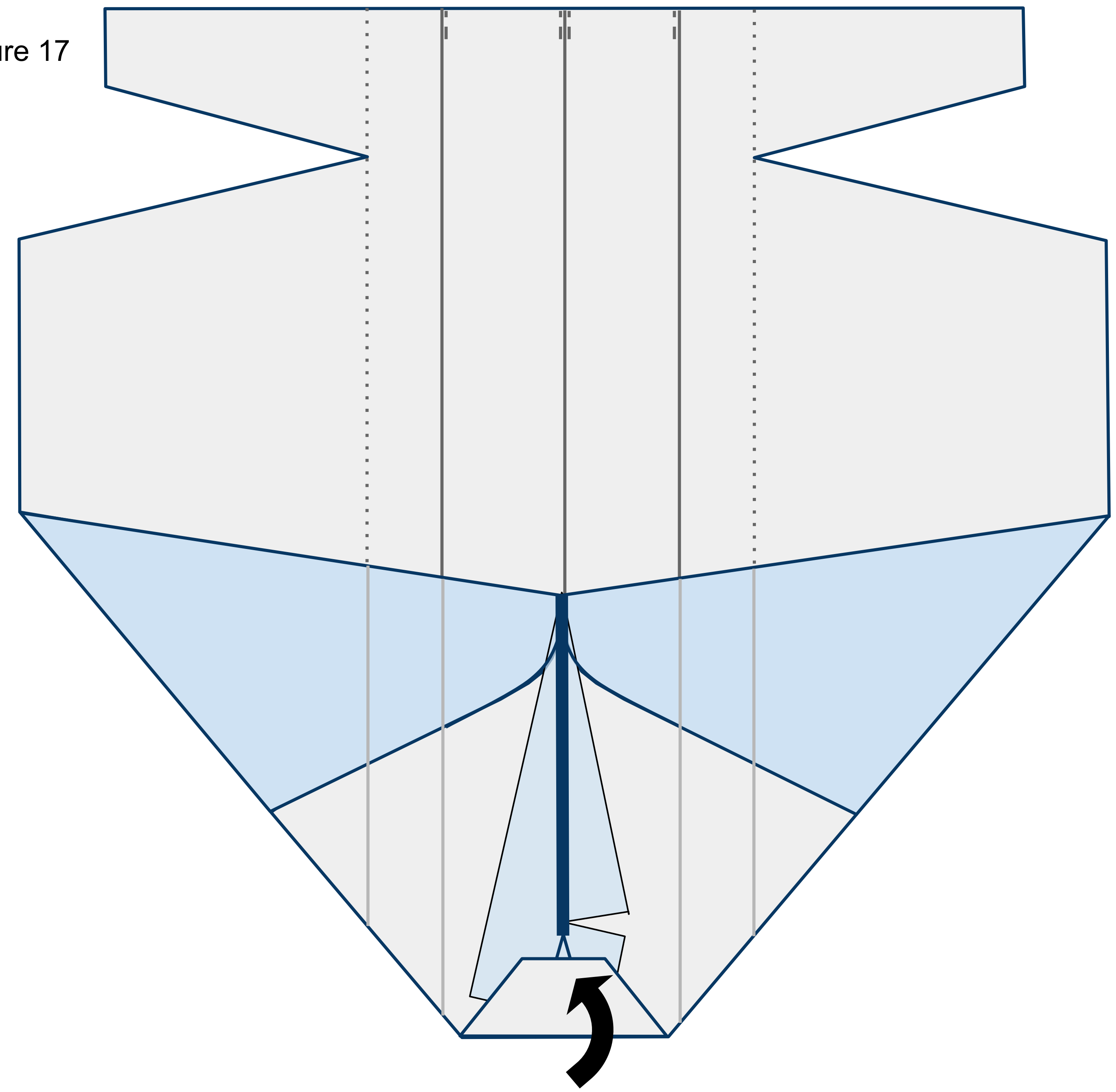
Figure 16



17

Turn the plane over to show the underside. The vertical standing keel is represented by the thick **blue** line. Fold the nose back as shown in Figure 17 below.

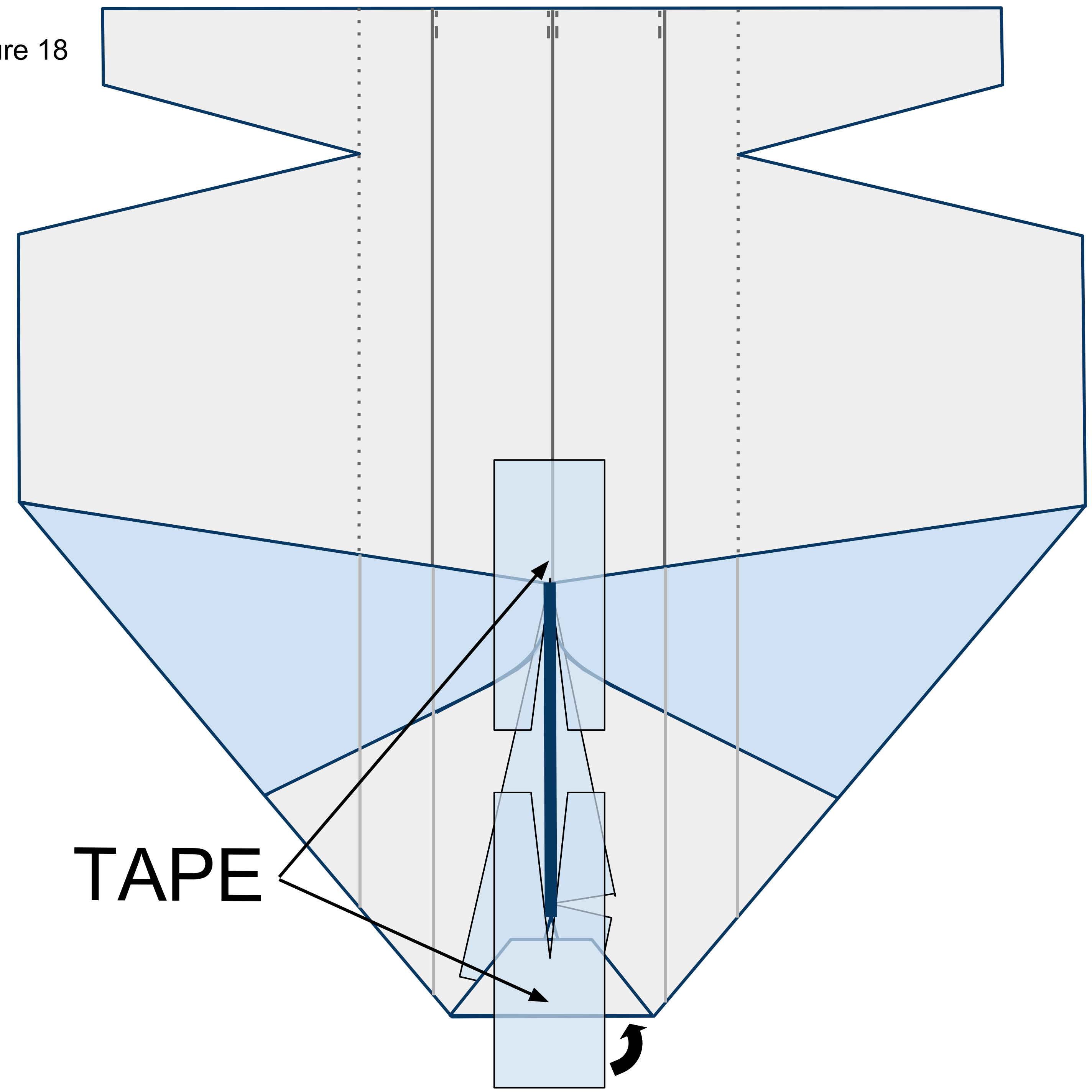
Figure 17



18

Place 2 pieces of tape with slices cut in them on the underside of the plane in front of and behind the keel as shown in Figure 18 below. Wrap the tape around the nose to the top side of the plane.

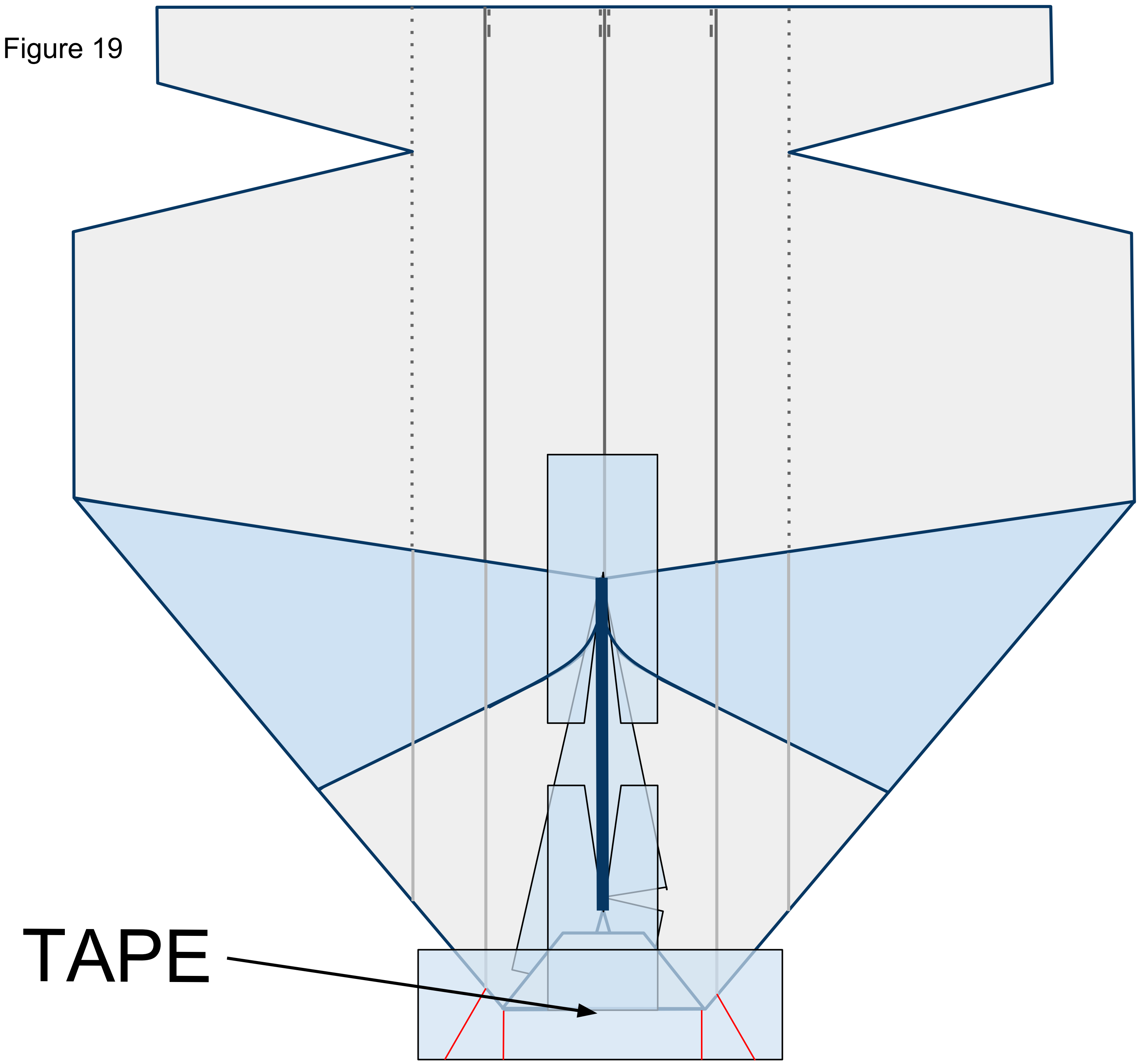
Figure 18



19

Place a piece of tape across the nose of the plane as shown in Figure 19 below. Cut slits along the red lines as shown. These slits will help you fold the tape over the plane in the next step.

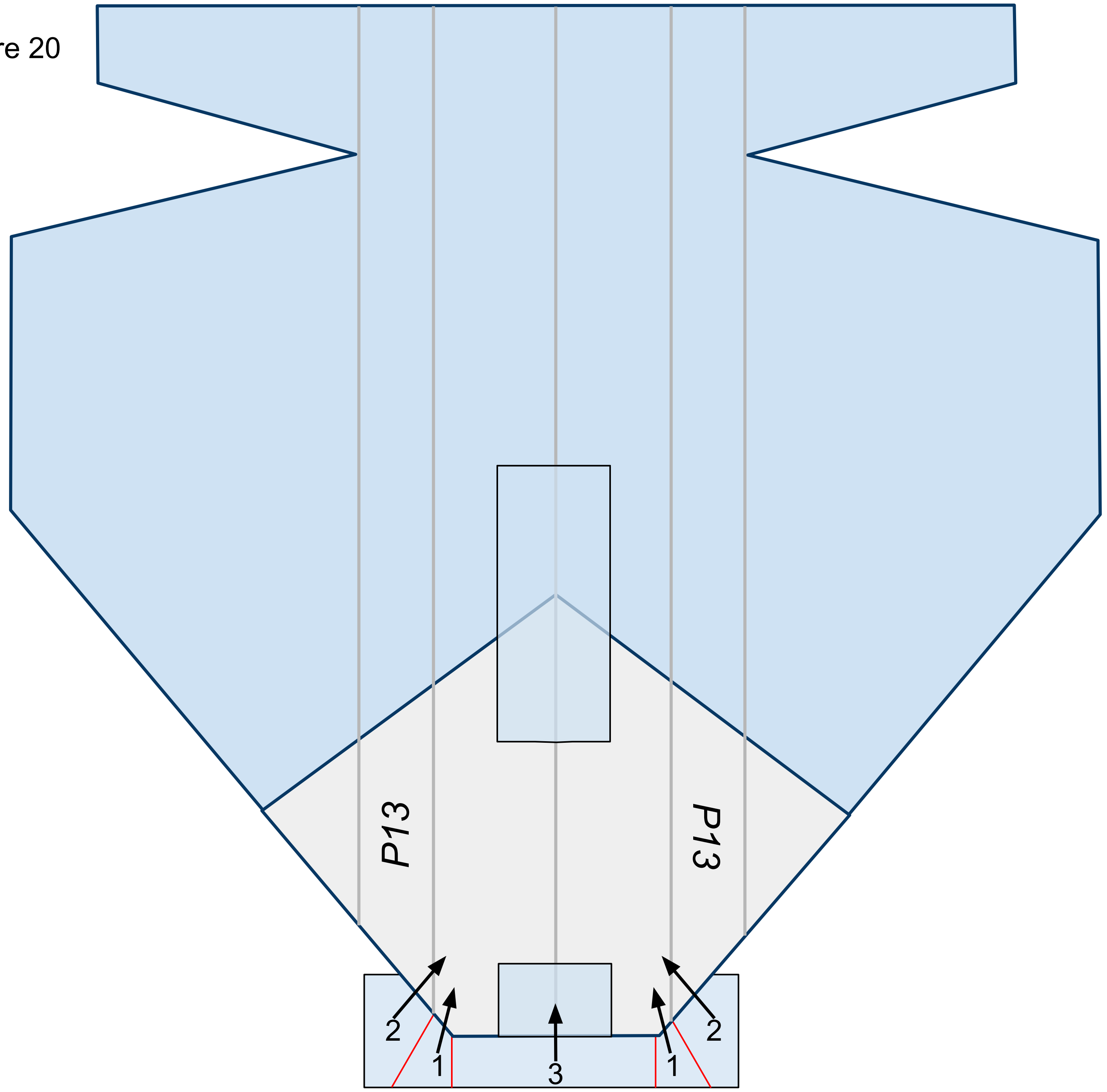
Figure 19



20

Flip the plane over so you can see the top side. Fold the flaps of tape, in number order, up and in as shown in Figure 20 below. They will overlap each other once folded.

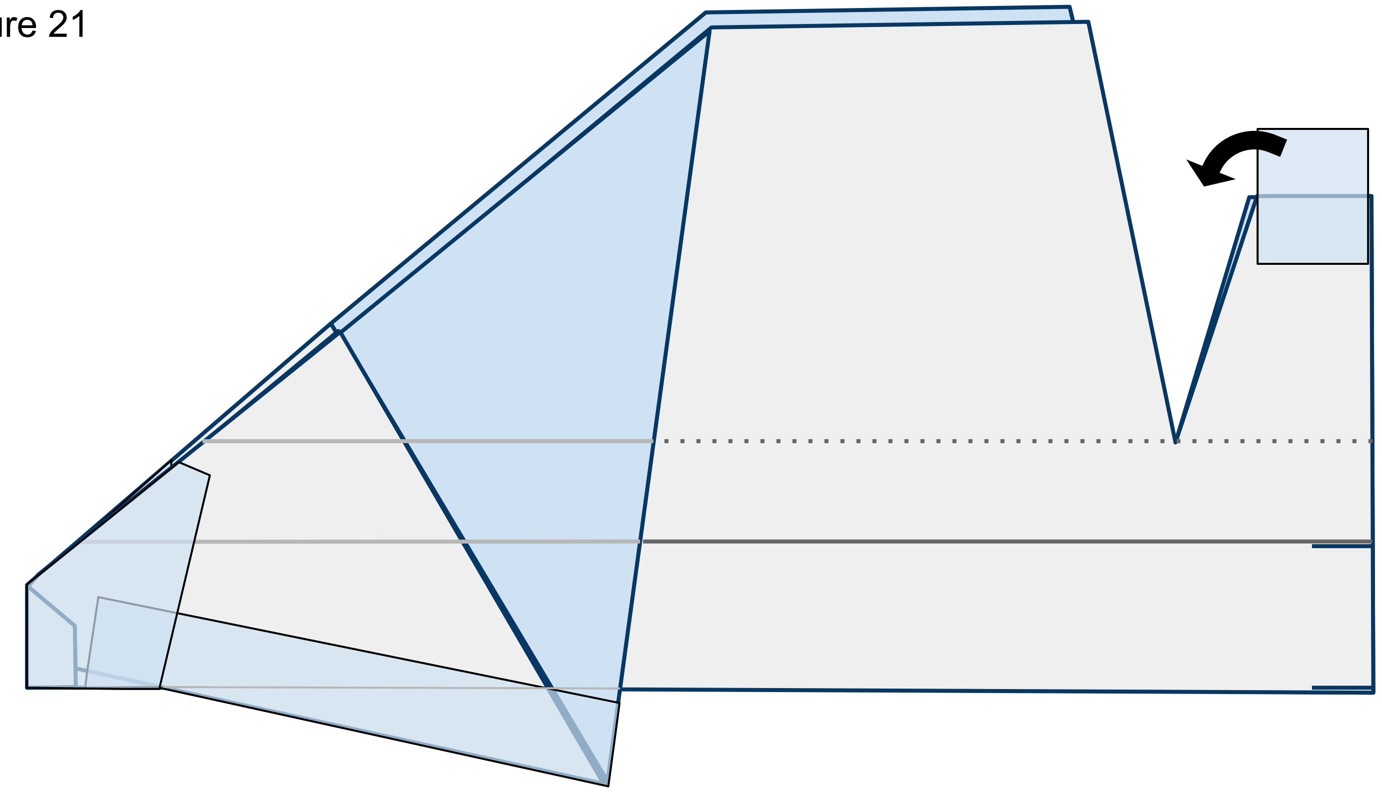
Figure 20



21

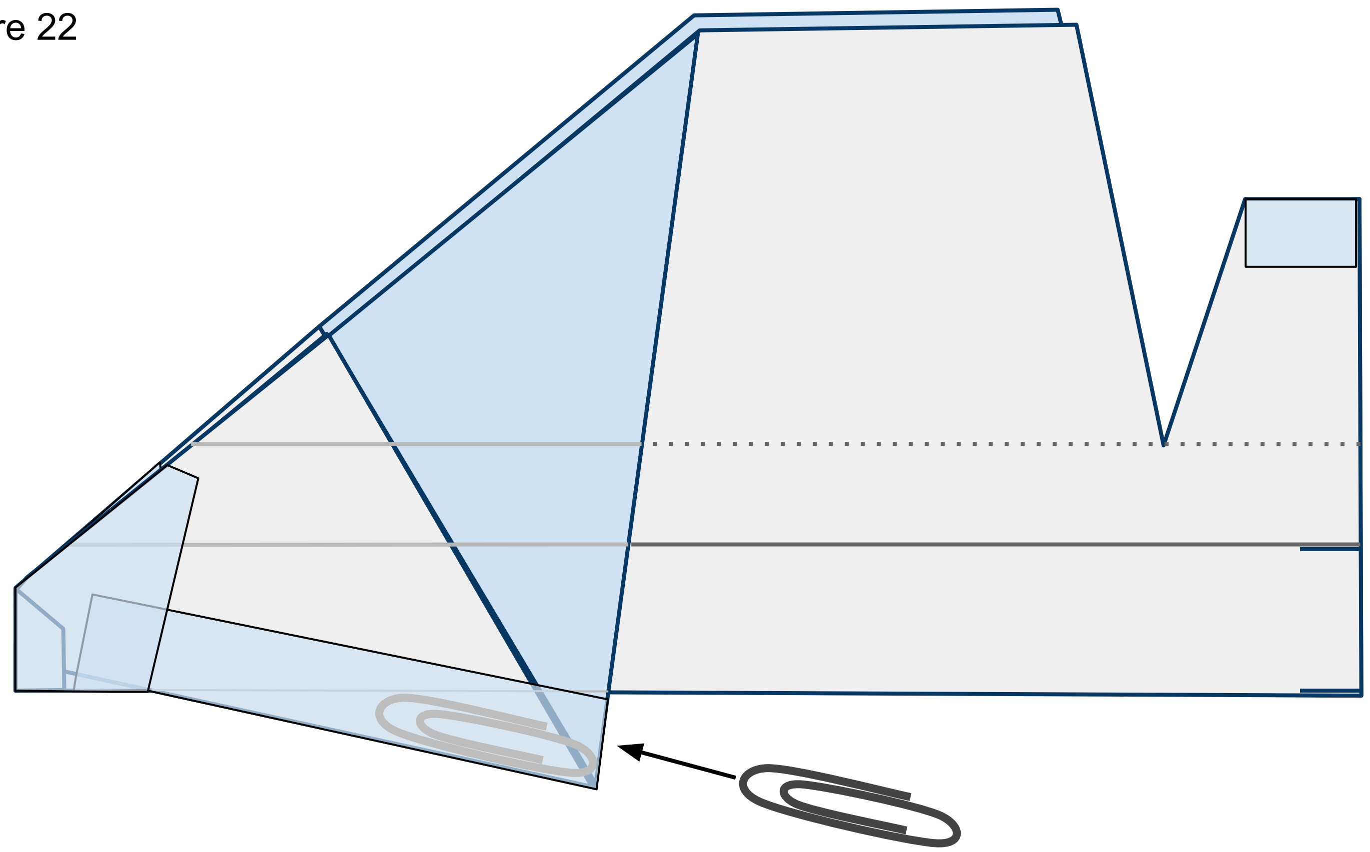
Turn the nose to the left and fold up the wings flat. Place tape across the tail section bringing the two tail sections together as shown in Figure 21 below.

Figure 21



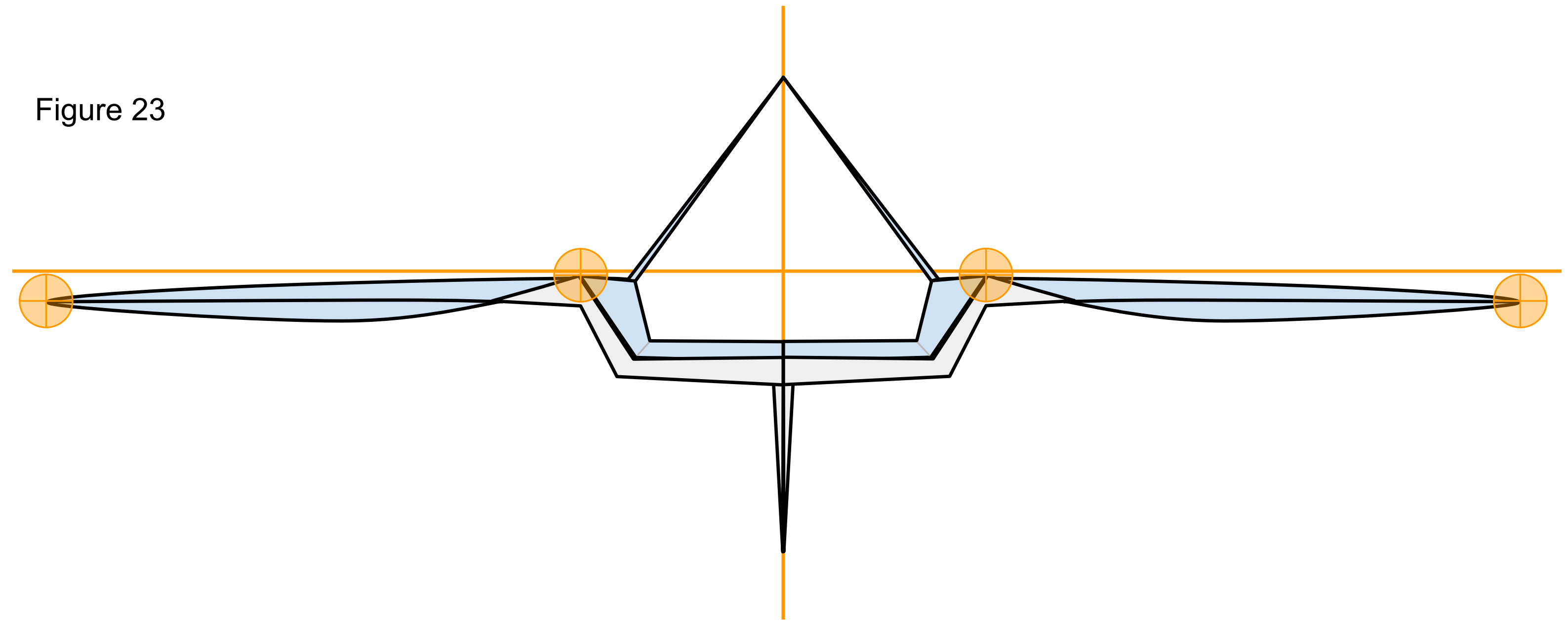
Insert a #1 size paper clip ***just*** inside the keel through the opening behind it as shown in Figure 21 below. Then seal the opening with a piece of tape.

Figure 22



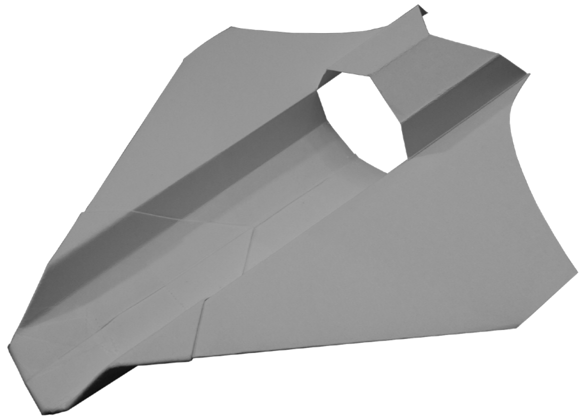
Now that you have constructed the plane ensure that it has the proper shape. Take a look at your plane head-on from the nose. The plane should be symmetrical. Take a look at Figure 23 below. Note the points of interest. Note the curved shape of the paper on the underside of the wings. To achieve this, carefully stick one side of a pair of scissors up into the pocket on the underside of the wing from behind and gently mold the paper into the proper shape. Don't crease it.

Figure 23



We are almost ready to launch! I recommend taking a quick look at the “PREFLIGHT” section to get some good tips. All paper planes fly erratically on their first flights. They require refining to make them great. To do this, take a look at the ”FINE-TUNING” section.

P14 SKYMASTER



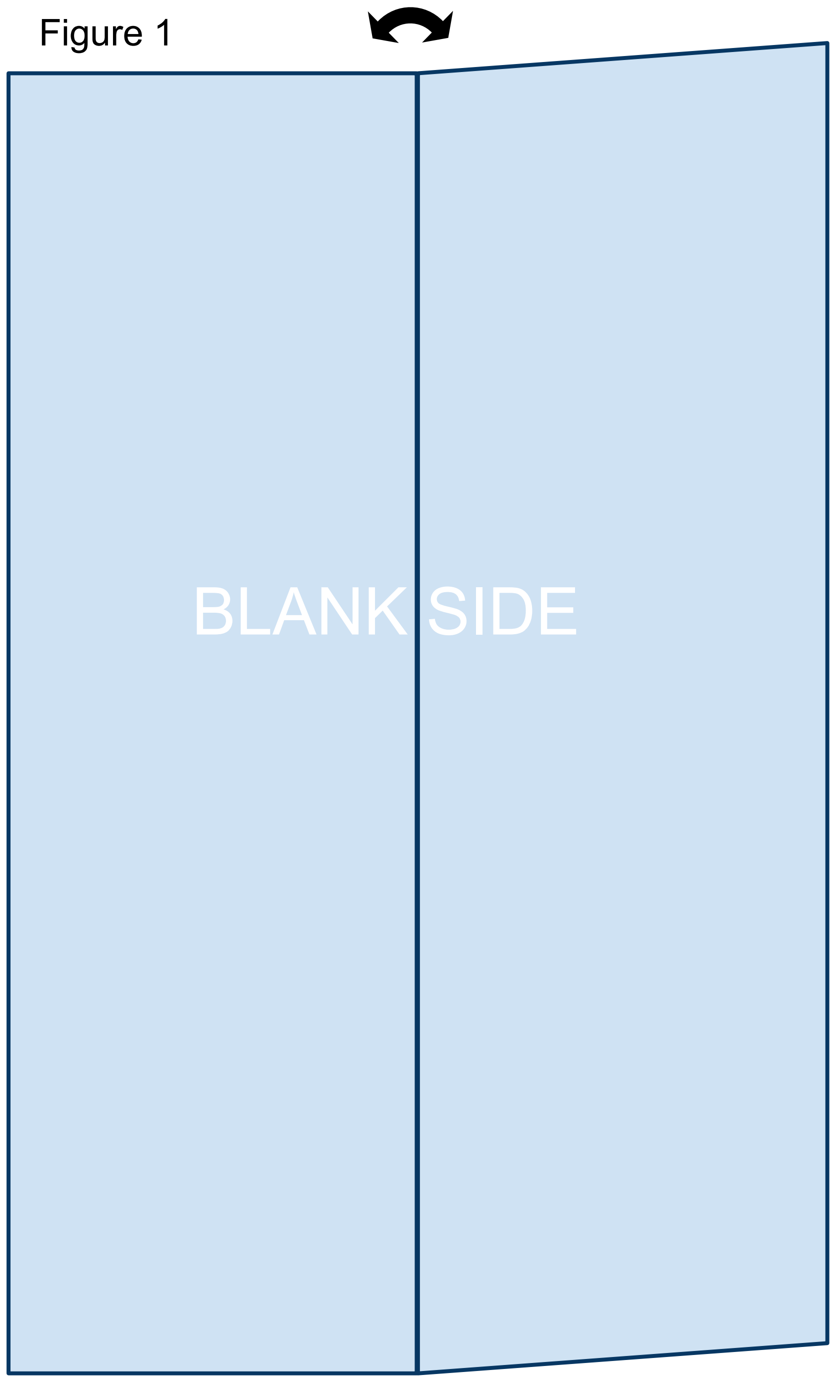
The printable template linked below is 4.25in x 5.5in. That is equivalent to half of an 8.5in x 11in sheet of printer paper. While the designs in this book will fly great with regular printer paper, I recommend 32lb/ream paper. The stronger paper will hold its shape better during flight allowing for higher and more exciting flights. Once you become comfortable making the planes from this template you can simply cut regular printer paper in half in order to start. Print the PDF template located at the bottom of the web page: https://sites.google.com/site/afspaperairplanes/p14

In case your reading device is not connected to the internet, doing a Google search for "AFS Paper Airplanes" should get you to the webpage.

1

Cut out the template and place the paper on the table so that the printed information is on the underside with the printed triangle facing away from you. You will be looking at the blank side. Bring the right side up and fold the paper in half the long way. Unfold leaving a visible crease as shown in Figure 1 below.

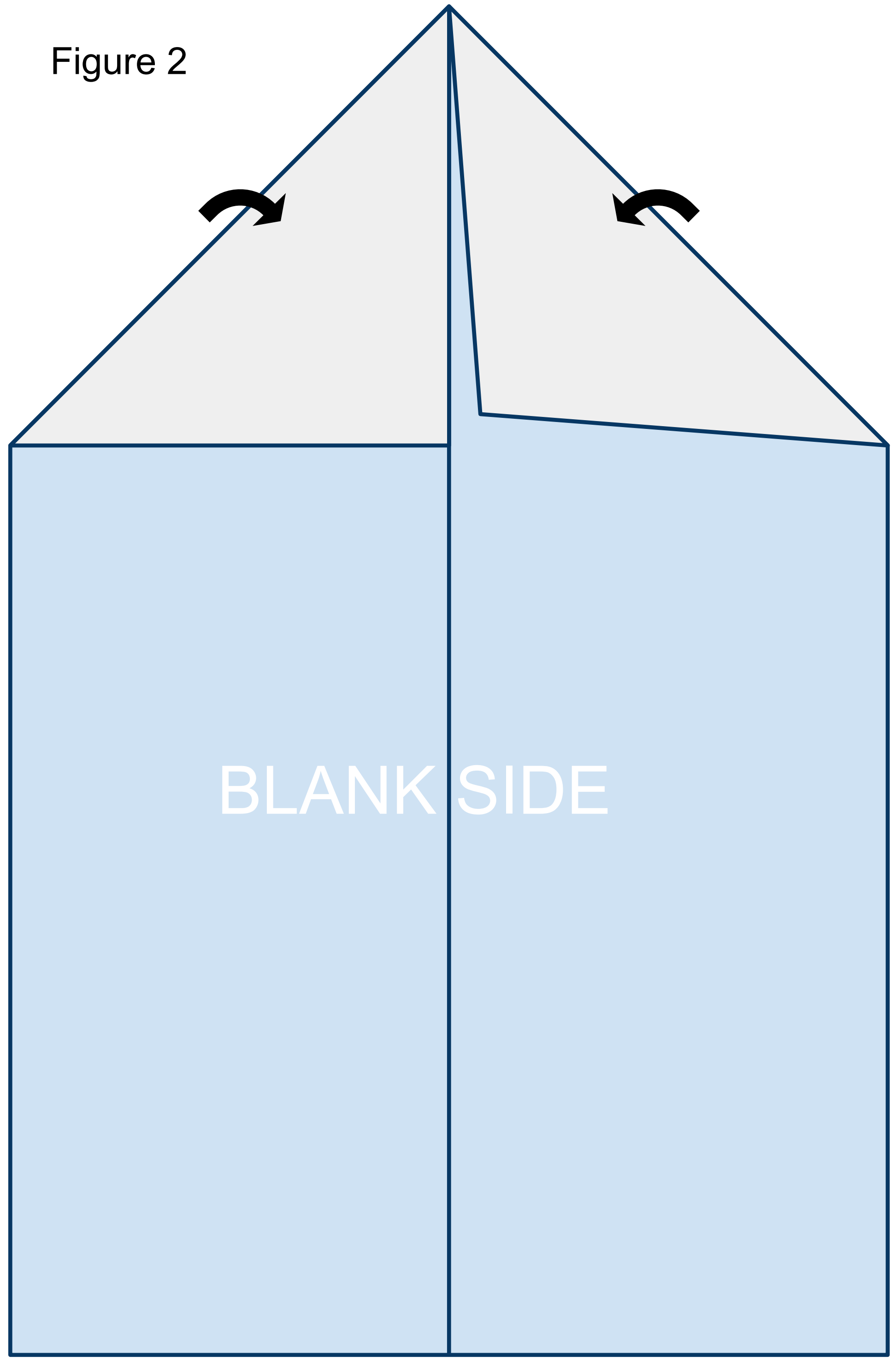
Figure 1



2

Fold the leading corners back. Line them up along the crease you created in step 1 as shown in Figure 2 below.

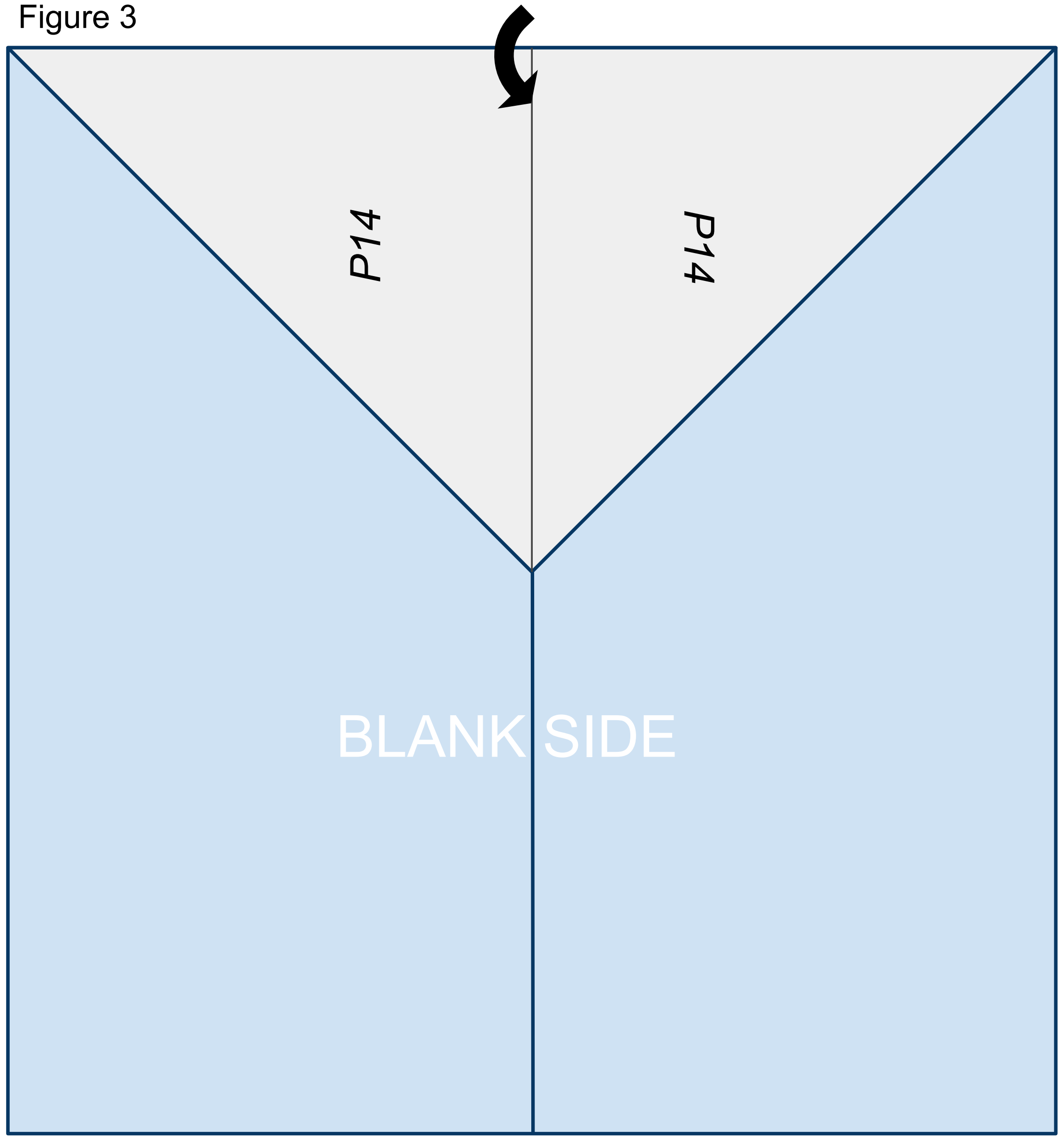
Figure 2



3

Fold the nose of the plane created in step 2 back along the solid line printed on the front of the paper so that it lines up on the center crease created in step 1 as shown in Figure 3 below.

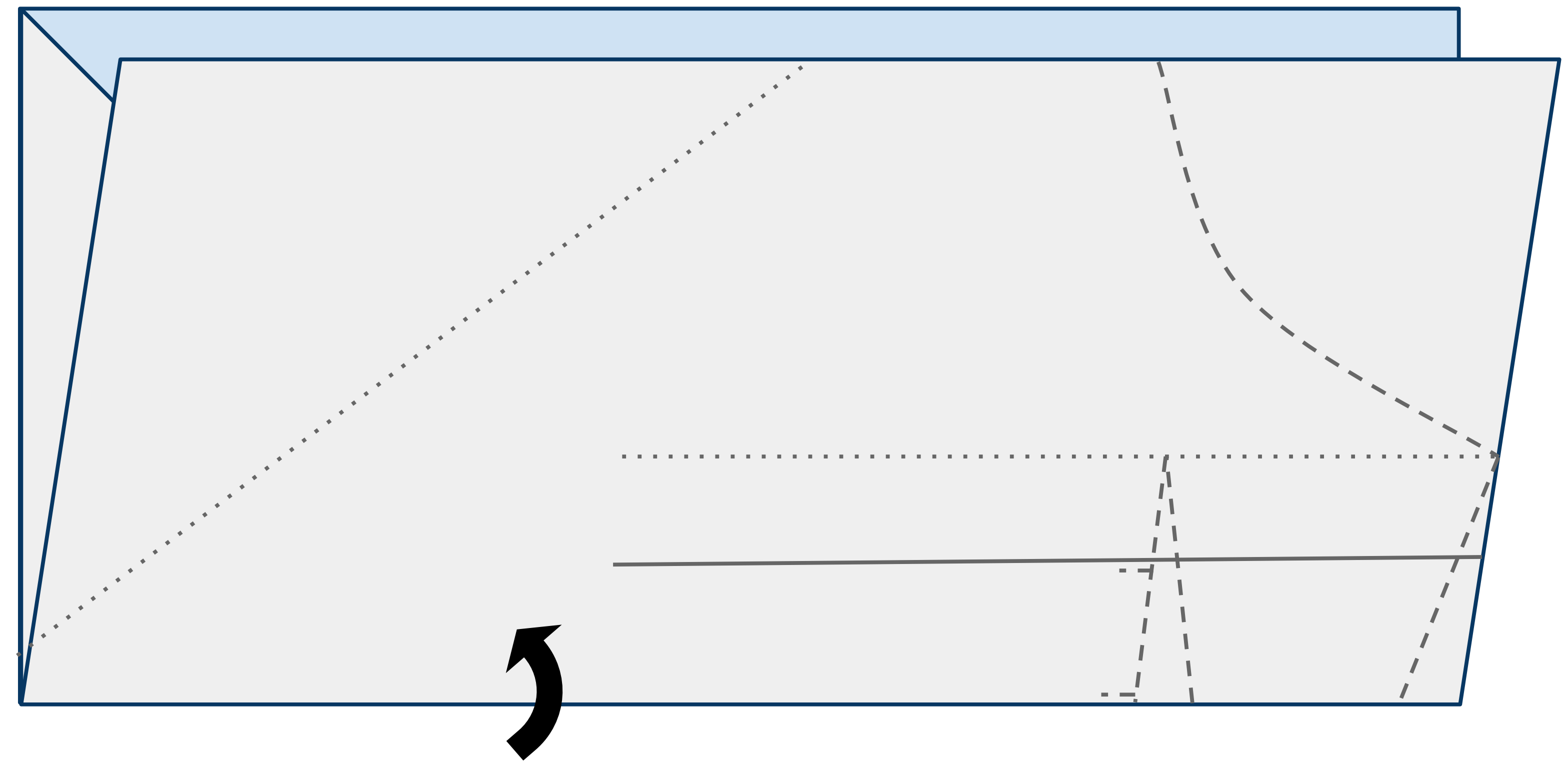
Figure 3



4

Turn the paper to the left (counterclockwise) and fold the paper upward in half hiding the triangle created in the previous steps refolding along the fold from step 1 as shown in Figure 4 below.

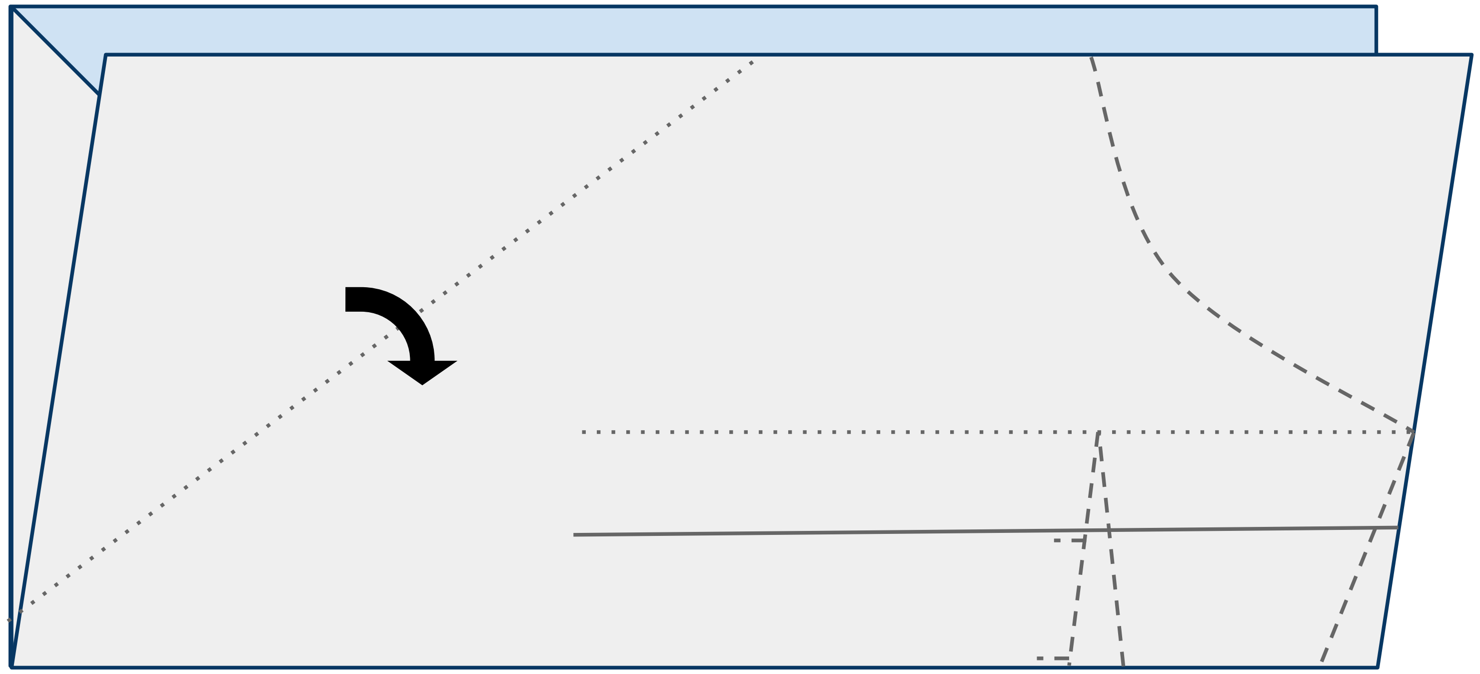
Figure 4



5

Fold the leading edge of the near side of the plane down along the dotted line as shown in Figure 5 below.

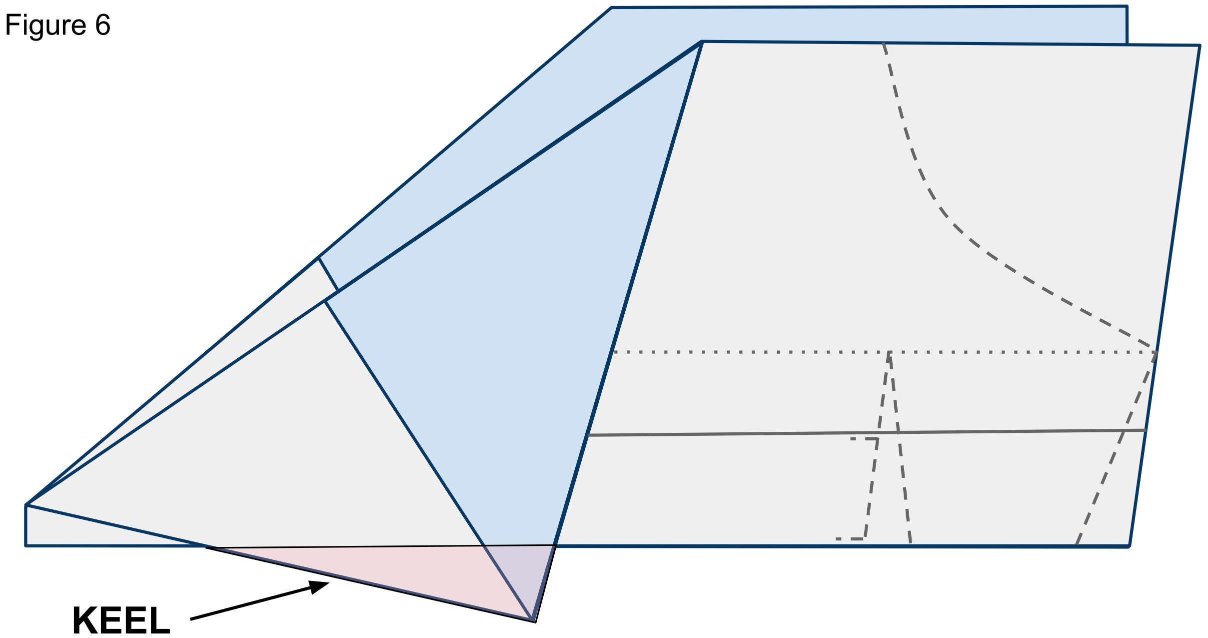
Figure 5



6

Do the same to the other side of the plane. It is important to match these 2 folds exactly so that your plane is symmetrical and will fly straight. The area that is created below the body and designated in pink will be called the keel.

Figure 6



7

Place tape along the keel as shown in Figure 7 below and cut a slit in the tape along the red line. Now fold the tape around and under both sides of the plane and keel securing them together. See Figure 7b for what the far side of the plane looks like.

Figure 7

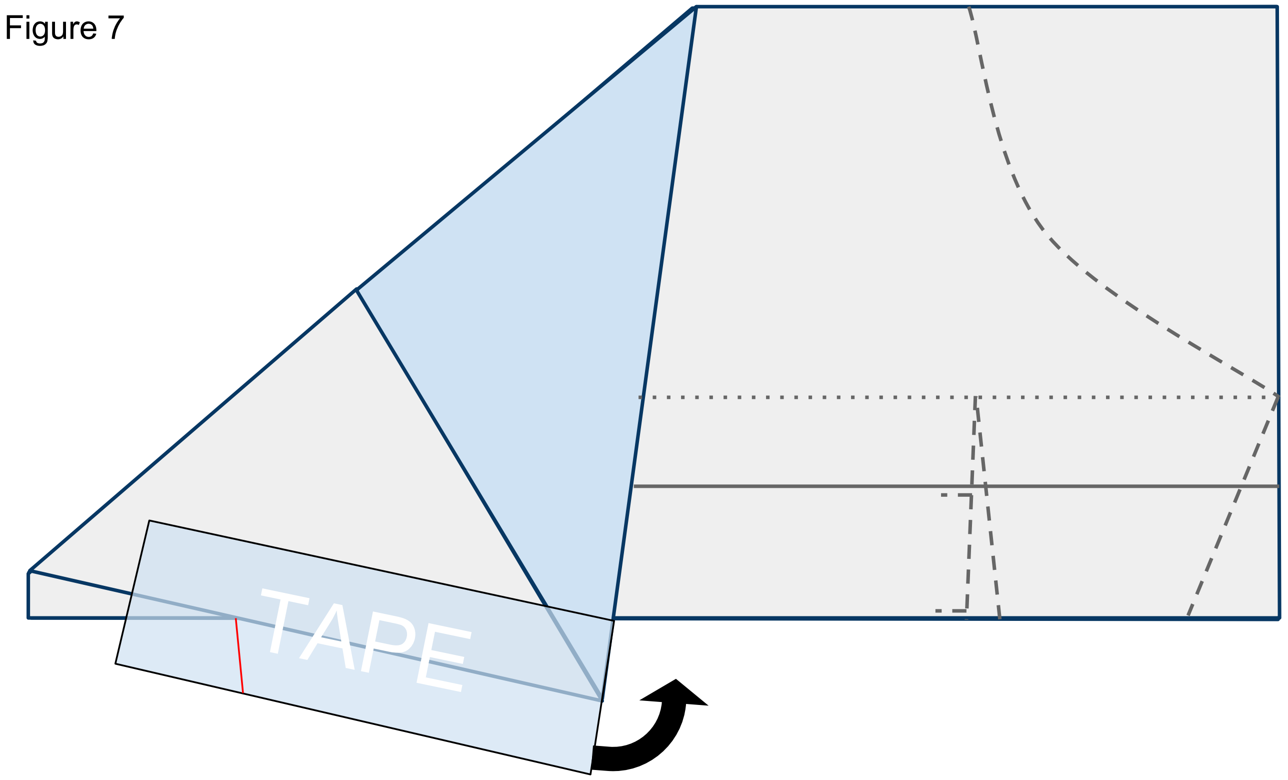
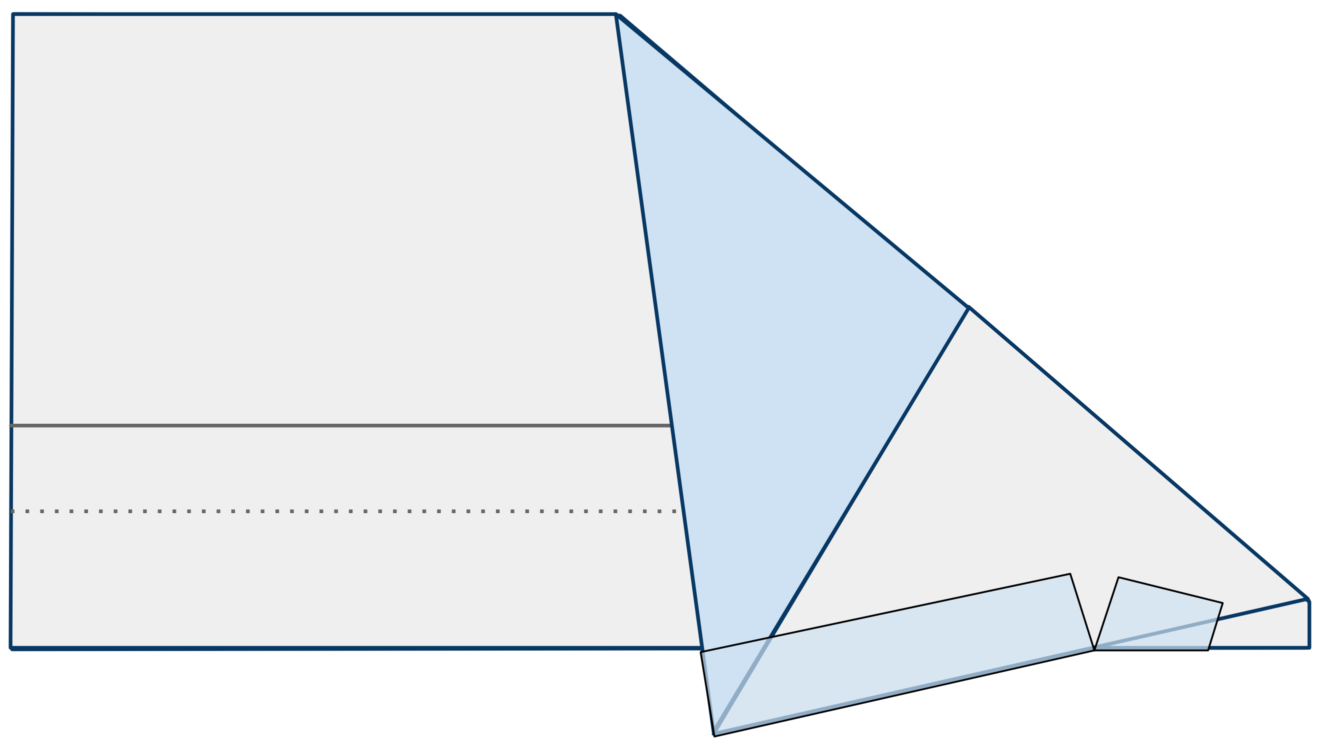


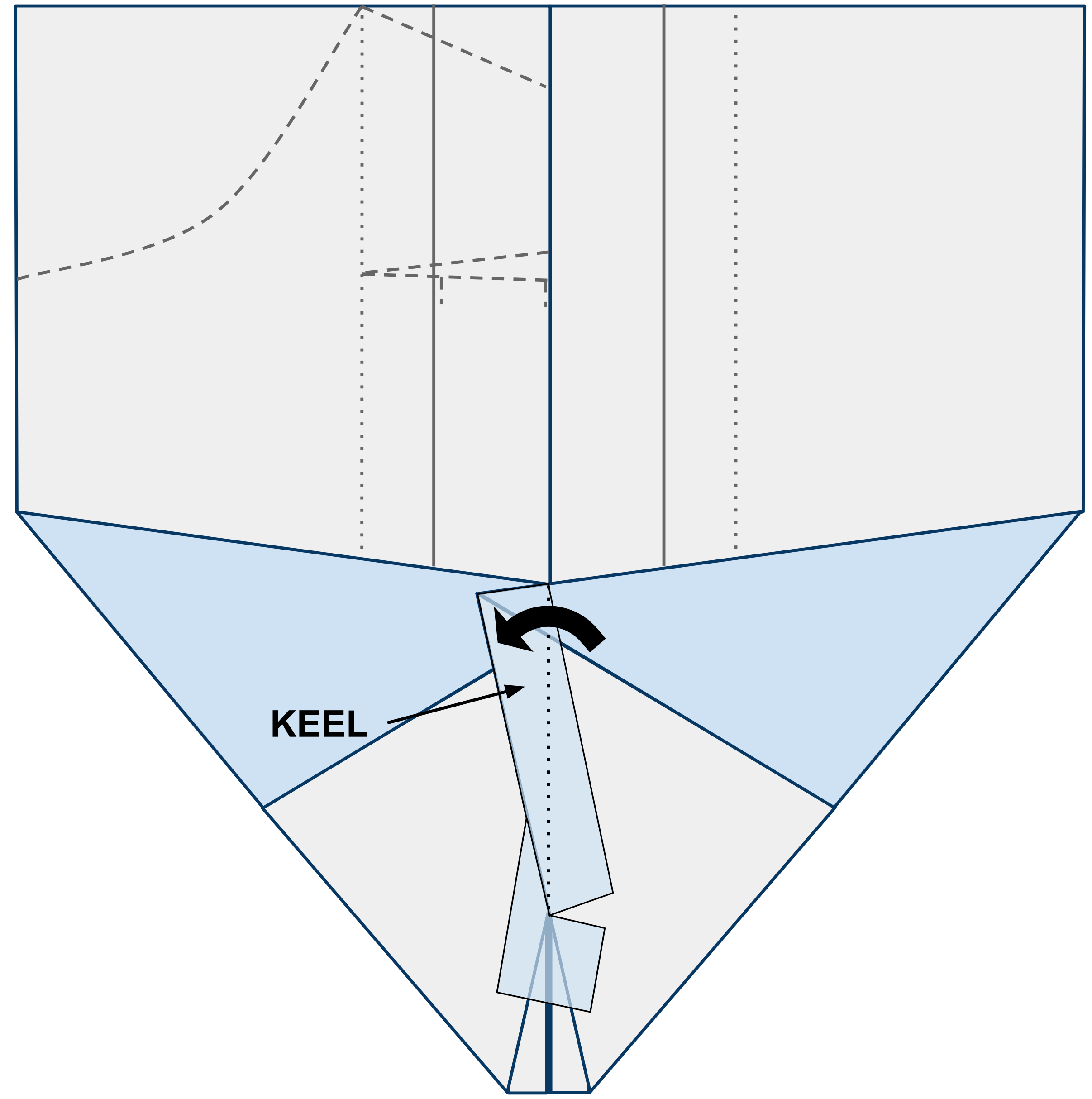
Figure 7b



8

Open the plane flat on your work surface with the nose toward you and looking at the underside of the body. When you do this the keel will want to stand up off the table. Bend it to the left along the center line and give it a strong crease as shown in Figure 8 below.

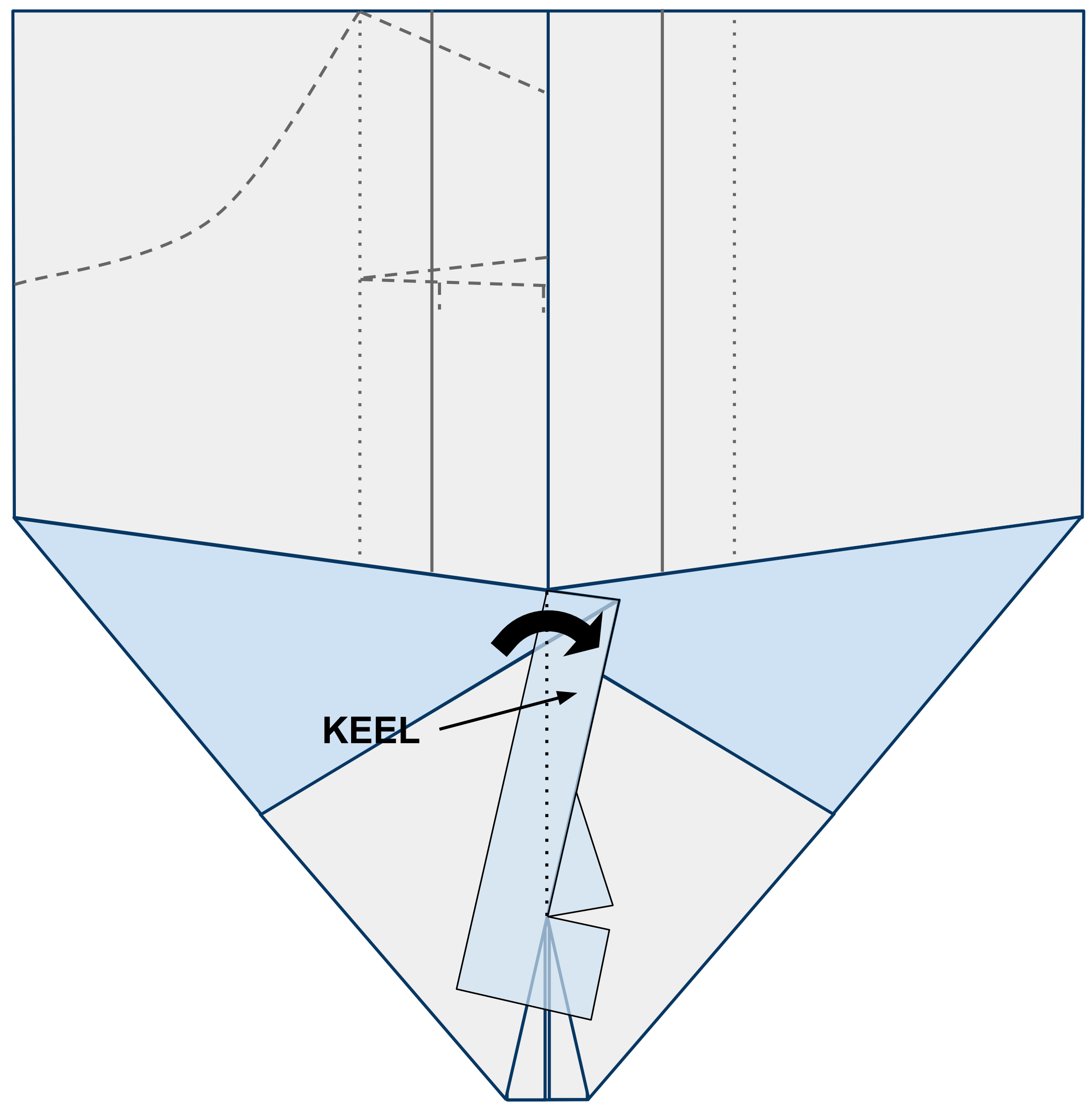
Figure 8



9

Now fold the keel to the right side creating a strong crease as shown in Figure 9 below. After making this crease, allow the keel to remain upright. This is where you will hold the plane when launching it.

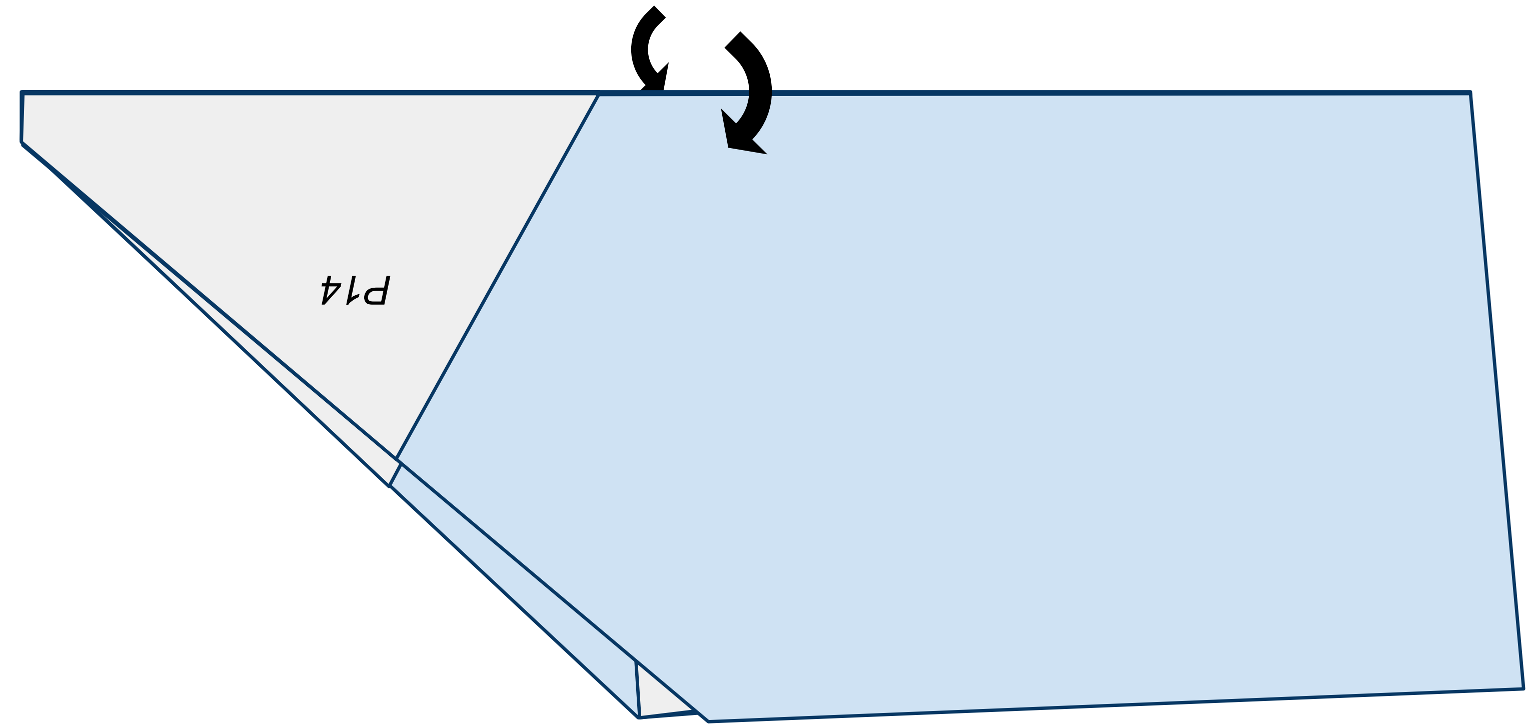
Figure 9



10

Rotate the nose of the plane to the left clockwise and fold the plane in half down the middle covering up the keel. Fold along the fold created in step 1 along the center line of the plane as shown in Figure 10 below.

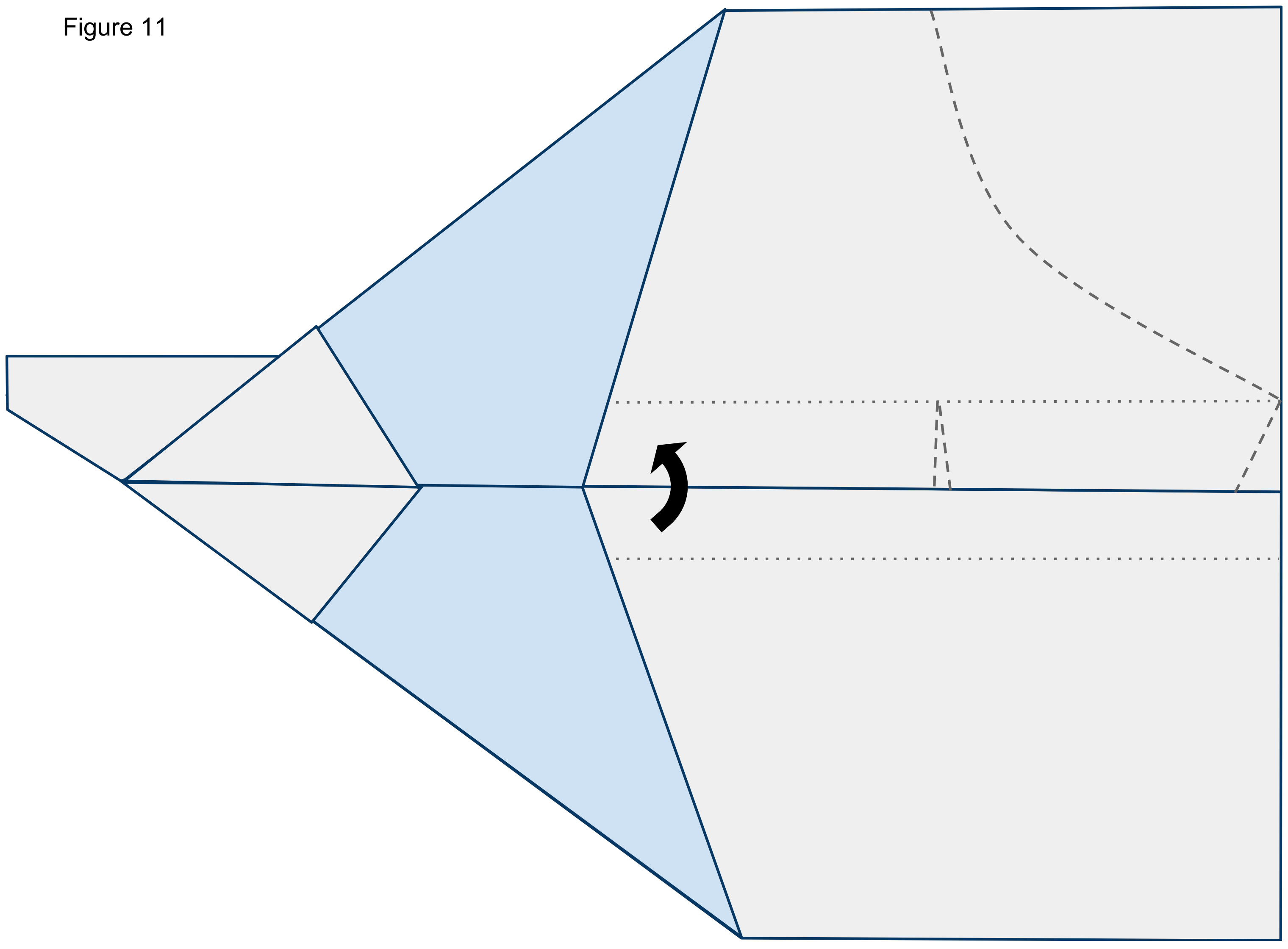
Figure 10



11

Now fold the near side wing back up along the solid line shown in Figure 11 below.

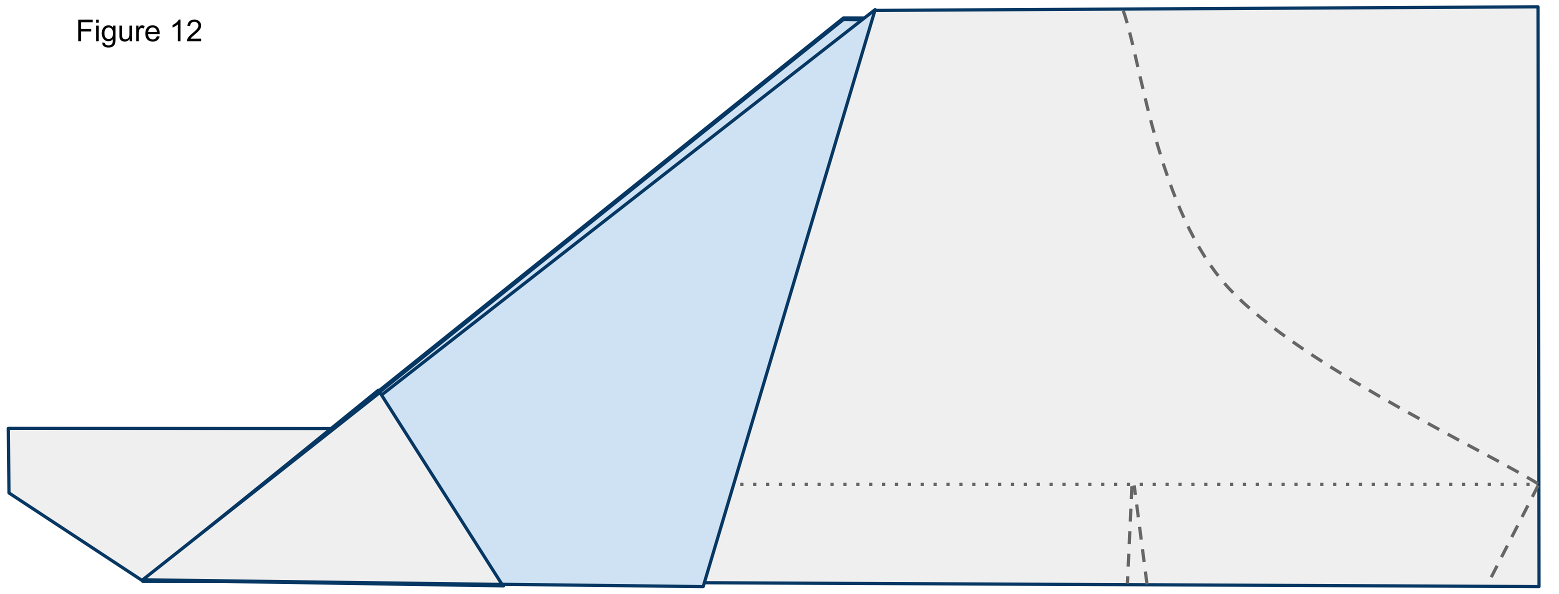
Figure 11



12

Do the same to the other side as shown in Figure 12 below. Remember that matching symmetry of the plane is more important that lining up the printed lines on the page.

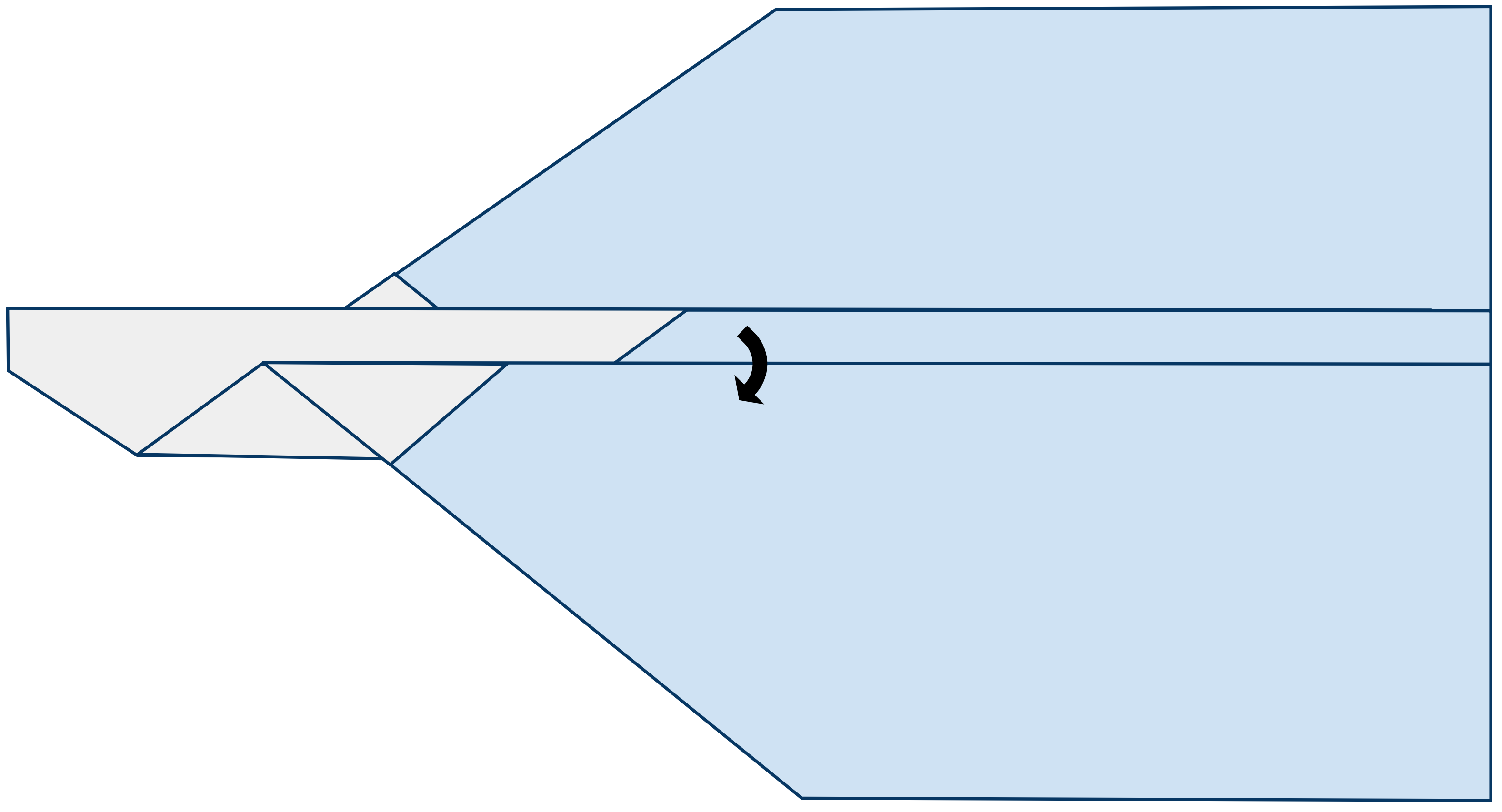
Figure 12



13

Now fold the near side wing down along the dotted line as shown in Figure 13 below.

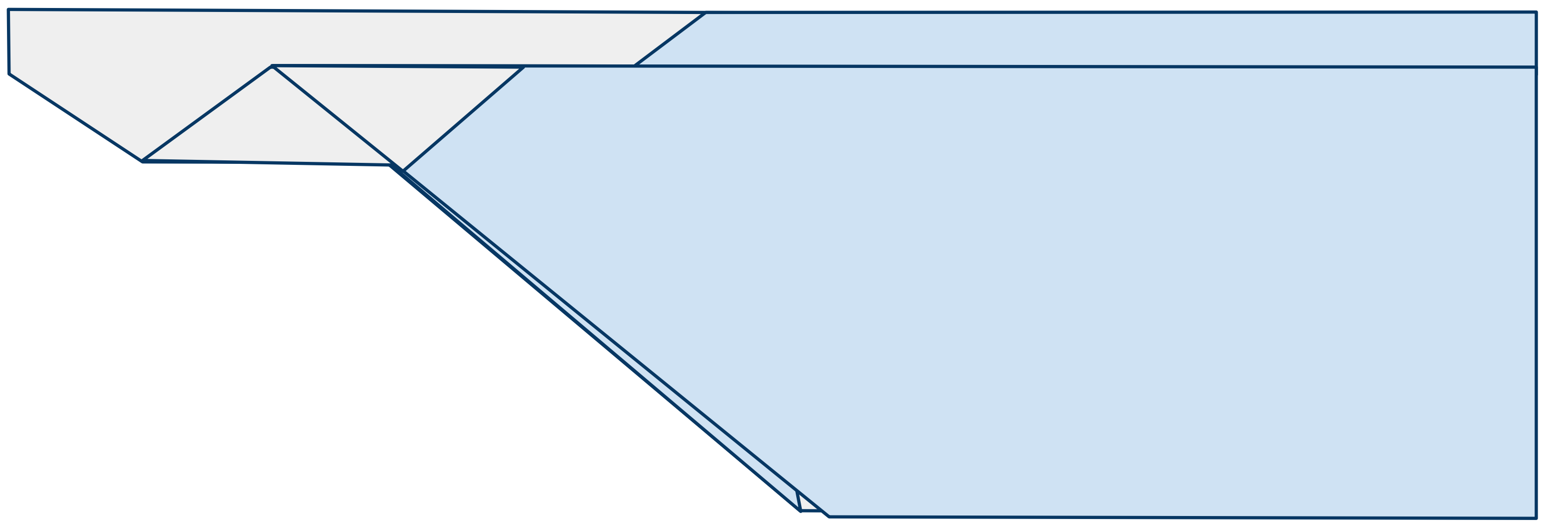
Figure 13



14

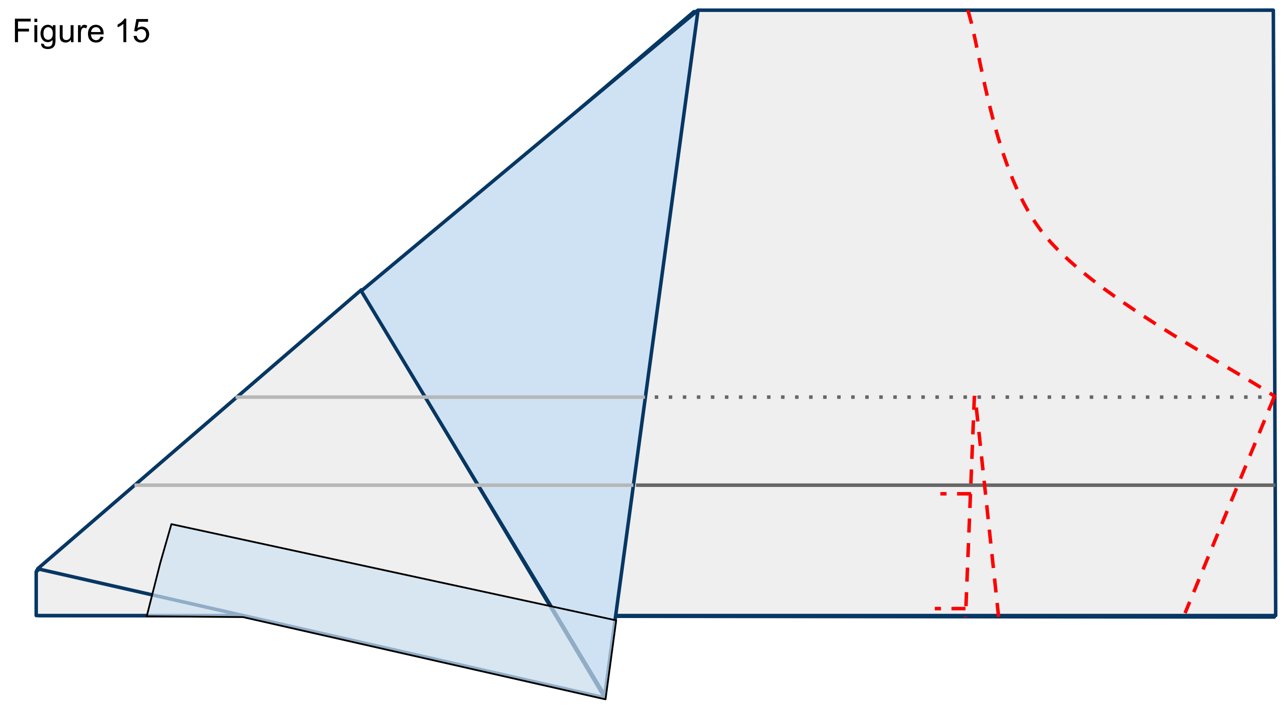
Do the same to the other side matching the plane symmetrically as shown in Figure 14 below.

Figure 14



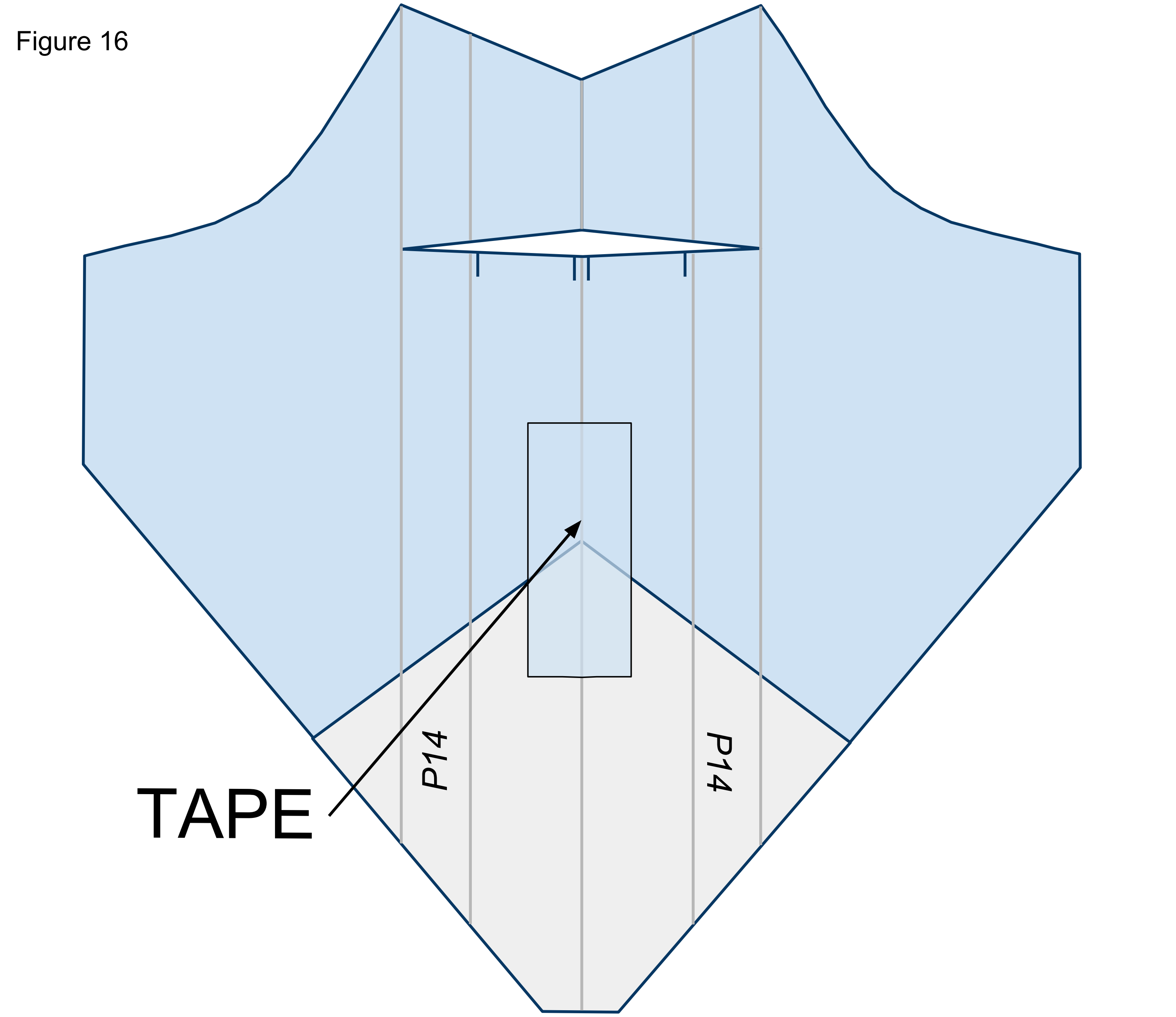
15

Open the wings all the way back up and lay the plane flat on the work surface. Now cut through both layers of paper along the **red** dashed lines as indicated by Figure 15 below. Don't forget the slits in order to create your flaps.



16

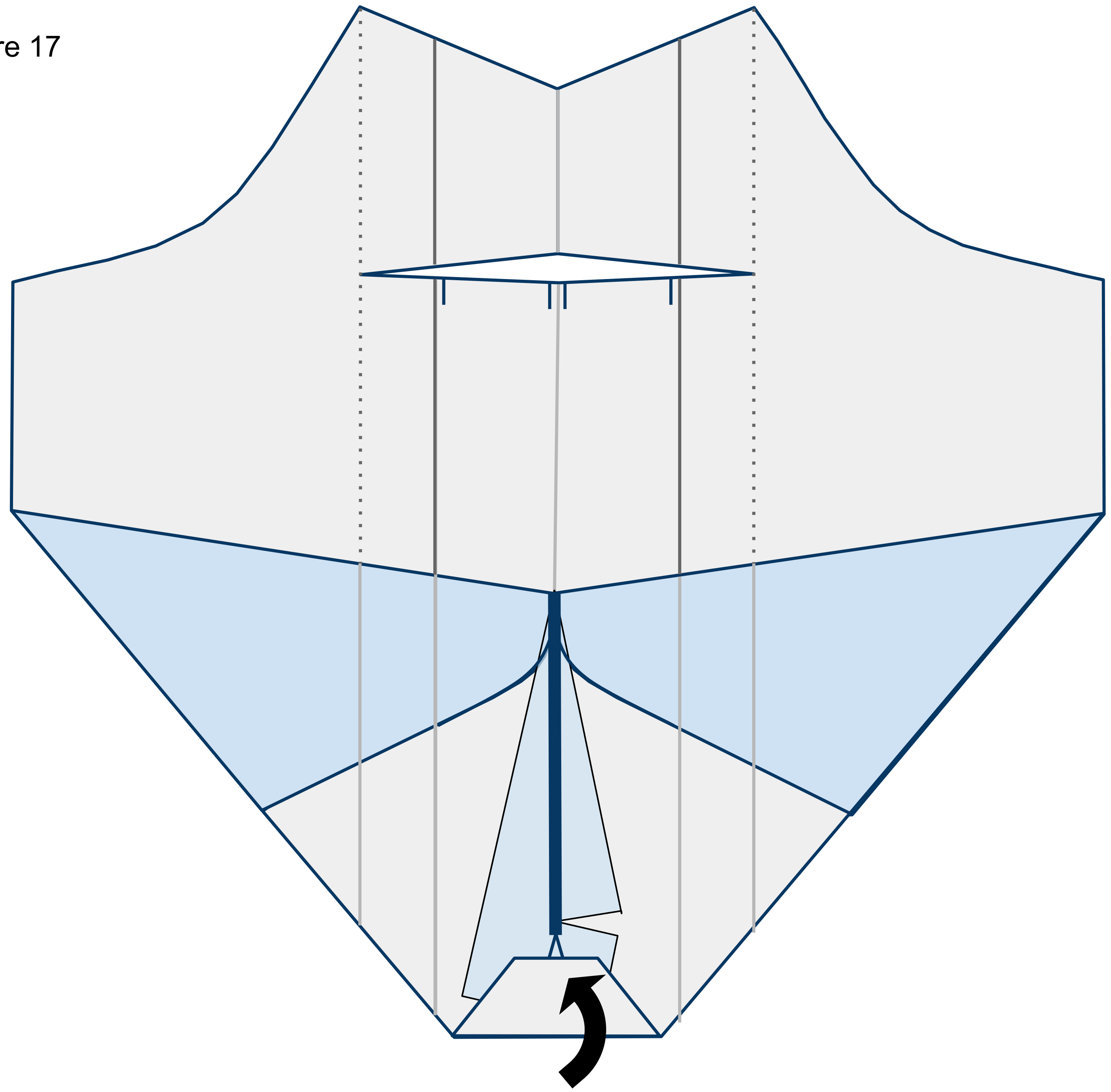
Turn the nose of the plane toward you counterclockwise and lay the plane flat with the keel on the underside. Lay the keel to one side or the other. Place tape along the fuselage as shown in Figure 16 below.



17

Turn the plane over to show the underside. The vertical standing keel is represented by the thick **blue** line. Fold the nose back as shown in Figure 17 below.

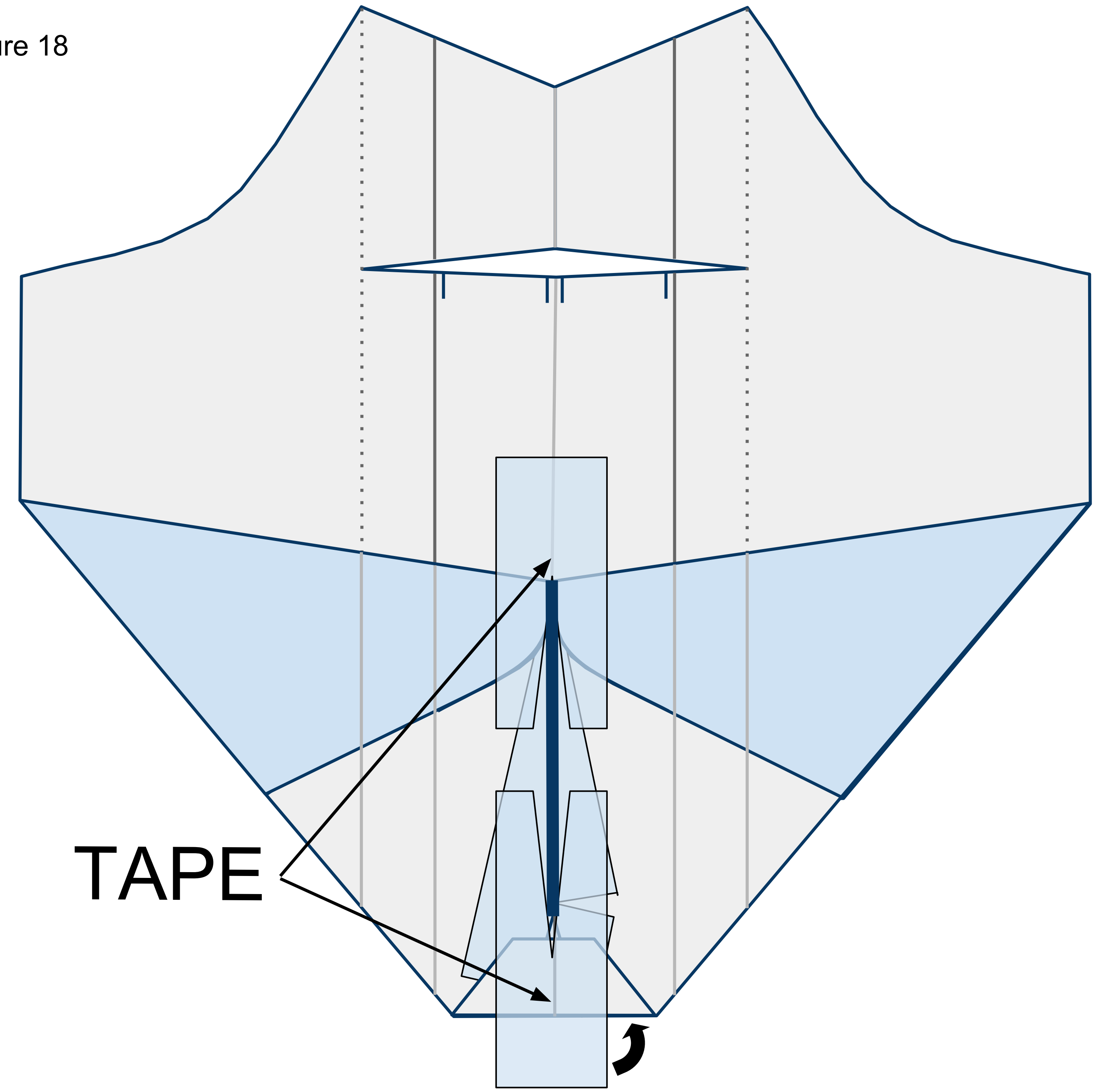
Figure 17



18

Place 2 pieces of tape with slices cut in them on the underside of the plane in front of and behind the keel as shown in Figure 18 below. Wrap the tape around the nose to the top side of the plane.

Figure 18



19

Place a piece of tape across the nose of the plane as shown in Figure 19 below. Cut slits along the red lines as shown. These slits will help you fold the tape over the plane in the next step.

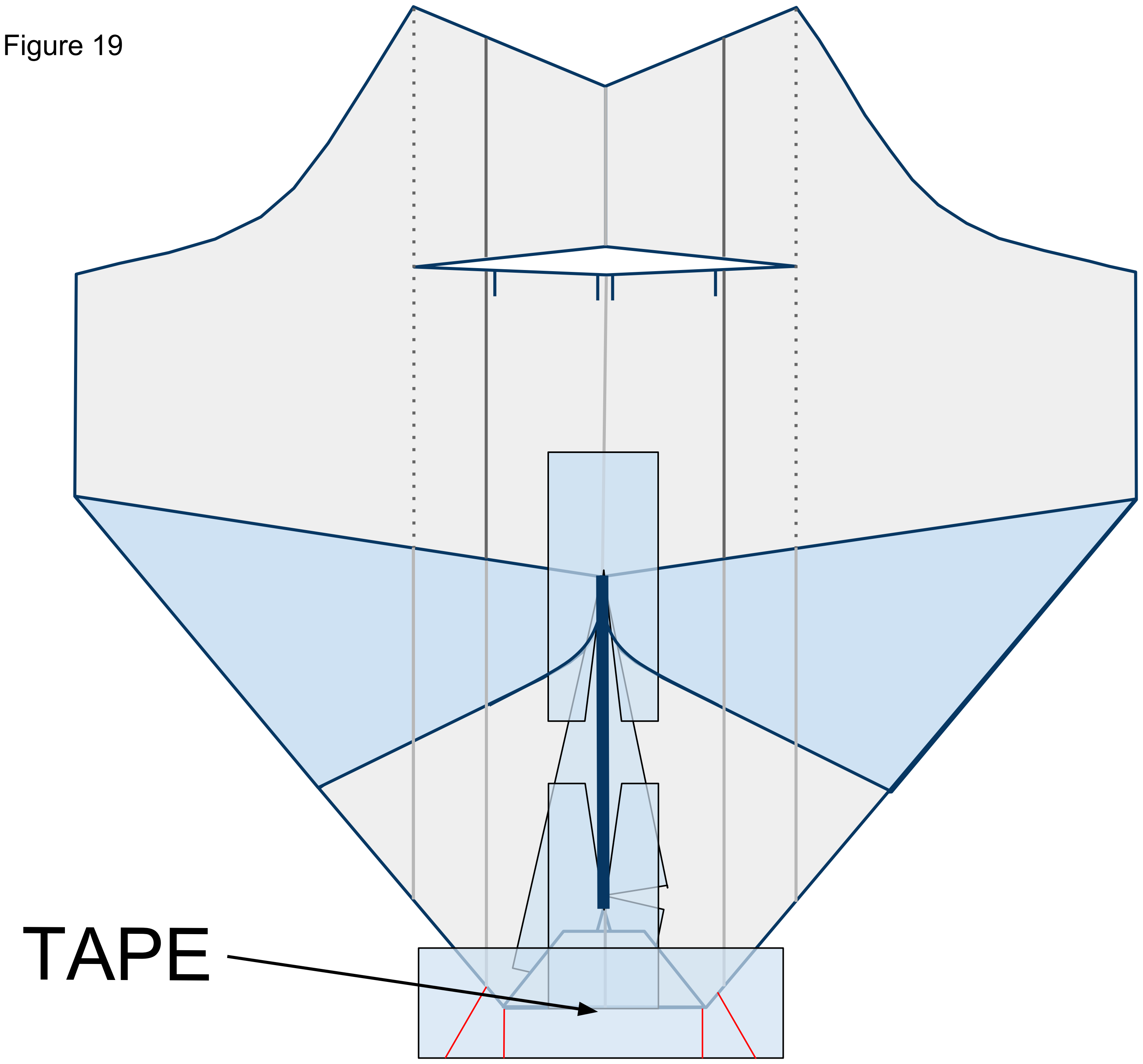


Figure 19

20

Flip the plane over so you can see the top side. Fold the flaps of tape, in number order, up and in as shown in Figure 20 below. They will overlap each other once folded.

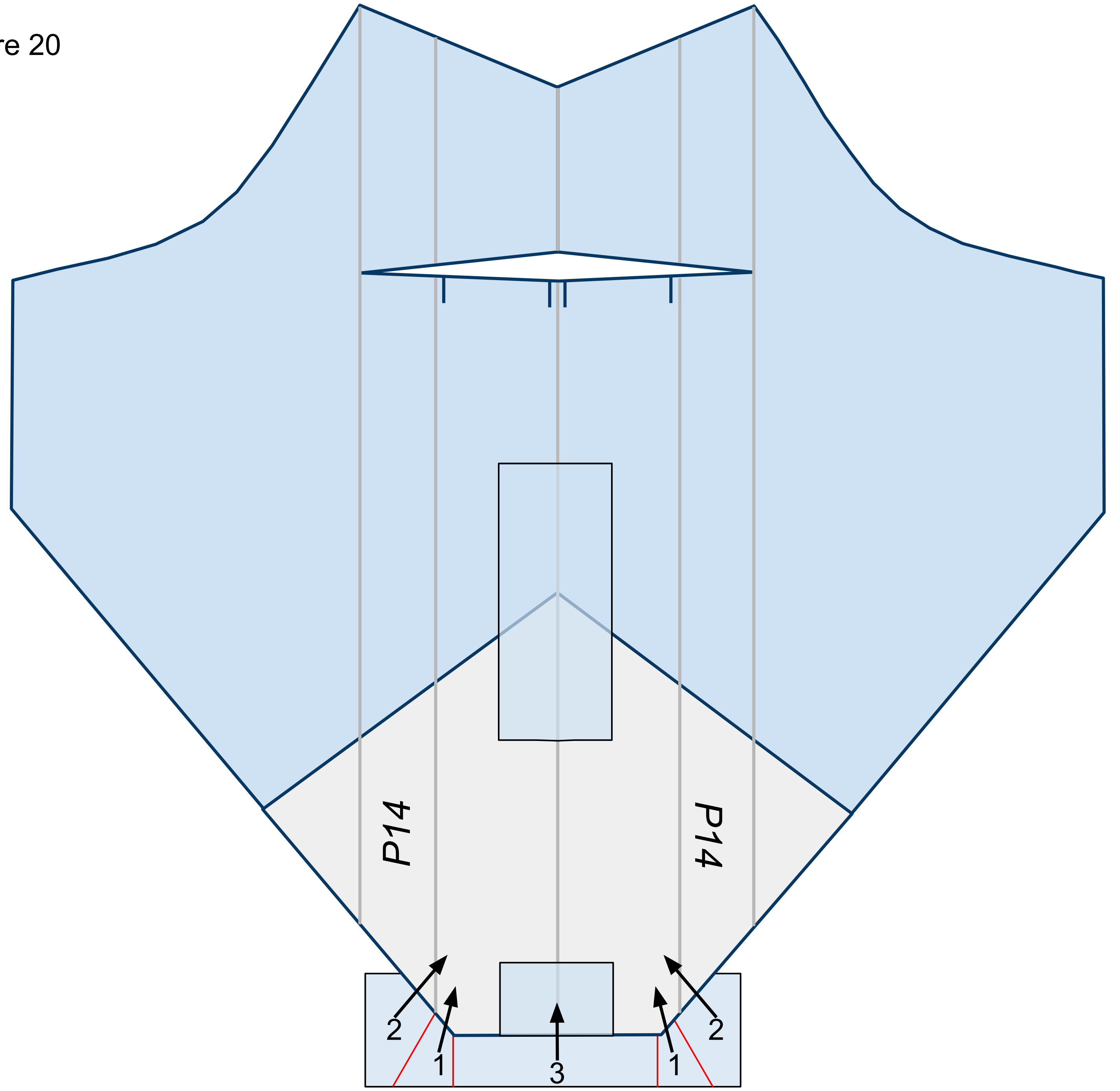
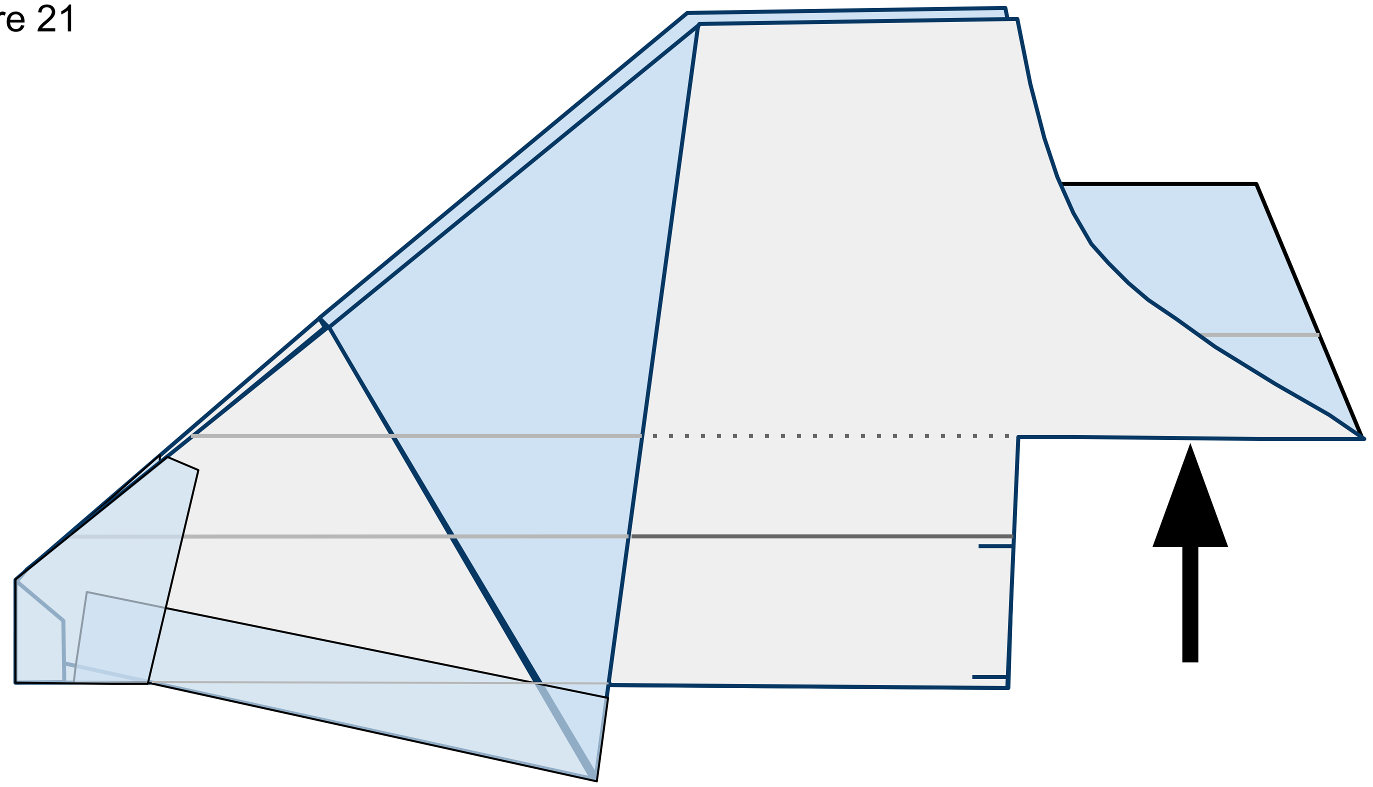


Figure 20

21

Turn the nose to the left and push the tail section up in between the two wings as shown in Figure 21 below. Reverse the creases across the tail. If you are having difficulty you can refer to "23" to get a head-on view.

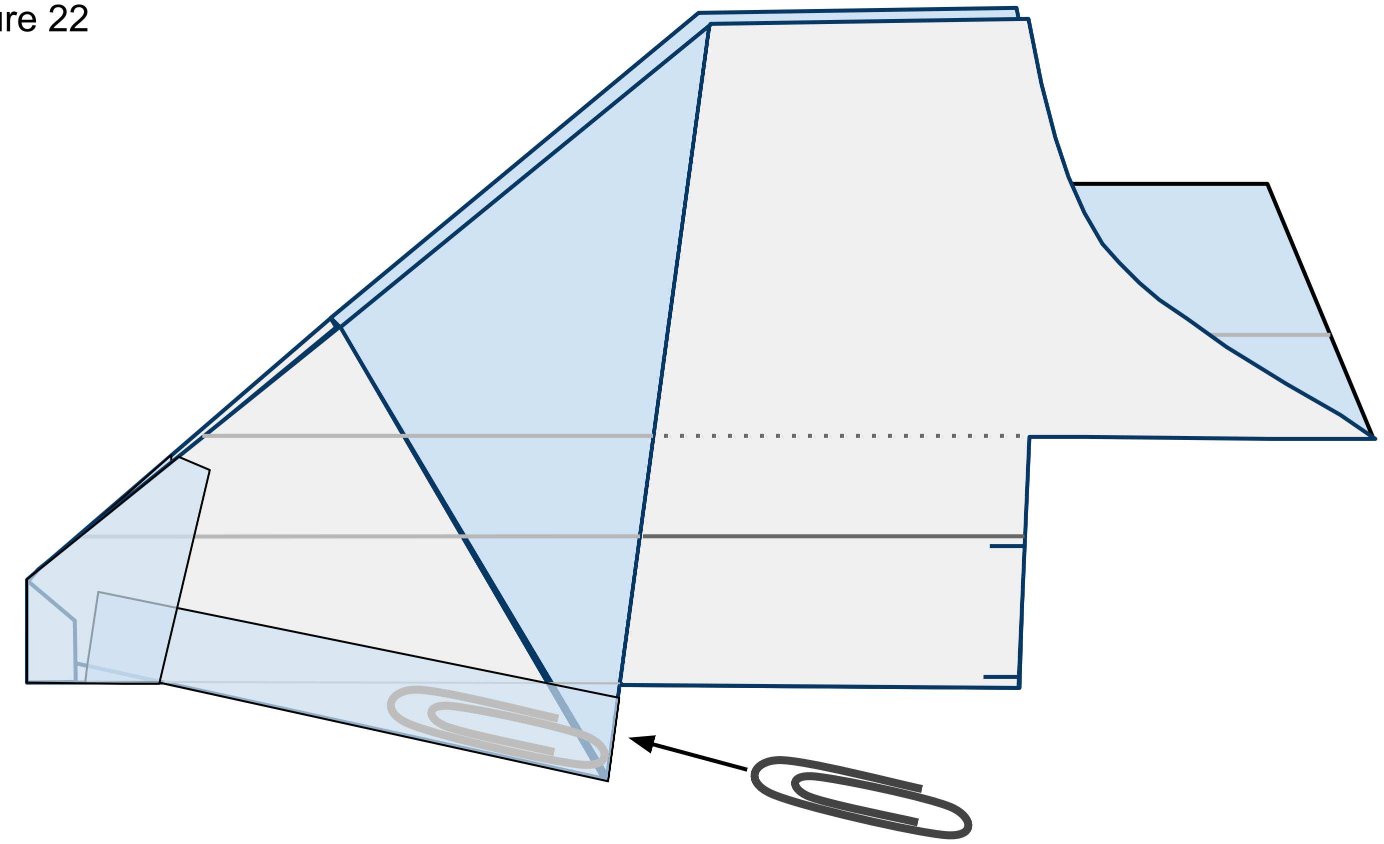
Figure 21



22

Insert a #1 size paper clip ***just*** inside the keel through the central opening behind it as shown in Figure 21 below. Then seal the opening with a piece of tape.

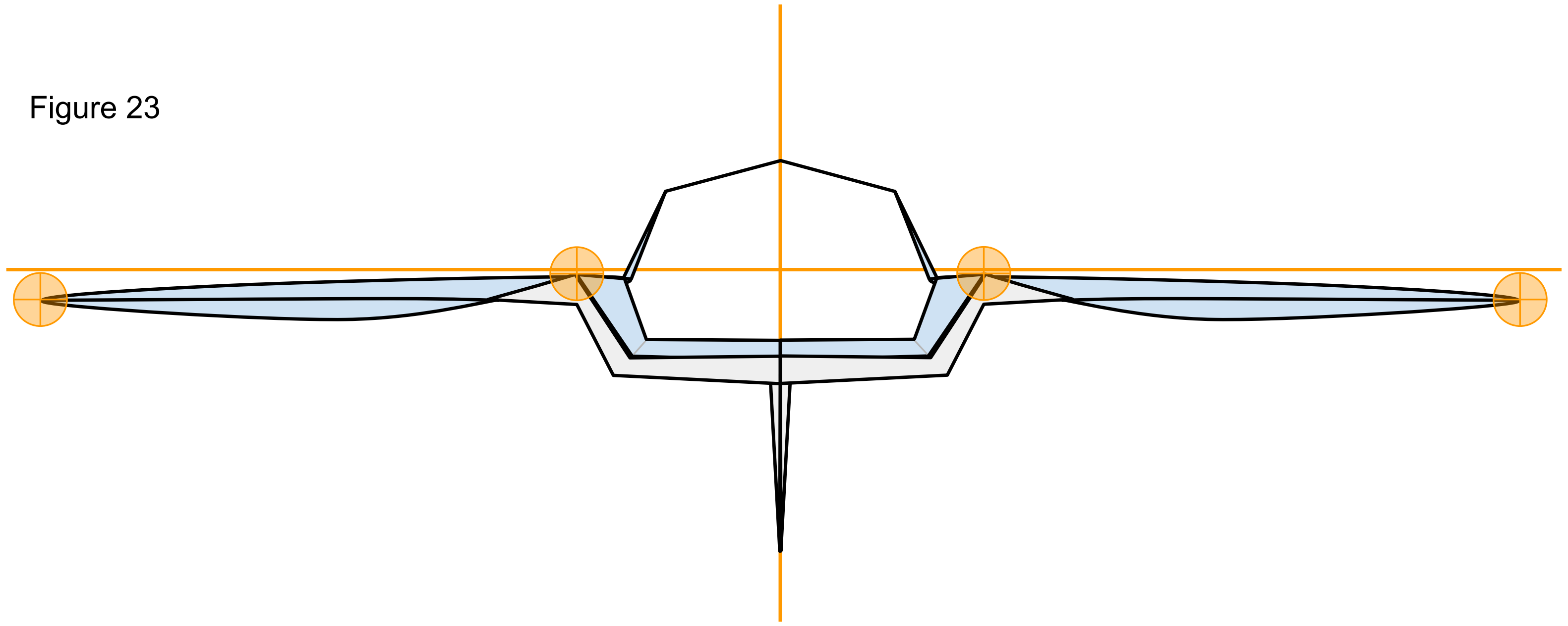
Figure 22



23

Now that you have constructed the plane ensure that it has the proper shape. Take a look at your plane head-on from the nose. The plane should be symmetrical. Take a look at Figure 23 below. Note the points of interest. Note the curved shape of the paper on the underside of the wings. To achieve this, carefully stick one side of a pair of scissors up into the pocket on the underside of the wing from behind and gently mold the paper into the proper shape. Don't crease it.

Figure 23



25

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P15 MANTA

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1

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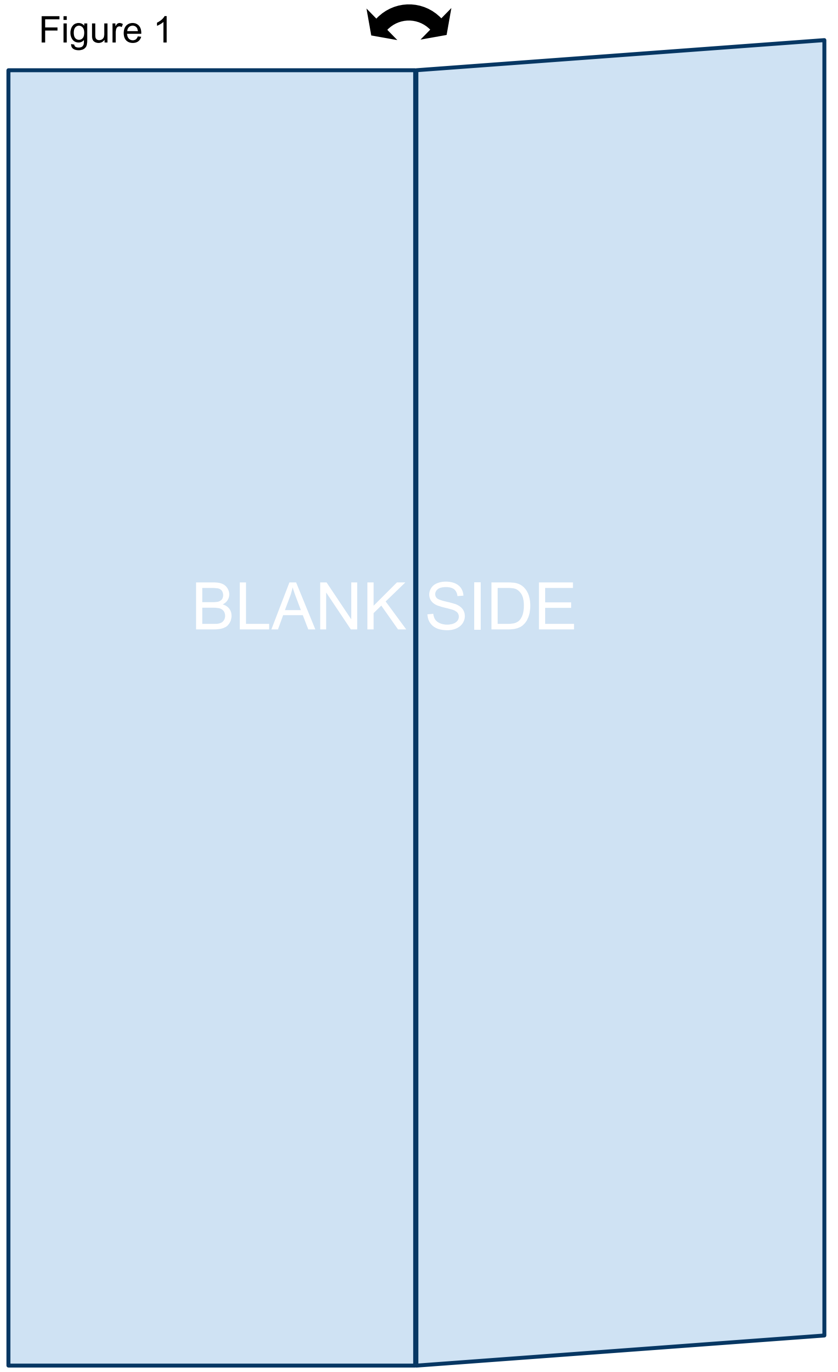
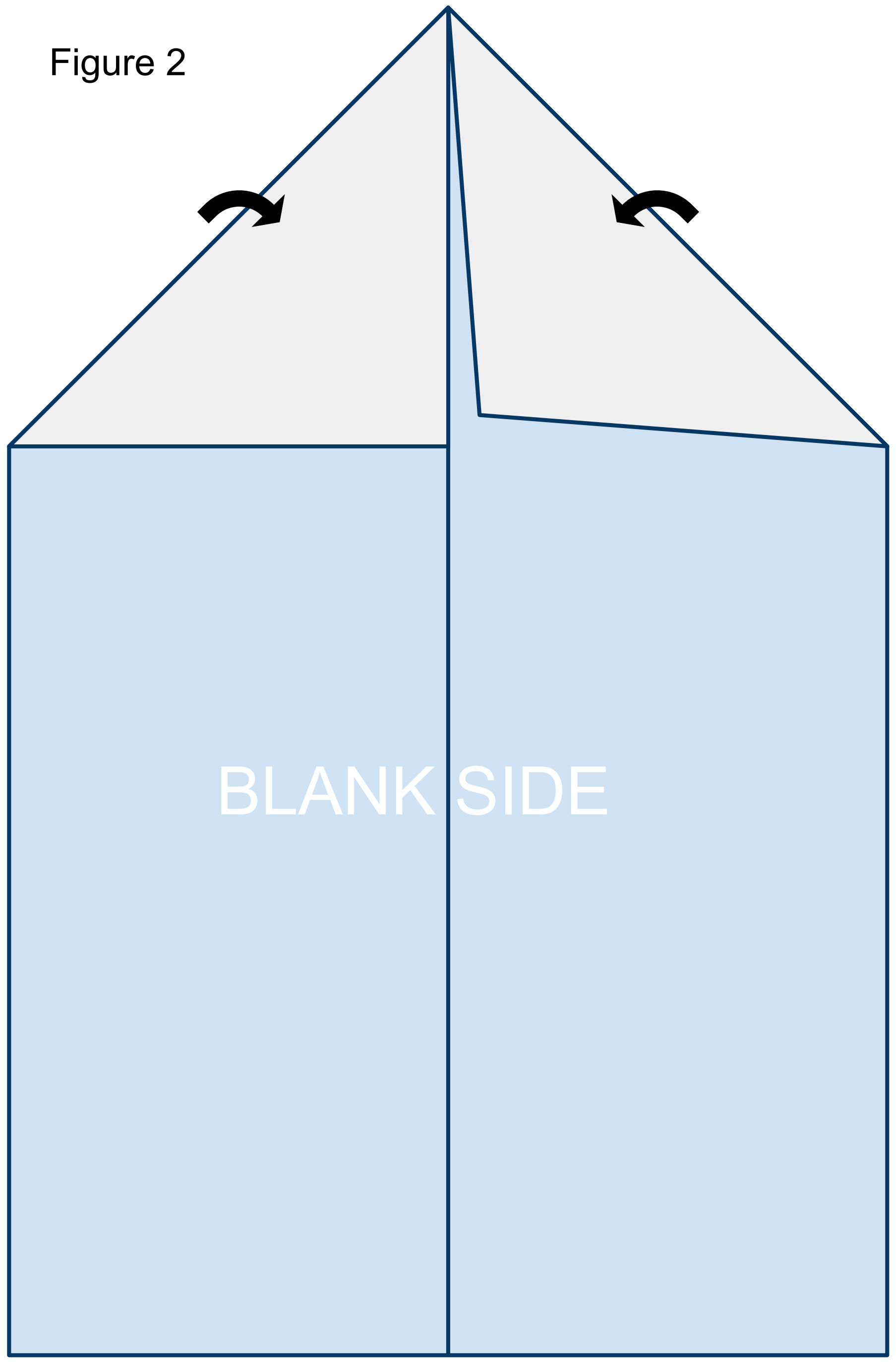
Figure 1



2

Fold the leading corners back along the printed solid lines. The edge of the paper should line up along the crease you created in step 1 as shown in Figure 2 below.

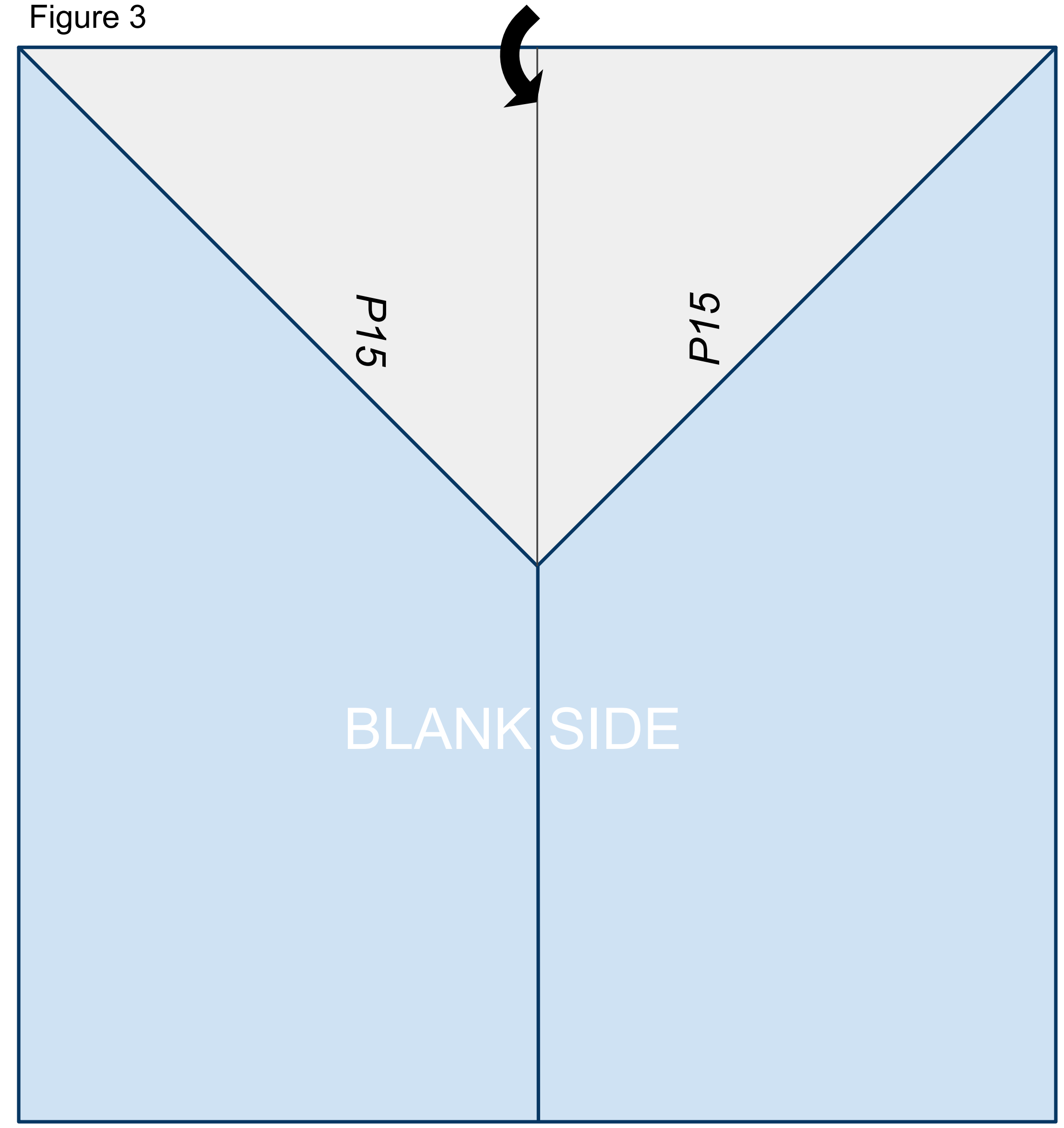
Figure 2



3

Fold the nose of the plane created in step 2 back along the solid line printed on the front of the paper so that it lines up on the center crease created in step 1 as shown in Figure 3 below.

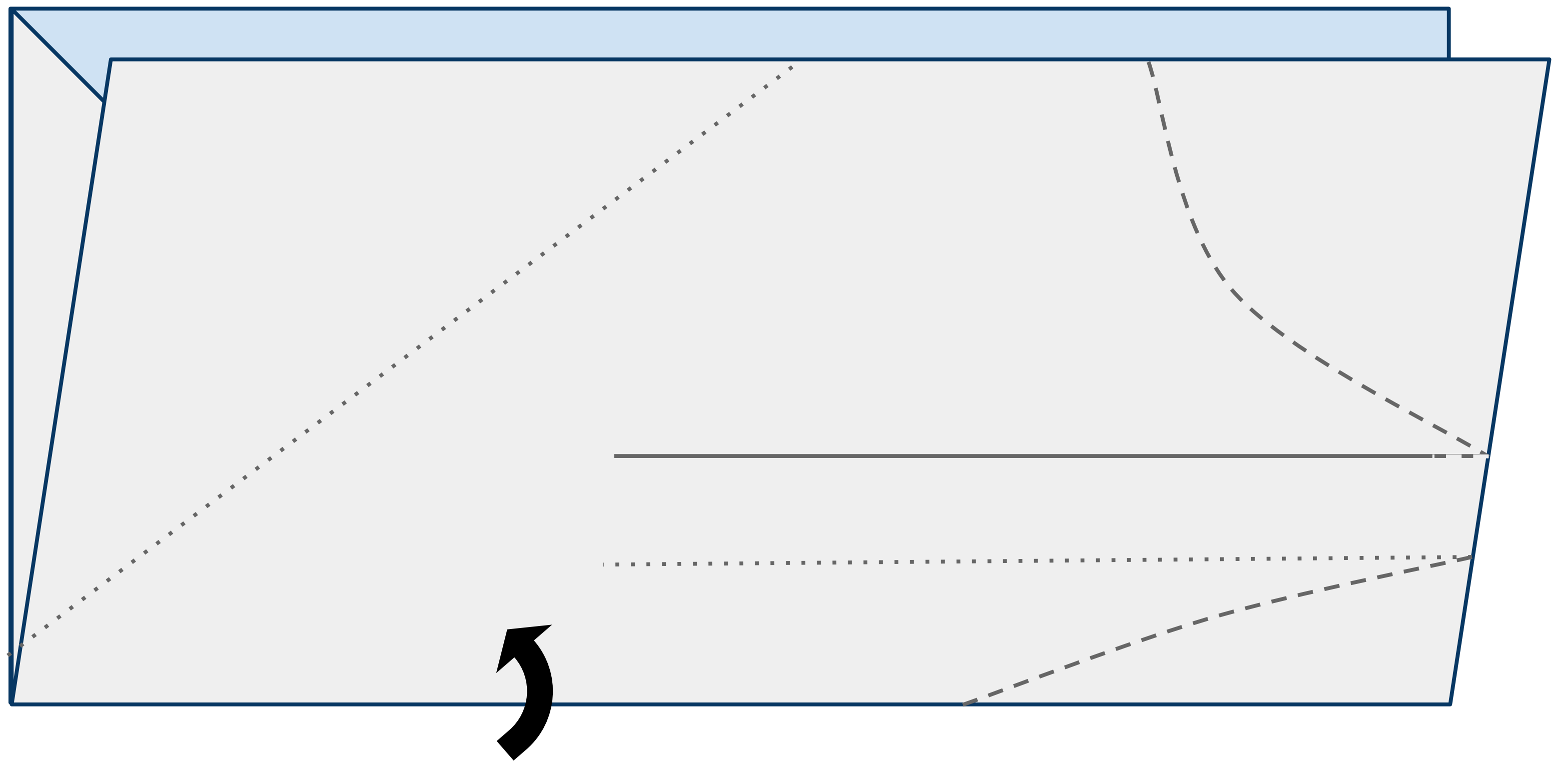
Figure 3



4

Turn the paper to the left (counterclockwise) and fold the paper upward in half hiding the triangle created in the previous steps refolding along the fold from step 1 as shown in Figure 4 below.

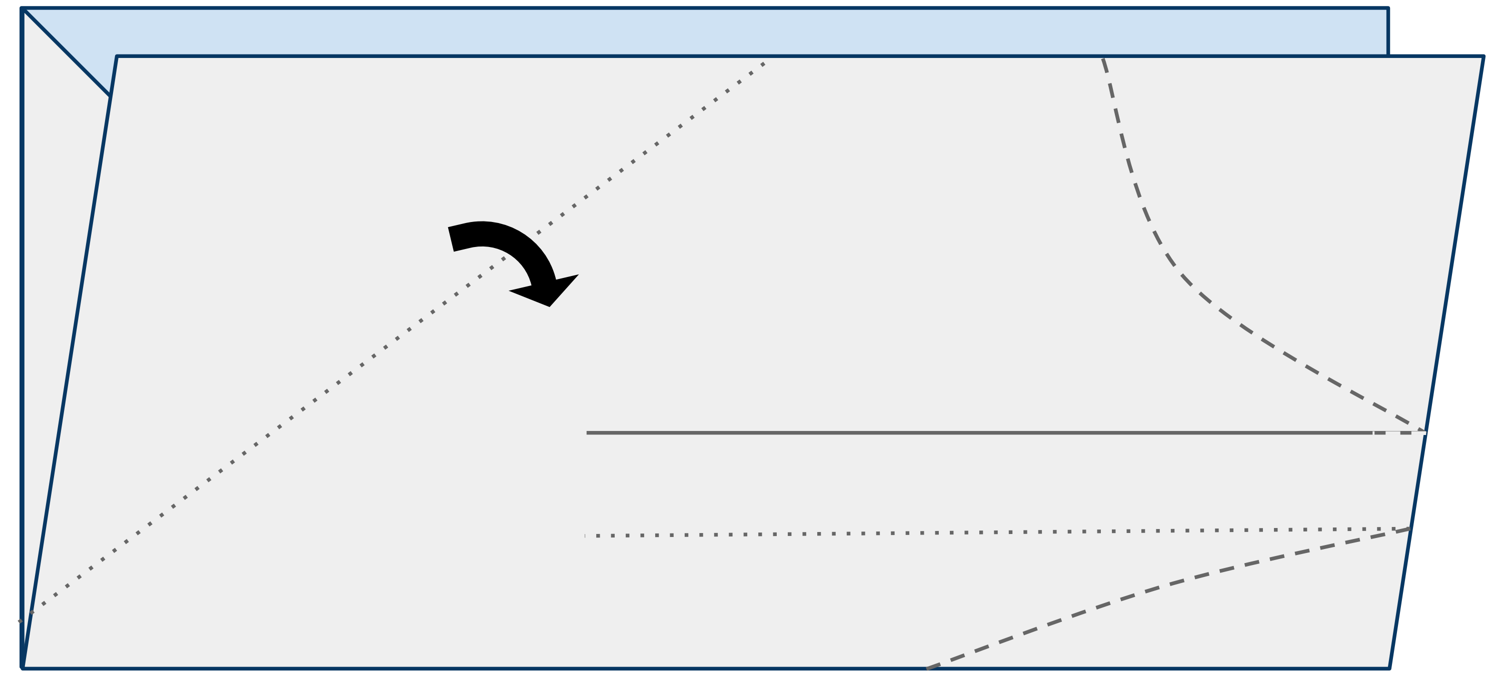
Figure 4



5

Fold the leading edge of the near side of the plane down along the dotted line as shown in Figure 5 below.

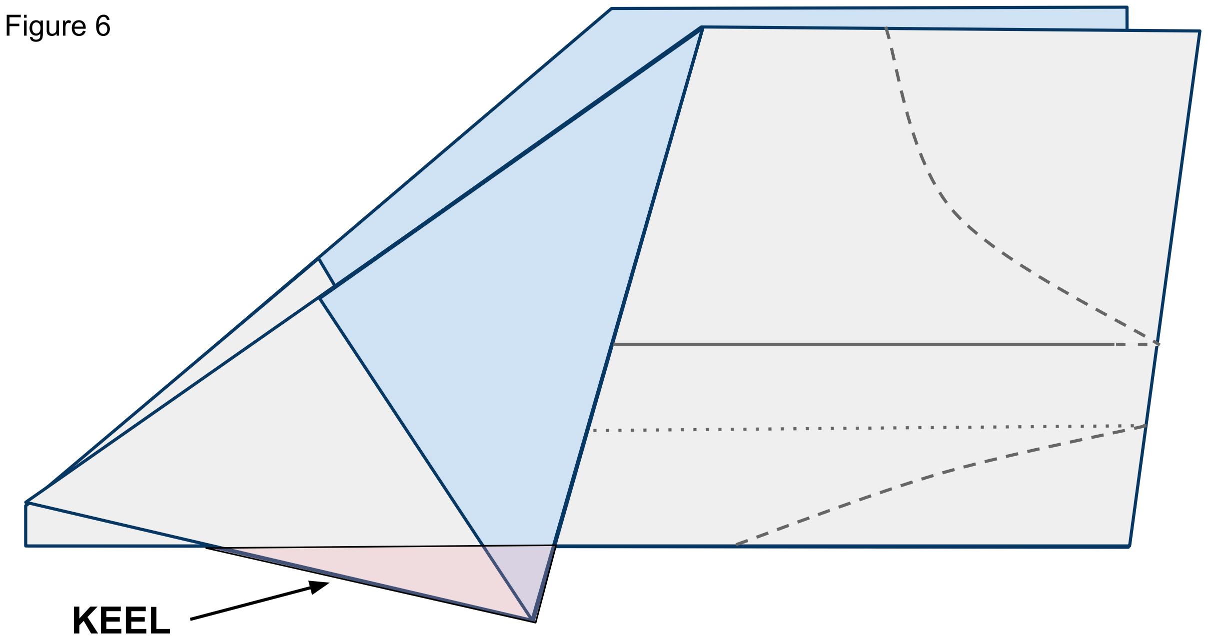
Figure 5



6

Do the same to the other side of the plane. It is important to match these 2 folds exactly so that your plane is symmetrical and will fly straight. The area that is created below the body and designated in pink will be called the keel.

Figure 6



7

Place tape along the keel as shown in Figure 7 below and cut a slit in the tape along the red line. Now fold the tape around and under both sides of the plane and keel securing them together. See Figure 7b for what the far side of the plane looks like.

Figure 7b

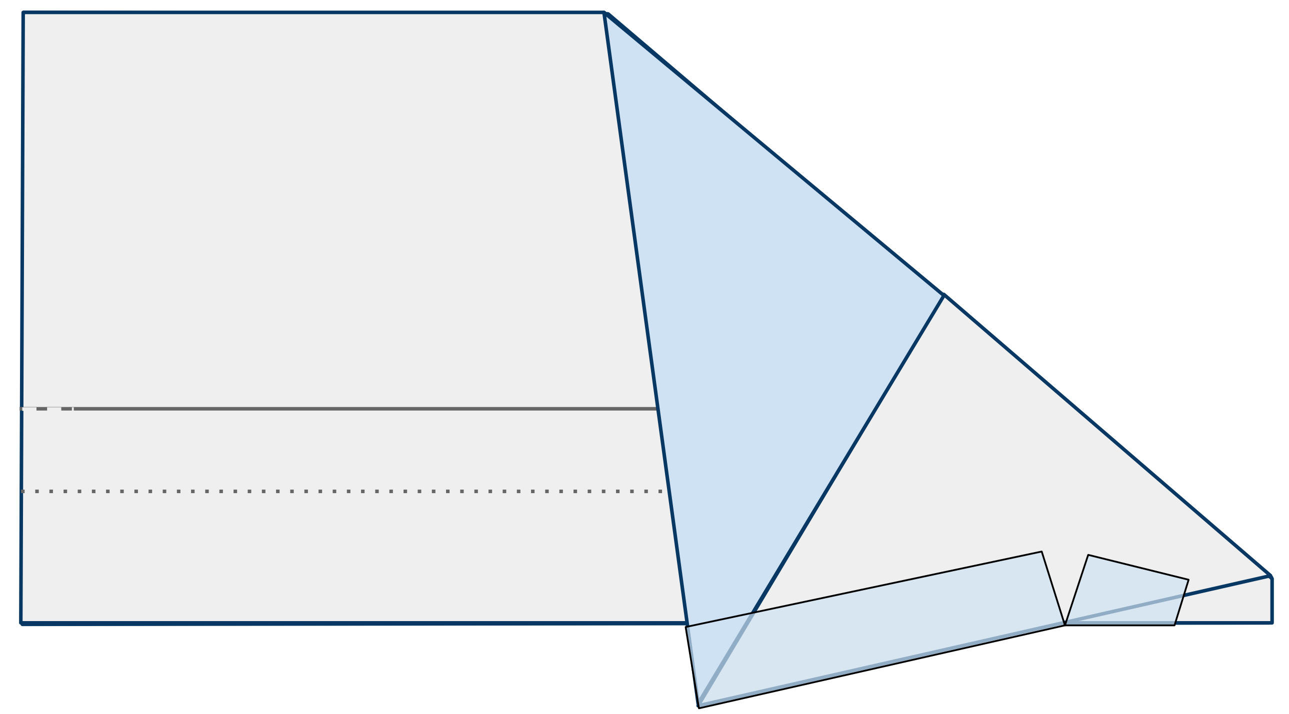
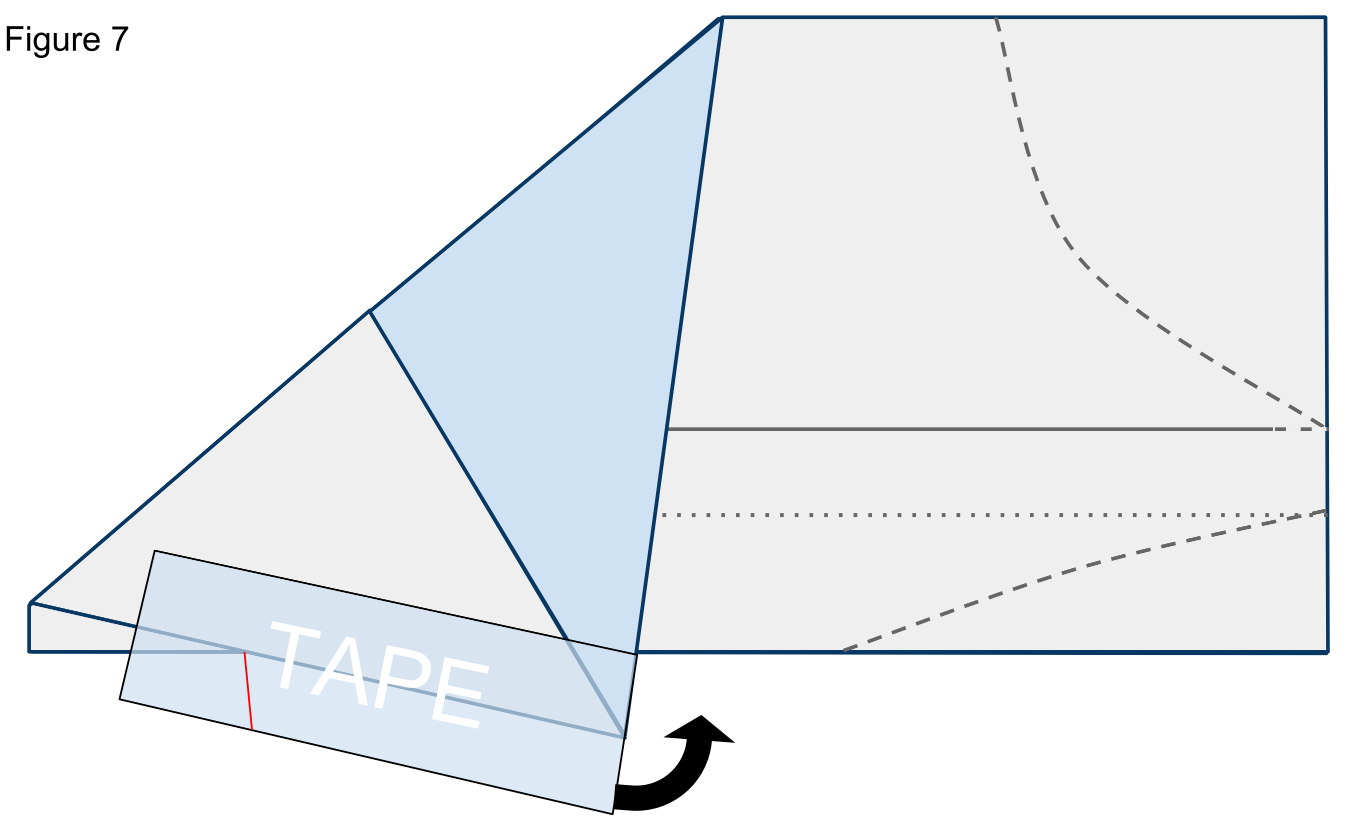


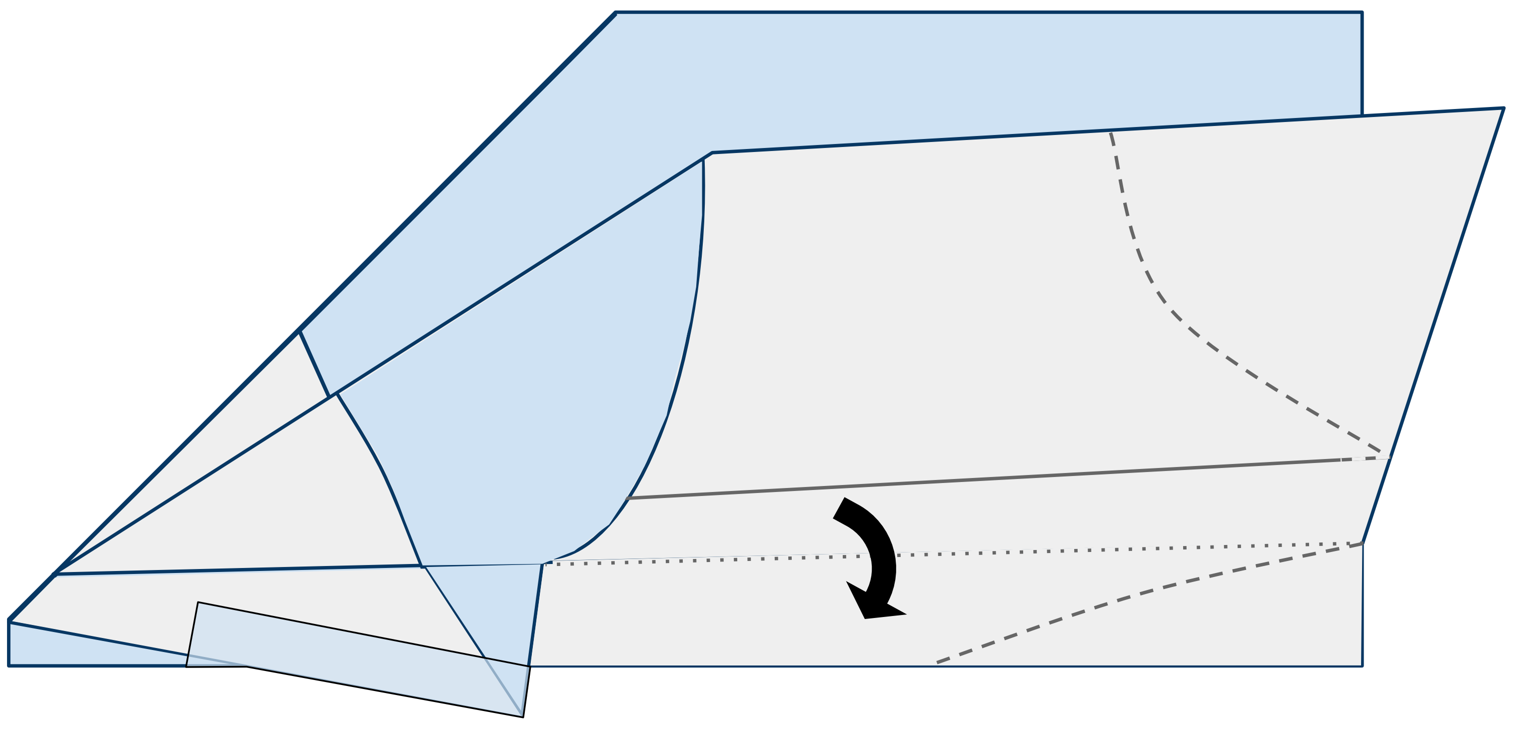
Figure 7



8

Fold down the near side wing along the dotted line as shown in Figure 8 below.

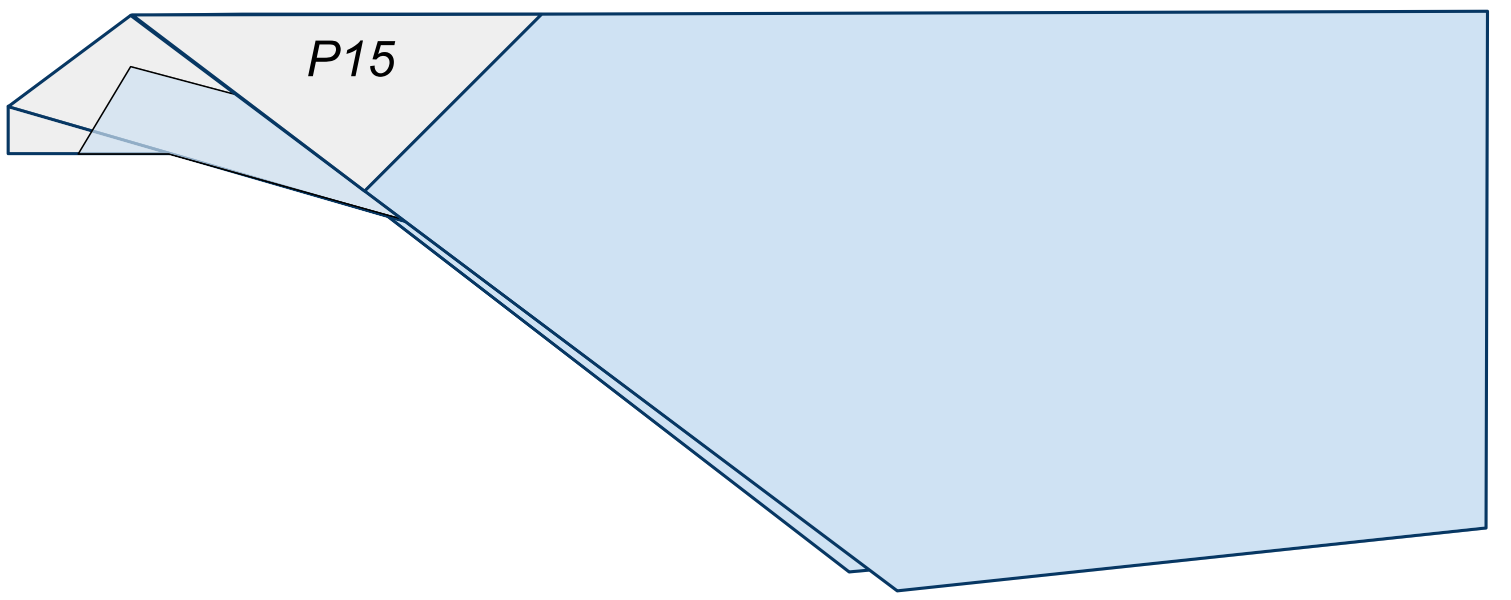
Figure 8



9

Fold down the far side wing along the dotted line also as shown in Figure 9 below. Insure that these folds are as symmetrical as possible. If you need to choose, always choose symmetry over following the lines printed on the paper.

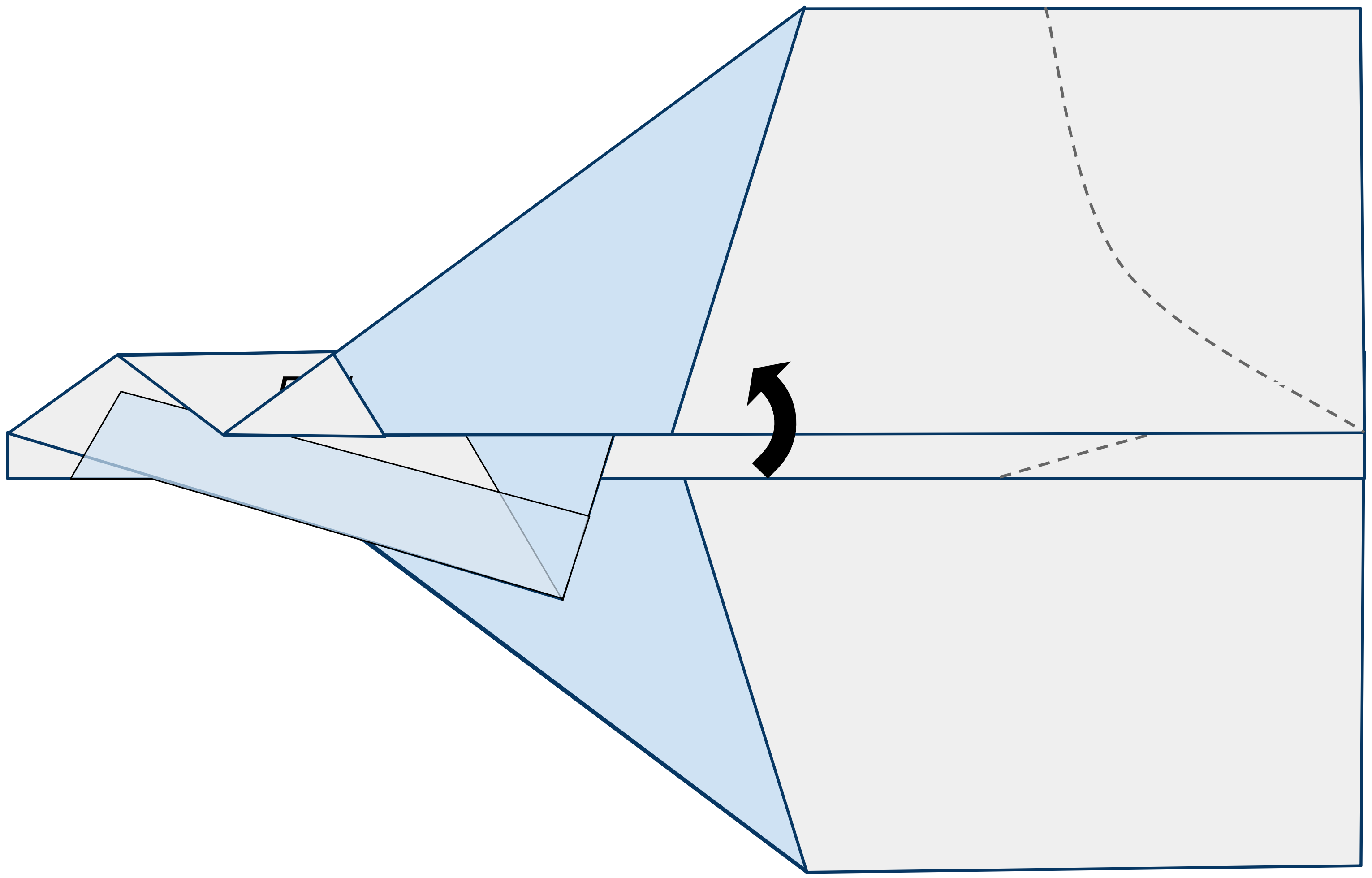
Figure 9



10

Fold the near side wing up along the printed solid line as shown in Figure 10 below.

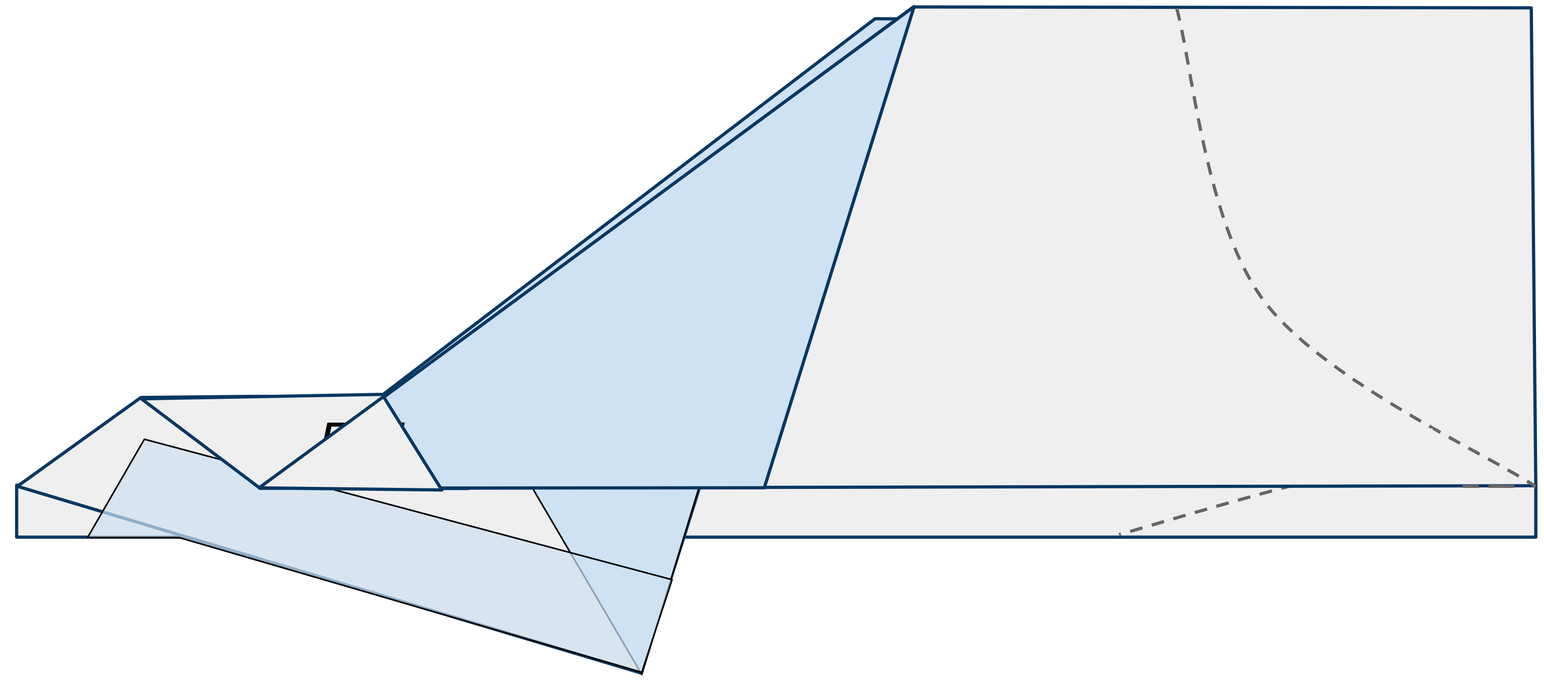
Figure 10



11

Do the same to the other side. Fold the far side wing up along the sold line on the other side of the plane as shown below in Figure 11 below.

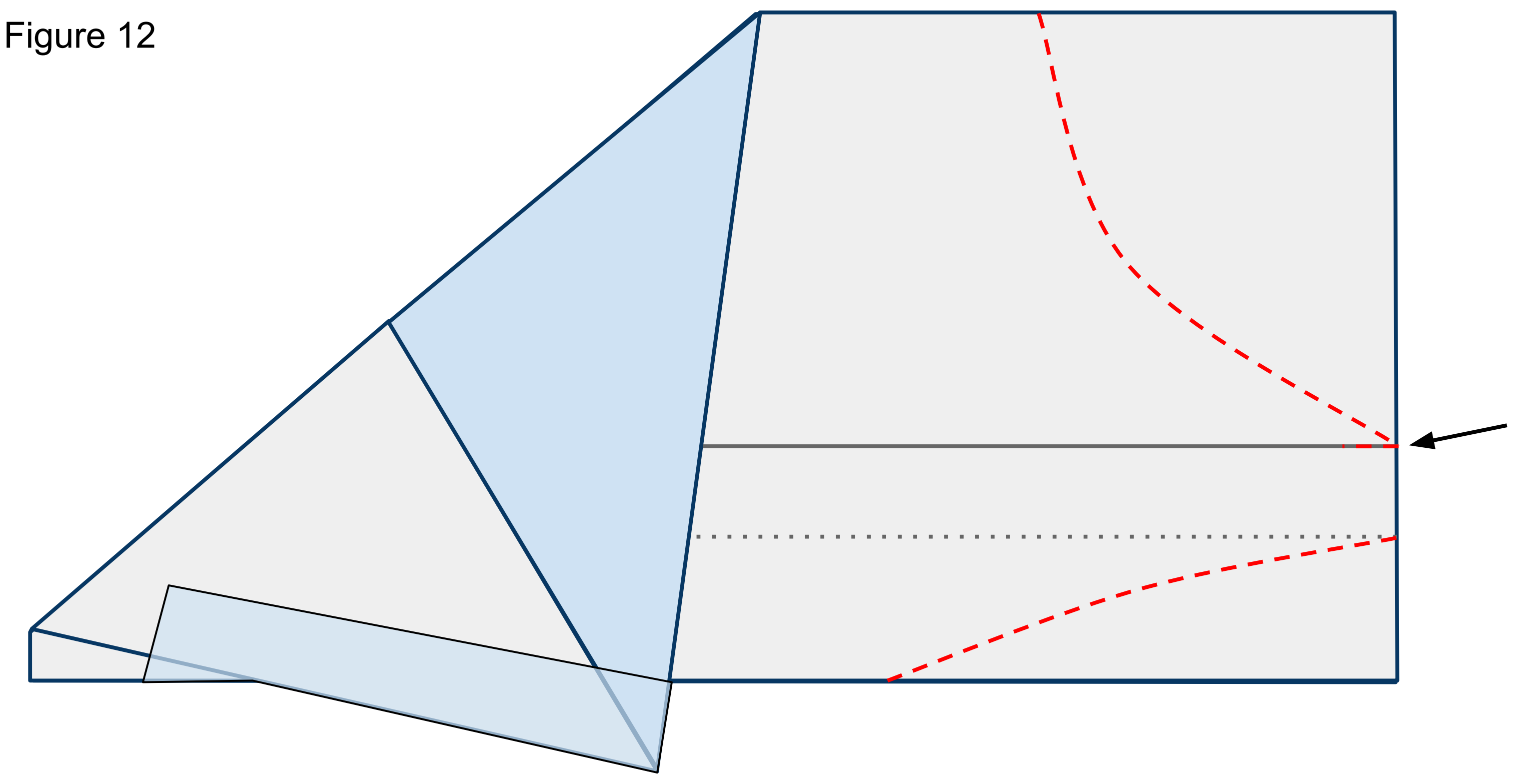
Figure 11



12

Unfold the wings that were folded in steps 8-11 and lay the plane flat on your surface as shown in Figure 12 below. Cut thought both layers of paper along the red dashed lines. Be sure to include the short horizontal cut line along the wing's fold. This cut will help create the plane's flaps.

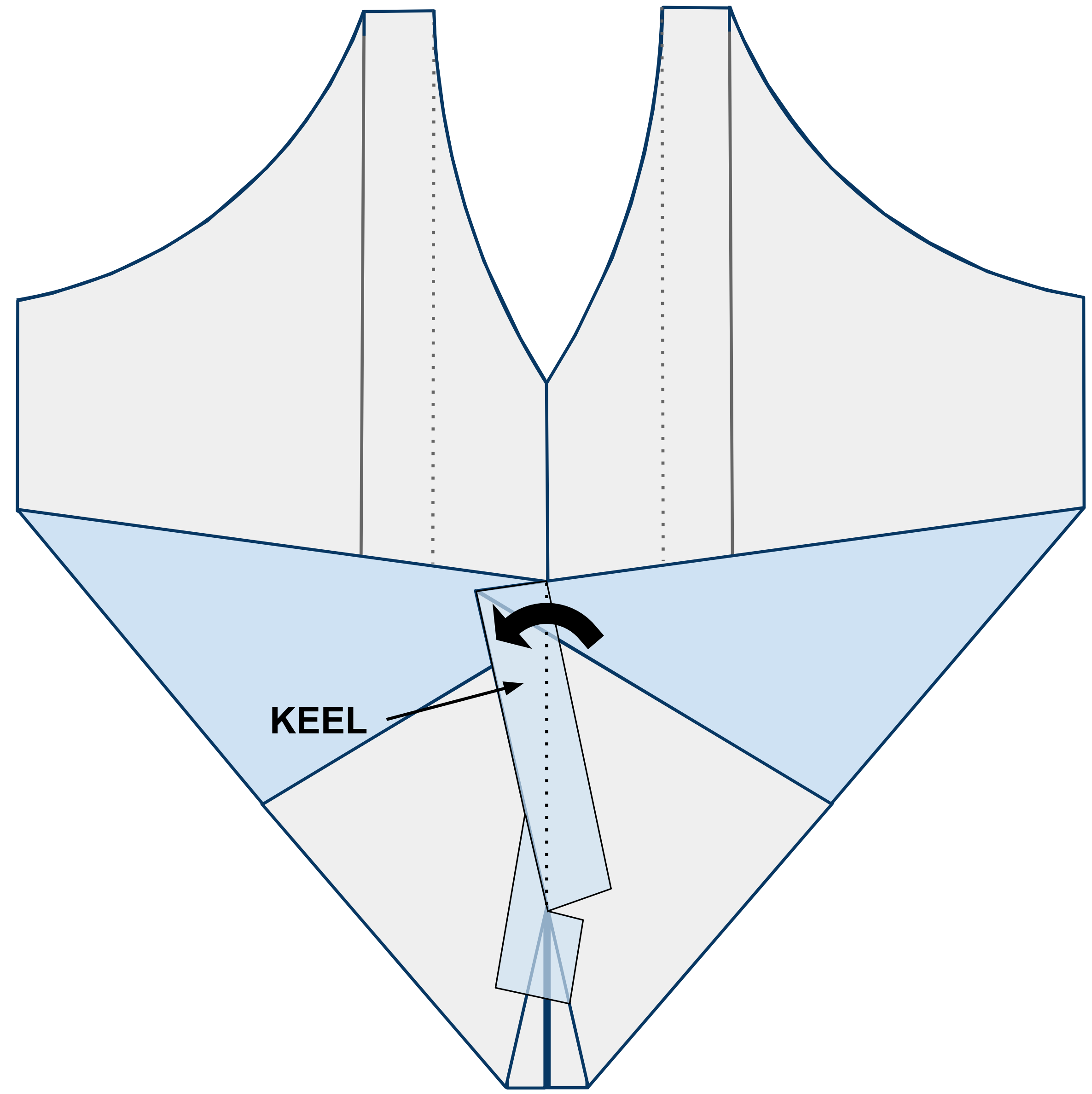
Figure 12



13

Open the plane up flat on your work surface with the nose toward you and looking at the underside of the body. When you do this the keel will want to stand up off the table. Bend it to the left along the center line and give it a strong crease as shown in Figure 13 below.

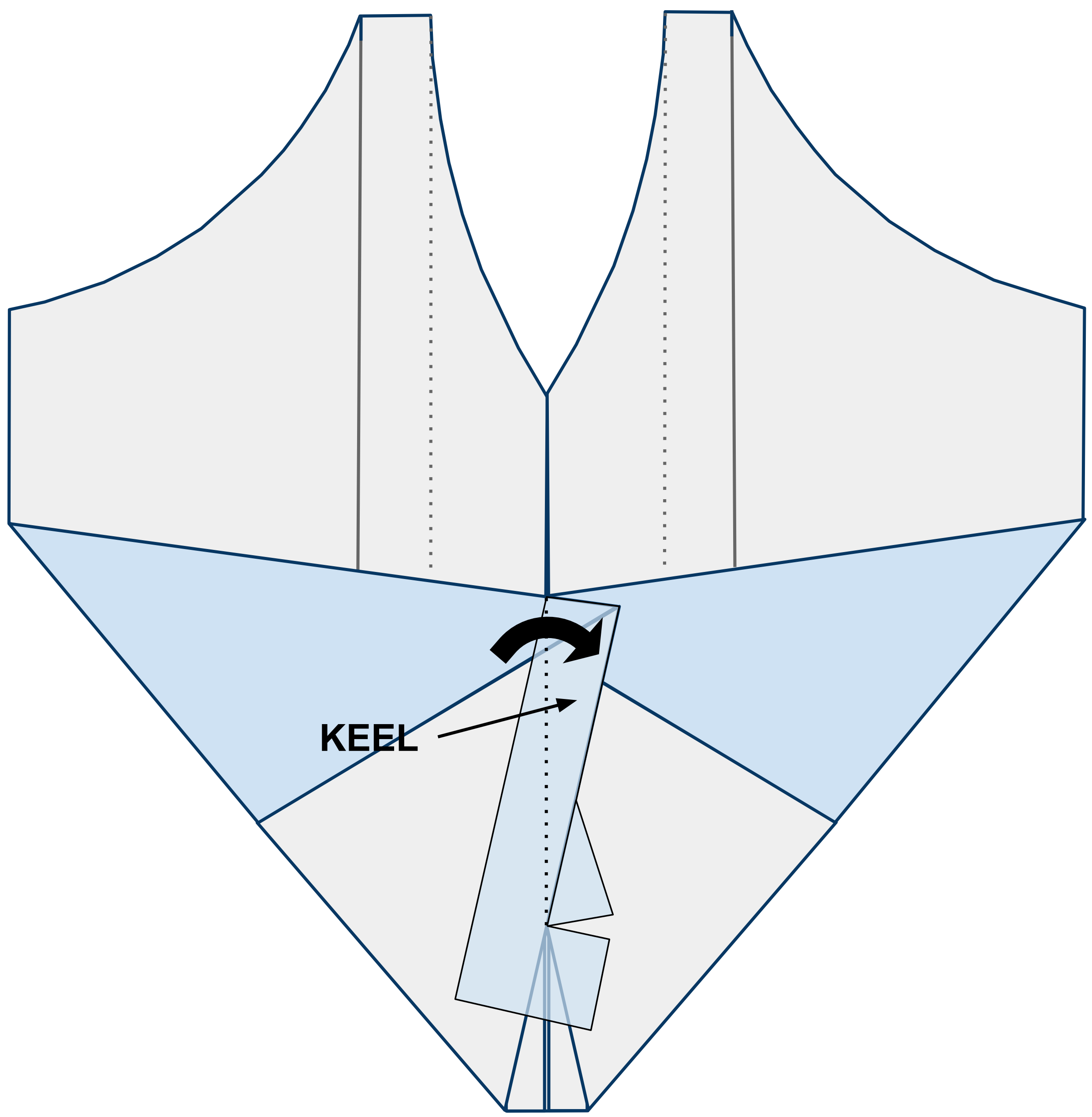
Figure 13



14

Now fold the keel to the right side creating a strong crease as shown in Figure 14 below. After making this crease, allow the keel to remain upright for the remainder of construction.

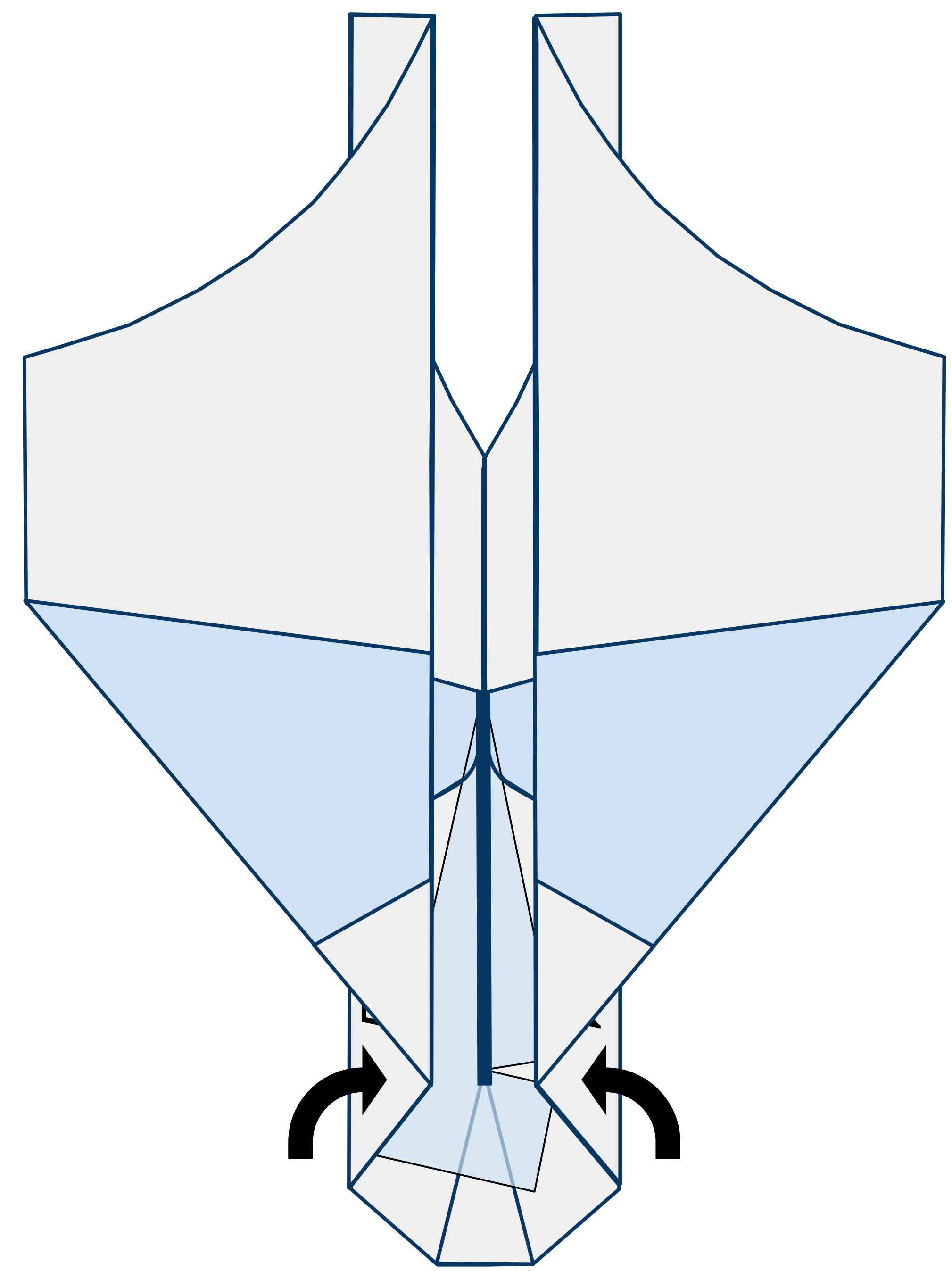
Figure 14



15

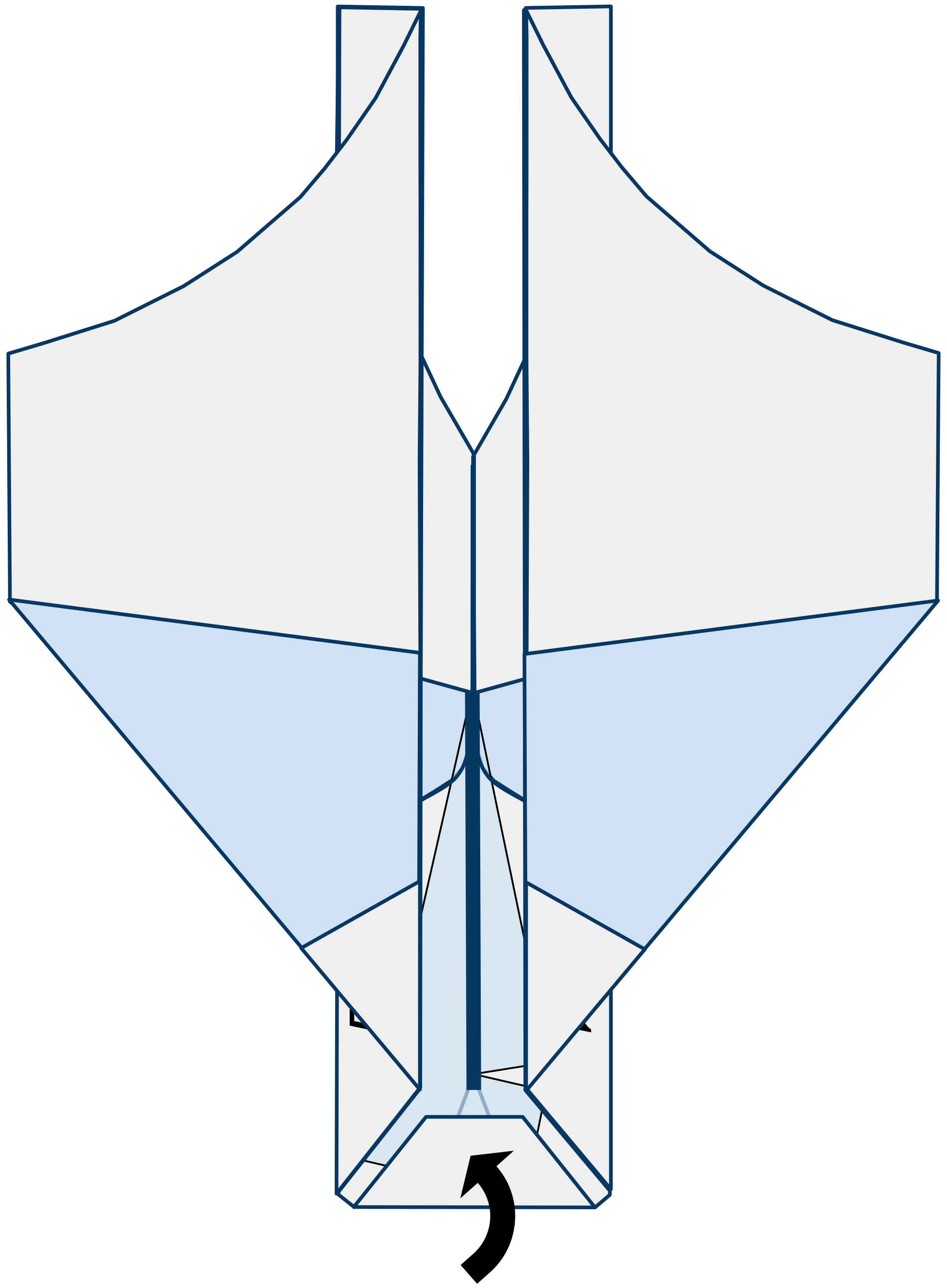
The keel is now represented by the thick **blue** line. Allowing the keel to remain vertical, bring the wings back in along their creases created in steps 8-11 as shown in Figure 15 below.

Figure 15



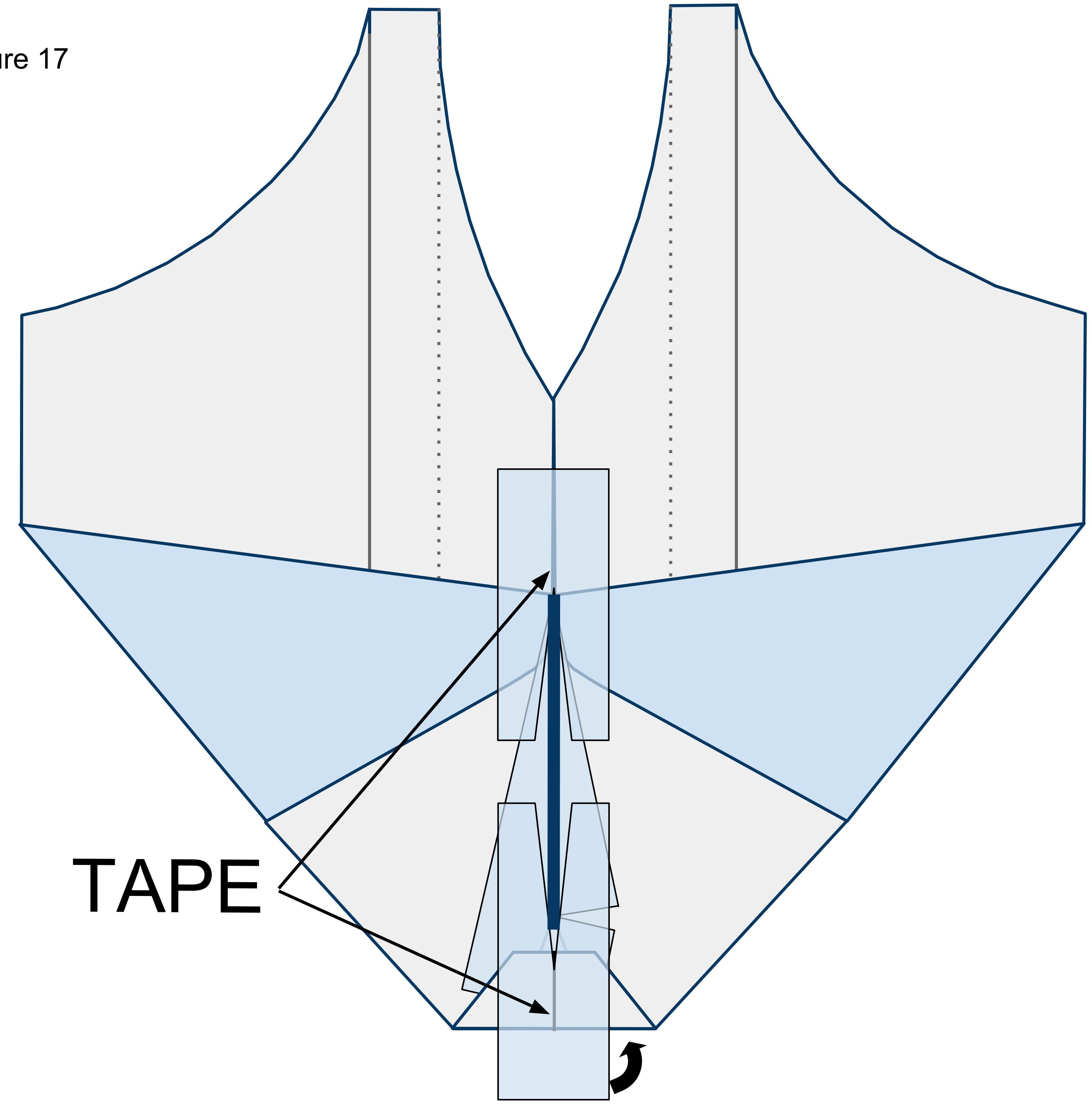
Fold the nose of the plane back and in as shown in Figure 16 below.

Figure 16



Push the wings back out and flat on the table. Place 2 pieces of tape with slices cut in them on the exposed underside of the plane in front of and behind the keel as shown in Figure 17 below. Wrap the tape around the nose to the top side of the plane.

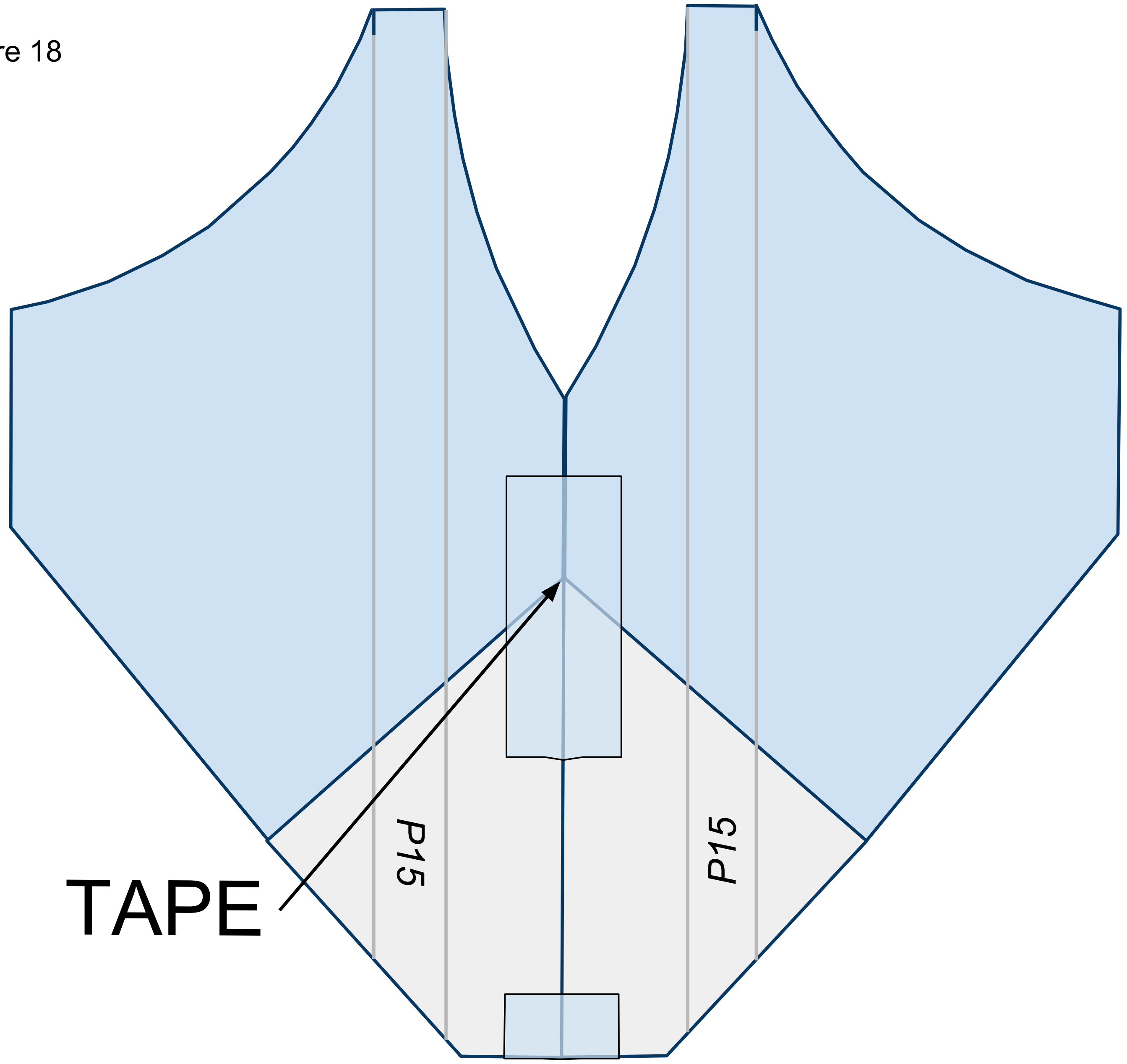
Figure 17



18

Turn the plane over laying the keel to one side. Place tape along the fuselage as shown in Figure 18 below.

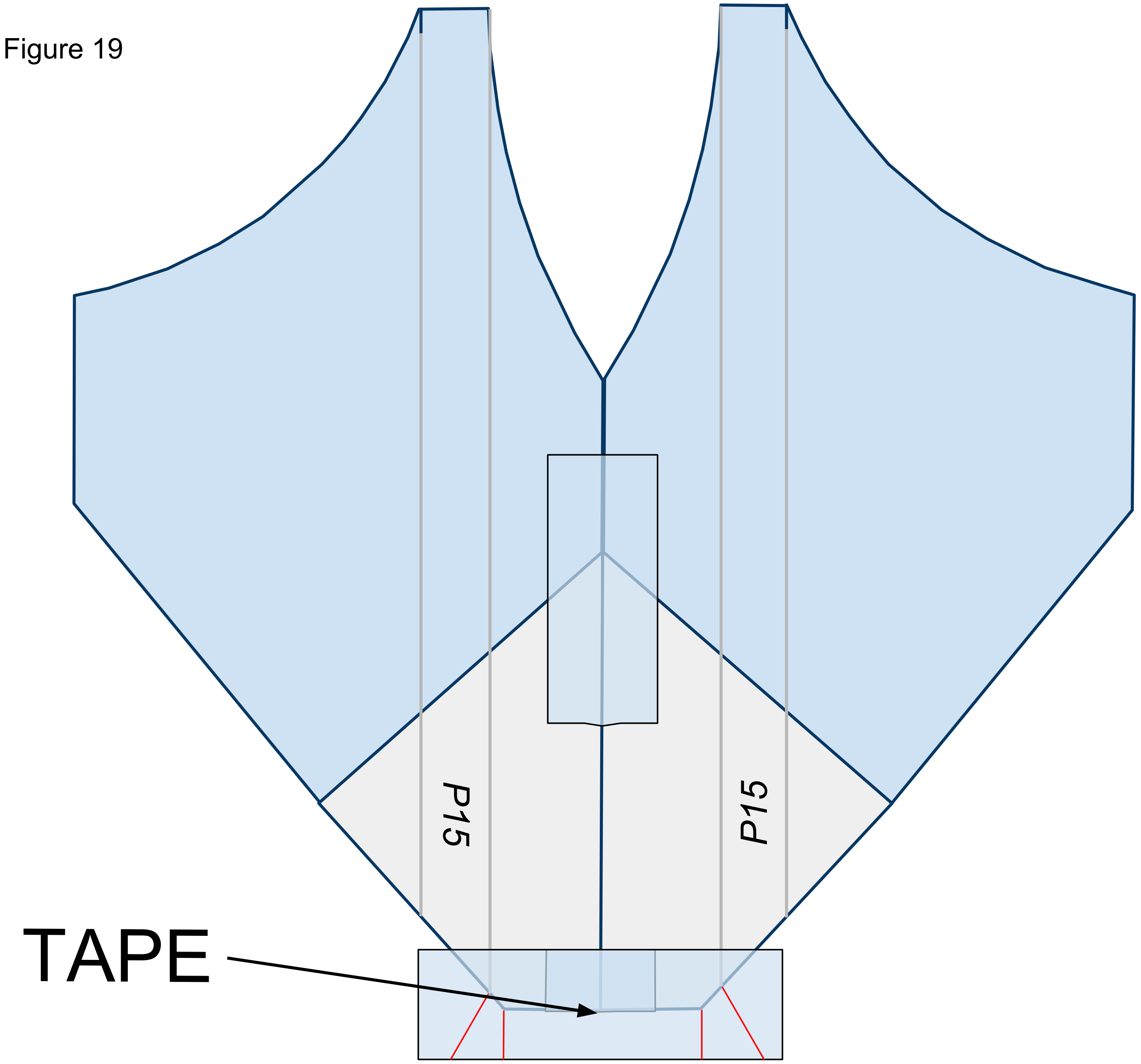
Figure 18



19

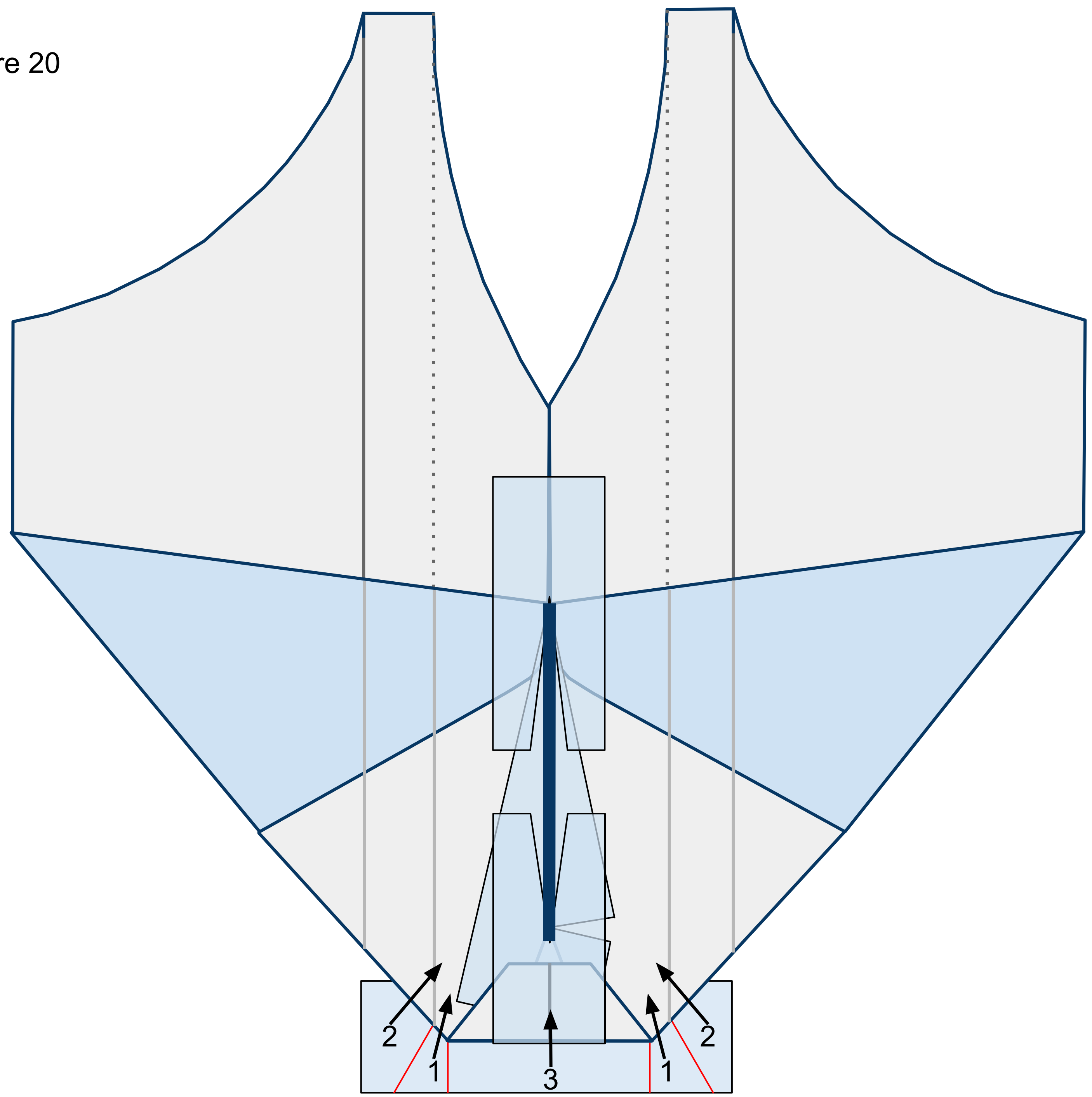
Place a piece of tape across the nose of the plane as shown in Figure 19 below. Cut slits along the **red** lines as shown. These slits will help you fold the tape under the plane in the next step.

Figure 19



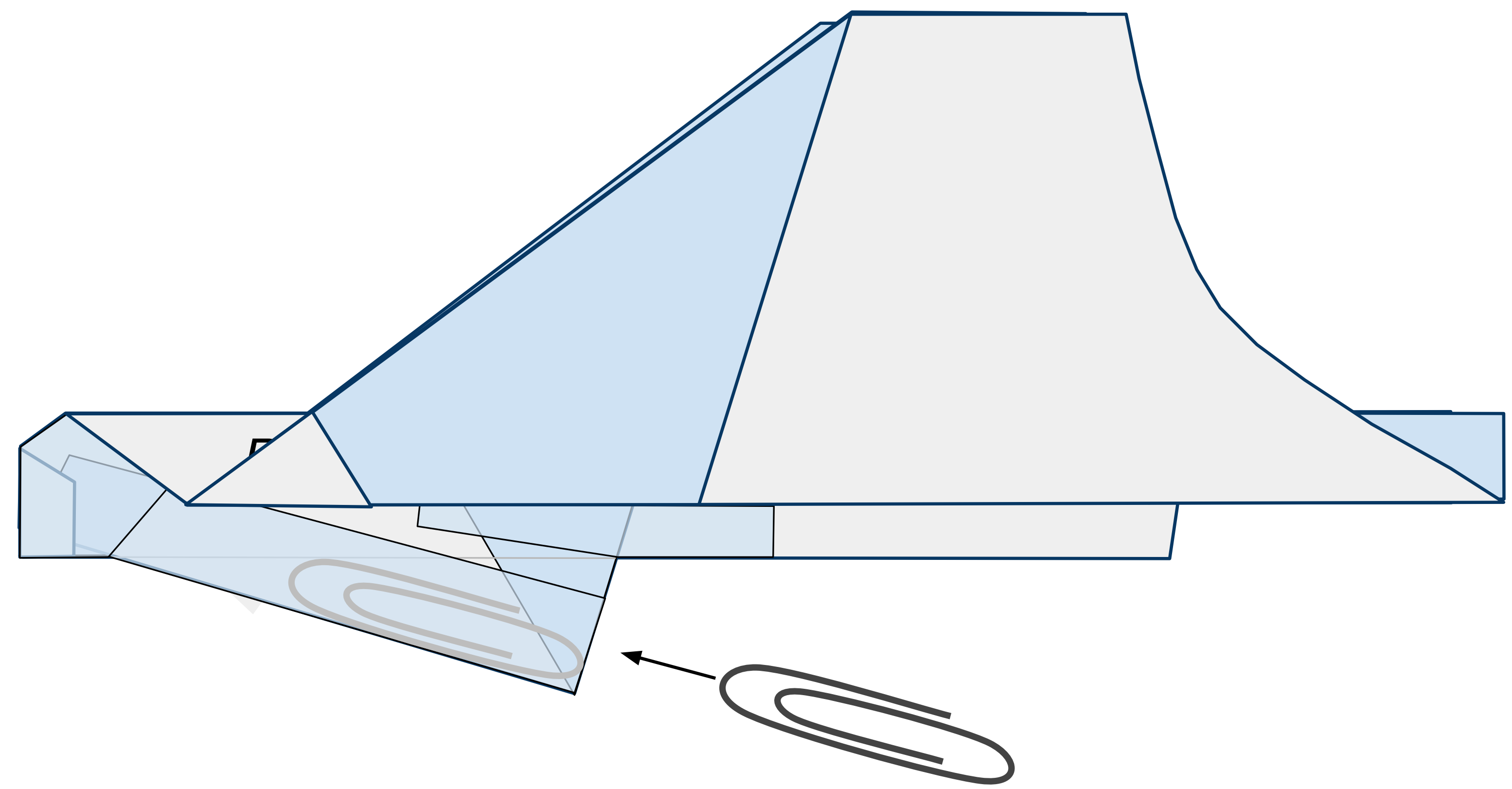
Flip the plane over so you can see the top side. Fold the flaps of tape, in number order, up and in as shown in Figure 20 below. They will overlap each other once folded.

Figure 20



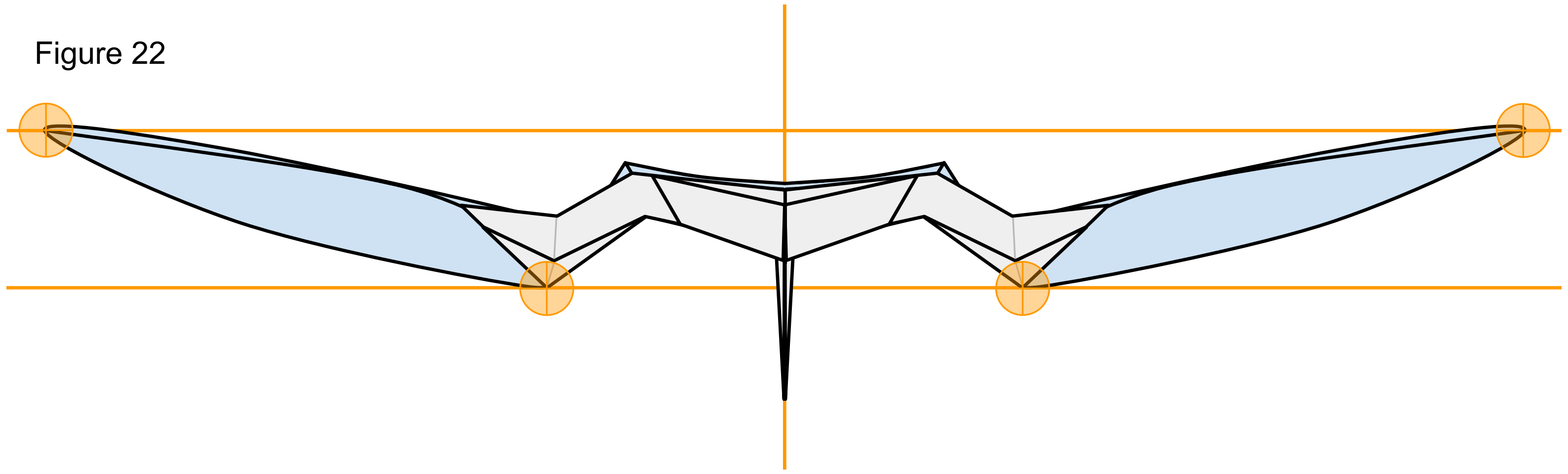
Turn the nose to the left and refold the wings keeping the tail up as shown. Insert a #1 size paper clip ***just*** inside the keel through the opening behind it as shown in Figure 21 below. Then seal the opening with a piece of tape.

Figure 21

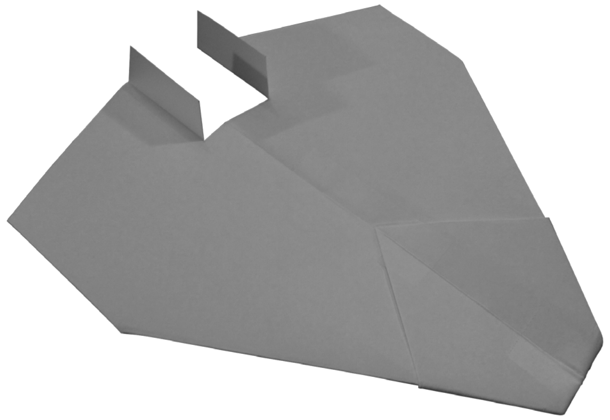


Now that you have constructed the plane ensure that it has the proper shape. Take a look at your plane head-on from the nose. The plane should be symmetrical. Take a look at Figure 22 below. Note the points of interest. Note the curved shape of the paper on the underside of the wings. To achieve this, carefully stick one side of a pair of scissors up into the pocket on the underside of the wing from behind and gently mold the paper into the proper shape. Don't crease it.

Figure 22



We are almost ready to launch! I recommend taking a quick look at the "PREFLIGHT" section to get some good tips. All paper planes fly erratically on their first flights. They require refining to make them great. To do this, take a look at the "FINE-TUNING" section.



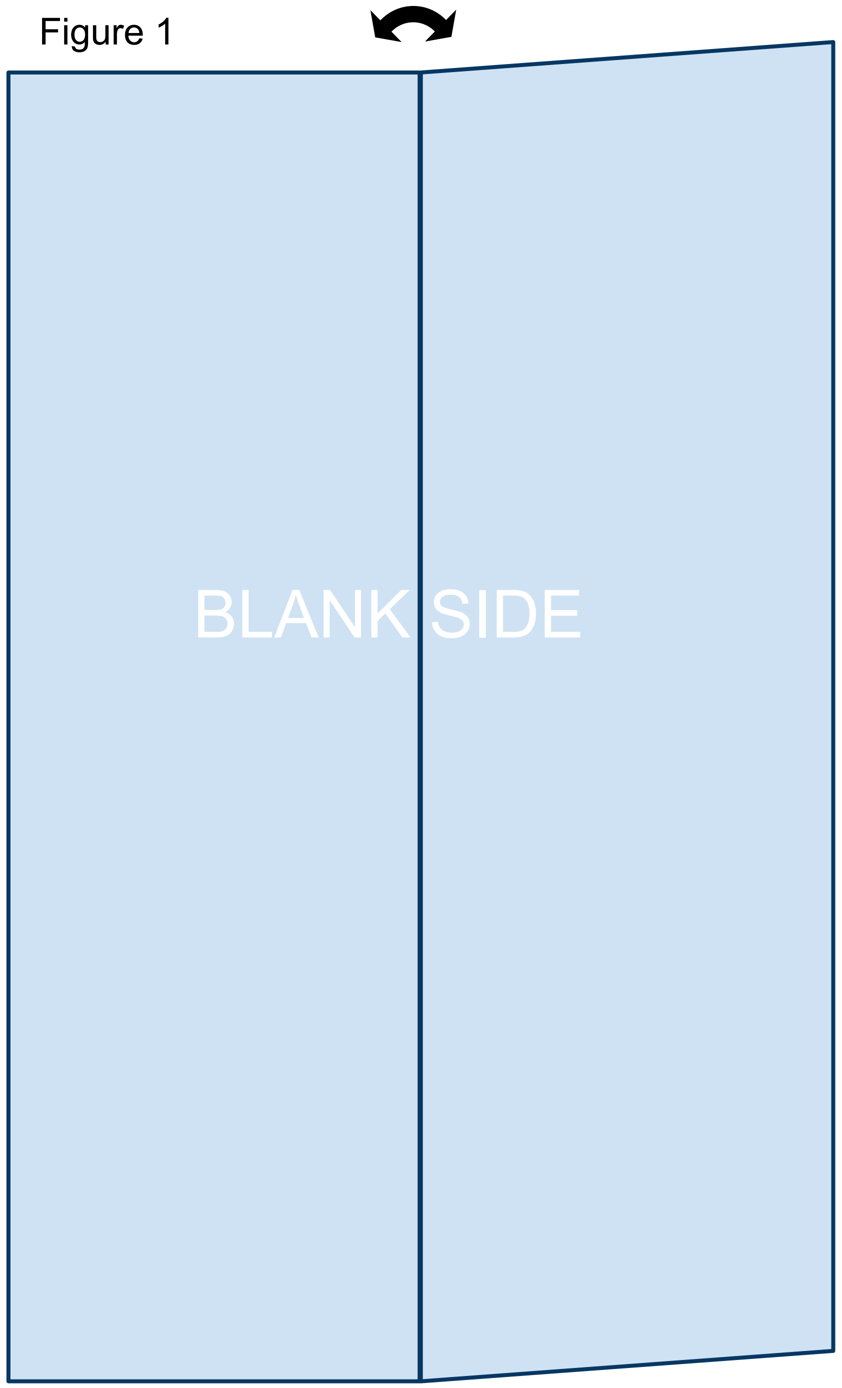
The printable template linked below is 4.25in x 5.5in. That is equivalent to half of an 8.5in x 11in sheet of printer paper. While the designs in this book will fly great with regular printer paper, I recommend 32lb/ream paper. The stronger paper will hold its shape better during flight allowing for higher and more exciting flights. Once you become comfortable making the planes from this template you can simply cut regular printer paper in half in order to start. Print the PDF template located at the bottom of the web page: https://sites.google.com/site/afspaperairplanes/p16

In case your reading device is not connected to the internet, doing a Google search for "AFS Paper Airplanes" should get you to the webpage.

1

Cut out the template and place the paper on the table so that the printed information is on the underside with the printed triangle facing away from you. You will be looking at the blank side. Bring the right side up and fold the paper in half the long way. Unfold leaving a visible crease as shown in Figure 1 below.

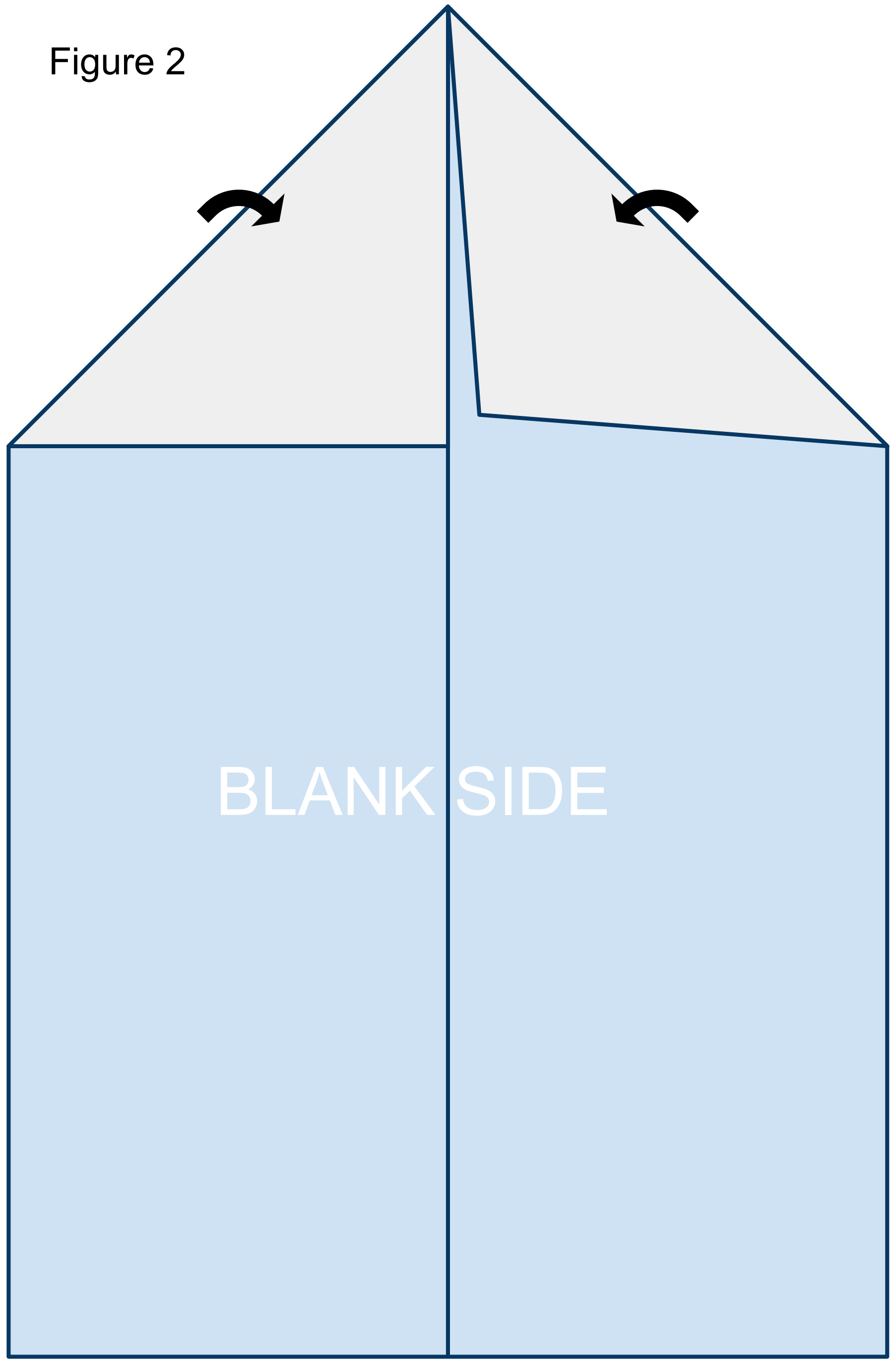
Figure 1



2

Fold the leading corners back along the printed solid lines. The edge of the paper should line up along the crease you created in step 1 as shown in Figure 2 below.

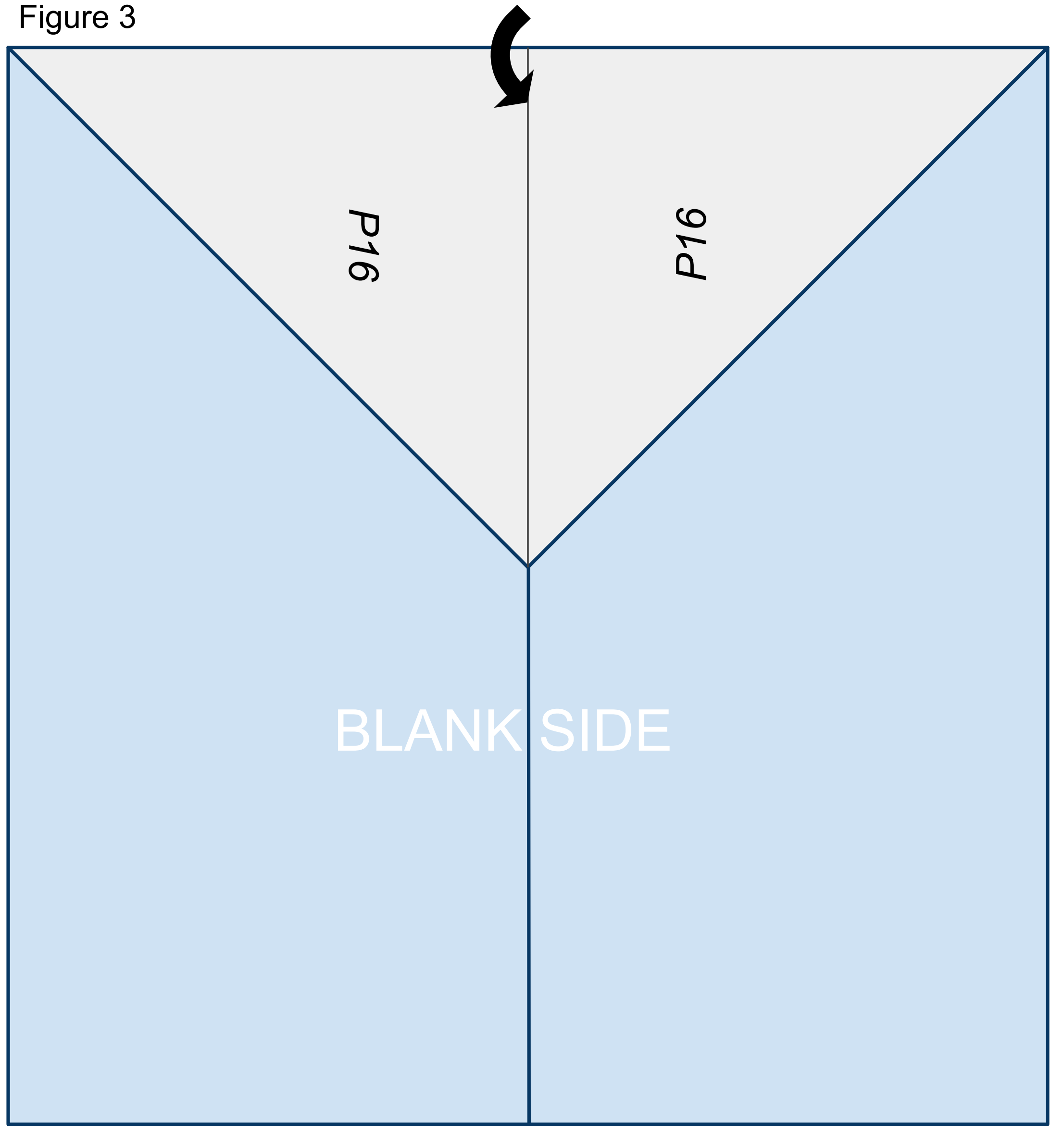
Figure 2



3

Fold the nose of the plane created in step 2 back along the solid line printed on the front of the paper so that it lines up on the center crease created in step 1 as shown in Figure 3 below.

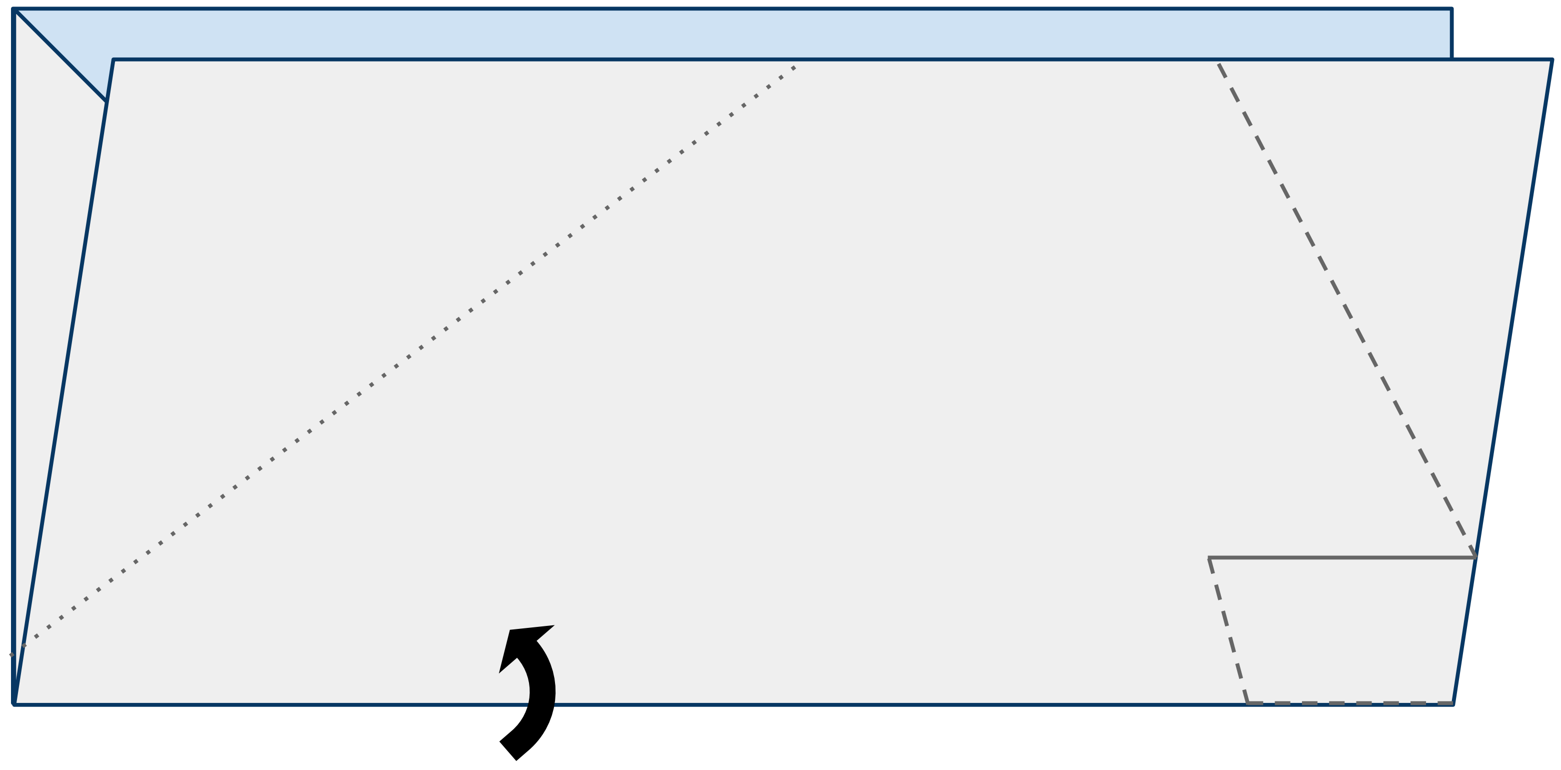
Figure 3



4

Turn the paper to the left (counterclockwise) and fold the paper upward in half hiding the triangle created in the previous steps refolding along the fold from step 1 as shown in Figure 4 below.

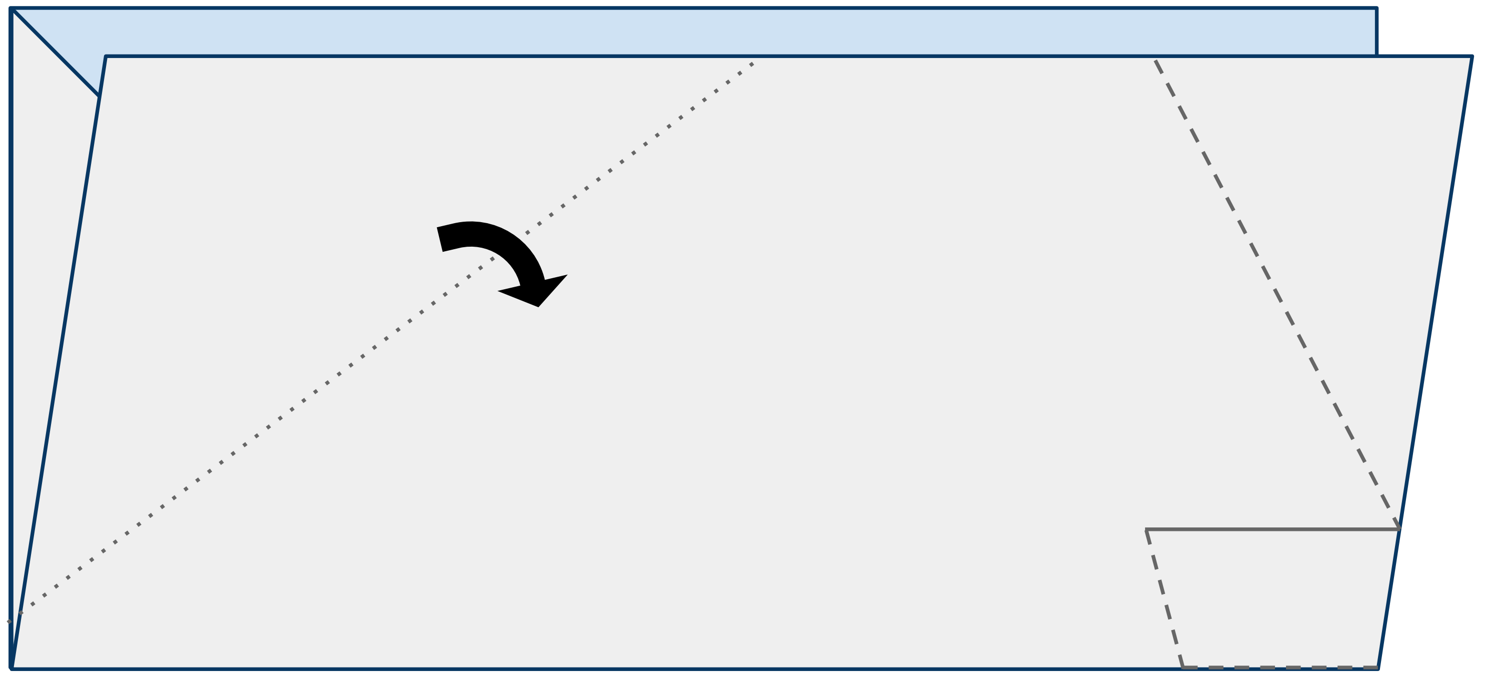
Figure 4



5

Fold the leading edge of the near side of the plane down along the dotted line as shown in Figure 5 below.

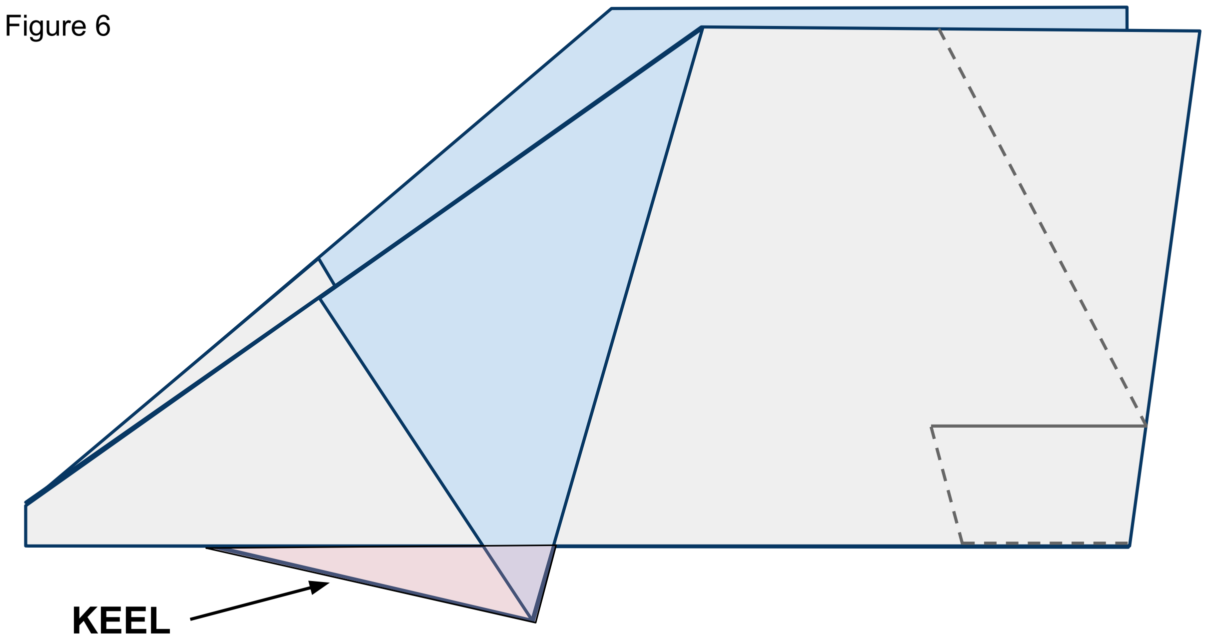
Figure 5



6

Do the same to the other side of the plane. It is important to match these 2 folds exactly so that your plane is symmetrical and will fly straight. The area that is created below the body and designated in pink will be called the keel.

Figure 6



7

Place tape along the keel as shown in Figure 7 below and cut a slit in the tape along the red line. Now fold the tape around and under both sides of the plane and keel securing them together. See Figure 7b for what the far side of the plane looks like.

Figure 7

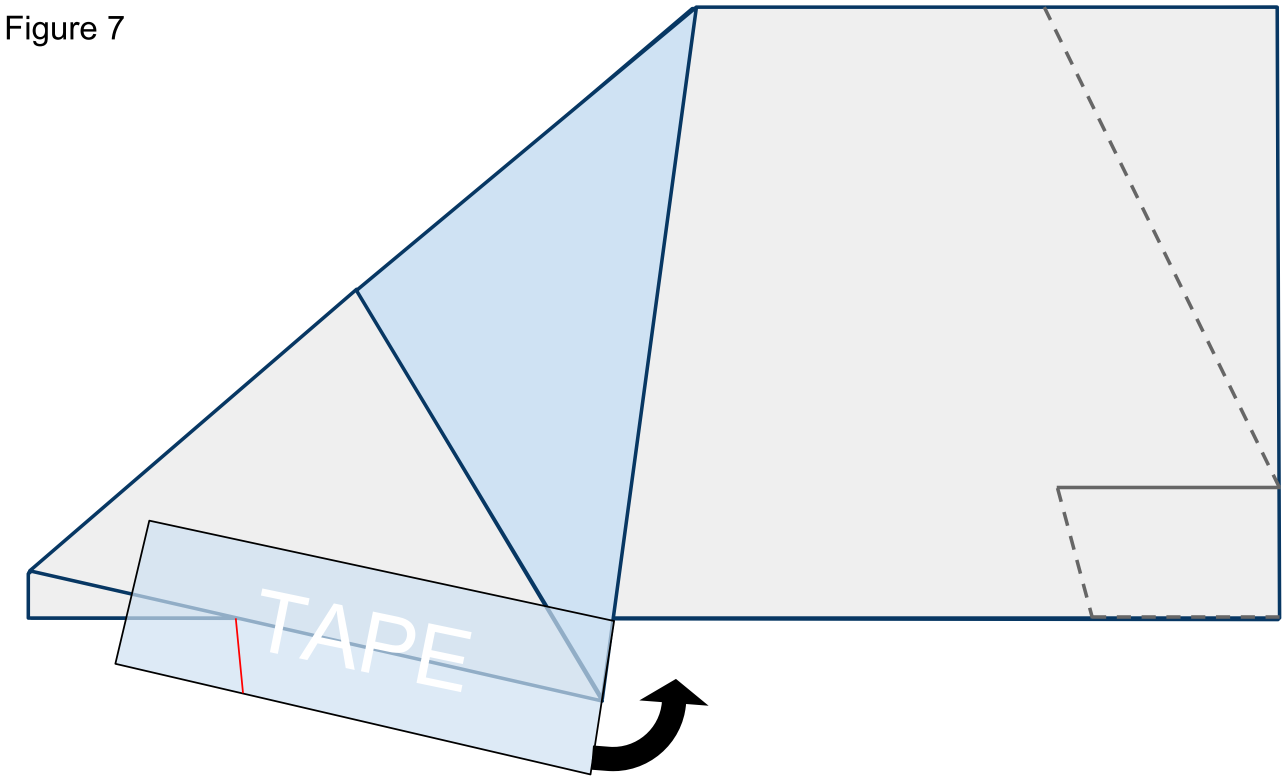
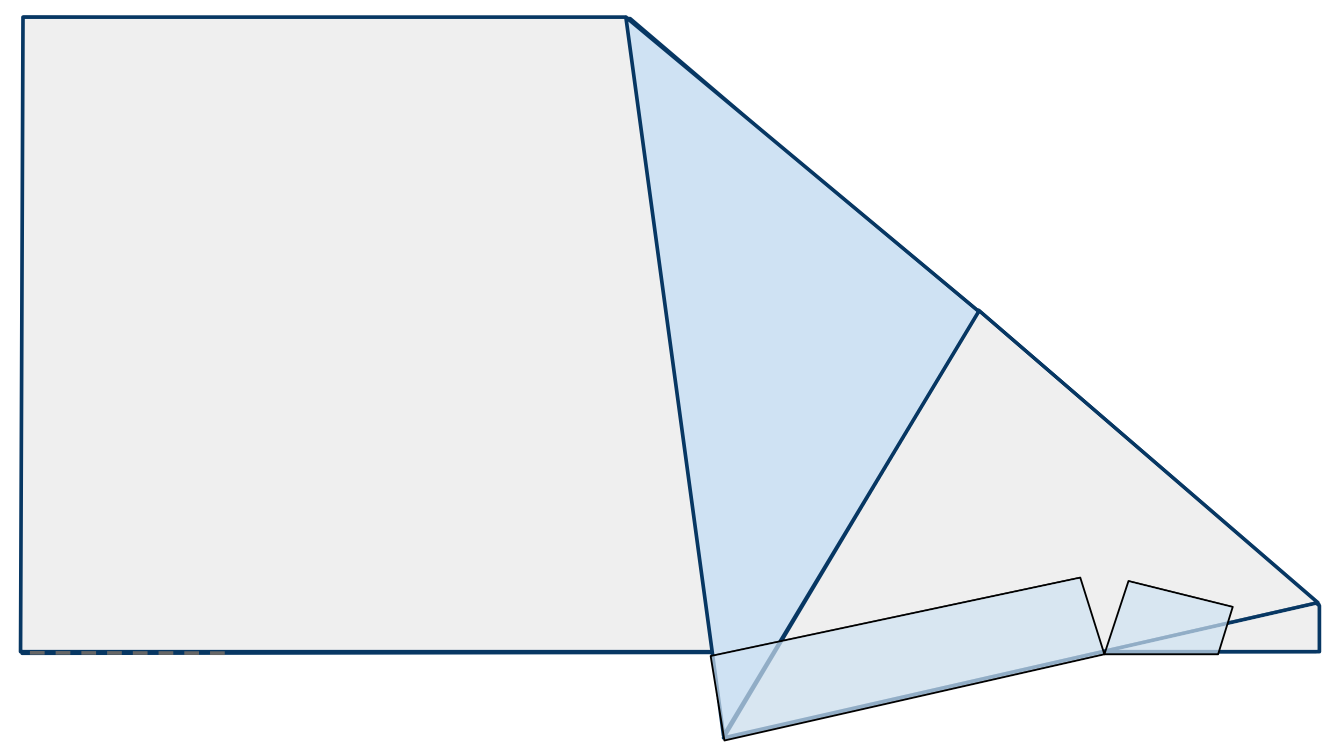


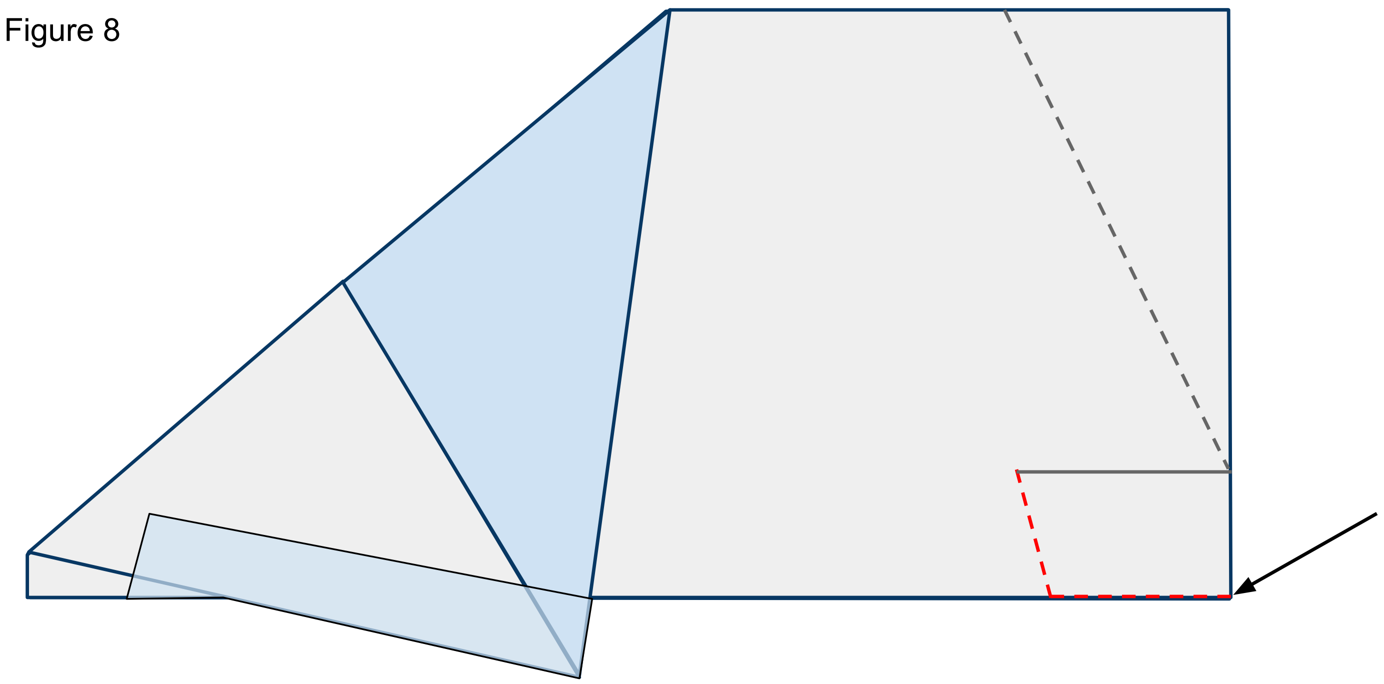
Figure 7b



8

Cut along the red dashed lines cutting through both sides of the plane as shown in Figure 8 below.

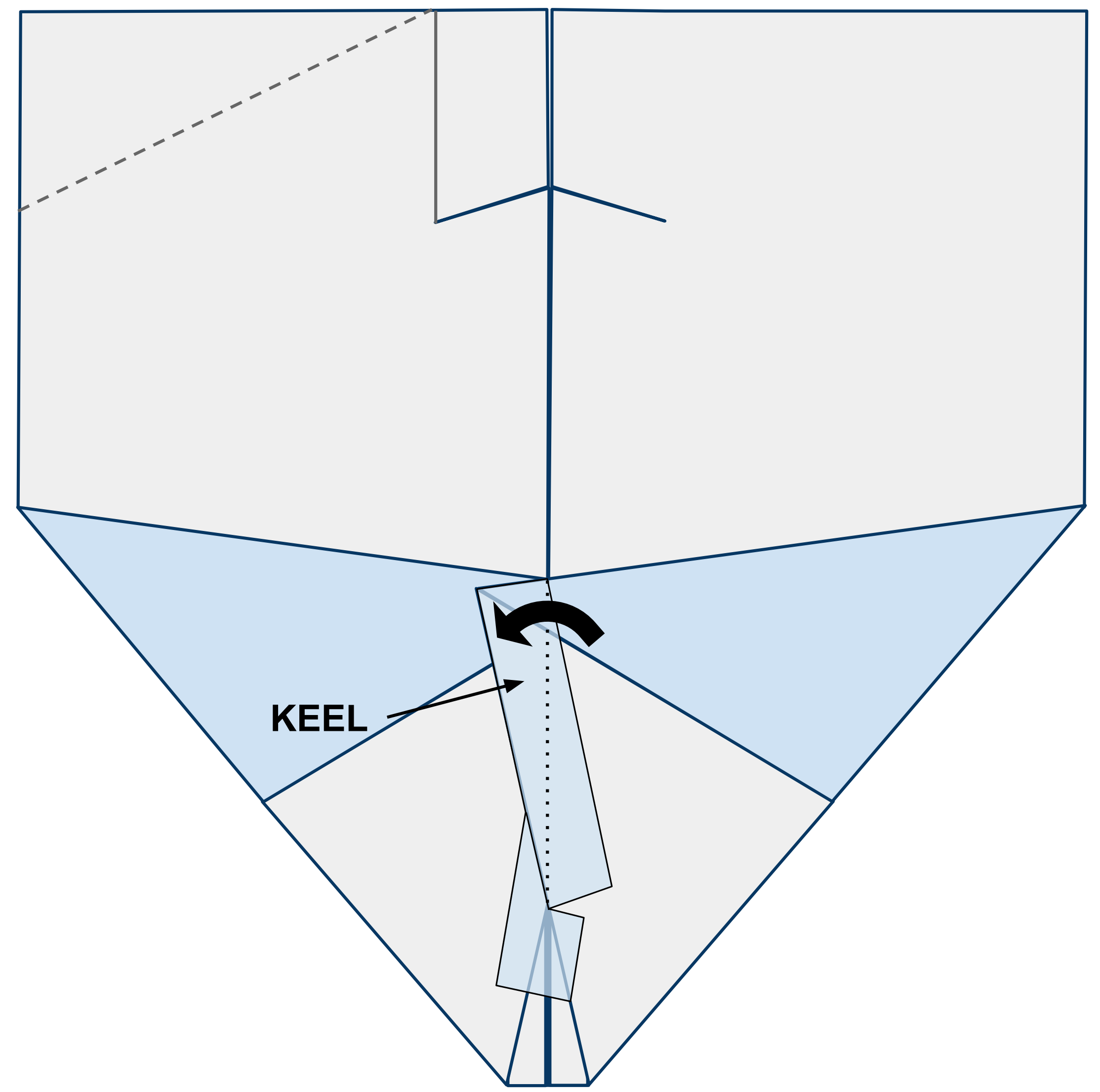
Figure 8



9

Open the plane up flat on your work surface with the nose toward you and looking at the underside of the body. When you do this the keel will want to stand up off the table. Bend it to the left along the center line and give it a strong crease as shown in Figure 9 below.

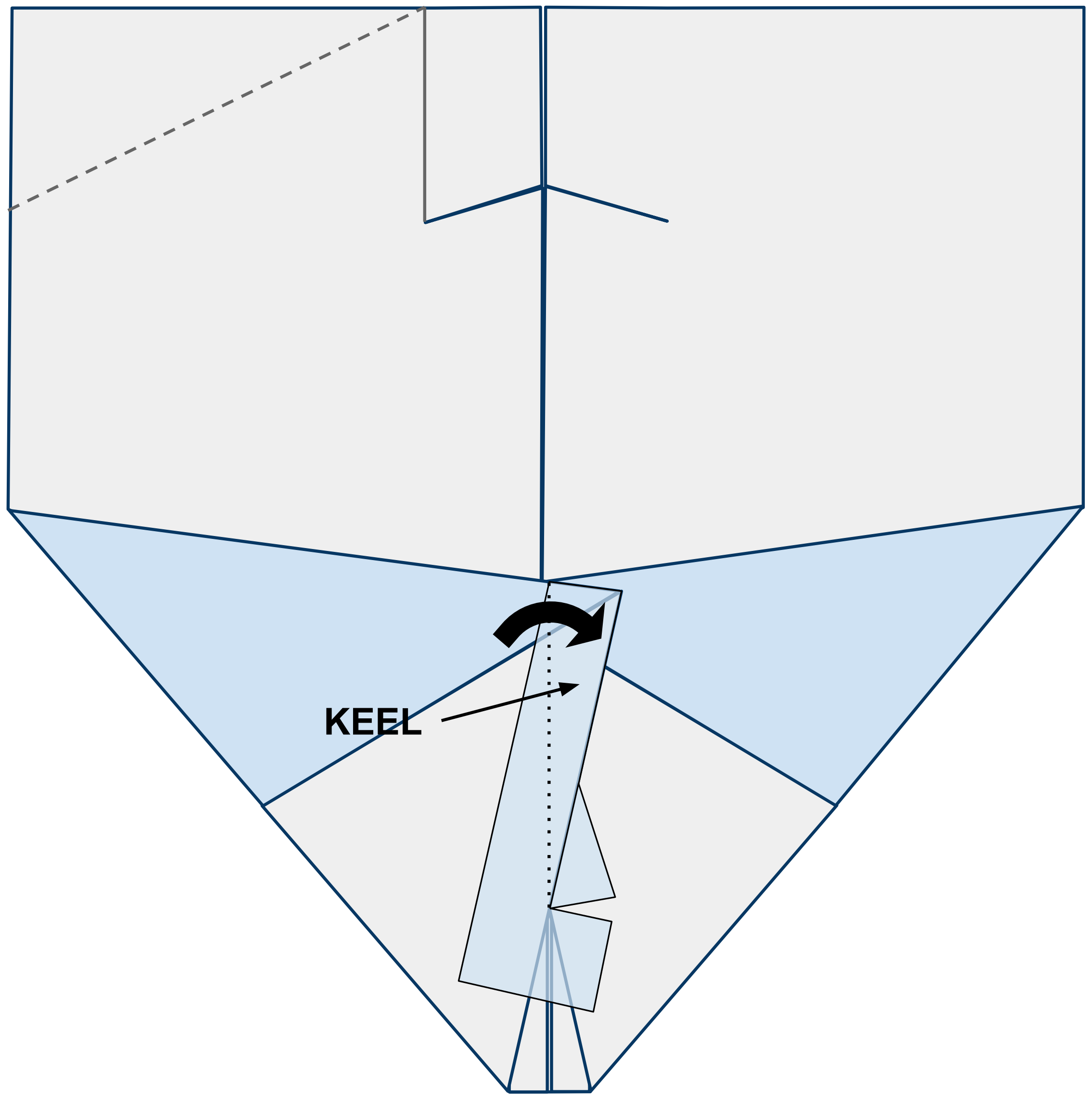
Figure 9



10

Now fold the keel to the right side creating a strong crease as shown in Figure 14 below.

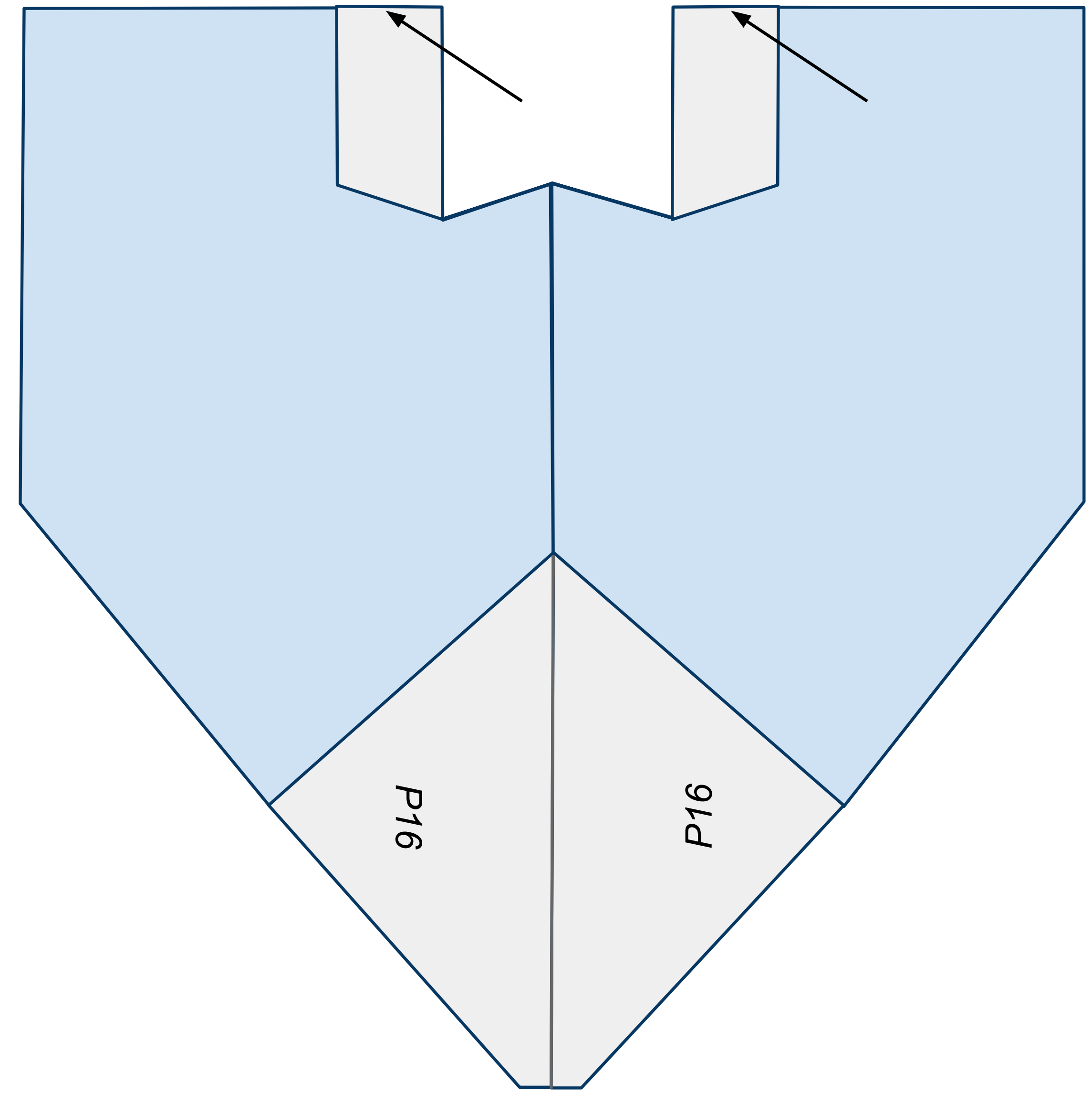
Figure 10



11

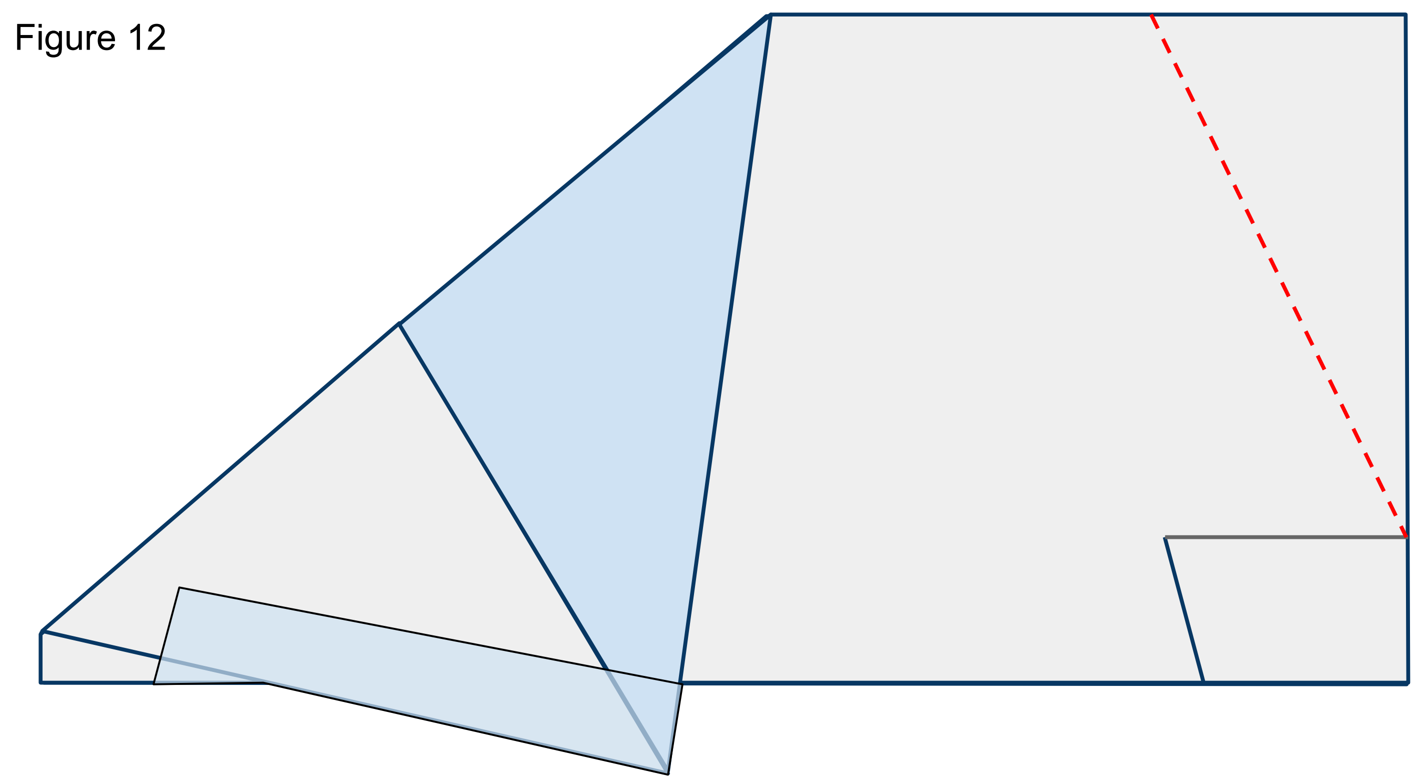
Turn the plane over and fold both tail sections out. Be sure to line up the end of the tail with the end of the wing as shown in Figure 11 below.

Figure 11



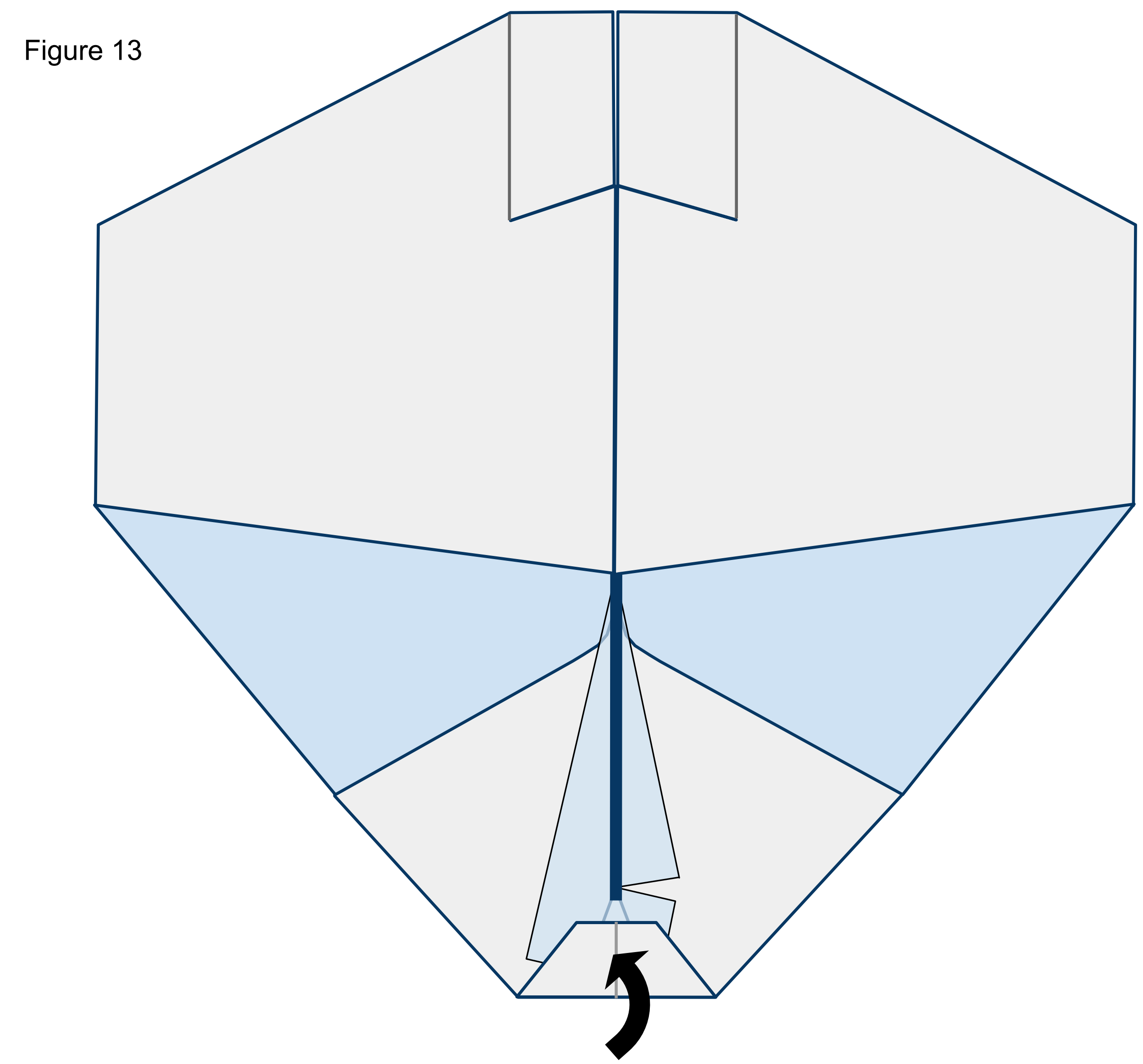
12

Turn the nose to the left and fold the plane back in half. Cut along the red dashed line including both wings as shown in Figure 8 below.



13

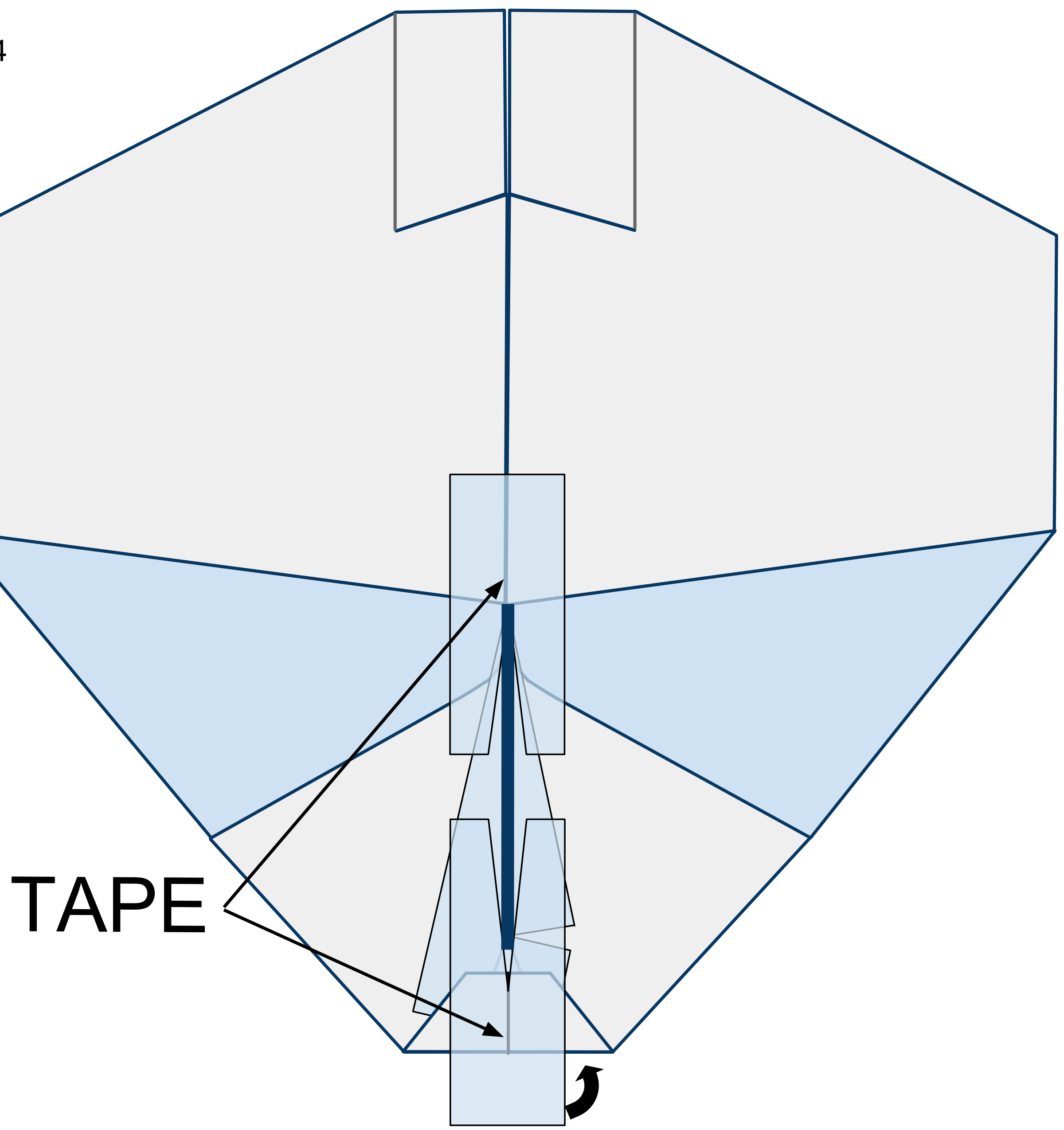
Turn the nose toward you. Open up the wings and lay them flat. Fold the nose of the plane back and in as shown in Figure 13 below. The vertical keel is represented by the thick blue line.



14

Place 2 pieces of tape with slices cut in them on the exposed underside of the plane in front of and behind the keel as shown in Figure 17 below. Wrap the tape around the nose to the top side of the plane.

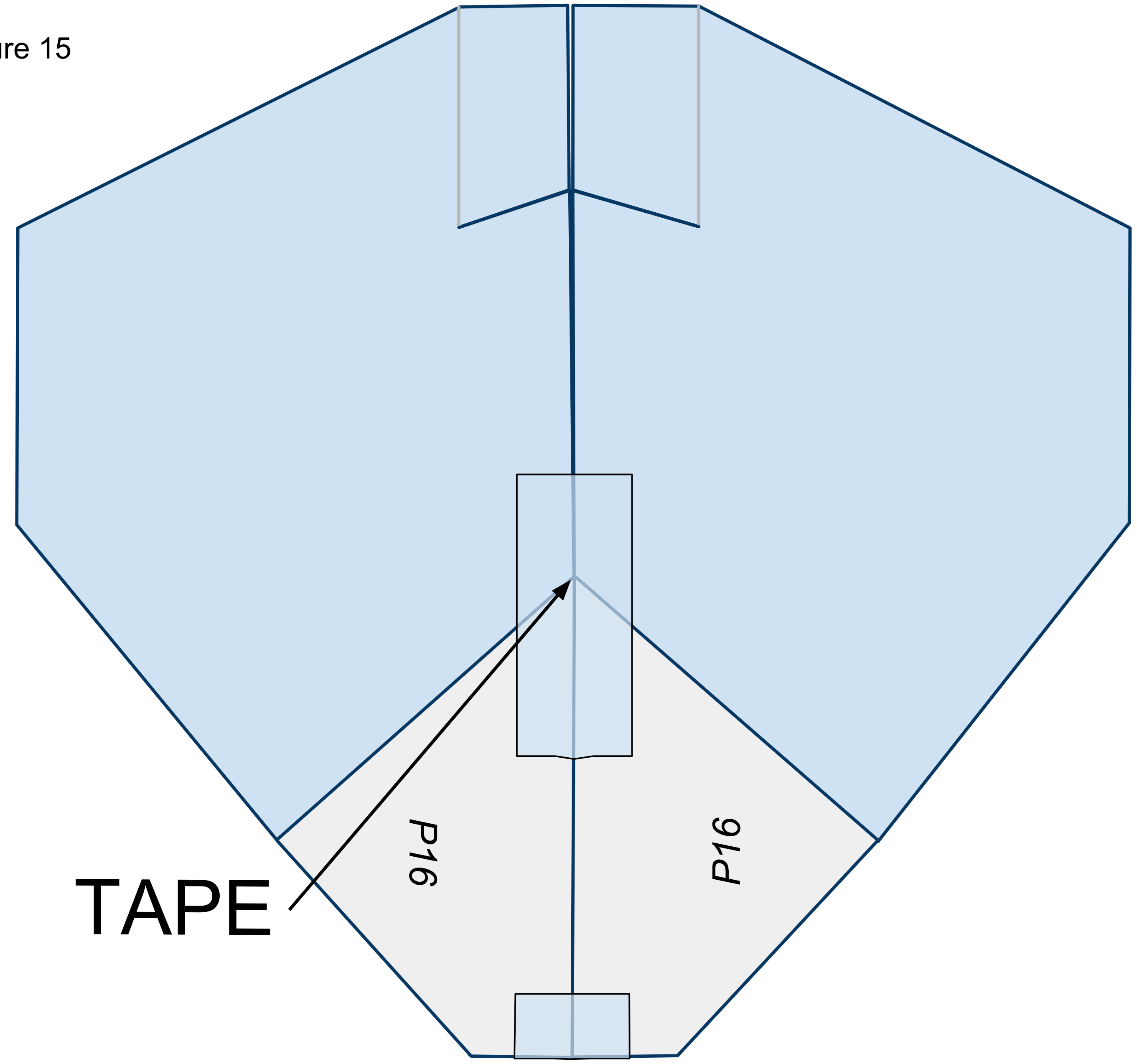
Figure 14



15

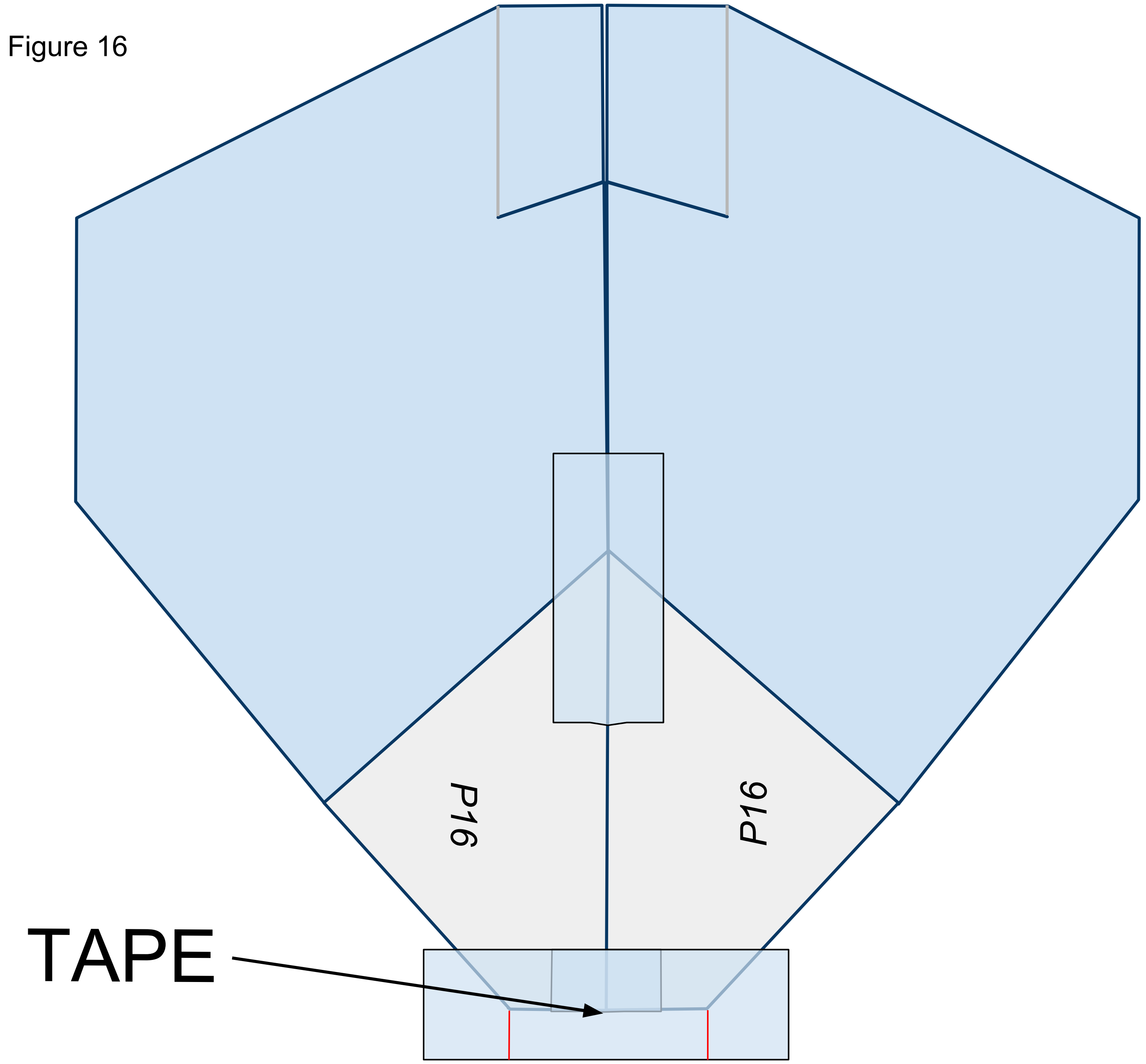
Turn the plane over laying the keel to one side. Place tape along the fuselage as shown in Figure 15 below.

Figure 15



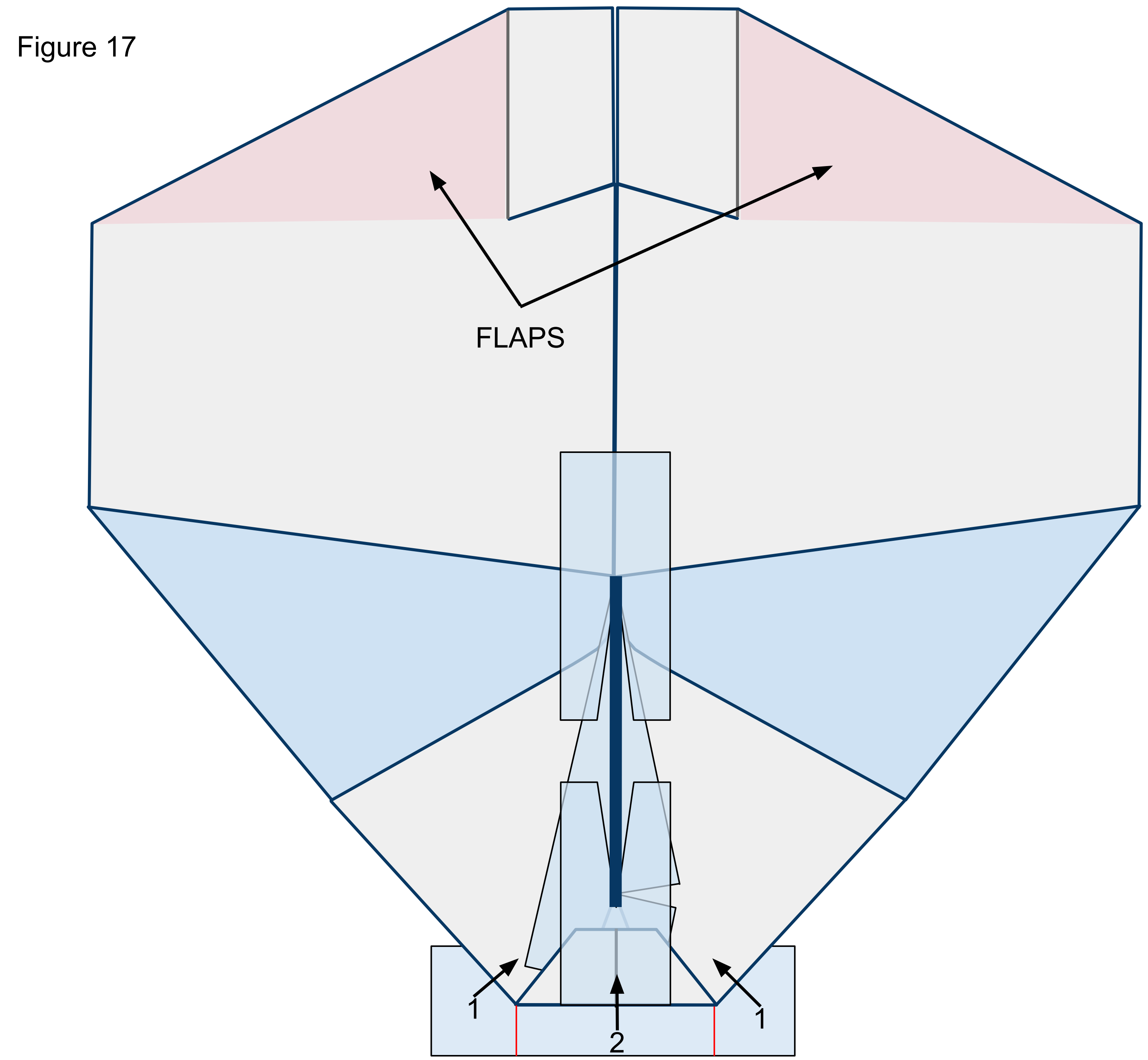
16

Place a piece of tape across the nose of the plane as shown in Figure 16 below. Cut slits along the **red** lines as shown. These slits will help you fold the tape under the plane in the next step.



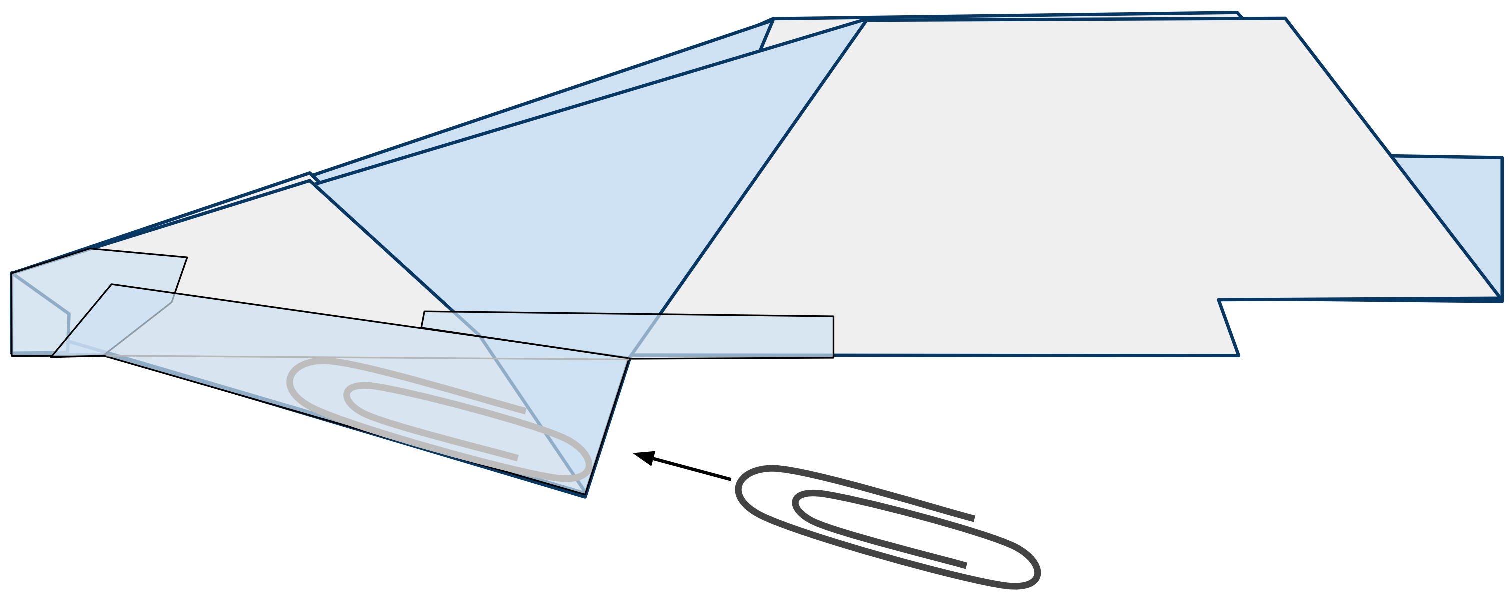
17

Flip the plane over so you can see the top side. Fold the flaps of tape, in number order, up and in as shown in Figure 17 below. They will overlap each other once folded. Also, note the area designated in pink as the flaps for this plane. Don't crease them but simply bend them up or down to control the plane.



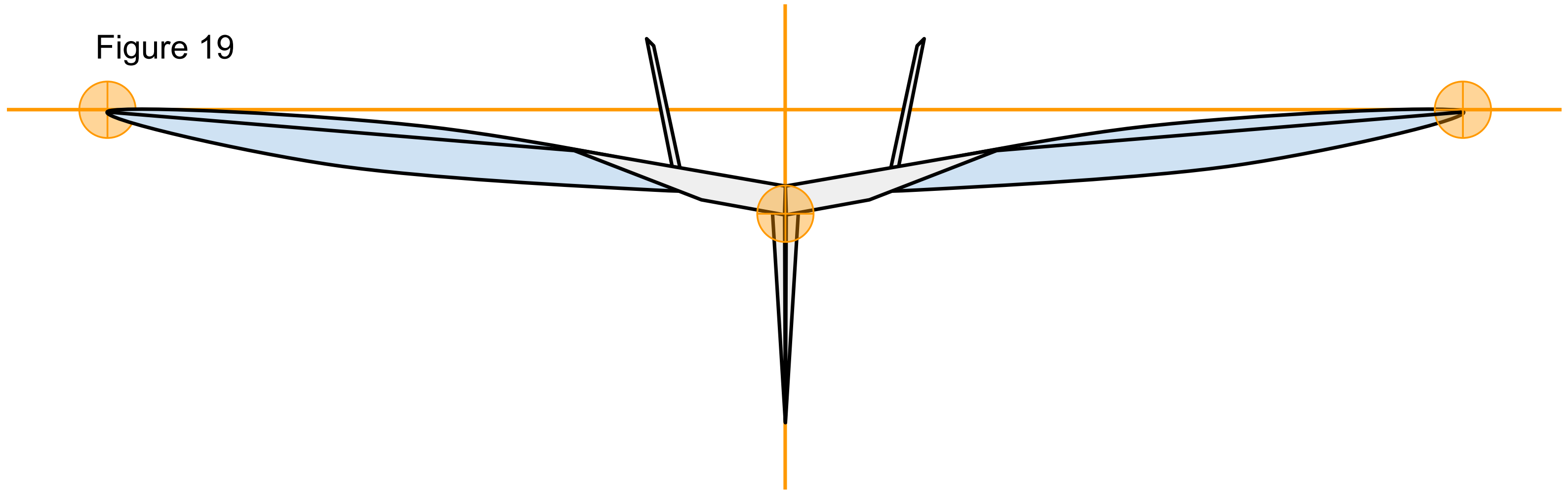
Turn the nose to the left. Insert a #1 size paper clip ***just*** inside the keel through the opening behind it as shown in Figure 21 below. Then seal the opening with a piece of tape.

Figure 18



Now that you have constructed the plane ensure that it has the proper shape. Take a look at your plane head-on from the nose. The plane should be symmetrical. Take a look at Figure 19 below. Note the points of interest. Note the curved shape of the paper on the underside of the wings. To achieve this, carefully stick one side of a pair of scissors up into the pocket on the underside of the wing from behind and gently mold the paper into the proper shape. The pocket under each wing will be stopped at the tape placed across the underside of the fuselage. Don't crease it.

Figure 19



We are almost ready to launch! I recommend taking a quick look at the "PREFLIGHT" section to get some good tips. All paper planes fly erratically on their first flights. They require refining to make them great. To do this, take a look at the "FINE-TUNING" section.

PREFLIGHT

To launch your plane hold it by the keel that protrudes out from under the plane. You can hold the keel in between your index finger and thumb or with your index finger placed on the keel's trailing edge. Give it a firm and level throw.

If you want your plane to fly straight symmetry is king. Take another look at your plane head-on from the nose. The plane should be exactly symmetrical. Paper airplanes are small. Any tiny imperfection is going to make a big difference in flight.

Now your plane should be ready to launch. No paper plane will fly perfect on its first few throws. Give it a few test flights inside and check out the "FINE-TUNING" section to get the best flights out of your plane.

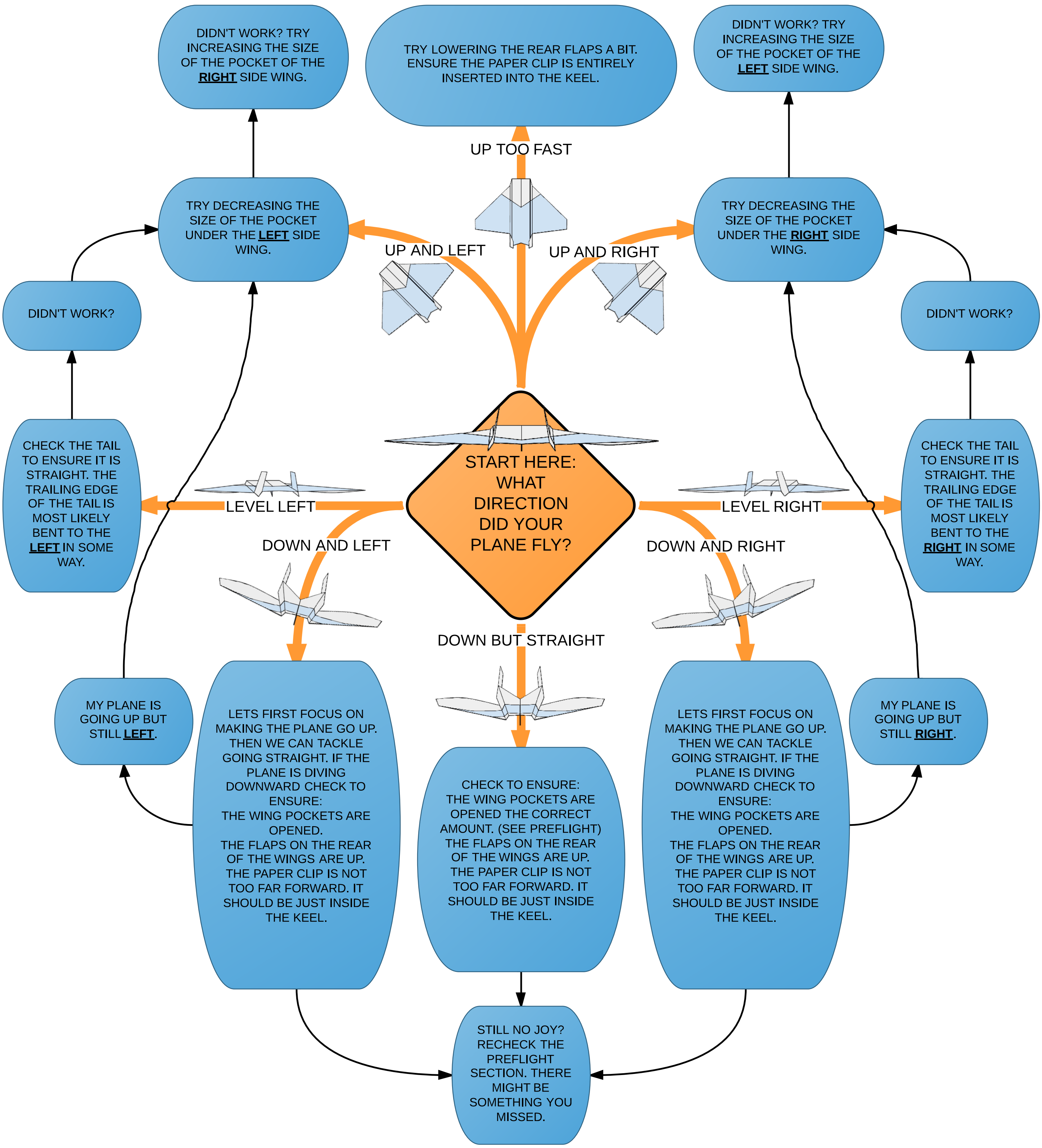
Here are a few important tips:

- Make sure you are in a big enough area. These planes will fly higher than a two story building on a good throw. If you're not careful you might end up spending more time getting your plane back than actually flying it.
- In general, the lower the wind the better. While high wind can make for exciting flights, it makes it difficult to make controlled adjustments to your plane.
- I recommend picking up the plane by the nose or keel to avoid damaging the paper surfaces of the wings or tail. Even slight wrinkles can make a big difference in flights.
- You are welcome to, but don't need to throw the planes straight up. If you throw them hard and at a slightly upward angle they will produce the lift required to reach a good height for a great flight.

FINE-TUNING

You can open this book on your smart phone or tablet so you can have it with you while flying the planes!

The purpose of this section is to help you get the best flights out of your plane. While it may not fix every issue, it should help you get the best out of your plane. The following diagram should help you diagnose and treat issues with your plane's flight. Start in the middle and follow the arrows to guide you to potential solutions. All directions are oriented as if you were holding the plane ready to throw it. Remember, the ***slightest adjustment*** can have a huge effect on your plane's flight!



A FEW GAMES

Here are a few examples of fun games to enjoy with the planes:

- **Distance King**: Designate an indoor or outdoor area as the throwing area (about the size of a hula hoop). Then compete with a friend to see how far away you can make the plane fly from that location. Measure it with steps and take turns throwing your planes. On a windy day you might get some very long throws!
- **Time King**: With a friend throw you planes into the air at the same time. Upon landing say "down." Try to be the last one down. Adjust your planes and try again.
- **Control King**: On one side of a large indoor room set up a target that you might be able to hit. From the far side of the room take turns with a friend to see how close you can come to hitting the target. First to hit wins, and then make the target smaller.

AFTERWORD

I hope that these designs have provided many hours of entertainment and have challenged you. I encourage you to experiment, adjust the planes and come up with your own designs. Also share your ideas and feedback with me and fellow paper plane enthusiasts at https://sites.google.com/site/afspaperairplanes/

P11 EAGLE

- Cut out the rectangular template below. Leave the Dashed Line cuts for later.
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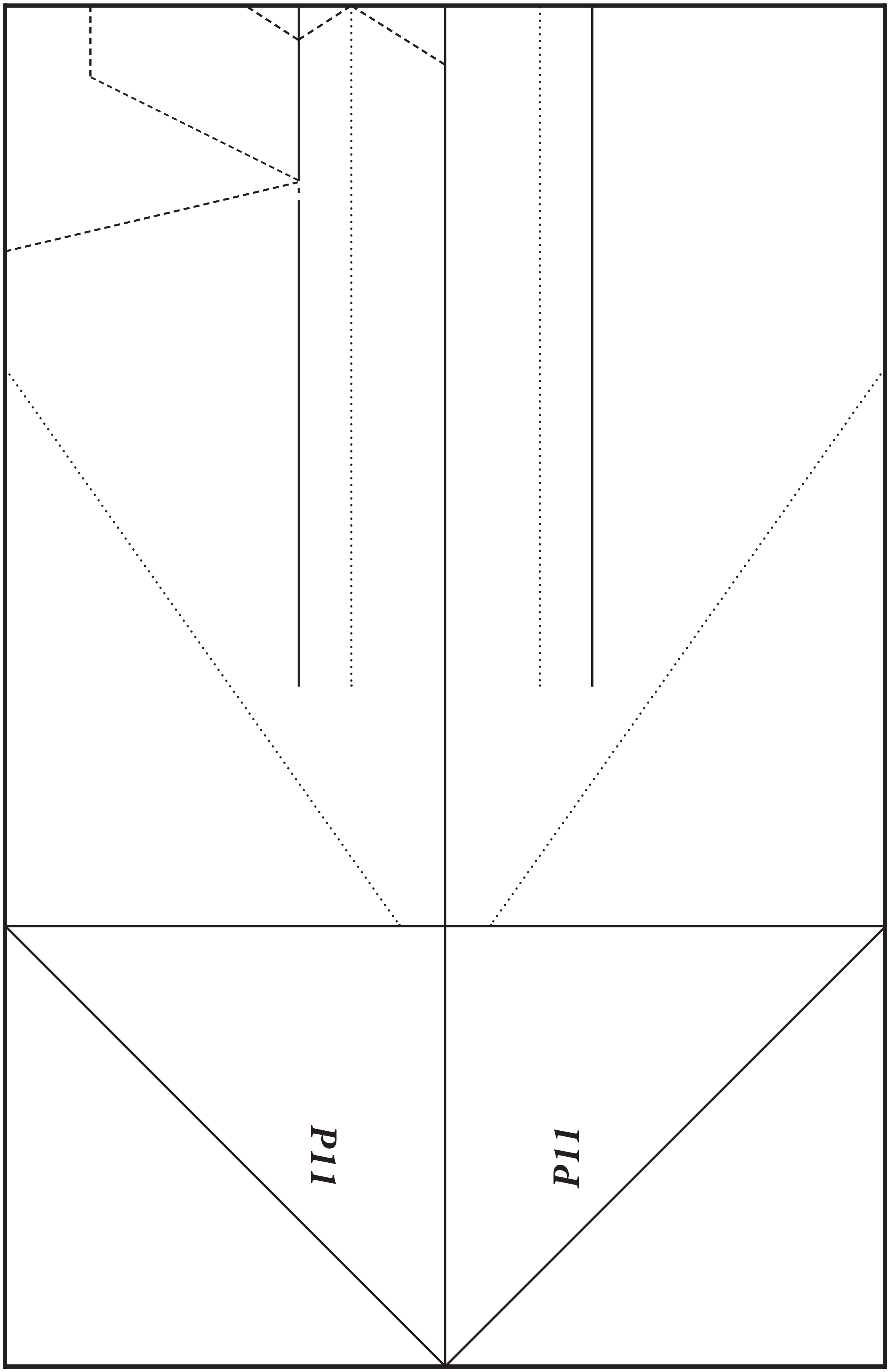
- Once you become comfortable making the planes from this template you can simply cut regular printer paper in half in order to start.
- While the printed fold lines should be very close to perfect for your folds, if you need to choose you should ***always choose symmetry*** of the plane over following the lines.

Solid Line = Mountain Fold ••••••••••••• Dotted Line = Valley Fold ----------Dashed Line = Cut



P12 RAPTOR

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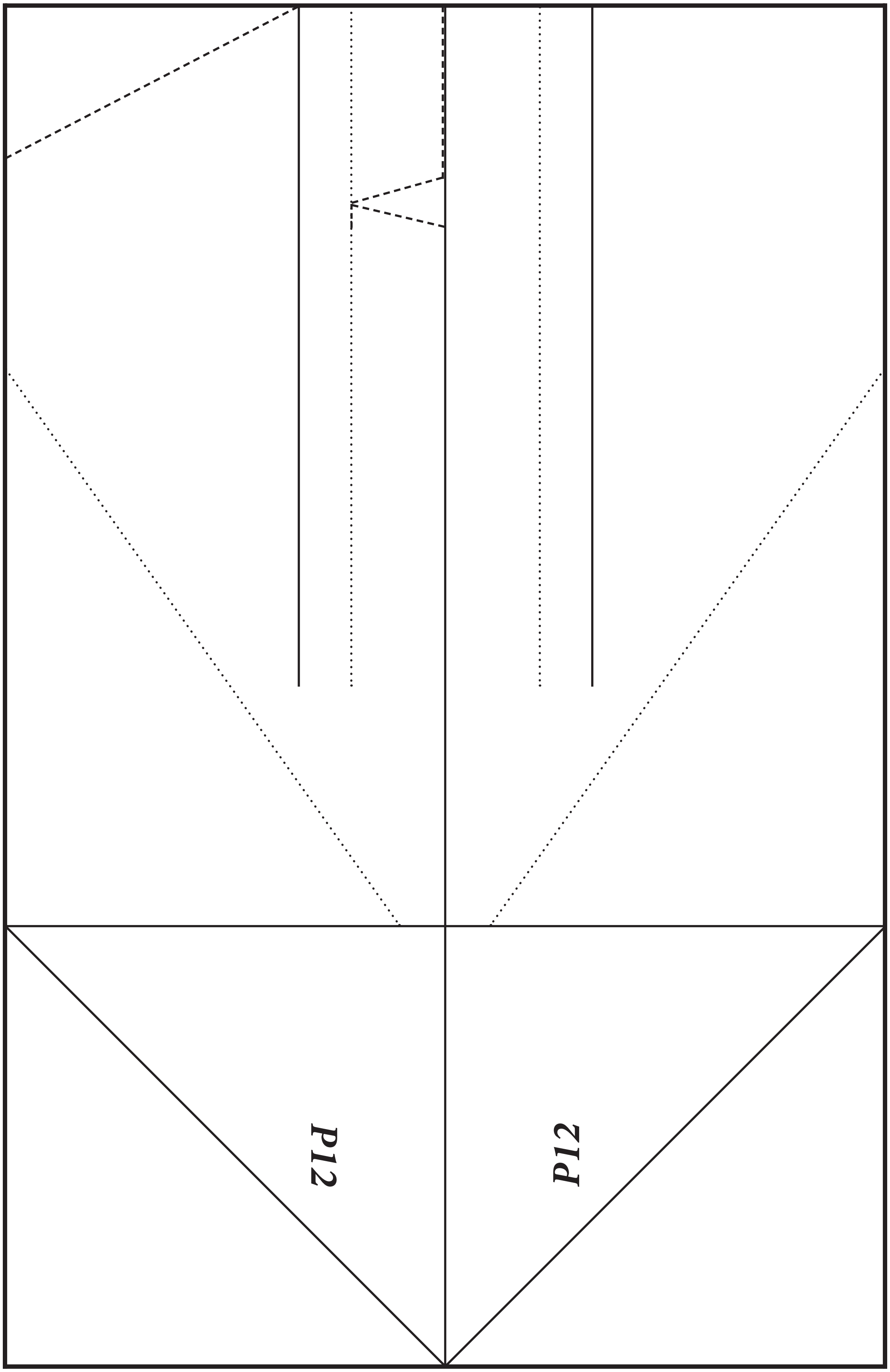

P13 SPARTAN

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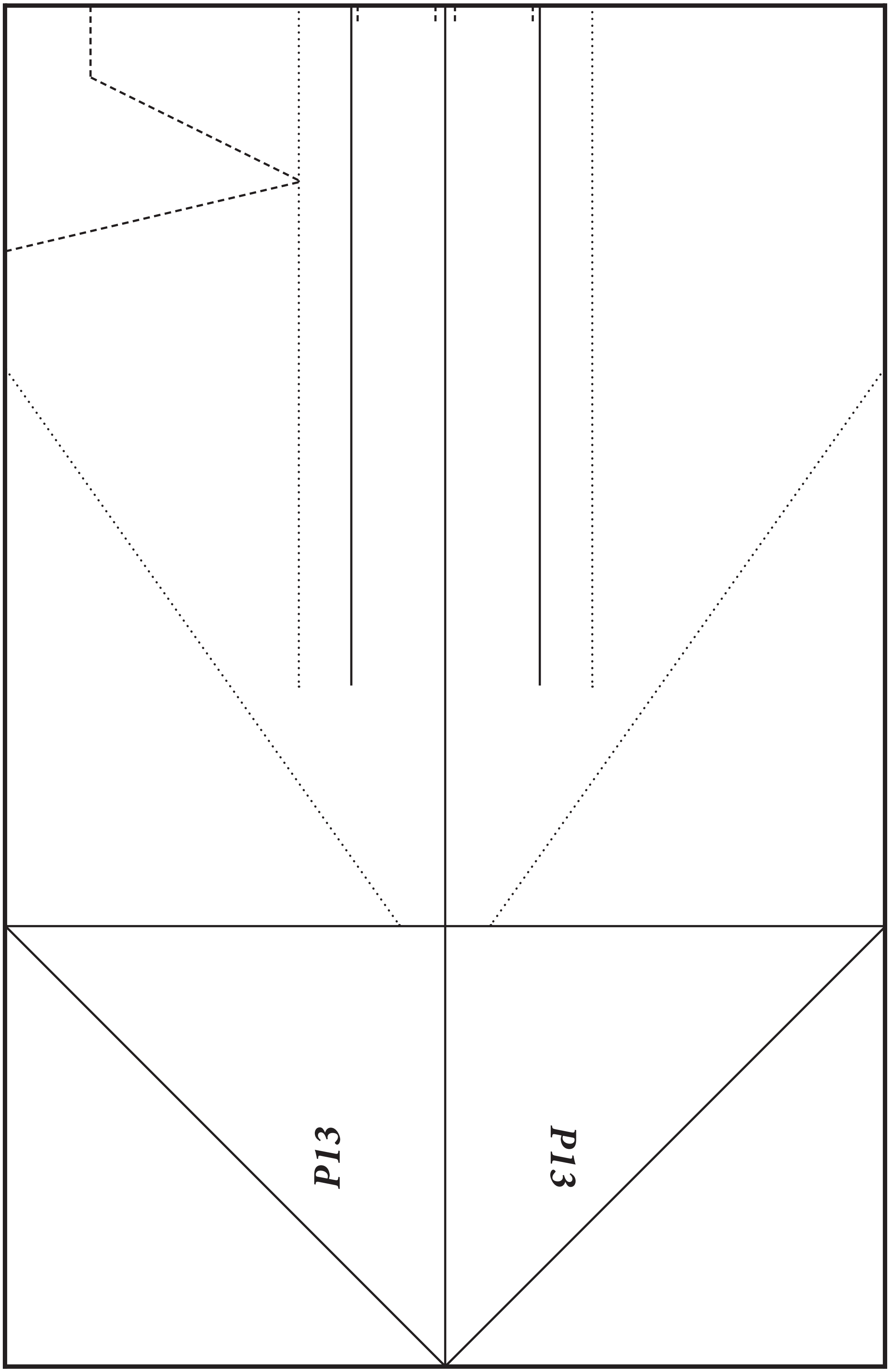

P14 SKYMASTER

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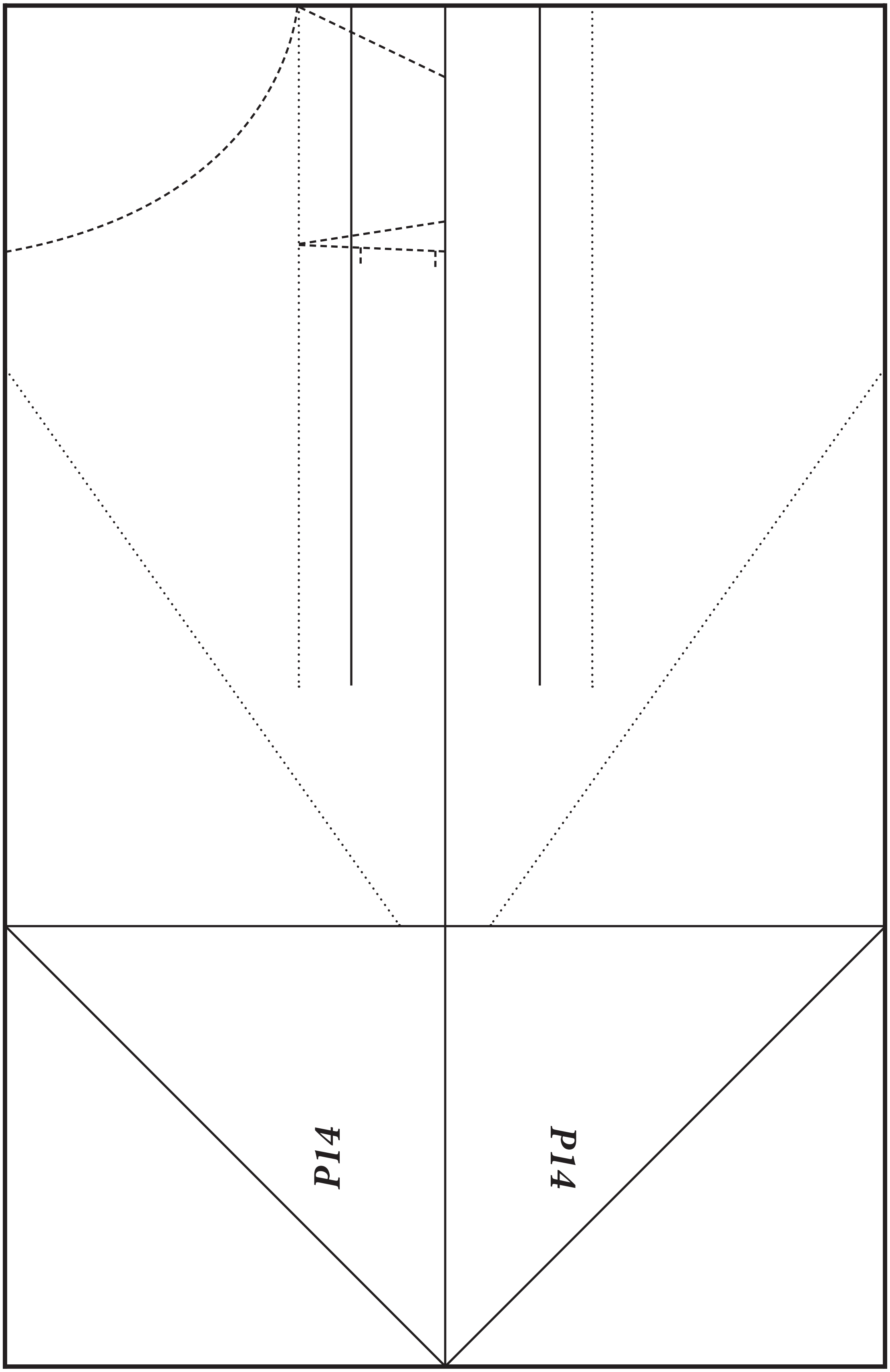

P15 MANTA

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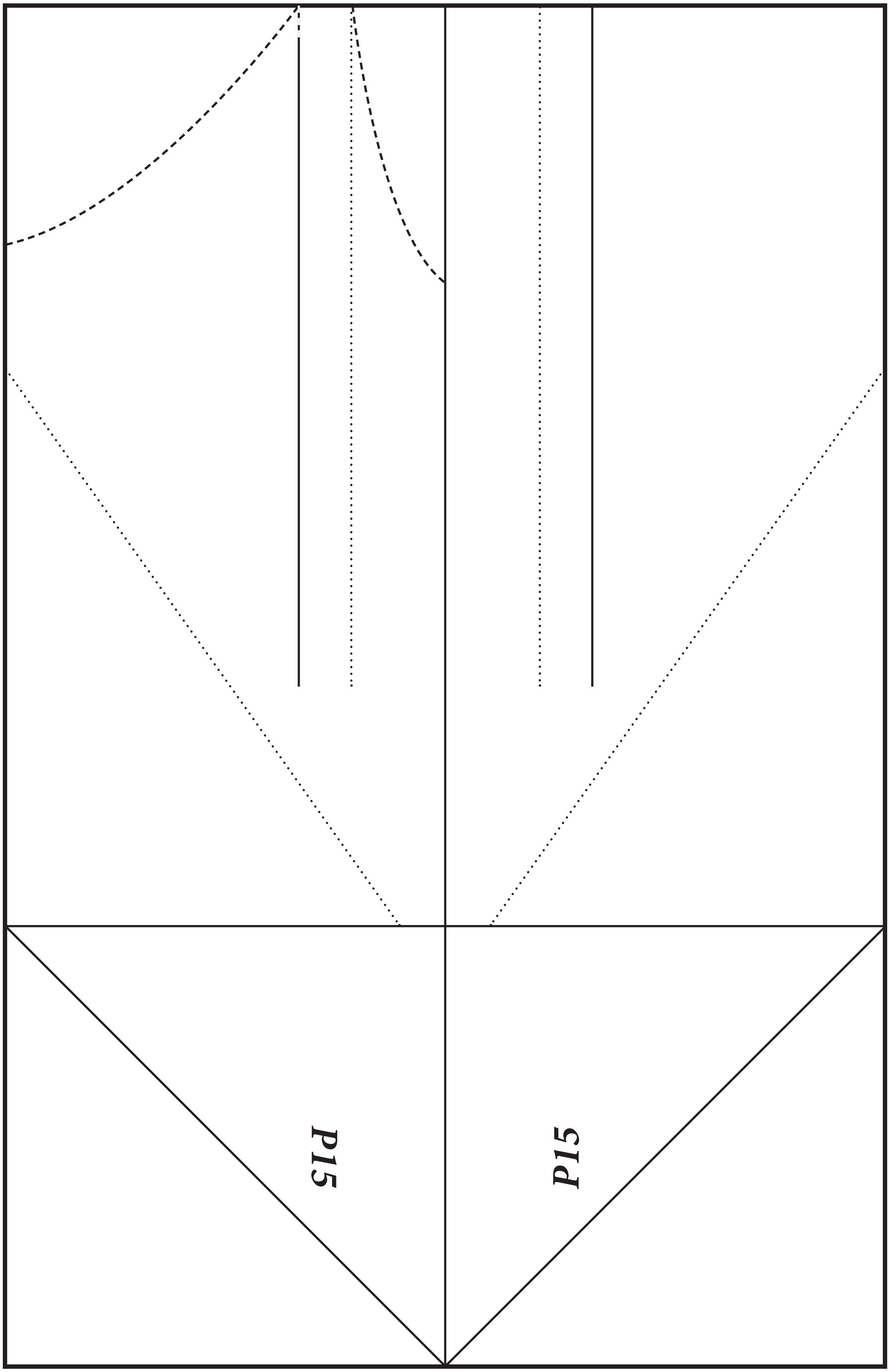
Solid Line = Mountain Fold   •••••••••••• Dotted Line = Valley Fold   ----------Dashed Line = Cut



P16 NASP

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